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MILITARY SPECIFICATION

COMPUTER SET, STANDARD AIRBORNE

AN/AYK-14(V)

GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for design, construction, performance and testing of the AN/AYK-14(V) general purpose, stored program, real-time digital computer, designated Standard Airborne Computer Set, referred to herein as the Computer and as the equipment.

1.2 Classification. The equipment described herein is not a standard computer but a group of computer configurations with common requirements (shop replaceable assemblies (SRAs)). This specification includes provisions to order the SRAs separately or combined in specific configurations in addition to the configuration listed in supplement 1 (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-B-5087	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Production and Acceptance for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment, General Requirement for
MIL-C-24308	Connectors, Electrical, Rectangular, Miniature, Polarized Shell, Rack and Panel, General Specification for
MIL-C-26482	Connector, Electrical, (Circular, Miniature, Quick Disconnect), Receptacles and Plugs, General Specifications for

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MIL-M-38510	Microcircuits, General Specifications for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removal Crimp Contacts
MIL-H-46855	Human Engineering Requirements for Military System and Facilities
MIL-C-81511	Connectors, Electrical, Circular, High Density, Quick Disconnect, Environmental Resisting
MIL-C-83723	Connector, Electrical, Receptacles and Plugs, General Specification for

See Supplement 1 for list of associated specifications.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-210	Climate Extremes for Military Equipment
MIL-STD-454	Electronic Equipments, General Requirements for
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-704	Aircraft Electrical Power Characteristics
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production

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MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1553A	Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-1562	List of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing, General Requirements for
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipments, General Requirements for
MIL-STD-2077	Test Program Sets, General Requirements for
MIL-STD-2084	Maintainability of Avionic and Electronic Subsystems and Equipment

2.1.2 Other Government documents, and publications. The following Government documents and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVSEA

0967-LP-598-2010	User's Handbook, Navy Standard 16-Bit Computers Support Software - Volume I (Level 1 and Level 2 Operating Systems)
0967-LP-598-2020	User's Handbook - Volume II (Standard Real Time Executive SDEX 20)
0967-LP-598-2030	User's Handbook - Volume III (Hardware, Assembler, Fortran)
0967-LP-598-2040	User's Handbook - Volume IV (CMS-2M)

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0967-LP-598-2050	User's Handbook - Volume V (MTASS 20/14)
0967-LP-598-2060	User's Handbook - Volume VI (MTASS/M Operational Guidelines)

Naval Avionics Center

TP-579	Interface Design Specification for AN/AYK-14(V) Support Equipment Interface
TP-591	Equipment Specification, Proteus Interface Module

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

EIA-STD-RS-232C	Interface Between Data Processing Terminal Equipment and Data Communications Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 I Street, N.W., Washington D.C. 20006.)

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.3 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Either microelectronic integrated circuits, or large scale integrated circuits are to be used for all arithmetic and logic functions.

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b. When previously produced models of this equipment did not use non-repairable subassemblies, the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity.

c. Non-repairable subassemblies shall be used in accordance with MIL-STD-2084 and MIL-E-5400.

d. All microelectronic devices and parts shall be readily available, off-the-shelf from at least two qualified sources. All second source microelectronic devices and components shall be mechanically and electrically interchangeable.

e. Circuits shall be designed to use a minimum of different materials, parts, subassemblies, cards and modules.

f. All parts and materials shall conform with applicable requirements of MIL-E-5400, or better.

g. All memory modules shall be readily available off-the-shelf from at least two qualified independent manufacturers. Memory modules shall be available for direct procurement by the Government. Interchangeability demonstration tests shall be conducted in accordance with 4.7.

3.3.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved as specified in MIL-STD-965, except that approval as a nonstandard part shall not be required for devices which are:

a. Listed in MIL-STD-1562;

b. Available from at least two sources which are listed in the appropriate QPL of MIL-M-38510; and

c. Procured to the requirements of MIL-M-38510 from a qualified source.

3.3.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965 and MIL-STD-883. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing circuits and discrete parts shall be easily removable without damage to the assembly.

3.3.3 Modules. The electronic portions of the equipment shall be functionally modularized. Detailed modularity requirements are specified in 3.6.1.1 of this specification.

3.4 Design and construction. The computer shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

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3.4.1 Total weight. The total weight of each computer configuration, excluding external cables, shall be as specified in the applicable specification sheet.

3.4.2 Dimensions. Dimensions of each computer configuration shall be as defined in the applicable specification sheet.

3.4.3 Reliability. The contractor shall conduct a reliability program in accordance with the requirements of Appendix B and utilizing MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued, unless otherwise specified in the contract (see 6.2.1).

3.4.3.1 Operational stability. The computer in any configuration shall operate with specified performance, continuously or intermittently, for a period of not less than 5,000 hours in the operational environment without the necessity for readjustment or scheduled maintenance (excluding air filter replacement).

3.4.3.2 Operating life. The computer shall have a minimum operating life of 10,000 hours with minimal servicing and replacement of parts.

3.4.3.3 Storage life. The computer shall be designed to meet the performance and service life requirements of the specification after five years storage under worldwide climate conditions specified in MIL-STD-210, when packaged in accordance with MIL-STD-794 Level A and MIL-P-116 Method 11A.

3.4.3.4 Reliability in mean-time-between-failure (MTBF). Each computer configuration, including BIT and fan, shall have a reliability (θ_0 of MIL-STD-781) as specified in its applicable specification sheet, when tested and accepted according to the requirements of 4.4.3.2 and 4.4.3.4. Reliability requirements for individual shop replaceable assemblies (SRAs) shall be as specified in 3.6.1.1.2.

3.4.3.5 Time totalizing meter. Each Weapons Replaceable Assembly (WRA) of the computer shall have a time totalizing meter which meets the requirements of MIL-M-7793. The meter shall be located such that it may be read without the removal of the cover or any part from the unit. The meter shall indicate "ON" time which is any operating mode including standby, and shall have a range from 0 to 9,999 hours, and shall not be capable of being reset.

3.4.4 Cabling and connections. The computer shall provide for the use of cables and connectors in accordance with MIL-E-5400, as specified herein. All external connectors, except fan connector, shall be scoop proof. The input power shall be brought into the computer through a separate connector.

3.4.4.1 Interconnection cabling. The equipment shall be capable of specified operation using external wiring in accordance with the applicable requirements of MIL-W-5088.

3.4.5 Interchangeability. In addition to the requirements of MIL-E-5400, all parts having the same part number shall be interchangeable with each other with respect to installation and performance. The computer itself shall be

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interchangeable with the same configuration of computer. Changes in vendor's part number shall be governed by the drawing number requirements of MIL-D-1000. Parts interchangeability shall be in accordance with MIL-I-8500.

3.4.6 Interference control. The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall meet the requirements of MIL-STD-461, with Notice 3, as modified herein (see 4.2.4) with all input/output (I/O) signal connectors capped. CE01, CE03, RE02, and RS03 Tests have been conducted with I/O cables connected. If specified in the contract, test results, including cable configurations will be provided by the procuring activity (see 6.2.1).

3.4.7 Maintainability. The contractor shall incorporate the design requirements for maintainability features, BIT features, construction, provisions for test points, and other maintainability parameters in accordance with MIL-STD-2084. A maintainability program in accordance with Appendix B and using MIL-STD-470 as a guide, shall be established.

3.4.7.1 Compatibility with automatic test equipment (ATE). The equipment shall be compatible with ATE in accordance with the requirements of MIL-STD-2076. An ATE test connector (see 3.6.2.1.1.6) which provides access to the computer, as necessary to meet the requirements of MIL-STD-2076 and MIL-STD-2084 shall be provided. When specified in the contract, ATE Test Program Sets (Shop Replaceable Assembly Isolation Diagnostics in accordance with Appendix C) shall be furnished in accordance with MIL-STD-2077. If ATE Test Program Sets exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program Sets shall be prepared as part of the equipment changes in accordance with MIL-STD-2077.

3.4.7.2 Built-in-test (BIT). Built-in-test features shall be incorporated to provide performance monitoring of the computer. BIT features shall provide automatic GO/NO-GO indication of equipment readiness via an electrical signal to a remote indicator (cockpit panel) and to the AN/ASM-607(V)7 Loader Verifier. These features shall be such as to energize a warning device (indicator) when an equipment failure occurs. An electro-mechanical failure location indicator shall be installed on the connector panel of the computer and shall meet all requirements of MIL-STD-2084. In the case when a detected fault is cleared upon computer restart (i.e., intermit-tent failure), the failure output signal to GO/NO-GO indicators shall indicate a GO status. However, the electromechanical failure location indicator on the computer shall remain set in a NO-GO state. The circuits and devices which provide BIT functions shall be designed in such a manner that failure of the BIT circuitry shall not degrade computer performance or cause damage to any associated equipment or ground support equipment or cause unintentional operation of the associated equipment which would cause a hazardous condition. BIT shall be capable of being initiated both automatically and manually.

3.4.7.3 In-Flight performance monitor. Monitoring and test routines as specified in Appendix C, operating in conjunction with BIT firmware/hardware, shall be interrupt driven or user callable and shall provide system GO/NO-GO status. These tests shall be performed without interfering with other functions. Detected failures shall cause an interrupt. The In-Flight Performance Monitor shall be demonstrated in accordance with section 40 of Appendix C.

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3.4.7.4 Diagnostics. Diagnostic software, as specified in Appendix C, shall be loaded from the Loader Verifier (L/V) specified by MIL-T-85413/7 or equivalent support equipment. The diagnostic software shall be capable of providing complete end-to-end checks and fault isolation of the computer to the faulty SRA. All diagnostic monitoring and fault indications shall be accomplished through the use of this support equipment using the protocol between the support equipment and the computer as defined in NAC TP-579. All diagnostic software shall meet the requirements of Appendix C. The diagnostic software shall be demonstrated in accordance with 4.5 and section 40 of Appendix C.

3.4.7.5 Intermediate level maintainability. The MTTR at the intermediate level maintenance shall be as specified in the applicable specification sheet and shall be demonstrated in accordance with 4.5. This time shall include:

a. Installation of unit for test;

b. Verification of fault;

c. Isolation of faulty SRA;

d. Removal/replacement of faulty SRA;

e. Verification of proper operation;

f. Removal of unit from test.

The maximum time to repair any failure (by SRA replacement) at the intermediate level shall not exceed sixty minutes.

3.4.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions, and for making laboratory bench tests:

a.	Temperature	Room ambient (25°C ± 10°C)
b.	Altitude	Normal ground (0 - 10,000 ft.)
c.	Vibration	None
d.	Humidity	Room ambient up to 90% relative humidity
e.	Input Power Voltage	115 ± 5.0 VAC 400 Hz ± 20 Hz 3-phase

3.4.9 Service conditions. The computer shall operate as specified under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2X equipment (external source conditioned cooling air supplied) or Class 1 (ambient air cooling supplied by a bolt-on fan), as required in the applicable specification sheet.

3.4.9.1 Vibration. The equipment shall operate as specified when subjected to the vibration requirements of Curve IIIb of MIL-E-5400, and the level of random vibration as specified in 4.2.5. Vibration isolators shall not be used.

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3.4.9.2 Cooling provisions. All configuration types shall use forced air cooling through plenums in the chassis structure. The computer shall be capable of operating continuously in the service environment specified in the applicable specification sheet. All SRAs shall be conduction cooled to the chassis plenums. No cooling air shall come in direct contact with any pluggable module SRA.

3.4.10 Thermal overload protection. The computer shall be capable of detecting a thermal overload condition which causes temperatures to rise above 95 ± 5°C at the power supply base plate, but does not constitute a safety of flight hazard. This condition shall not occur within the continuous operation temperature limits of MIL-E-5400 for Class 2 equipment. The computer shall take corrective action depending upon the state of the Thermal Override Input. If the Thermal Override Input is not set, automatic shutdown shall occur. If the Thermal Override Input is set, shutdown shall not occur unless the power supply base plate temperature is elevated to 115 ± 5°C causing a possible safety-of-flight hazard which shall cause automatic shutdown. When automatic shutdown occurs at the lower of these two temperatures, the computer shall be capable of resuming normal operation when temperature conditions return to a safe operating level. If the temperature is allowed to rise beyond the lower shutdown temperature, damage to components may result. If damage occurs, normal operation can not be resumed. The computer shall output discretes as specified in 3.4.19.8 and 3.4.19.9 to indicate these overtemperature conditions.

3.4.11 Warm-up time. The time required for the computer to warm up prior to operation or program execution shall be kept to a minimum and shall be not greater than 30 seconds at extreme service conditions. During this warm-up time, no damage to the equipment or alterations in any of the contents of the memory subsystem shall occur. The computer shall prevent a BIT indication or interrupt from occurring during warm-up. A fixed time-out period shall not be used to determine minimum warm-up time.

3.4.12 Input power. The computer shall meet all applicable requirements of MIL-STD-704 and shall be designed to provide specified performance from AC power sources with characteristics as specified in MIL-STD-704. Specific operating power requirements are provided in the applicable appendix. The computer shall be capable of providing filtered AC power, with characteristics as specified in MIL-STD-704, directly from the computer power input connector filter assembly to the maintenance connector for powering the Loader/Verifier as specified in 3.6.2.1.1.6 and MIL-T-85413/7. When power is applied to the computer, it shall force a Master Clear signal as specified in 3.4.19.1 to be applied to the computer. Orderly power shutdown shall be permitted, as defined by computer content protection in 3.4.12.2, for all voltage transients from 0 to 0.5 second duration as defined in MIL-STD-704, except performance shall not be affected by short pulses (75 microseconds or less) of loss of the three phase input power. Overvoltage transients, as defined in MIL-STD-704, shall not affect performance. Computer operation shall return to normal when the voltage transient expires. Loss of one or more input power phases shall cause the computer to shut down as specified in 3.4.12.2.

3.4.12.1 Undervoltage protection. The computer shall not be damaged by voltage below the minimum specified in MIL-STD-704 and shall automatically

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resume normal operation when the voltage returns within limits as specified in 3.4.12.2. Computer programs, data, and register contents shall be capable of being saved during under-voltage conditions as specified in 3.4.12.2.

3.4.12.2 Power failure protection. Power monitoring and auto restart circuits shall be provided. In configurations containing nonvolatile memory, these circuits shall provide computer content protection in the event of power failure by delaying the computer shutdown, as specified in 3.6.2.4, until the present memory cycle has been completed and the computer state has been saved under software control, in nonvolatile memory. When power returns to normal, the computer shall reset those controls required to commence orderly processing and shall be capable of automatically returning to normal operation at the restart location defined by Master Clear (see 3.4.19.1). In configurations containing volatile memory, saving of memory and register data shall not be required. The computer shall not cause spurious output signals to other interfacing equipment nor recognize any input signals during shutdown or start-up.

3.4.12.3 Circuit protection. Overload protection shall be provided in the computer. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. Fuses or circuit breakers shall not be used. Power supplies of the current and voltage limiting type shall be used. Protection of output DC voltages shall be sufficient to prevent the computer from causing faults in, or spurious outputs to, other interfacing equipment or aircraft wiring. No damage to the computer shall occur when any signal pin is shorted to ground or is open.

3.4.12.3.1 Internal protection. The computer shall contain the necessary practical internal safeguards to ensure that failure of individual segments of the equipment shall not result in overall damage to the computer or any associated equipment.

3.4.12.4 Power normal indication. The computer shall output a discrete which indicates a power normal condition.

3.4.13 Nomenclature, nameplates, and identification marking. Nomenclature, and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307 except that the computer serial number shall be contained on the removable nameplate vice a separate serial number plate.

3.4.14 Human performance/human engineering. Human performance/human engineering design criteria for the computer shall be in accordance with MIL-STD-1472. MIL-H-46855 shall be used as a guide for the contractor's program when required by the contract.

3.4.15 Hazardous conditions. Under no circumstances, shall the computer be permitted to operate to the point at which it represents a safety hazard to surrounding equipment, personnel, or aircraft.

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3.4.16 Thermal design,. Thermal design shall be as specified in MIL-E-5400. Cooling design performance limits and evaluation as specified in MIL-E-5400 shall be in accordance with MIL-T-23103.

3.4.17 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the equipment, except for those modules (memory and Input/Output) which take advantage of the AN/AYK-14 reconfigurability features.

3.4.18 Case mounts. Case mounts shall be provided such that the equipment does not sustain damage when the fasteners are subjected to torque of up to 100 inch-pounds during installation.

3.4.19 System control signals and status indicators. The following discretes shall be provided and shall be available on one of the operational connectors unless otherwise specified herein or in the supplement to this specification:

a. Master Clear Input (CPU/IOP, DIFF)

b. Master Clear Input (PWR, SWCL)

c. Reset Input

d. Thermal Override Input

e. Initiate Program Load (IPL) Input

f. Power On/Off Input

g. IPL Fail Output

h. Thermal Overtemperature Output

i. Thermal Safety Shutdown Output

j. Power Normal Output

k. BIT Error (CPU/IOP) Output

l. BIT Error (PWR) Output

m. BIT Flag Driver Output

n. Bank Interleave Force Input (BIFI) (Available only with MCM)

3.4.19.1 Master clear input (CPU/IOP, DIFF). Master Clear Input (CPU/IOP,DIFF) shall be a TTL compatible differential input which, when present, shall hold a reset on the control logic on all modules (including the CCU/LV interface of the GPM and IOP); clear all interrupts (except for DIM and DIOM external interrupts, and IPL EVENT, which shall be sampled at the end of the master clear signal); and force the GPM and IOP micromemory addresses to 0. Upon release, the CPU and IOP shall begin executing microcommands at address 0 in their respective micromemory. The microcode executed by the CPU

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or IOP starting at micromemory address 0 shall perform BIT and initialization of all internal registers for orderly start-up. This microcode shall also determine the state of the IPL input signal and determine if an external equipment is connected to the maintenance connector interface. From these definitions, initial Program counter values and RUN/STOP software conditions shall be established. If the CPU and IOP both exist in a given configuration, the microcode for each shall sequence in an orderly, coordinated fashion to perform these functions. The maximum time from release of Master Clear until computer control is passed to the software shall not exceed five milliseconds. This signal shall be input to the GPM and IOP modules via one of the operational connectors.

3.4.19.2 Master clear input (PWR, SWCL). Master Clear Input (PWR, SWCL) shall be a TTL compatible switch closure input which performs the functions specified in 3.4.19.1. This signal shall be input to the Power Converter Modules via one of the operational connectors except on the Type 4 chassis.

3.4.19.3 Reset input. The Reset Input shall be a TTL compatible differential input which shall generate an interrupt (not maskable) to the CPU/IOP to reset those controls required to commence orderly processing under the control of the operational executive program.

3.4.19.4 Thermal override input. The Thermal Override Input shall be a TTL compatible switch closure input which shall override computer shutdown due to a thermal overtemperature condition as described in 3.4.10 and 3.4.19.8.

3.4.19.5 Initiate program load (IPL) input. The IPL input shall be a TTL compatible differential input which shall initiate a program bootstrap load via logical input/output channel number 1.

3.4.19.6 Power on/off input. The Power ON/OFF Input shall be a TTL compatible switch closure input which shall cause the computer to execute a power down sequence as defined in 3.4.12 and 3.6.2.4, regardless of thermal overload or power input conditions.

3.4.19.7 IPL fail output. The IPL Fail Output shall be a TTL compatible differential output which shall indicate the inability of the processor to successfully perform the bootstrap function.

3.4.19.8 Thermal overtemperature output. The Thermal Overtemperature Output shall be a TTL compatible differential output that shall indicate an overtemperature condition which is not a safety of flight condition. This signal shall remain as long as the overtemperature condition is present and primary power is supplied to the computer, and shall be available on the maintenance and one of the operational connectors.

3.4.19.9 Thermal safety shutdown output. The Thermal Safety Shutdown Output shall be a TTL compatible differential output which shall indicate a safety of flight overtemperature condition. This signal shall remain as long as the safety of flight overtemperature condition is present and primary power is supplied to the computer, and shall be available on one of the operational connectors.

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3.4.19.10 Power normal output. The Power Normal Output shall be TTL compatible differential output which shall indicate a power normal condition.

3.4.19.11 BIT error (CPU/IOP) output. The BIT Error (CPU/IOP) Output shall be a TTL compatible differential output which shall provide a GO/NO-GO indication of equipment readiness. This indication will not be valid after a power subsystem power down sequence as specified in 3.4.12 and 3.6.2.4 has been performed. It shall be valid, however, as long as the Power Normal Output as specified in 3.4.19.10 indicates a power normal condition. When this indication is valid and set, it shall be reset via either of the master clear inputs as specified in 3.4.19.1 or 3.4.19.2. Upon power subsystem power up sequence or release of either of these master clear inputs, a NO-GO indication of maximum duration of 1 microsecond (pulse) may be generated.

3.4.19.12 BIT error (PWR) output. The BIT Error (PWR) Output shall be a TTL compatible differential output which provides the same information as the BIT Error (CPU/IOP) Output, but remains active as long as main power is supplied to the computer. This GO/NO-GO indication shall include readiness monitors within the power subsystem as well as readiness monitors within the CPU/IOP. It shall be valid as long as the Power Normal Output in accordance with 3.4.19.10 indicates a power normal condition. If a NO-GO indication results from CPU/IOP readiness monitors, it shall be reset via the Power On/Off Input as specified in 3.4.19.6 or either of the master clear inputs as specified in 3.4.19.1 or 3.4.19.2. If a NO-GO indication results from the power subsystem readiness monitors, main input power must be cycled to reset this indication. Upon power subsystem power up sequence or release of either of the master clear inputs, a NO-GO indication of maximum duration of one microsecond (pulse) may be generated.

3.4.19.13 BIT flag driver output. The BIT Flag Driver Output shall be a TTL compatible single-ended output used to drive the BIT Flag.

3.4.19.14 Bank interleave force input (BIFI). The BIFI shall be a TTL compatible input to the MCM (see 3.6.2.3.3) such that jumpering pins on a front panel connector shall force the computer into the bank interleave mode regardless of the memory configuration.

3.4.20 Electrical bonding. All external metallic parts of equipment, not intentionally above ground potential, shall be bonded to provide a continuous, uninterrupted, low impedance electrical path to equipment cases or other metal enclosures. Equipment cases or other metallic enclosures shall provide electrically clean, conductive surfaces devices or other designs that provide for easy, ready and reliable installation and low impedance bonding to the structure, and be commensurate with meeting a 2.5 milliohm direct current resistance between the equipment and its mounting structure. MIL-B-5087, which specifies requirements for radio frequency bonding and shock hazard prevention, shall be used as a design guide.

3.4.21 Exterior (touch) temperature. The exterior (touch) temperature shall be not greater than 65°C under any condition of operation specified when exposed to an ambient temperature of 25°C with the normal cooling airflow rates specified in Appendices and an inlet cooling air temperature of 25°C.

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3.4.22 Design loads. The equipment, when installed in the aircraft shall be designed for normal operating and ultimate load factors.

3.4.22.1 Normal operating load factors. The equipment, when installed in the aircraft, shall meet all performance requirements during application of normal operating load factors of any combinations up to and including the limits set forth below. No physical distortion or permanent set of the equipment is permitted.

a. Normal operating load factors:

(1) Vertical: + 11.3/-4.4 G - Forces from positive vertical load factors act down on the computer.

(2) Lateral: ± 5.4 G - Forces from positive lateral load factors act toward the starboard side of the computer.

(3) Longitudinal: ± 2.4 G - Forces from positive longitudinal load factors act aft on the computer.

3.4.22.2 Ultimate load factors. The equipment shall withstand ultimate load factors specified below with no failure of the structural supporting elements and the equipment shall remain in place. Distortion and permanent set are permitted, and equipment operation is not required during or after application of ultimate load factors.

a. Ultimate load factors:

(1) Vertical: +16.95/-6.6 G - Forces from positive vertical load factors act down on the computer.

(2) Lateral: ± 8.1 G - Forces from positive lateral load factors act toward the starboard side of the computer.

(3) Longitudinal: ± 3.6 G - Forces from positive longitudinal load factors act aft on the computer.

3.4.23 Acoustic noise susceptibility. The equipment shall be capable of meeting the acoustic noise susceptibility requirements specified in 3.4.23.1 and 3.4.23.2. All sound pressure levels are referenced to 0.0002 Nt/m^2.

3.4.23.1 Overall performance sound pressure level. The equipment shall meet all performance requirements and not incur physical damage when exposed to a wide band acoustic noise environment of an overall sound pressure level of 134 dB and relative power distribution of Figure 1 for 30 minutes.

3.4.23.2 Overall endurance sound pressure level. The equipment shall be capable of operation but not be required to meet performance requirements when exposed to the endurance sound pressure level of 140 dB for 45 minutes. The equipment shall incur no physical damage and shall meet all performance requirements upon completion of the 45-minute exposure. The relative power distribution of Figure 1 shall apply.

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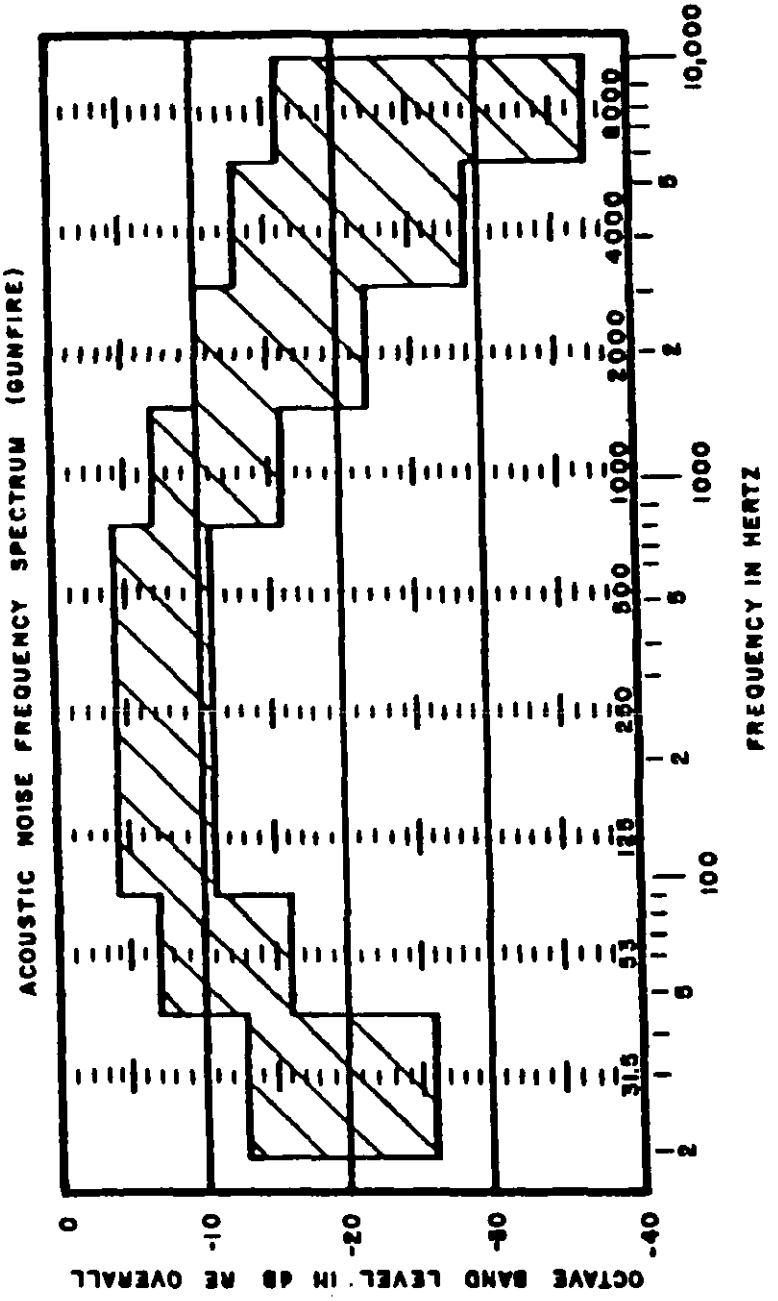


FIGURE 1. Acoustic Noise Frequency Spectrum (Gunfire).

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3.4.24 Air leakage. The computer chassis cooling air plenum system shall have a cooling air leakage less than 10 percent of the design flow rate when operating at 80°F inlet temperature, 160°F ambient, at the appropriate design flow rate, at an ambient pressure corresponding to maximum service altitude.

3.5 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and input power conditions.

3.5.1 Operation. The computer shall be capable of executing both 16- and 32-bit instructions. The CPU and IOP shall execute the instructions listed in Tables XX, XXI, XXII, and XXIII of Appendix A. Except where noted, all instructions common to the AN/UYK-20(V) Computer are defined in the User's Handbook for Navy Standard 16-bit Computers Support Software, Volume III. Modified AN/UYK-20(V) and unique AN/AYK-14(V) instructions are specified in Tables XXI and XXII, of Appendix A to this specification. The computer shall correctly execute programs developed using the support software described in the User's Handbook for Navy Standard 16-Bit Computers Support Software, Volumes I through IV (NAVSEA documents 0967-LP-598-2010, -2020, -2030, and -2040). Assembly and load media generation shall be performed by MTASS/M support software specified in 0967-LP-598-2050.

3.6 Detail requirements.

3.6.1 Computer system configurability. The computer system shall be designed for flexibility to enable construction of different system configurations without redesign or modification, except for mounting RXM adapter to the RXM module, to any of the subsystems specified below (excluding chassis design).

a. Processor subsystem (see 3.6.2.1)

b. Input/Output (I/O) subsystem (see 3.6.2.2)

c. Memory subsystem (see 3.6.2.3)

d. Power supply subsystem (see 3.6.2.4)

e. Chassis subsystem (see 3.6.2.5)

3.6.1.1 System modularity. The computer shall be designed such that all SRAs are of the Quick Replaceable Assembly type as specified by MIL-STD-2084. All subsystems of the computer shall be functionally partitioned, as specified herein. The subsystems shall be configured using the following SRAs:

a. Processor Subsystem

(1) CPU
- General Processor Module (GPM)
- Processor Support Module (PSM)

(2) I/O Processor (IOP)
(3) IOP Jumper Module
(4) Extended Arithmetic Unit (EAU)

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b. I/O Subsystem

(1) NTDS Slow (NIM-A)
(2) NTDS Fast (NIM-B)
(3) (Not Used)
(4) NTDS Expanded ANEW (NIM-C1)
(5) NTDS Serial (NIM-S)
(6) MIL-STD-1553A (SIM-A)
(7) DISCRETE (DIM)
(8) DIM Cable (part of Chassis Subsystem)
(9) PROTEUS (PIM)
(10) RS-232 (RIM)
(11) Parallel Input/Output - Serial Output Channels (PIC/POC/SOC)
(12) Special DIM (DIOM)
(13) Bus Extend (BEM)

c. Memory Subsystem

(1) Core Memory (CMM) (32K AND 64K)
(2) Semiconductor Memory (SMM) (32K)
(3) Read/Write Expansion Memory (RXM)
(4) Memory Control (MCM)
(5) RXM Adapter

d. Power Supply Subsystem

(1) Power Conversion Module-1 (PCM-1)
(2) Power Conversion Module-2 (PCM-2)

e. Chassis Subsystem

(1) Chassis
    (a) Type 1 Chassis Assembly
    (b) Type 3 Chassis Assembly
    (c) Type 4 Chassis Assembly
    (d) Type 5 Chassis Assembly
(2) Fan
(3) External Cooling Adapter Plate

All active components of the computer shall be located on pluggable module SRAs and none shall be part of the chassis. The SRAs shall be designed in a manner which permits them to be utilized as Government Furnished Equipment (GFE) to vendors for incorporation into other subsystems which require AN/AYK-14(V) modules. The Processor Subsystem SRAs shall be designed in a manner which permits them to be utilized as an "embedded processor" as Government Furnished Equipment (GFE) to vendors for incorporation into subsystems which require self-contained processing capability.

3.6.1.1.1 SRA timing. Each logic SRA shall contain its own time base. Asynchronous communication between SRAs shall be accomplished through the Systems Communication Structure.

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3.6.1.1.2 SRA power, weight, and reliability. The maximum power and weight and reliability (Q, of MIL-STD-781, when tested in accordance with 4.4.3.2) shall be as specified in Table I.

3.6.1.2 Systems communication structure. Internal data and control flow between subsystems shall be accomodated by two separate bi-directional electrically and functionally identical buses.

a. The IOBUS

b. The CPUBUS

Both buses shall communicate with subsystems as request - acknowledge interface systems. Both buses shall be independent, 24-bit, bi-directional buses to provide communication throughout the computer/control system.

3.6.1.2.1 Conflict resolution logic. Conflict resolution logic shall be contained on the PSM to permit up to three users on the IOBUS and two users on the CPUBUS. Bus operation shall be initiated by the user and shall consist of the following parallel word transfers:

a. A 24-bit control word transferred from the GPM to address a particular module and to provide control information.

b. A 16-bit output data, input data, output status, or input status word as specified by the control word.

The IOP shall have priority over the CPU when simultaneous requests to memory occur on the IOBUS. External requests (DMA) on the IOBUS to the memory via the BEM shall have equal priority with the CPU and shall be locked out by setting bit 0 of Status Register 1 as specified in 3.6.2.1.1.8.2. Under no circumstances shall a CPU bus request be locked out of memory due to any bus conflicts for more than one memory cycle.

3.6.1.2.2 Bus interface signals. The standard bus interface signals shall be as shown in the Bus Signals portion of Figure 2. The 24-bit parallel bi-directional bus shall be used to transfer the control and data or status words. The Request signal shall be generated by the user which shall indicate the presence of the control word on the bus. The Acknowledge signal shall be generated by the addressed module when the control word has been received and shall allow the user to remove the control word from the bus. The Resume signal shall be generated by the addressed module to indicate that the requested input or output data/status word has been transferred.

3.6.1.2.3 Output word format. The 24-bit output word format shall be as illustrated in the Bus Control Word portion of Figure 2. The first 16 bits shall specify a memory address or I/O function word and shall also be used for the data word transfer. Bits 16 and 17 shall specify the type of bus operation: input data, output data, input status, or output status. Bit 18 shall specify whether this bus operation is a memory word transfer or an I/O module transfer. The last five bits shall specify an I/O module channel number or the type of memory reference.

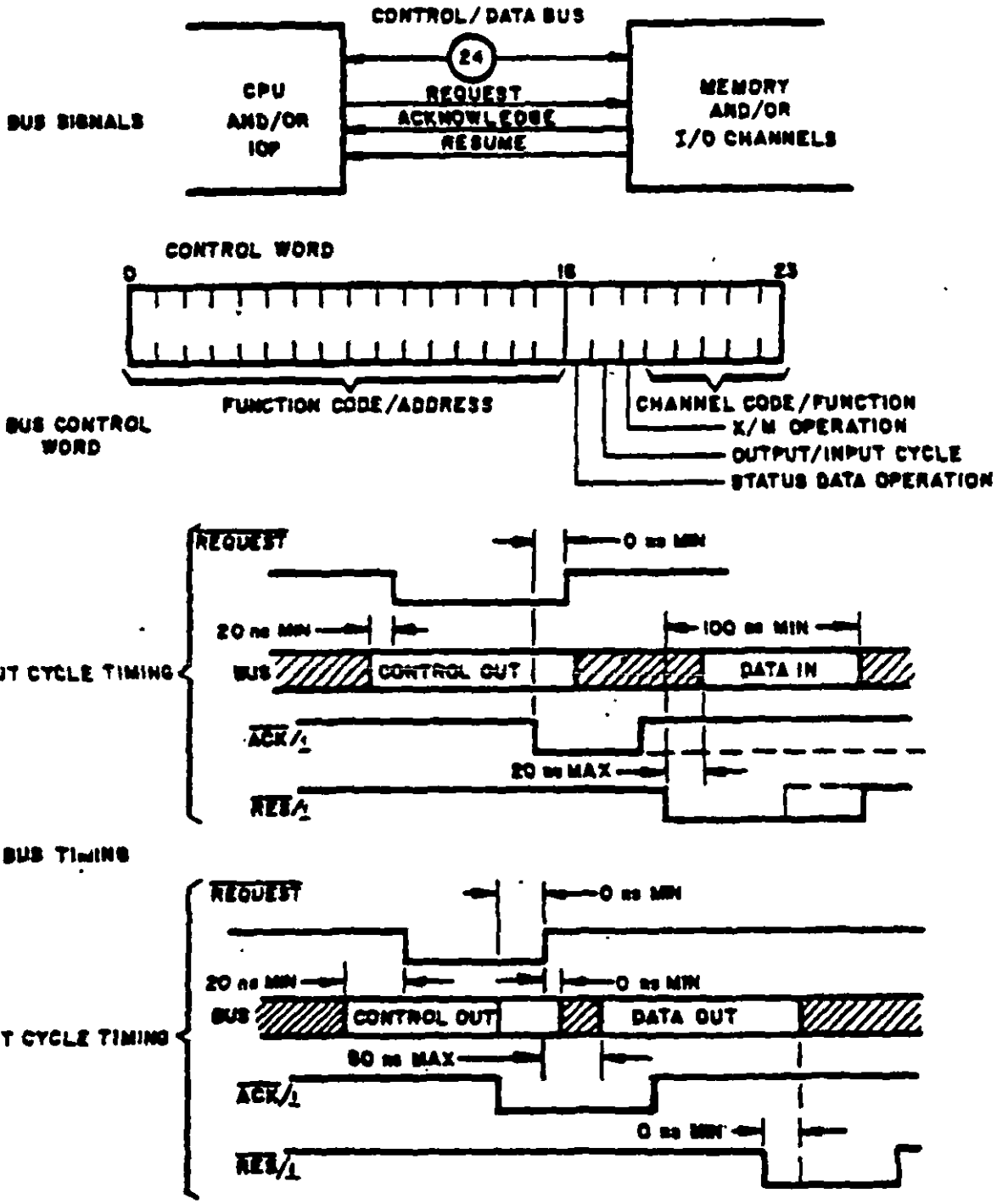
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Table I. SRA power, weight and reliability requirements.

SRA	Maximum Power (watts)	Maximum Weight (lbs)	MTBF (hrs)
(Processor Subsystem)			
GPM	47.0	2.14	9,780
PSM	42.0	2.13	9,030
IOP	44.0	2.20	10,230
IOP Jumper	NONE	0.15	1,112,000
EAU	46.0	2.10	9,530
(Input/Output Subsystem)			
NIM-A (Slow)	14.2	1.06	14,010
NIM-B (Fast)	11.8	1.06	14,620
NIM-C1 (Anew)	11.2	1.05	15,470
NIM-S (Serial)	15.0	1.06	15,470
SIM-A	18.0	1.12	5,970
DIM	15.0	1.02	24,020
RIM	9.0	1.06	23,000
PIM	14.0	1.06	18,700
PIC/POC/SOC	20.0 at 5V 23.0 at 15V	1.02	20,600
DIOM	28.5	1.90	14,360
BEM	25.0	1.05	25,230
(Memory Subsystem)			
CMM (32K)	38.6 1/	3.20	8,190
CMM (64K)	50.9 1/	4.60	4,550
SMM	20.0	3.40	3,600
RXM	12.0	1.07	10,050
RXM Adapter	NONE	0.38	N/A
MCM	36.0	1.94	13,160
(Power Supply Subsystem)			
PCM-1	(efficiency 68% min)	9.60	5,510
PCM-2	(efficiency 68% min)	12.00	5,160
(Chassis Subsystem)			
XN-1 Chassis	NONE	15.90	16,320
XN-3 Chassis	NONE	11.90	17,770
XN-4 Chassis	NONE	18.30	16,000
XN-5 Chassis	NONE	11.80	16,660
Fan Assembly	See 3.6.2.5.2	1.75	16,980

1/ Power for a single CMM Module. When more than one CMM is used, total power may be less than a direct multiple of power listed.

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1/ $\overline{ACK}$ AND $\overline{RES}$ PULSE WIDTHS ARE 80 ns MINIMUM.

FIGURE 2. CPUBUS and IOBUS.

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3.6.1.2.4 Bus timing. The bus signal timing relationships for input and output operations shall be as indicated in the Bus Timing portion of Figure 2. If no response is received by the GPM or IOP within 35 microseconds maximum after a request, a Hardware Fault interrupt shall be generated and the BIT indicator shall be set.

3.6.1.3 Event structure. Internal interrupt requests, service requests, attention requests, and other event requests between subsystems and functional sections within subsystems shall be accomodated by an event interface termed the event structure. All event generation logic shall communicate the existence of a given event via this event structure. The PSM within the CPU shall provide an event monitor function for this event structure. All event priority encoding shall be performed by this event monitor and an event vector shall be generated to the GPM for use by the microcode in vectoring to the required service routine for the given highest priority active event. The IOP shall also provide an equivalent monitor function for the event structure. The design of the structure shall allow all possible system configurations defined without redesign or modification of any logic. In the total system configuration, the event structure shall be separated into two sections. The event monitor logic on the IOP shall interface all IOP internal functions, I/O channels, RXM memory, and external BEM functions. The event monitor logic on the PSM shall interface all remaining internal CPU, memory subsystem, and unit power subsystem events. All software interrupts, not automatically trapped by the microcode (i.e., arithmetic faults), shall originate as an event and utilize the event structure and priority encoding. This event structure shall provide for eight different priority classes of events. Four of these classes shall be defined as non-I/O classes and the other four shall be I/O classes. Within each of the four non-I/O classes there shall be up to eight prioritized discrete events. Within each of the four I/O classes, events shall be prioritized into eight groups or channel pairs. And finally, within each group or channel pair there shall be eight prioritized discretes with four for each channel of the pair.

3.6.1.3.1 Event structure organization. The Event structure shall consist of the following three items:

a. Event Monitor Logic

b. Event Interface Signals

c. Event Generation Logic

The event monitor logic shall be located on the PSM (with a duplicate on the IOP) and shall operate in a periodic polling sequence manner. This polling sequence shall be independent of CPU (or IOP) firmware execution and shall utilize the event interface signals defined below to determine if any system events are active and, if they are, which is of highest priority. There shall be a total of 24 event interface signals to the Event monitor logic as shown in Figure 3. There shall be 8 signal lines (EMON) used to drive event monitor logic signals to all event generation logic and 16 more signal lines which interfacing event generation logic shall utilize to drive back to the event monitor in the CPU (or IOP). The 8 signal lines from the event monitor logic are grouped into three groups as follows:

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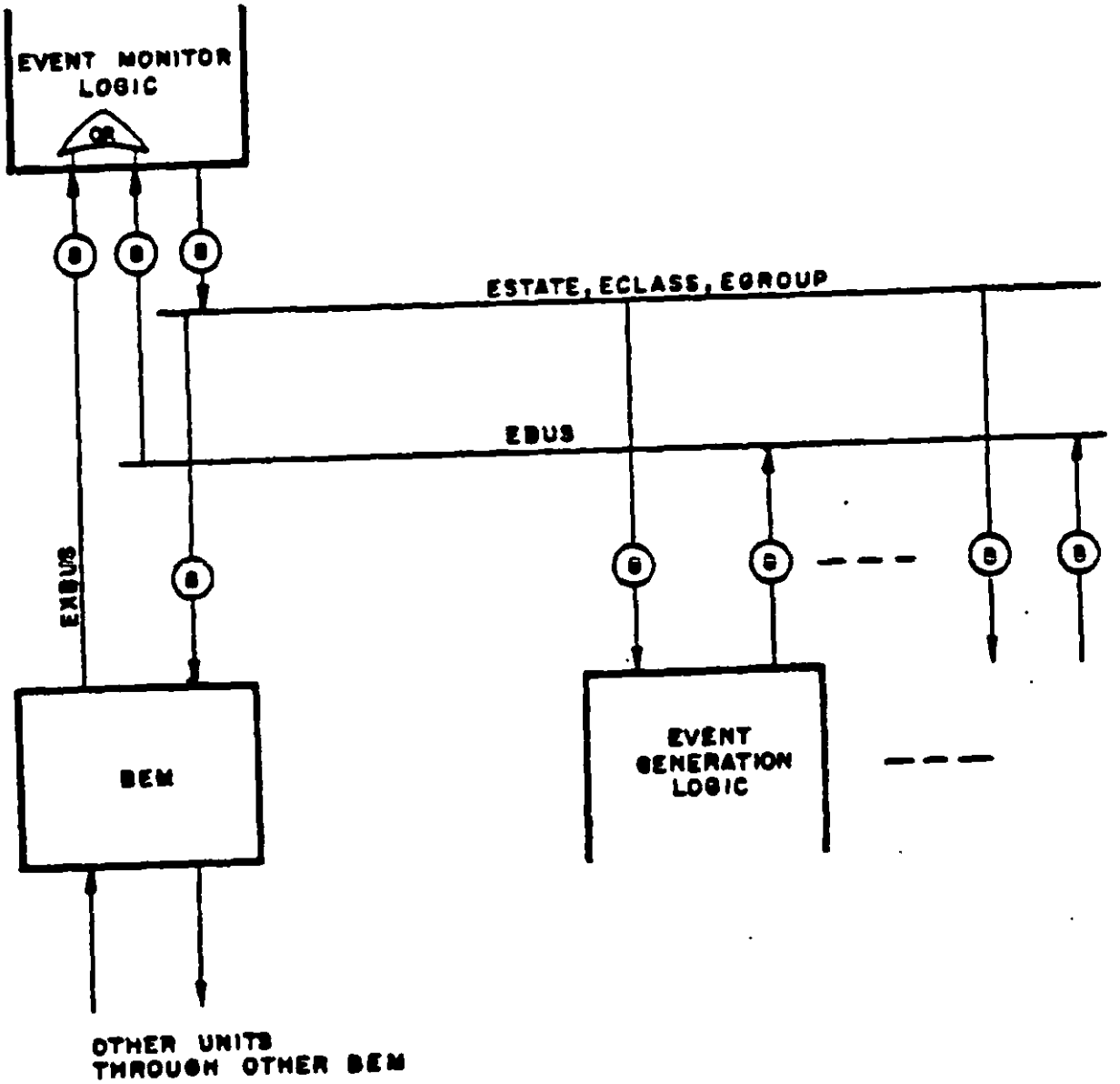


FIGURE 3. Event interface structure and lines.

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	NUMBER	FUNCTION	NAME
a.	2 signals	Polling State	ESTATE: 0,1
b.	3 signals	Polling Class	ECLASS: 0,1,2
c.	3 signals	Polling Group	EGROUP: 0,1,2

The 16 lines feeding signals back from the event generation logic to the event monitor logic shall be grouped into two sets of 8 lines, one termed the EBUS and the other termed the EXBUS. The EBUS lines and EXBUS lines shall be logically "ORed" one for one at the event monitor logic to provide a single set of 8 lines for event monitor logic interrogation. The EBUS lines shall be driven by TTL open collector circuits and shall be terminated at the event monitor logic with pull-up resistors. The EXBUS lines shall be driven by TTL totem pole circuits at the BEM and function specifically to receive external events through a BEM from other units with expanded I/O and power supplies. The EBUS and EXBUS, therefore, are functionally identical, although electrically driven differently, with the EBUS used for event generation logic within the given chassis and the EXBUS used by the BEM for event generation logic outside of the given chassis.

3.6.1.3.2 Event structure operation. The polling sequence performed by the event monitor logic shall maintain a 2-bit state code to define the intervals of the sequence. When no event within the event structure is active, the event monitor logic shall sample the EBUS lines for an event to become active. All event generation logic sections within the event structure shall respond on the EBUS to indicate any events which become active. If, at a given sample time, the event monitor detects one or more active classes of events, it shall establish a 3-bit code defining the highest priority class with an active event. If the class is an I/O class, all event generation logic on I/O channels shall decode the ECLASS lines and determine if they have one or more active events in that established class. If so, they shall decode the ECLASS lines and determine if they have one or more active events in that established class. If so, they shall report the existence of such by indicating their channel pair priority code on the appropriate EBUS bit. Any active events not in the I/O class established by the ECLASS lines shall not be reported and shall remain pending until all higher priority events have been serviced. The event monitor logic shall then sample the EBUS lines and encode a 3-bit code defining the highest priority group or channel pair with an active event in the particular established class. The event monitor logic shall then establish the group code on the EGROUP lines. All non-I/O event generation logic shall interrogate the ECLASS lines and detemine if they have one or more active events in that established class. If so, they shall report the existence of such on the appropriate EBUS bit corresponding to their particular active discrete event. If the established class is an I/O class, they shall not respond. If the established class is an I/O class, then all I/O event generation logic shall decode not only the class, but also the established group. If a given I/O event generation logic section has an event in the established class and its channel priority code matches the established group or channel pair, then it shall report all of the referenced discrete events on the EBUS lines. Each I/O channel can, therefore, generate up to 4 unique prioritized events in up to 4 unique prioritized classes, for a total of 16 uniquely prioritized events. The priority hierarchy is, therefore:

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a. Class first
b. Channel Priority Code second
c. Discrete last

The event monitor logic shall then sample the EBUS lines and encode a 3-bit code defining the highest priority discrete within the established class (and channel pair if an I/O class). The 3-bit class code, 3-bit group code and 3-bit discrete code shall be concatenated into an address and used to define, via a PROM look-up table, the required micro jump vector for firmware processing of the highest priority active event. If such an active event is defined, the "event active" condition to the firmware shall be set to indicate such. After sampling the EBUS, the event monitor shall sample the EBUS again for class information, and start the polling process over again. If the event monitor detects that, in the polling sequence, the event being polled has become inactive, then the event monitor logic shall clear the event active condition to the firmware and restart the polling sequence. In addition, if the event monitor detects that no events are active, then the event monitor logic shall clear the event active condition to the firmware, and begin high frequency sampling for class information until an event is reported again.

3.6.1.4 Interrupt recognition. The computer can be interrupted during its execution of programs. Any interrupt will place the computer in the executive mode. Some interrupts are generated by events within the CPU, some within the IOP, and some as interrupt requests by peripheral equipment via I/O channels. Computer interrupts are classified in three priority levels (Class I, Class II, and Class III). Interrupts within a class are assigned a priority rank within that class and an identifying code that is used by CPU logic to select the appropriate processing subroutine from memory. Refer to Table II for the interrupt priority classifications, priority rank, and identifying codes. Higher priority is given to the class and the interrupt within the class that has the lower number (Class I has highest priority). As each interrupt is honored, its class and all lower prioity classes can be locked out by the reloaded status register 1 until released by the processor program. Power Fault, Thermal Overload, Instruction Faults, Executive Mode Instruction Fault and System Reset cannot be locked out by the lockout designator in status register 1. The power up master clear clears the lockout bits in the status register, thus disabling most interrupts until the program is ready to handle them. The program, when ready, must execute a Load Status Reg 1 instruction to enable them. Some individual interrupts within the classes are enabled and disabled by executing particular instructions. Each I/O channel shall employ a single mask bit associated with priority 2, 3, and 4 interrupts for that given channel. Each of the possible 16 I/O channels within a given computer configuration may generate any or all of the four Class III interrupts listed in Table II. These Class III interrupts shall be prioritized first on the basis of channel physical location assignment within a given chassis and secondly as shown in Table II. Any Class III, priority 1, I/O abnormal interrupts incorporated in a given I/O channel type shall be unmaskable at the individual I/O channel level. An EIE (External Interrupt Enable), if incorporated in a given I/O channel type design, as indicated in the applicable paragraphs, allows separate enabling and disabling of the EII interrupt via the enable and disable external interrupt instructions. One

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TABLE II. CPU and IOP interrupt summary (minimum).

Software Class	Priority Within Class	Interrupt	INT/EXT	Binary Interrupt Code Generated	Paragraph Reference 3.6.1.4 and
Class I, Hardware Errors	1 3/	Power Fault	EXT	0000	3.6.2.4
	2	Memory Resume	EXT	0010	3.6.2.3.3
	3	Memory Parity	EXT	0100	3.6.2.3.3
	4 2/	Hardware Fault Warning	EXT	0110	3.6.2.1.1.13.3, 3.6.2.1.2.9
	5	I/O Failure	EXT	1010	3.6.2.2.1
	6 3/	Thermal Overload (2 Levels)	EXT	1100	3.4.10
	7	Hardware Fault (Bit)	EXT	1110	3.4.7.2, 3.6.2.1.1.13.3
Class II, Software Interrupts	1 3/	CPU Instruction Fault	INT	00000	3.6.2.2.1c
	2 3/	IO Instruction Fault	INT	00010	3.6.2.2.1d
	3	1/ Arithmetic Error (overflow/ underflow)	INT	00100	3.6.2.1.1.15.5, 3.6.2.1.1.15.6
	4	Executive Return Instruction	INT	00110	3.6.2.1.1.10
	5 3/	1/ Executive Mode Instruction Fault	INT	10000	3.6.2.1.1.10
	6	1/ Memory Protect Fault	EXT	11000	3.6.2.3.3
	7	RTC Overflow	EXT	01000	3.6.2.1.1.13.1, 3.6.2.1.2.8
	8	1/ Monitor Clock	EXT	01010	3.6.2.1.1.13.2
	9 3/	System Reset	EXT	10110	3.4.19.3
	10	External Interrupt 0	EXT	01100	Table IV Initiate External Interrupt
	11	External Interrupt 1	EXT	01110	

1/ Not in IOP.
2/ Not maskable on the fourteenth successive occurrence for the CPU and the seventh successive occurrence for the IOP.
3/ Not maskable.

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TABLE II. CPU and IOP interrupt summary (minimum). - Continued

Software Class	Priority Within Class	Interrupt	INT/EXT	Binary Interrupt Code Generated	Paragraph Reference 3.6.1.4 and
Class III I/O Interrupts	1	Intercomputer Time-Out (IO Abnormal)	EXT	110	3.6.2.2.1e
	2	External/Discrete	EXT	000	
	3	Output Chain Interrupt	EXT	100	
	4	Input Chain Interrupt	EXT	010	

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level of these Class III interrupts shall remain pending in the given I/O channel when disabled, masked off, or are otherwise pending recognition. During interrupt processing, the contents of the status registers and Program address register are stored in memory locations specified in Table III.

TABLE III. Assigned memory locations.

Function	CPU						IOP					
	I		II		III		I		II		III	
	Hex	Octal	Hex	Octal	Hex	Octal	Hex	Octal	Hex	Octal	Hex	Octal
Store P	58	130	50	120	48	110	68	150	70	160	48	110
Store SR1	59	131	51	121	49	111	69	151	71	161	49	111
Store SR2	5A	132	52	122	4A	112	6A	152	72	162	4A	112
Store RTC lower	5B	133	53	123	4B	113	6B	153	73	163	4B	113
P reload	5C	134	54	124	4C	114	6C	154	74	164	4C	114
SR1 reload	5D	135	55	125	4D	115	6D	155	75	165	4D	115
SR2 reload	5E	136	56	126	4E	116	6E	156	76	166	4E	116
Store RTC upper	5F	1137	57	127	4F	117						
I/O Command cell	$60\text{-}61_{16}$, $140\text{-}141_8$						$62\text{-}63_{16}$, $142\text{-}143_8$					
Auto start entrance	$7F_{16}$, 177_8											
External interrupt word storage	$80\text{-}8F_{16}$, $200\text{-}217_8$						$80\text{-}8F_{16}$, $200\text{-}217_8$					
NDRO	$0\text{-}3F_{16}$ and $C0\text{-}3FF_{16}$, $0\text{-}77_8$ and $300\text{-}1777_8$											

3.6.2 Subsystems.

3.6.2.1 *Processor subsystem.* The processor subsystem of the computer shall be its principal arithmetic, logic and control elements. The processor subsystem shall consist of combinations of five SRAs:

a. GPM
   CPU
b. PSM
c. IOP
d. IOP Jumper
e. EAU

The central processor shall consist of the GPM and PSM SRAs. The processor subsystem shall be functionally configurable as CPU only (with or without EAU), CPU and IOP (with or without EAU), or IOP only.

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3.6.2.1.1 CPU. The CPU shall be implemented as a 16-bit processor and shall consist of the GPM and PSM SRAs, as shown in Figure 4, and shall be microprogrammable and capable of executing the instructions specified in Table XX of this specification. The CPU shall have the following functional structure:

a. 16-bit Arithmetic Logic Unit (ALU)

b. Two sets of 16 x 16-bit General Purpose Registers (addressable)

c. Microprogram Control Unit

d. Interrupt Logic

e. Read Only Memory (ROM) Bootstrap

f. Microprogrammed 32-bit Floating-Point Arithmetic

The CPU shall be capable of interfacing with the CPUBUS, IOBUS, and Events BUS (EBUS).

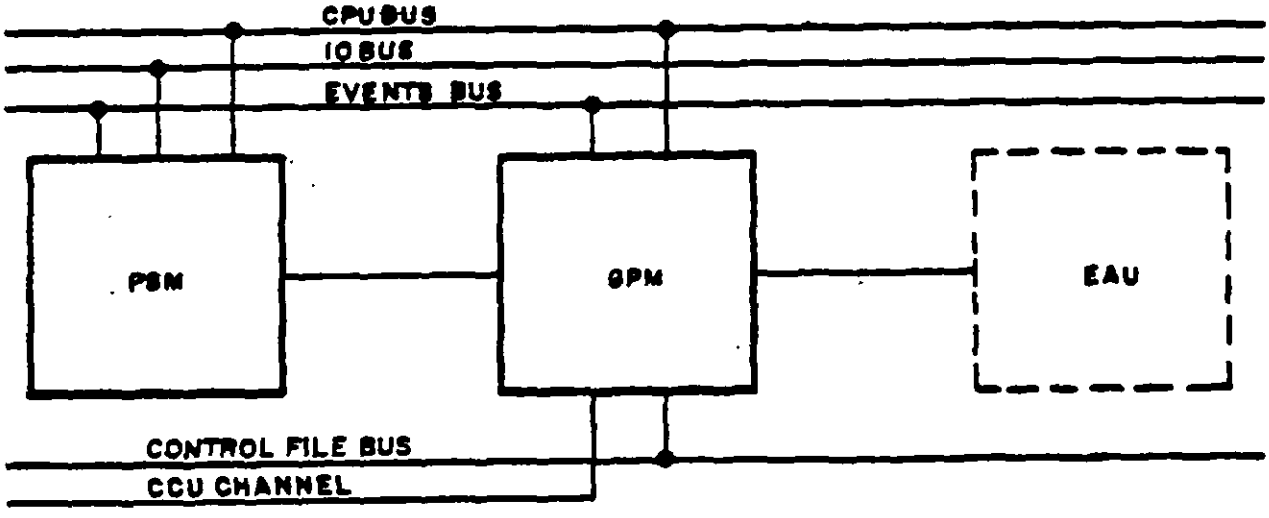


FIGURE 4. CPU.

3.6.2.1.1.1 Arthmetic logic unit. The CPU Arithmetic Logic Unit shall execute arithmetic operations on 16-bit 2's complement fixed point data.

3.6.2.1.1.2 CPU word size. The CPU shall be required to process the following data types:

a. One bit

b. 8-bit

c. 16-bit fixed point

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d. 32-bit fixed point

e. 4-bit literal

f. 32-bit floating - point

3.6.2.1.1.3 CPU performance. Fixed Point execution speeds (as a minimum) shall be as listed in Table IV when executing an instruction mix consisting of 20 percent multiplication and 80 percent load, store, and add type instructions. Floating-point execution speeds without an EAU shall be as listed in Table V when executing an instruction mix consisting of 80 percent floating-point add, 10 percent floating-point multiply, and 10 percent floating-point divide.

3.6.2.1.1.4 Microcontrol logic. The microcommands for the GPM shall be 48 bits in length. The current microcommand shall control one machine cycle and shall specify the address of the next microcommand. The current microcommand execution shall be overlapped with the next microcommand access. The micromemory address control shall supply the 12-bit micromemory address (4096 words) of the microcommand. Micromemory breakpoint capability shall be provided. Sufficient controls and micromemory address and data buses shall be available on a CPU connector to allow use of an auxiliary microprogram memory of up to 4096 words as a substitute for the internal microprogram memory.

3.6.2.1.1.5 Control file (C-File). The 256-word by 16-bit control file shall be included in the GPM architecture for use as control memory for I/O channels. Access to the C file via the CONTROL FILE BUS shall be controlled by the PSM. The PSM priority logic shall allow up to four users to access the C-file.

3.6.2.1.1.6 Support equipment interface (maintenance connector). An interface shall be provided which will provide the necessary signals and 115 volts ac, 3-phase, 400 Hertz power to the Loader/Verifier (L/V) MIL-T-85413/7 via the maintenance connector to load software programs, including the diagnostics programs of 3.4.7.1, into the computer. This interface shall also provide for controlling the computer and passing fault isolation information to the support equipment from the diagnostics programs as required in NAC TP-579. This interface shall provide a thermal overload signal (see 3.4.10) to the support equipment (SE). The interface signals shall be isolated and protected against internal part failures due to external shorts. When an SE is connected to the equipment, the GPM shall receive an SE connected input signal. The SE shall communicate with GPM by transmitting control messages, some of which generate events. The GPM shall have two SE event code outputs (SE event and SE E-code) which indicate to the CPU subsystem event monitor that an SE event has occurred. Communication between the GPM and the SE shall be with a 150K baud serial interface channel. The GPM shall be capable of receiving 8-bit control/data words from the SE over the serial interface channel. The control word format and functions shall be as described in Figure 5. The GPM shall transmit a differential SE output signal when doing an output operation, and shall receive a differential SE input signal during input operations. It shall be possible via the Diagnostic Jump instruction for the software to determine if an SE is connected. Software communicates

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with the SE via the SCI and SCIO instruction (see Tables XX and XXIII). There is no interrupt to notify the computer of SE completion of the SCI and SCIO instructions.

TABLE IV. Minimum fixed point execution speeds for AN/AYK-14(V).

Instruction Type	Data Length	Memory Technology	KOPS 1/ (Interleaved Memory)	KOPS 1/ (Non-InterTeaved Memory)
16 bit				
reg-reg	16 bits	core/semi	650/675	630/675
reg-memory	16 bits	core/semi	520/575	405/560
reg-reg	32 bits 2/	N/A	395	395
reg-memory	32 bits 2/	core/semi	295/300	250/295
32 bit (reg-memory)	16 bits	core/semi	390/430	300/380
32 bit (reg-memory)	32 bits 2/	core/semi	265/290	210/280

1/ Memory not extended via the BEM

2/ A 32x32 bit multiply shall yield a 64 bit product.

TABLE V. Minimum floating point execution speeds for AN/AYK-14(V) (without EAU).

Instruction Type	Residue	Memory Technology	KOPS 1/ (Interleaved Memory)	KOPS 1/ (Non-Interleaved Memory)
16 bit				
reg-reg	Not selected	core/semi	39.8/39.8	39.8/39.8
reg-memory	Not selected	core/semi	38.0/38.0	37.0/38.0
reg-reg	Selected	core/semi	39.0/39.0	39.0/39.0
reg-memory	Selected	core/semi	37.4/37.5	36.4/37.3
32 bit				
reg-memory 2/	Not Selected	core/semi	34.7/35.2	33.6/34.9
reg-memory 2/	Selected	core/semi	34.0/34.5	33.0/34.3

1/ Operands normalized, interrupts disabled.

2/ Assuming one level of indirect addressing with indexing.

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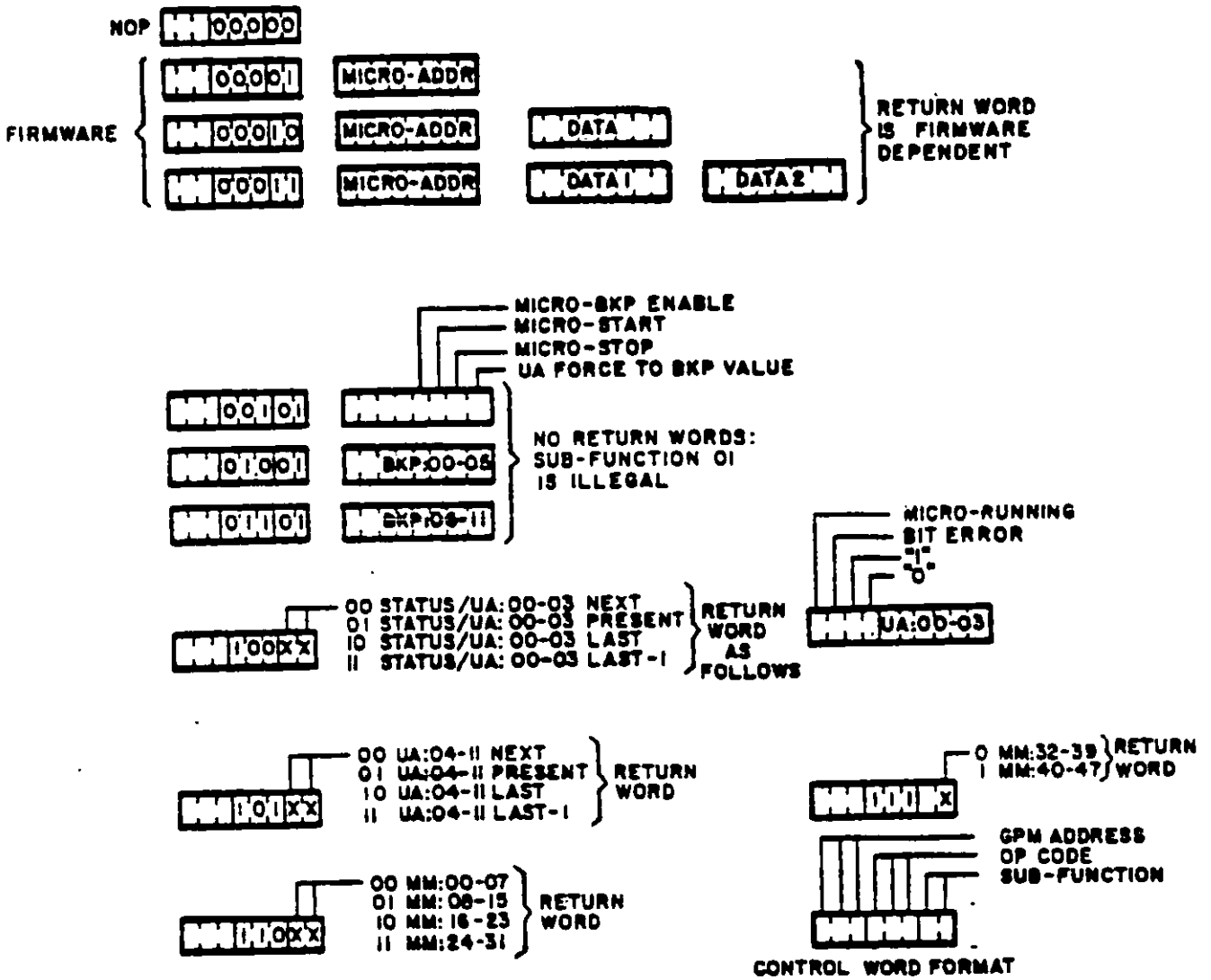


FIGURE 5. CCU control word format.

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3.6.2.1.1.7 EAU interface. The CPU shall be capable of interfacing with an EAU to perform high-speed floating point arithmetic. The CPU shall transfer instructions and operands to the EAU. The CPU shall output control signals to the EAU indicating whether information on the bus is data or an instruction. Provisions shall be made to prevent hangup of the CPU/EAU interface.

3.6.2.1.1.8 Addressable registers. The CPU shall contain the following addressable registers:

a. General Purpose

b. Status

c. Program Address

d. Breakpoint

e. C-File

3.6.2.1.1.8.1 General purpose registers. The CPU shall provide two sets of 16 general purpose, nondedicated registers 16 bits each in length. Each register shall have all capabilities provided to any register with respect to arithmetic, logical, shift, or addressing operations. With respect to microprogramming, the only functional distinction between registers shall be their address designation. For operations requiring 32-bit registers, register pairs shall be utilized as required. A designated bit in status register 1 shall indicate which set of 16 registers is in use at any given time.

3.6.2.1.1.8.2 Status registers. The CPU shall contain two Status Registers with formats as specified in Figure 6.

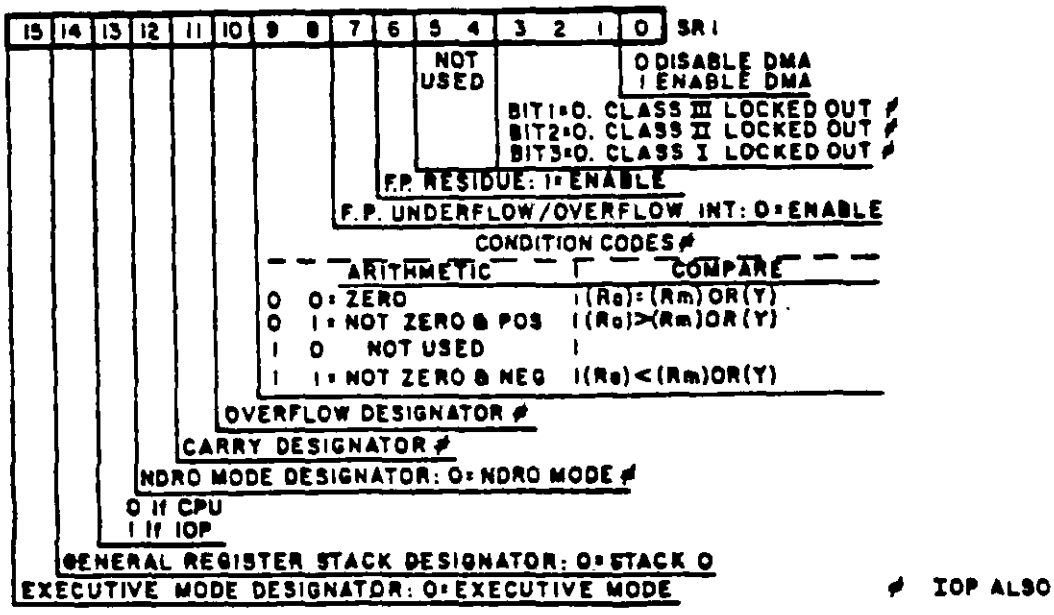
3.6.2.1.1.8.3 Program address register. The program address (P) register shall hold the address of the next instruction to be executed in a program sequence. Execution of a 16 bit instruction shall increment the contents by one, while execution of a 32-bit instruction shall increment the contents by two. Jump instructions shall load the register with the address of the program instruction that is to receive control.

3.6.2.1.1.8.4 Breakpoint registers. Two breakpoint registers shall be provided. One breakpoint register shall be used to halt CPU instruction execution when the contents of the register matches the contents of the P register. The second breakpoint register shall be used to halt the CPU when the contents of the register matches the contents of the location addressed by either of the operands of the instruction being executed. The breakpoint registers shall be loaded and set manually via the Support Equipment Interface of 3.6.2.1.1.6. When the P Breakpoint is enabled, the computer shall stop operation after completion of the previous instruction and before execution of the instruction located at the breakpoint address. When the Operand Breakpoint is enabled, the computer shall stop after completion of the instruction which referenced the operand located at the breakpoint address.

3.6.2.1.1.8.5 C-File. The CPU shall contain a C-File as specified in 3.6.2.1.1.5.

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STATUS REGISTER NO.1 FORMAT



STATUS REGISTER NO. 2 FORMAT

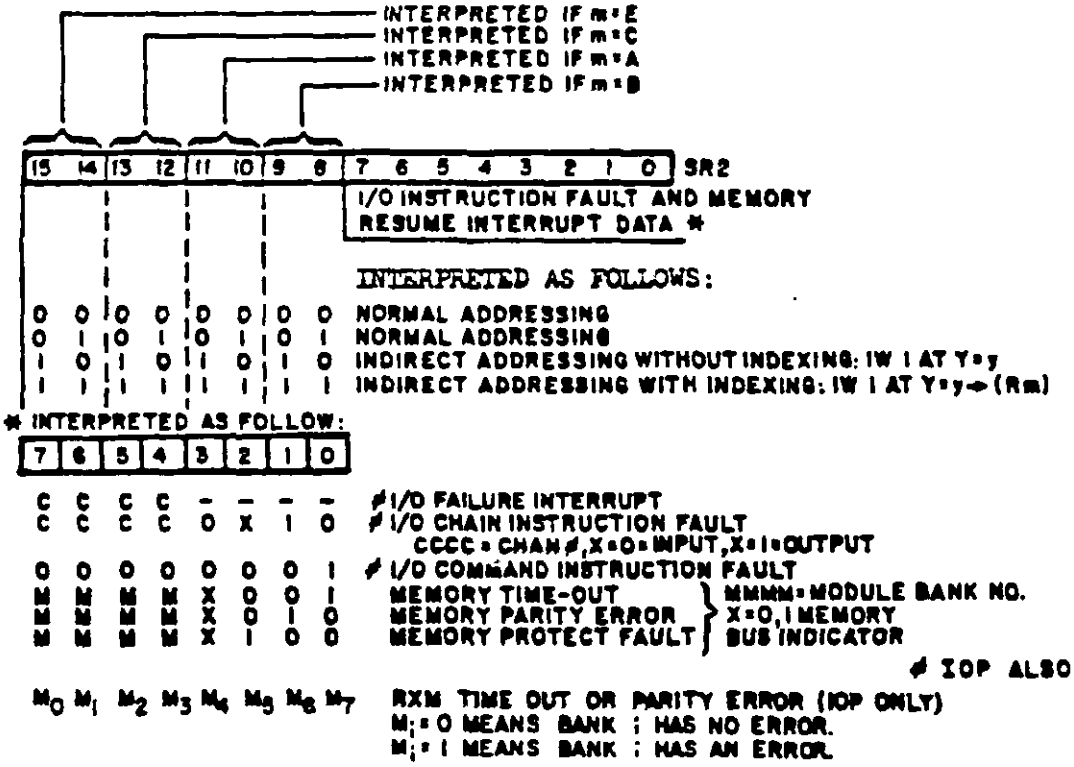


FIGURE 6. Status register formats.

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3.6.2.1.1.9 CPU addressing modes. The CPU shall provide the addressing modes defined in Table XX. Using the page registers (see 3.6.2.3.3) the CPU shall be capable of addressing a maximum memory of 524,388 words of 18 bits. The CPU shall have the capability to address consecutive 18-bit word locations in memory.

3.6.2.1.1.10 Execution modes. The CPU shall have two executions modes: executive mode and task mode. All instructions may be executed in the executive mode. Any attempt to execute Executive mode instructions in the task mode shall result in an Executive mode Instruction Fault interrupt. An Executive Return instruction shall be provided which shall generate an Executive Return Instruction interrupt and place the computer in the executive mode. Executive mode instructions are defined in Table XX.

3.6.2.1.1.11 Micromemory. All CPU instructions shall be implemented with microprogram memory located on the PSM SRA. The micromemory shall contain 4096 words by 48 bits. The storage medium shall be Read Only Memory (ROM) or Programmable Read Only Memory (PROM). Twelve address lines are presented to the PSM. The address corresponding to these 12 lines shall be read from memory and placed on the output lines. No other control shall be required. In addition, sufficient controls and micromemory address and data buses shall be made available on the CPU SRA(s) connector(s) to allow the CPU to use an auxiliary microprogram memory (ROM simulator) of up to 4096 words as a substitute for the internal microprogram memory.

3.6.2.1.1.11.1 Micromemory design option. As an option, provisions shall be made to allow either the upper 1024 or lower 1024 words of micromemory to be supplied from the GPM.

3.6.2.1.1.12 Input/Output channel bootstraps. Program load bootstrap shall be provided to allow bootstrap loading of the computer via an input/output channel from an external bulk storage device. The PSM shall provide 1024 words by 16 bits of non-volatile PROM or ROM for this purpose. Access to this memory shall be via the CPUBUS or IOBUS. When the appropriate function is received, the PSM shall reference the memory at the address specified by bits 06 through 15 of the control word. The resulting 16-bit data word shall be placed on the bus (CPUBUS or IOBUS) from which the request was received. The Bootstrap memory shall provide, as a minimum, an initialization routine, an interrupt handler, and a system status check. The program load bootstraps shall be capable of loading an absolute load file generated in the following format:

a. The Header shall consist of four control words; (1) the definition of the transfer mode and number of transfers, (2) initial load address, (3) program entry address and (4) physical page number and control bit indicating a record to follow.

b. The set of 16 bit data words.

c. The Trailer shall consist of a page control word followed by a check sum.

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Bootstrap loading, via the input/output channel, shall be initiated by setting the Initiate Program Load discrete specified in 3.4.19.5. In the event that the computer is unable to successfully complete the bootstrap load function, the IPL fail discrete, (see 3.4.19.7), shall occur. The input/output channel shall always be assigned to computer logical channel number one. The PROM shall be addressable in the NDRO mode as addresses 0-$03F_{16}$ and $00C0_{16}$ - $03FF_{16}$. Bootstrap memory shall contain the following: contents of (0000) equals $857F_{16}$ and contents of (0001) equals 8100_{16}. The computer shall be capable of bootstrapping from the following channel type: MIL-STD-1553A (SIM-A) - External device must be assigned remote terminal address 31.

3.6.2.1.1.13 Clocks. The CPU shall contain four counting registers for the purpose of time control. Two of these registers, each 16 bits in length, shall form the real-time clock (RTC), one 16-bit register shall be the monitor clock (MON), and one 4-bit register shall be the BIT timer.

3.6.2.1.1.13.1 Real time clock. The 32-bit RTC shall be an incrementing clock running at a 1-MHz rate. When its lower half (RTCL) overflows an RTC overflow interrupt shall be generated. The RTC shall be capable of being loaded, read, enabled or disabled under program control. The RTC shall not run when the software is stopped via the support equipment interface.

3.6.2.1.1.13.2 Monitor clock. The MON shall be a decrementing clock running at a 10 KHz rate. The clock shall be capable of being set, read, enabled or disabled under program control. The starting count may be any value up to its maximum capacity of $2^{16} - 1$. When enabled it shall count down and when zero is reached, the MON clock interrupt shall be generated. MON shall not run when the software is stopped via the support equipment interface.

3.6.2.1.1.13.3 Bit timer. The four-bit wide BIT timer shall be incremented one count every 2.097 seconds. Master Clear and the Reset Bit Timer instruction shall set the BIT timer to zero. Each count shall generate a Hardware Fault Warning interrupt which may be locked out by software for 13 occurrences. The fourteenth shall not be locked out and shall provide hardware fault warning to the software. When the BIT timer reaches its maximum count, i.e. 15, without being reset, a hardware fault shall be presumed. A Hardware Fault interrupt shall be generated to both CPU and IOP and the external bit indicator shall be set.

3.6.2.1.1.14 Fast multiplier. A fast multiplier shall be provided on the PSM which shall multiply a 32-bit multiplier and a 32-bit multiplicand and shall result in a 64-bit product. The multiplicand and multiplier are supplied to the PSM via the CPUBUS or IOBUS. The 64-bit product shall be broken into four 16-bit words and stored until requested by the CPUBUS or IOBUS.

3.6.2.1.1.15 Floating - point arithmetic. The processor shall perform floating-point addition, subtraction, multiplication and division. All floating-point operations shall be capable of being performed "with residue" and shall use a floating-point two-word format.

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3.6.2.1.1.15.1 Floating-point format. Floating-point numbers shall be represented in scientific notation. The hexadecimal number $\pm f_1, f_2, f_3, f_4, f_5, f_6 \times 16^e$ shall be represented by the two-word format illustrated in Figure 7. Word 1 must be contained in an even-numbered general register of a two-register pair or at an even-numbered memory address Y. Word 2 is contained in the odd-numbered general register of the two-register pair or at the odd-numbered memory address, Y+1.

	15	14 8	7 0
WORD 1	S	Characteristic	Mantissa
WORD 2		Mantissa	

where:

Word 1: S - is the sign of the number (1 for minus, 0 for plus)

Characteristic - formed by adding 64 (40_{16}) to the exponent e

Mantissa - consists of the two most significant fractional digits f_1, f_2

Word 2: Mantissa - consists of the fractional digits f_3, f_4, f_5, f_6

FIGURE 7. Two word format.

3.6.2.1.1.15.2 Data definition. The following definitions shall be utilized in the floating-point algorithms:

a. Floating-point zero - any data value whose mantissa is zero.

b. Machine zero - any data value whose characteristic and mantissa are both zero.

c. Machine infinity - any data value whose characteristic and mantissa are both all ones.

d. Machine zero and machine infinity shall both be considered valid normalized floating-point numbers.

3.6.2.1.1.15.3 Normalization definition. A floating-point number is defined to be normalized when the most significant hexadecimal digit is non-zero. Normalization consists of shifting the fraction left in hexadecimal fashion, four bits at a time, until the high-order hexadecimal digit is non-zero, and reducing the characteristic by one for each 4-bit shift. Input floating-point numbers shall not have to be normalized; however, the results of all floating-point operations shall be normalized (excluding residue) in all cases.

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3.6.2.1.1.15.4 Floating-point residue. The floating-point residue is defined as follows:

a. As a result of the floating-point add, subtract, and multiply operations, the residue represents the significant bits of the result that cannot be contained within the 24-bit (6 hexadecimal-digit) fraction result. The residue has the same sign as the result for multiply.

b. As a result of a floating-point divide operation, the residue is formed from the remainder and has the same sign as the dividend.

Floating-point residue shall be generated when bit 6 of SR1 is one and shall not be generated when bit 6 of SR1 is zero. The result shall be truncated in either case.

3.6.2.1.1.15.5 Normalization and residue generation. The following characteristics of these instruction algorithms shall be implemented:

a. A normalized result shall always be generated and placed into the specified result registers.

b. If the correct normalized result is too small to represent due to characteristic underflow, then a machine zero with zero sign bit shall be generated and underflow indicated.

c. If the correct normalized result is too large to represent due to characteristic overflow, then a machine infinity with appropriate sign bit shall be generated and overflow indicated.

d. If specified by SR1 bit 06, a residue shall always be generated and placed into the specified residue registers. No residue normalization shall be required.

e. If the residue is floating-point zero or has a characteristic too small to represent, or if result underflow is indicated, then a residue of machine zero with zero sign bit shall be generated.

f. If result overflow is indicated, then a residue of machine infinity with appropriate sign bit shall be generated.

These characteristics shall be maintained, regardless of normalized or unnormalized source operands. The implemented algorithms shall not cause overflow or underflow indications when it is possible to represent the correct normalized result (i.e., no early termination shall occur due to algorithm implementation which produces overflow/underflow of intermediate calculations if the final result is representable as a correct normalized number).

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3.6.2.1.1.15.6 Undefined operations. The indicated results shall be generated for the following special cases:

a. Floating-point zero divided by floating-point zero shall produce a result of machine zero.

b. Machine infinity divided by machine infinity shall produce a result of one.

c. Floating-point zero multiplied by machine infinity shall produce a result of machine zero.

d. Any value added to or subtracted from machine infinity shall produce a result infinity.

All of the above operations shall cause an overflow indication to be given.

3.6.2.1.1.15.7 Status register bit setting. The floating-point operations shall modify the condition code in status register 1 to reflect positive, negative, zero, non-zero, and overflow results. Upon a floating-point overflow, bit 10 in status register 1 shall be set and bit 11 in status register 1 shall be cleared. Upon a floating-point underflow, both bits 10 and 11 in status register 1 shall be set. Then bit 7 of status register 1 shall be tested. If bit 7 is a 1, no interrupt shall occur; if bit 7 is a 0, the arithmetic fault interrupt shall be generated, contingent upon bit 2 of status register 1.

3.6.2.1.2 IOP. The IOP is a single SRA which shall employ a microprogrammable processor, and shall monitor, supervise and manipulate data entering and exiting the computer. It may perform such functions as data formatting, data assignment, peripheral handling, data packing, data checking, data annotation, data queuing, and external interrupt management. The IOP shall be designed for the following modes of operation: stand-alone mode and combined CPU/IOP modes. Chassis constraints are specified in 3.6.2.5.

a. IOP stand-alone mode. In the stand-alone configuration (Figure 8), the IOP module shall be capable of being configured with up to 16 I/O modules, 32K of main memory plus 4K RXM, or 64K of main memory. The stand-alone IOP shall execute software programs stored in memory and perform I/O channel programs and data transfer functions associated with the I/O channels (see note B of Figure 48). The IOP shall recognize the IPL signal and function as specified in 3.6.2.1.2.7 for bootloading.

b. CPU/IOP Modes.

1) I/O Controller Mode - this mode (Figure 9) requires a configuration which includes a CPU. In this mode, the CPU shall execute all software programs, including initiation of IOCR instructions. The IOP shall perform only I/O channel instructions and associated I/O data transfer activity. Note that in this configration two

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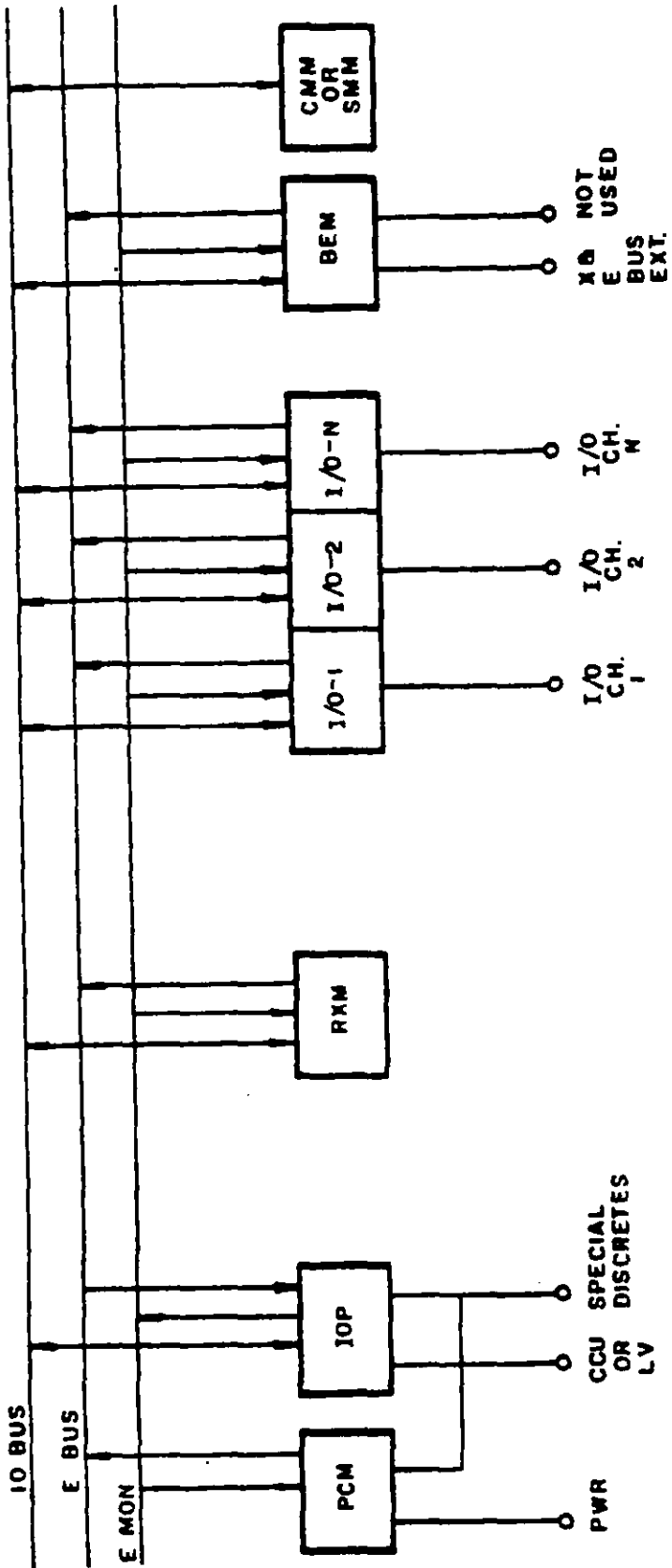


FIGURE 8. Stand-alone AN/AYK-14(V) IOP configuration.

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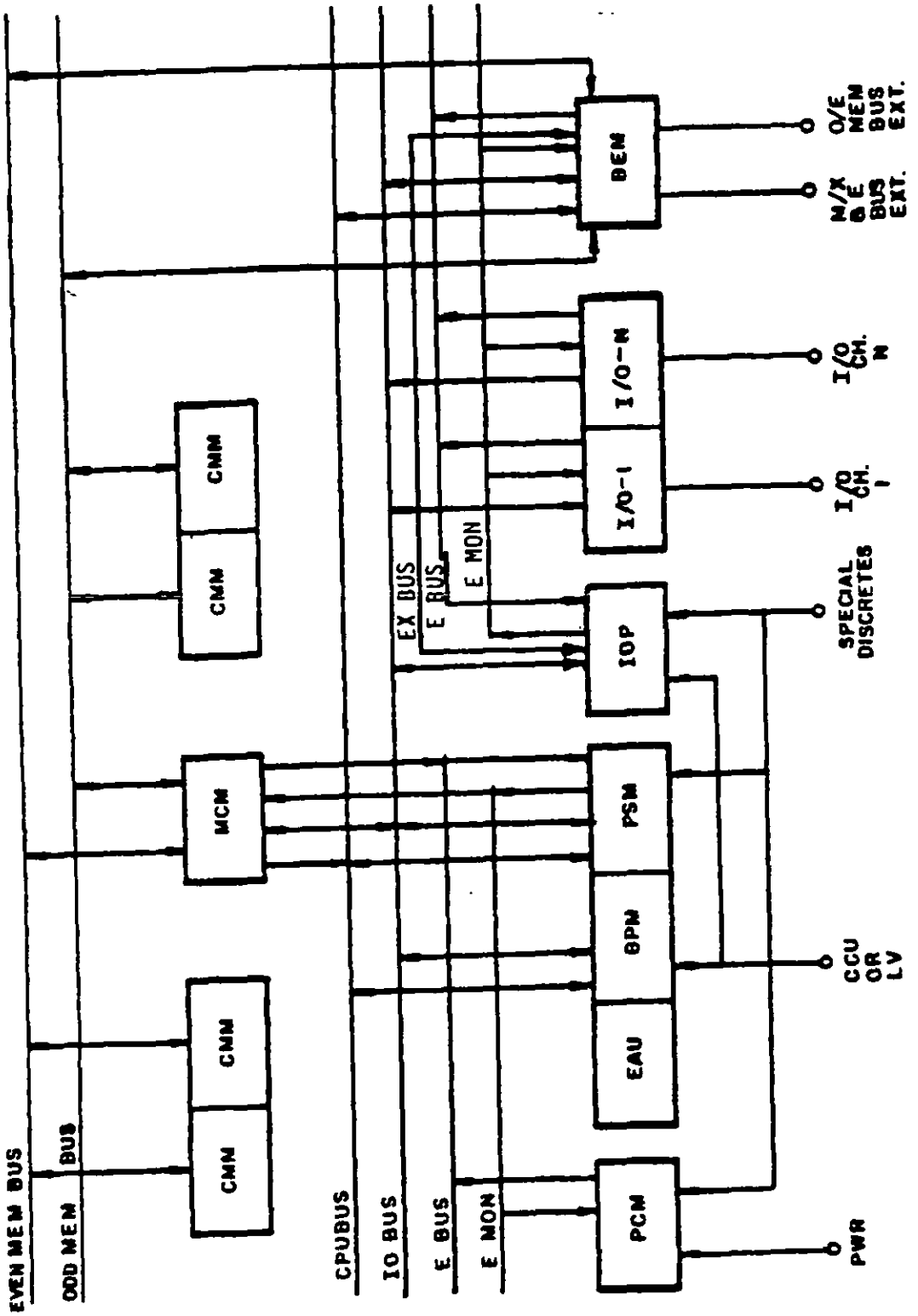


FIGURE 9. AN/AYK-14(V) Configuration with CPU and IOP.

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separate events (or interrupt) buses are required. This allows the IOP event monitor system to receive all I/O-channel related events, and the CPU event monitor system to receive interrupts relating to memory and the power supply. This dual event system also allows CPU-to-IOP and IOP-to-CPU events as required for CPU/IOP communication. This includes the capability for the IOP to transfer I/O channel software-level interrupts to the CPU for processing.

2) Multi-Processor Mode - This mode (see Figure 9) shall also require a configuration with a CPU. In this mode, the CPU and IOP shall each execute separate software programs residing in the same memory subsystem. The IOP shall also execute I/O channel instructions and perform I/O channel data transfers. IOCR instructions may also be initiated by the CPU. In this mode, the IOP shall process all I/O channel-related events, including software-level interrupts. During bootstrap loading, the IOP shall revert to the I/O controller mode.

3.6.2.1.2.1 IOP performance. Minimum execution speeds (when using direct addressing) shall be as listed below when executing an instruction mix consisting of 20 percent multiplication and 80 percent load, store, add type instructions.

Execution Speed (KOPS)

	Instruction Type	CMM	RXM
a.	16-Bit Reg-Reg	434	434
b.	16-Bit Reg-Memory	298	298
c.	32-Bit Reg-Memory	259	285

3.6.2.1.2.2 IOP requirements. The IOP requirements shall be identical to that specified in 3.6.2.1.1 through 3.6.2.1.1.15, with the following exceptions:

a. One set of 16 x 16-bit general purpose registers are required.

b. Microprogrammed floating point arithmetic is not required.

c. The IOP shall not be required to operate as a two-state (Executive and Task) system.

d. I/O command cells shall be 0062_{16} and 0063_{16}.

e. No operand breakpoint.

f. No indirect addressing.

g. No monitor clock.

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h. Free running 16 bit MHz system clock and program-controlled 16-bit, 0.9765 KHz real-time clock.

i. Interfaces to IOBUS only.

j. Only 2048 words of micromemory containing a minimum growth capability of 40 words.

k. No fast multiplier.

l. No trigonometric instruction firmware interface.

m. No control C file or EAU interface.

n. Interrupt set not identical to CPU (See Table II).

o. Performance requirements as specified in 3.6.2.1.2.

p. No Bootstrap memory.

3.6.2.1.2.3 IOP word size. The IOP shall be required to process, as a minimum, the following data and instruction types:

a. 1-bit per datum

b. 8 bit byte data.

c. 16-bit fixed point data

d. 16 and 32-bit instructions

3.6.2.1.2.4 IOP addressing modes. The addressing modes of the IOP shall be identical to those specified in 3.6.2.1.1.9, except that no indirect addressing shall be provided. The maximum memory addressing capability of the IOP shall be 65,536 words of 18-bit memory. In CPU/IOP configurations using the memory controller (3.6.2.3.3) the IOP shall have access to the 65K of memory defined by the page registers. Note that when the IOP is used as a stand-alone processor in the XN-3 chassis, only one 32K main memory module shall be addressable.

3.6.2.1.2.5 Instruction set. The IOP shall execute, as a minimum, the instructions specified by Table XXIII. The instruction set shall be a subset of that of the CPU, and shall provide upward software compatibility with the CPU except as noted in Tables II and III.

3.6.2.1.2.6 Support equipment interface. The IOP shall communicate with the Memory Loader Verifier (or equivalent support equipment) and ROM Simulator (2048 words of memory) in the same manner as the CPU. In the CPU/IOP modes both the CPU and the IOP shall be capable of being controlled by the Memory Loader Verifier (or equivalent support equipment).

3.6.2.1.2.7 Bootstrap loading. In CPU/IOP configurations, the IOP shall be initialized to the I/O controller mode regardless of its mode of operation prior to initialization of IPL. In this mode the IOP shall execute all the

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I/O functions in the bootstrap. In the IOP only configuration, recognition of the IPL signal shall cause the IOP to execute a bootstrap loading program stored in PROM on the RXM, providing that a RXM is present and the bootstrap program is in the RXM optional PROM.

3.6.2.1.2.8 IOP clocks. The IOP shall contain two 16-bit clocks. These clocks shall comprise the system clock and the real-time clock. The System clock shall have a 1.0 microsecond resolution and shall be used as the time base for the real-time clock. This clock (system clock) shall be readable under software control. The Real-time clock shall be software readable and loadable. The Real-time clock shall have a resolution of 1.024 milliseconds. Usage of the clocks shall be as follows:

a. System clock - the system clock shall be a free-running (cannot be disabled) clock which when bit 10 of the clock toggles shall cause an event to occur which shall cause the real-time clock to be incremented by 1 count. The system clock shall be stored when a Stored Real-Time Clock Double instruction is executed.

b. Real-Time Clock - The real-time clock shall be a loadable, readable clock with a resolution of 1.024 milliseconds. When enabled by the enable clock instructions, the real-time clock shall count up at a 1.024 millisecond rate. When the real-time clock overflows (change from all ones to all zeros), a real-time clock interrupt shall be generated. The real-time clock shall continue to count up until disabled by the disable clock instruction. If the Real-Time Clock Overflow interrupt is locked out (disable clock interrupt), one Real-Time Clock Overflow interrupt shall be held pending.

3.6.2.1.2.9 BIT counter. The 8-bit built-in-test (BIT) counter shall be updated by the system clock each 65.536 milliseconds. The BIT counter shall generate a Hardware Fault Warning interrupt every 2.097152 seconds except that the first hardware fault warning interrupt shall be 1.048576 seconds after a hardware Master Clear (except for a master clear entered via the support equipment interface) or the execution of the RESET BIT TIMER instruction. If the BIT counter is not reset via the Reset Bit Timer instruction every 15.72864 seconds (maximum), the counter shall halt at a value of 00FF (hex), stop generating Hardware Fault Warning interrupts, generate the Hardware Fault interrupt to both CPU and IOP, and set the bit indicator.

3.6.2.1.2.10 IOP modularity. The IOP shall be implemented in a single SRA.

3.6.2.1.3 IOP jumper. The computer shall provide an IOP jumper module to be used in the IOP location in a chassis when the IOP is not required. This module shall be used to tie together the I/O and CPU event bus systems when the IOP is not present.

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3.6.2.1.4 Extended arithmetic unit (EAU). The 32-bit floating-point modular hardware Extended Arithmetic Unit shall be provided. The EAU shall retain the same numerical convention as the microprogrammed floating-point capability of the CPU. The EAU execution speeds as a design goal shall be 250 KOPS when executing an instruction mix consisting of 80 percent floating-point add instructions, 10 percent floating-point multiply, and 10 percent floating-point divide instructions. This performance shall apply for RR instruction format with no residue selected and with normalized input operands. This performance shall apply for core and semiconductor memories in interleaved or non-interleaved configurations. EAU execution speeds for EAU unique operations shall not exceed 30 microseconds for nomalized input arguments not requiring argument reduction.

3.6.2.1.4.1 Extended arithmetic unit floating point. The operation codes 50, 51, 52 and 53 octal shall perform the operations of floating-point subtract, add, multiply, and divide, respectively. These instructions shall be performed via microcode in the CPU when the optional EAU module is not included. The identical results and residue shall be obtained (including overflow and underflow indications) with improved performance when the EAU module is utilized with the CPU subsystem.

3.6.2.1.4.2 Trigonometric instructions. The EAU shall perform the floating-point instructions in addition to the add, subtract, multiply, and divide instructions as shown in Table VI. The accuracy of these instructions shall be as shown in Table VII.

3.6.2.2 Input/Output (I/O) subsystem. The input/output subsystem shall provide the principal communication path between the computer and user subsystems. The I/O subsystem shall consist of functionally independent input/output channels which communicate internally through the Input/Output Bus (IOBUS), and which communicate externally as defined by the applicable paragraphs. Both the CPU and IOP shall be capable of providing input/output control functions. Specific input/output configurations are specified in the applicable appendices. Communication with the various I/O channels shall be independent of channel types. The channel priority of a given channel shall be defined by back-panel wiring in the interconnect chassis. The I/O subsystem shall also provide for an interface to extend the internal buses of the computer to the external devices (see 3.6.2.2.2).

3.6.2.2.1 Input/Output channel. The I/O channel structure of the computer shall provide for communication between the computer and peripheral equipment and other AN/AYK-14(V) computers. This software and hardware structure shall functionally accommodate up to 16 I/O channels per computer. Each I/O channel in a given computer configuration may be assigned a unique 4-bit channel logical number which the software uses to address the given channel. A second unique 4-bit channel priority number shall be specified for each I/O channel in a given configuration by the physical I/O channel module locations. The computer hardware shall utilize this channel priority assignment in resolving conflicts among the several I/O channels. This conflict situation may occur when data transfer with main memory is required, or when various event servicing is required by multiplexed hardware common to several I/O channels. An I/O Failure interrupt is generated when the channel is requested to perform an I/O operation on a channel not in the system configuration.

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TABLE VI. EAU Trigonometric Instructions. 1/

Octal Format	Hexa-decimal Format	Coding Format	Instruction	Operation	Time 2/ (usec)
04 0 a 05	10 a 5	SQRT a	Square Root	3/ (Ra,Ra⊙1)←Ra,Ra⊙1	14.0
37 1 a 00	7D a 0	SIN a	Sine	4/ (Ra,Ra⊙1)←Ra,Ra⊙1	26.9
37 1 a 01	7D a 1	COS a	Cosine	4/ (Ra,Ra⊙1)←Ra,Ra⊙1	26.9
37 1 a 02	7D a 2	TAN a	Tangent	4/ (Ra,Ra⊙1)←Ra,Ra⊙1	26.3
37 1 a 03	7D a 3	ASIN a	Arcsine	5/ (Ra,Ra⊙1)←Ra,Ra⊙1	24.9
37 1 a 04	7D a 4	ACOS a	Arccosine	5/ (Ra,Ra⊙1)←Ra,Ra⊙1	27.1
37 1 a 05	7D a 5	ATAN a	Arctangent	5/ (Ra,Ra⊙1)←Ra,Ra⊙1	20.2
37 1 a 06	7D a 6	EXP a	Exponential	6/ (Ra,Ra⊙1)←Ra,Ra⊙1	26.5
37 1 a 07	7D a 7	ALOG a	Natural Log	7/ (Ra,Ra⊙1)←Ra,Ra⊙1	28.5

1/ The standard 32-bit floating-point format as specified in 3.6.2.1.1.15.1 shall be used to express input arguments and results of each EAU trigonometric function.

2/ Typical execution time with interrupts disabled and no events active (for interrupts active add 1.0 usec). Input Operands are normalized, not equal to zero or infinity, and no argument reduction required.

3/ The normalized square root of (Ra,Ra + 1) shall be stored in Ra,Ra + 1. Negative input numbers shall set the overflow designator.

4/ The indicated operation shall be performed on the floating-point value in (Ra,Ra + 1), and the result returned to Ra,Ra + 1. The input value, 0, shall be expressed in radian units.

5/ The indicated operation shall be performed on the floating-point x value in (Ra,Ra + 1), and the floating-point result returned to Ra,Ra + 1. Input values of magnitude greater than 1 for ASIN x and ACOS x shall result in overflow. The output value shall be expressed in radian units.

6/ The exponent of the floating-point x value in (Ra,Ra + 1) shall be returned to Ra,Ra + 1.

7/ The natural logarithm of the floating-point x value in (Ra,Ra + 1) shall be returned to Ra,Ra + 1.

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Table VII. EAU Trigonometric Instruction Accuracy.

Instruction	Input Range	Max Absolute Error
SQRT	$(-\infty, +\infty)$	1 Bit
SIN	$(2\pi, +2\pi)$	1 Bit
	$(-1.6 \times 10^7, +1.6 \times 10^7)$	6 Bits
COS	$(-2\pi, +2\pi)$	1 Bit
	$(-1.6 \times 10^7, +1.6 \times 10^7)$	6 Bits
TAN	$(-2\pi, +2\pi)$	1 Bit 2/
	$(-1.6 \times 10^7, +1.6 \times 10^7)$	6 Bits
ASIN	(-1, +1)	1 Bit
ACOS	(-1, +1)	1 Bit
ATAN	$(-\infty, +\infty)$	1 Bit
EXP	(-16, +16)	1 Bit
	(-174, +174) 1/	3 Bits
ALOG	$(0, +\infty)$	1 Bit

These figures are derived by exercising the functions over a set of 10,000 randomly selected input operands. The standards for comparison are the results obtained from a large mainframe computer, rounded to 24-bit mantissas. The absolute error is calculated as:

1-Bit error: Expected Mantissa - Calculated Mantissa = 1
2-Bit error: Expected Mantissa - Calculated Mantissa = 2 or 3
3-Bit error: Expected Mantissa - Calculated Mantissa = 4, 5, 6, or 7

1/ For exponential function, input operands which are greater than +174 produce a result of machine infinity, and those which are less than -174 produce a result of machine zero.

2/ The accuracy of the TAN function will degrade to an error which is theoretically equal to all 24 bits as the value of the input operand approaches $\pi/2$ radians. The accuracy begins to degrade when a value of 1.56 radians (89.6 degrees) is reached.

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a. Interchangeability - All I/O channel module types, unless otherwise specified in the applicable paragraph, shall be physically and electrically interchangeable in the I/O channel module positions or card slots in the chassis, within the constraints specified in 3.6.2.5. DIM interchangeability is specified in 3.6.2.2.1.3. Each I/O channel type, unless otherwise specified in the applicable paragraph, shall be constructed as a single modularly-independent subsystem.

b. I/O Control Memory - Each I/O channel shall have associated with it a 16-word by 16-bit control memory as part of the CPU subsystem. This control memory shall contain the parameters which direct and control the operation of the associated I/O channel. Chain addresses, data buffer addresses, data buffer word counts, data word length definitions, channel mode definitions, and parity bit polarity control are examples of the type of information contained in the control memory parameters. The format and definition of each location in a given I/O channel control memory shall be dependent on the given I/O channel type. Although the basic architecture of the I/O channels shall incorporate this 16-word control memory for each I/O channel in a given configuration, the IOP shall support only the first 12 words of these control memory locations.

c. I/O Channel Operation - I/O channel operation shall be initiated by execution of a processor instruction defined as INPUT/OUTPUT COMMAND (Op Code 74 hexadecimal, mnemonic IOCR). This processor instruction, when encountered in the software sequence during program execution, shall cause the I/O channel instruction located at CPU address 0060 (IOP address 0062) hexadecimal in main memory to be executed. I/O channel instructions (any instruction whose Op Code begins with value 111 binary) are otherwise illegal and will cause a CPU Instruction Fault interrupt when encountered in a processor software program execution, unless encountered in this "out-of-sequence" manner via the IOCR processor instruction. Subsequent to initiation by the processor, further I/O channel operations on the affected channel shall be controlled by I/O channel instructions in an I/O channel program in main memory. Once initiated by the processor, execution of the I/O channel program on the active I/O channel or channels shall require no further intervention by the processor software, but shall operate independently of it. Note that several I/O channels may be simultaneously active in execution of I/O channel programs. However, under certain conditions, I/O channel types may dictate relative channel priority. When an I/O channel is active (has an I/O channel program in execution), two forms of I/O channel activity may be in progress:

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(1) I/O information transfer activity

(2) I/O chaining activity

Under certain conditions, various I/O channel types may have both forms of I/O channel activity in progress at the same time.

d. I/O Chaining - I/O chaining activity exists on an I/O channel, when sequential I/O channel instructions are being executed on that channel. These I/O channel instructions normally provide for transferring parameters between the main memory and the I/O channel control memory, and for initiation of transfer of blocks or buffers of data and/or control words on the channel interface lines. When an I/O channel instruction that initiates a buffer transfer on the interface lines is encountered in I/O chaining activity on a given I/O channel, I/O chaining activity is normally halted on that I/O channel, and I/O information transfer activity begins. Once this instruction is encountered, no further software shall be necessary to control information transfers until the buffered information transfers are complete. The I/O channel shall transfer this information under the control of various control memory locations which must be set up by software. Upon completion of I/O information transfer activity (buffer termination), I/O chaining activity shall be re-initiated. I/O channel instruction shall also exist to provide branching or jumping within the I/O channel program and halting of I/O channel operations. Where specified, I/O channel types shall be able to execute two I/O channel operations at the same time, one which performs I/O information output transfers, and one which performs I/O information input transfers. Execution of instructions whose operational codes are not in the $00E0_{16}$ to $00FE_{16}$ range shall cause an I/O Instruction Fault interrupt if encountered in the chain program. Note that all I/O information input transfers shall be subject to any existing main memory page write protection defined by the CPU processor program. Several different I/O channel instructions may be executed "out-of-sequence", via the IOCR processor instruction, to perform additional communication between the processor software and the I/O channel. Among the capabilities provided shall be the following:

(1) Initiate I/O channel operation.

(2) Halt I/O channel operation.

(3) Enable/Disable I/O channel software interrupts.

(4) Load/Store I/O channel control memory parameters.

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(5) Set/Clear discrete lines on certain I/O channel types.

(6) Read I/O channel status.

These capabilities shall enable processor software to start, stop, monitor, and modify I/O channel operation. In addition, the processor software shall be capable of modifying any given I/O channel program, even while in execution by an I/O channel.

e. I/O Communication - Communication in the other direction (I/O channel to processor software programs) shall be performed primarily via the interrupt system in the computer. Interrupts shall exist to provide "Check-pointing" notification to processor software, via execution in an I/O channel program, of an I/O channel instruction which generates a chain interrupt. Interrupts also shall exist to indicate occurrence of error or abnormal conditions on specified I/O channel types. I/O channel instructions that facilitate I/O channel to processor software communications shall exist to provide the following capabilities:

(1) Generate I/O channel chain interrupts.

(2) Store I/O channel status in main memory.

(3) Set/Clear flag bits in main memory.

I/O channel interrupts shall be Class III type. Each of the 16 possible I/O channels of the computer shall be capable of generating up to 4 unique Class III interrupts. These 64 possible Class III interrupts shall be prioritized within Class III by channel priority first, and priority level among the 4 interrupts of an I/O channel secondly. Channel priority shall be defined by physical location of a given I/O channel type in a computer chassis configuration. Regardless of this priority, each I/O channel shall be assigned a logical channel number which is used by all instructions which reference the given channel. Within a channel, priority level 1 interrupt shall be highest priority, and 4 shall be lowest. The 4 Class III interrupts, per I/O channel, shall be named as follows:

(1) Priority 1 ERI Error or Timeout

(2) Priority 2 EII External

(3) Priority 3 OCI Output Chain

(4) Priority 4 ICI Input Chain

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Not all I/O channel types shall be required to generate all 4 Class III type interrupts. Those which are implemented on a given channel shall follow this priority scheme. When a Class III interrupt is generated, not masked out or locked out, and is recognized by the processor, an interrupt recognition sequence shall be performed in the processor to allow for resuming processor program operation at the point of interruption, after interrupt processing, and shall enter the interrupt processing routine.

3.6.2.2.1.1 NTDS interface modules (NIM). The computer shall provide four single SRA NTDS I/O channel options as defined in MIL-STD-1397 including one NTDS channel with expanded capabilities. They shall be the NTDS Slow (Type A), the NTDS Fast (Type B), and an expanded version of the NTDS ANEW (referred to herein as Type C1) parallel channels; and the NTDS serial channel. NTDS channels shall be capable of masking the EII interrupt specified in 3.6.1.4. The control memory for NTDS channel operation shall be as defined in Figure 10. The buffer control word format for input (word 0) and output (word 4) operation shall be as in Figure 11.

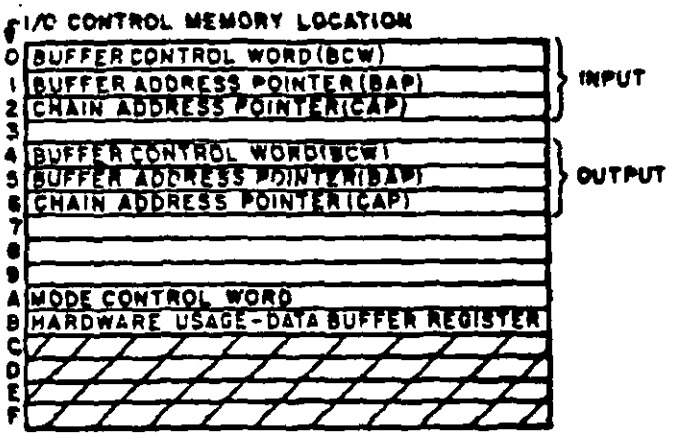


FIGURE 10. NTDS control memory usage.



TM: 00 ABORT TRANSFER (INPUT DATA ONLY)
01 BYTE TRANSFER (8 BITS)
10 SINGLE-LENGTH TRANSFER (16 BITS)
11 DUAL CHANNEL (32 BIT DOUBLE-LENGTH TRANSFER)
* NOT USED
B: BYTE POINTER

FIGURE 11. NTDS buffer control word.

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3.6.2.2.1.1.1 NTDS parallel channels. The three NTDS parallel channel types, Slow (Type A), Fast (Type B), and expanded ANEW (Type C1) shall have identical operating characteristics from the software viewpoint, and differ only in interface electrical signal characteristics and data rates. The NTDS, types A, B, and C1 (parallel), I/O channels shall provide the following modes of operation (modes g and h are applicable only to Type C1):

a. Computer to peripheral - 16-bit

b. Compter to computer - 16 bit

c. Test mode

d. Computer to peripheral - 32-bit

e. Computer to computer - 32-bit

f. Externally specified addressing - 32-bit

g. Peripheral to computer - 16 bit

h. Peripheral to computer - 32 bit

The channels shall be capable of full-duplex operation. The parallel channels, which are normally 16-bit channels, shall be capable of being combined in pairs to form a single 32-bit dual channel, with the following restrictions: both channels shall be the same interface type, and the two channels shall be located in adjacent physical channel slots with the least significant 16-bit channel located in an odd numbered slot. The mode control word shall be as defined in Figure 12. Mode status shall be maintained in the I/O mode status in the I/O channel hardware status word (Figure 13).

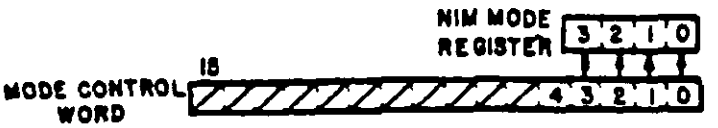
3.6.2.2.1.1.1.1 NTDS Slow. This interface shall be characterized by nominal values of 0 VDC and -15 VDC to represent binary 1 and binary 0, respectively.

3.6.2.2.1.1.1.2 NTDS Fast. This interface shall be characterized by nominal values of 0 VDC and -3 VDC to represent binary 1 and binary 0, respectively.

3.6.2.2.1.1.1.3 NTDS expanded ANEW. This interface shall be characterized by nominal values of 0 VDC and +3.5 VDC to represent binary 1 and binary 0, respectively.

3.6.2.2.1.1.1.4 NTDS parallel test capabilities. The NTDS Parallel I/O channels shall provide for a wrap-around test capability. This wrap-around shall be internal to the NTDS I/O channel and shall connect the data and control transmitter inputs to the circuitry normally fed by the data and control receiver outputs. It shall exercise the data and control handling logic on this I/O channel without transmitting or receiving control signals from external equipment. Data transmitters shall not be disabled.

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MODE REG 3 2 1 0	MODE OF OPERATION
0 0 0 0	HARDWARE DEFAULT MODE
0 0 0 1	
0 0 1 0	
0 0 1 1	
0 1 0 0	
0 1 0 1	
0 1 1 0	
0 1 1 1	
1 0 0 0	COMPUTER TO PERIPHERAL 16 BIT
1 0 0 1	COMPUTER TO COMPUTER 16 BIT
1 0 1 0	PERIPHERAL TO COMPUTER 16 BIT
1 0 1 1	TEST MODE
1 1 0 0	COMPUTER TO PERIPHERAL 32 BIT
1 1 0 1	COMPUTER TO COMPUTER 32 BIT
1 1 1 0	EXTERNALLY SPECIFIED ADDRESSING
1 1 1 1	PERIPHERAL TO COMPUTER 32 BIT

FIGURE 12. NTDS parallel mode control word.

3.6.2.2.1.1.2 NTDS serial channel. The NTDS serial channel shall provide asynchronous full-duplex communication. The NTDS serial I/O channel shall provide the following modes of operation:

a. Computer to peripheral - 16-bit

b. Computer to computer - 16-bit

c. Test mode

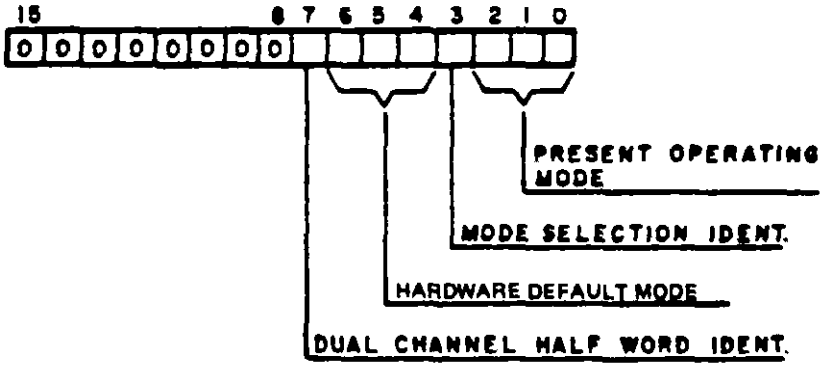
d. Computer to peripheral - 32-bit

e. Computer to computer - 32-bit

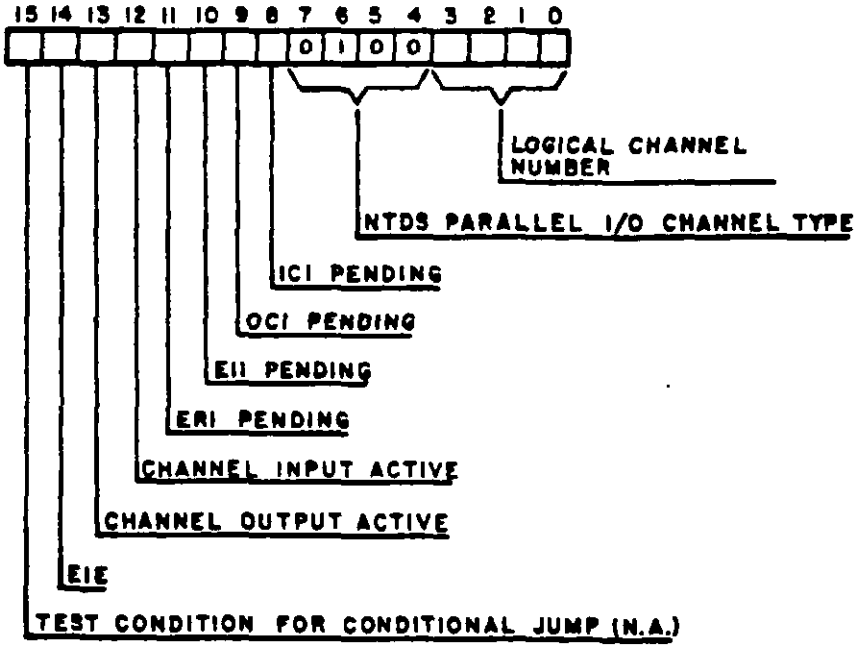
Two software programmable modes of 32-bit operation shall be provided to maintain compatibility between parallel and serial NTDS channels. One mode shall require the dedication of two adjacent I/O slots. The second 32-bit mode shall require the dedication of only one I/O slot. The NTDS Serial Mode Control Word shall be as defined in Figure 14. Mode status shall be maintained in the NTDS serial channel mode status word and hardware status in the NTDS serial I/O channel hardware status word (see Figure 15).

3.6.2.2.1.1.2.1 NTDS serial test capabilities. The NTDS Serial test mode shall provide for the internal wrap around of data to and from the module input and output subchannels. This wrap-around shall connect the control logic for the transmitter of the output channel to the circuitry normally fed by the input channel receiver logic, and the control logic for the transmitter

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NTDS PARALLEL I/O MODE STATUS WORD



NTDS PARALLEL CHANNEL HARDWARE STATUS WORD

FIGURE 13. NTDS parallel I/O status words.

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MODE REG 3 2 1 0	MODE OF OPERATION	
0 0 0 0	HARDWARE DEFAULT MODE	
0 0 0 1		
0 0 1 0		
0 0 1 1		
0 1 0 0		
0 1 0 1		
0 1 1 0		
0 1 1 1		
1 0 0 0	OFF	
1 0 0 1	CP TO CP, LOOP TEST	16 BIT
1 0 1 0	CP TO PERIPHERAL	16 BIT
1 0 1 1	CP TO CP	16 BIT
1 1 0 0	CP TO PERIPHERAL	32 BIT
1 1 0 1	CP TO CP	32 BIT
1 1 1 0	CP TO PERIPHERAL DUAL CHANNEL	32 BIT
1 1 1 1	CP TO CP DUAL CHANNEL	32 BIT

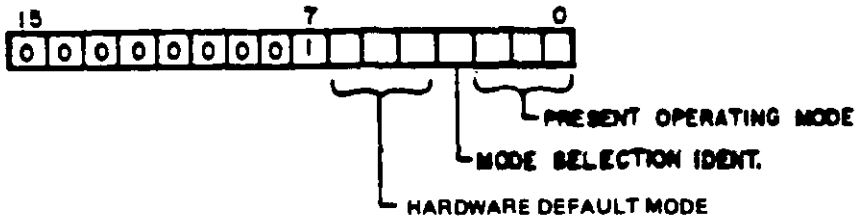
FIGURE 14. NTDS serial mode control word.

of the input channel to the circuitry normally fed by the output channel receiver logic. All logic except the transmitter and receiver circuitry shall be tested without receiving data from external equipment. Data transmitters shall not be disabled during testing.

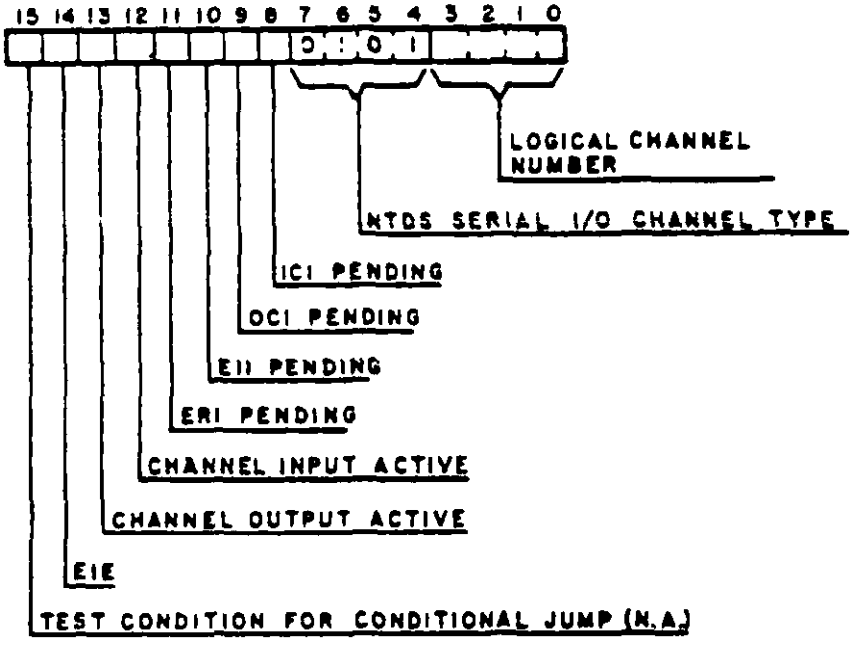
3.6.2.2.1.2 Serial interface module (SIM-A). The computer shall provide bi-directional serial multiplex I/O channels per MIL-STD-1553A. Each channel shall contain two multiplex data receiver/transmitters and sufficient logic and circuitry, as necessary, to detect, process, and participate in communication on two data buses. Channel setup, mode selection, initiation of I/O operations, and monitoring of control storage and status shall be under the control of the CPU, IOP, or combination of both. Each channel shall operate independently, such that bi-directional transfer of control information and multiplex data to memory shall occur in a nonconflicting manner. Each channel shall be capable of operating either as bus controller (BC), or as a remote terminal (RT). Note that the SIM operates in a synchronous (i.e., non-handshaking control) manner during message sequences. Once a message sequence has been initiated, information transfer must be continuous within that message. Therefore, attention must be given to the priority of the SIM channels and their simultaneous utilization with other Input/Output channel utilization in a given configuration.

3.6.2.2.1.2.1 SIM control memory definition. Figure 16 defines the control memory format and usage for the SIM I/O channel type. The specific function and format of certain of the control memory locations shall be

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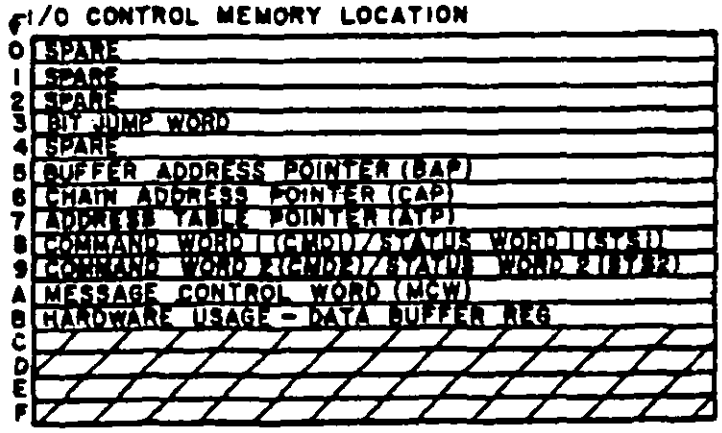
NTDS SERIAL I/O CHANNEL MODE STATUS WORD



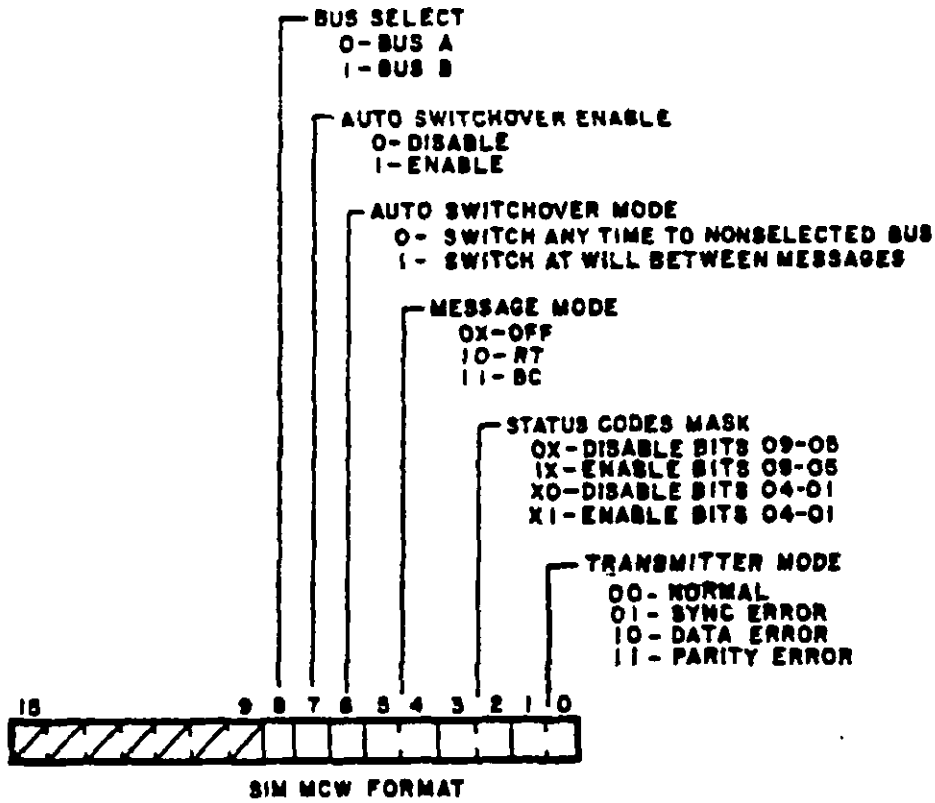
NTDS SERIAL CHANNEL HARDWARE STATUS WORD

FIGURE 15. NTDS serial I/O status words.

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SIM CONTROL MEMORY DEFINITION MAP



SIM MCW FORMAT

FIGURE 16. SIM control memory usage.

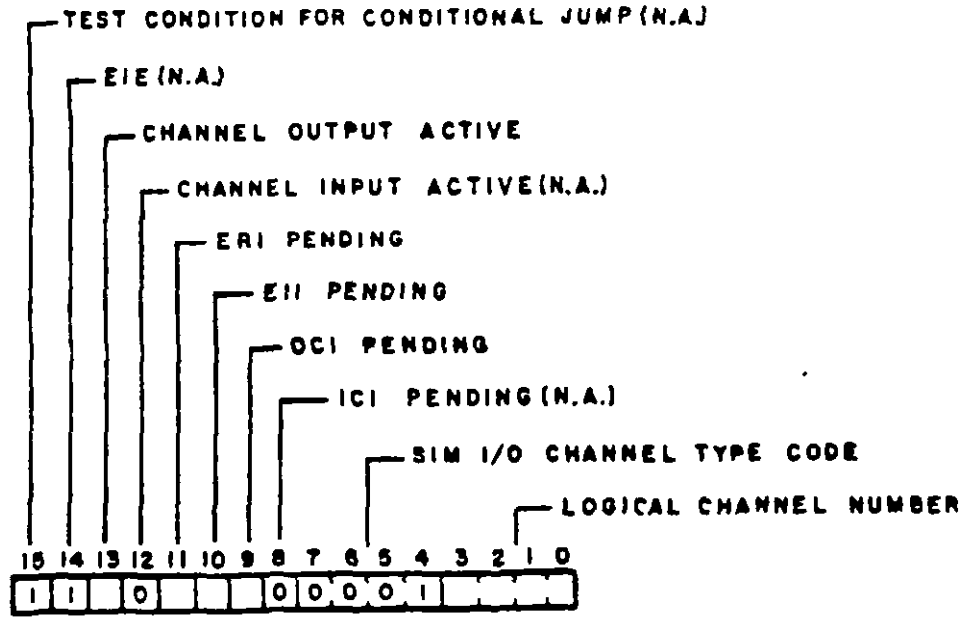
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different for different modes (BC or RT) of operation. However, locations 3, 5, 6, A and B hexadecimal shall always remain identical. Location 3 shall always be used for the BIT Jump instruction. Location 5 shall always contain the address of the main memory location where the next sequential Data Word is to be input or output during a message sequence. This address value shall be termed the Buffer Address Pointer (BAP), and shall be incremented by 1 for each Data Word transferred during I/O information transfer activity. Control memory location 6 shall contain the address of the main memory location where the next I/O channel instruction is located, and shall be termed the Chain Address Pointer (CAP). This address value shall be updated appropriately during I/O chaining activity and shall be the I/O channel program address register for the channel. The Message Control Word (MCW) in control memory location A hexadecimal shall define the mode and control of the SIM. The format of the MCW is shown in Figure 16. The Address Table Pointer (ATP) shall be used in the RT mode to point to an address in memory of a list of addresses corresponding to one of 31 input and 31 output possible message types. Control memory locations 8 and 9 contain the message COMMAND WORD/ STATUS WORD 1 and 2, respectively, for use in controlling message sequences and reporting message status. SIM I/O channel status words 0 and 1 are defined in Figure 17, and contain fields designating current channel status. The SIM channel shall have ERI (abnormal) interrupts, EII interrupts, and OCI interrupts.

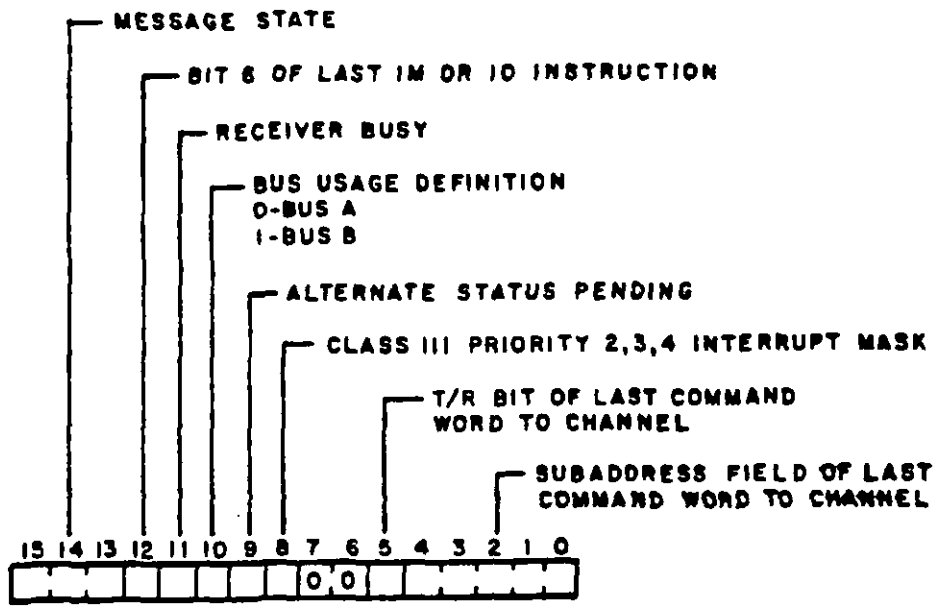
3.6.2.2.1.2.2 BUS selection. The active bus shall be selectable under program control. In the BC mode, this shall be the only bus used, but in the RT mode, this selection may or may not be used at any given instant in time, depending upon bits 7 and 6 of the MCW. In the RT mode, two auto switch-over modes shall exist. In the first mode, the SIM I/O channel hardware shall continually search for the non-selected bus for a valid command word SYNC pattern, and when one is detected, automatic reset and switch-over to that non-prime bus shall occur in order to receive command words and data. This shall be permitted at any point during any phase or message state, even during a message sequence on the selected or prime bus. This switch-over mode shall allow a redundant bus scheme in which the BC within a MIL-STD-1553A network utilizes a primary bus and switches all RTs over automatically by simply beginning operation of the alternate bus. This would be necessary if the prime bus were to experience continuous or extremely noisy traffic, and once switching has been accomplished, all RTs would remain on the alternate bus, regardless of any noise or erroneous traffic on the primary bus. The inactive receiver/transmitter shall reflect MIL-STD-1553A "receive-mode" impedance to the bus. In the second mode, the SIM I/O channel shall continually search the nonused bus at any instant in time for a valid command word SYNC pattern. If a valid SYNC is detected and a message sequence with the SIM is not presently in progress, then a switch-over shall be experienced and monitoring or searching for a valid command word SYNC shall begin on the other bus. In this mode, alternate switching back and forth shall be permitted between messages with the SIM channel. This mode enables the BC in a network to alternately switch back and forth between buses of the redundant pair in the case where some of the RTs are addressable only on one bus and some on the other. The inactive receiver/transmitter shall reflect MIL-STD-1553A "receive-mode" impedence to the bus. The bus in use at any time can be determined by examining bit 10 of status word 1 (Figure 17).

3.6.2.2.1.2.3 Message sequence. I/O information transfer activity shall be controlled by message sequencing logic on the SIM I/O channel type module.

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STATUS WORD 0



STATUS WORD 1

FIGURE 17. SIM status word formats.

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3.6.2.2.1.2.3.1 BC mode. In the BC mode, the SIM I/O channel type shall provide for initiation of all the following message formats:

a. BC to RT data transfer

b. RT to BC data transfer

c. RT to RT data transfer

d. BC to RT mode command transfer

When a channel is operating as a bus controller, it shall be capable of communication with 32 remote terminals on each bus. A message sequence shall be initiated by execution of an I/O channel instruction INITIATE I/O MESSAGE or INITIATE I/O TRANSFER in an I/O channel program. I/O chaining activity shall be halted and I/O information transfer activity shall be initiated on that I/O channel. Upon initiation, the specified message sequence shall begin. The SIM shall wait, if necessary, until a 7 µs gap time is detected on the selected bus, and shall then initiate the transmitter circuitry. Once initiated, the transmitter circuitry shall wait, if necessary, to assure 10 µs gap time on the selected bus, and shall then begin transmission of the positive SYNC pattern for the message initial command word. Further action shall depend upon the particular message sequence required, as described below.

3.6.2.2.1.2.3.1.1 BC to RT. Immediately following the Command Word, the SIM shall transmit all of the specified number Data Words, each with a negative SYNC pattern and parity bit. No gap time shall be inserted during the Command Word and Data Word transmission period. The Data Words shall be fetched from sequential main memory addresses, starting at the address specified by the Buffer Address Pointer (BAP) in the control memory. This value shall be incremented during the transmission sequence. After a 2 to 2.5 µs gap has been defined on the bus following the last Data Word the SIM shall begin monitoring for the reception of the Status Word response from the addressed RT. The SIM shall continue to monitor for the status word response for a minimum of 5 µs. A message monitoring timer shall also run during the message sequence to prevent hang-up. Expiration of this timer (1.0 to 1.5 msec) shall force the SIM to abort the message sequence with an ERI interrupt generated to the processor and I/O operation halted. Any message abnormalities detected during the message shall cause an ERI interrupt to the processor and halt SIM I/O channel operation. The Status Word received shall be stored in the SIM I/O channel hardware and is available for reading upon message sequence termination via I/O channel instructions. This Status Word may also be read by the processor, via the IOCR instruction, at any time without interfering with SIM I/O channel operation and activity. Also note that reception of a status word with any ME, unmasked Status Codes, or TF field bits set shall cause an EII interrupt to the processor and shall cause I/O chaining activity to remain halted (not reinitiated) at the end of the I/O information transfer activity. Masking of status code bits shall be determined by the MCW (Figure 16).

3.6.2.2.1.2.3.1.2 RT to BC. The SIM shall wait until the Transmit Command Word has been inserted onto the bus and until a 2 µs gap time has been defined on the bus following this word. Once the 2 µs gap has been defined,

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the SIM shall begin monitoring for the reception of the Status Word response from the addressed RT. The SIM shall continue to monitor for the Status Word response for a minimum of 5 µs. Upon error-free reception of the Status Word, the SIM shall receive the remainder of the Data Word sequence of the message. A message monitoring timer shall run during the message sequence to prevent hang-up. Expiration of the message timer (1.0 to 1.5 msec) shall force the SIM to abort the message sequence with an ERI interrupt generated to the processor and I/O operation halted. The received Data Words shall be stored in main memory, starting at the address defined by the Buffer Address Pointer (BAP) in the control memory. The Status Word received shall be stored in SIM I/O channel hardware and is available for reading upon message sequence termination via I/O channel instructions. This Status Word may be read by the processor, via the IOCR instruction, at any time without interfering with SIM I/O channel operation and activity. Also note that reception of a Status Word with any ME, unmasked Status Codes, or TF field bits set shall cause an EII interrupt to the processor and shall cause I/O chaining activity to remain halted (not reinitiated) at the end of the I/O information transfer activity.

3.6.2.2.1.2.3.1.3 RT to RT. Once the first Command Word has been inserted on the bus, the second Command Word shall be transmitted. The SIM shall wait until this second Command Word has been inserted on the bus and until a 2 µs gap time has been detected following this second Command Word. Upon determination of the 2 µs gap, the SIM shall begin monitoring for the reception of the first Status Word response from the addressed transmitting RT. The SIM shall continue to monitor for the status word response for a minimum of 5 µs. Once this is received, the SIM shall wait for the 2 µs gap time on the bus following all of the Data Word transmissions from RT to RT. When this second 2 µs inter-message gap is detected, the SIM shall begin monitoring for the reception of the second and final Status Word response from the receiving RT. The SIM shall continue to monitor for the status word response for a minimum of 5 µs. A message monitoring timer shall run during the message sequence to prevent hang-up. Expiration of the message timer (1.0 to 1.5 msec) shall force the SIM to abort the message sequence with an ERI interrupt generated to the processor and I/O operation halted. Both Status Words received shall be stored in SIM I/O channel hardware and are available for reading upon message sequence termination via I/O channel instructions. Either of these Status Words may be read by the processor, via the IOCR instruction, at any time without interfering with SIM I/O channel operation and activity. Also note that a reception of either Status Word with any ME, unmasked Status Codes, or TF field bits set shall cause an EII interrupt to the processor and shall cause I/O chaining activity to remain halted (not reinitiated) at the end of the I/O information transfer activity.

3.6.2.2.1.2.3.1.4 BC to RT mode command. For this message sequence, the SIM shall wait until the transmitter circuitry has inserted the entire Command Word, with parity bit, onto the bus and until a 2 µs gap time has also been detected following this Command Word. After this 2 µs gap time has been detected, the SIM then shall begin monitoring for the reception of the Status Word response from the addressed RT. The SIM shall continue to monitor for the status word response for a minimum of 5 µs. A message monitoring timer shall run during the message sequence to prevent hang-up. Expiration of the message timer (1.0 to 1.5 msec) shall force the SIM to abort the message sequence with an ERI interrupt generated to the processor and I/O operation halted. The Status Word received shall be stored in SIM I/O channel hardware

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and is available for reading upon message sequence termination via I/O channel instructions. This status word may be read by the processor, via the IOCR instruction, at any time without interfering with SIM I/O channel operation and activity. Also note that reception of a Status Word with any ME, unmasked Status Codes, or TF field bits set shall cause an EII interrupt to the processor and shall cause I/O chaining activity to remain halted (not reinitiated) at the end of the I/O information transfer activity.

3.6.2.2.1.2.3.2 RT mode. In the RT mode, the SIM I/O channel type provides for responding to the following message formats:

a. BC to RT data transfer

b. RT to BC data transfer

c. RT to RT data transfer

d. BC to RT mode command transfer

In addition, a monitoring function may be performed which receives all bus traffic words and inserts them into main memory in a sequential address list. I/O information transfer activity shall be initiated on the SIM, when in RT mode, in the same manner as is a message sequence when in the BC mode. This is via the execution of an I/O channel instruction IM or IO in an I/O channel program. When initiated via the IM or IO instruction, the SIM shall monitor the bus for a 2 μs gap time and then begin monitoring the selected bus (or buses, if AUTO SWITCH-OVER is enabled). The IM or IO instruction shall not initiate a message sequence as in the BC mode, but only instruct the SIM channel to respond to message sequences, when addressed on the bus(es). I/O chaining activity shall be halted, and I/O information transfer activity shall be enabled. Until master cleared, halted, or switched to BC mode or OFF, the SIM shall remain enabled to perform I/O information transfer.

3.6.2.2.1.2.3.2.1 Receive command word. All error-free Data Words shall be received and stored in memory at sequential addresses, starting at the established BAP address. If at any time during this sequence a 7 μs gap is detected, the SIM shall stop inputing and shall not return a status or generate any interrupts. Note that any positive SYNC words which are received shall be ignored so as to support RT to RT message sequences. Also note that any data word parity or Manchester code error detected shall be stored in memory and then shall terminate the data reception and set the ME bit. Should abnormalities other than parity or Manchester code error be detected, the SIM shall immediately reset and advance to begin monitoring for a new Command Word. After all Data Words specified by the word count field of the Receive Command Word have been received, the SIM shall monitor for a 2 μs gap time on the bus. When this gap has been detected, the transmitter circuit shall be initiated to respond on the bus with the first Status Word in the control memory. If the ME bit was set, then this Status Word transmitted shall have the ME bit a logic 1 value. Once initiated, the transmitter circuitry shall wait to assure a 2 to 5 μs gap time and then shall begin transmission of the Status Word response with a positive SYNC pattern. When the Status Word transmission has been completed the SIM shall resume monitoring the bus for Command Words.

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3.6.2.2.1.2.3.2.2 Transmit command word. After receiving a Transmit Command Word, the SIM shall monitor the bus for a 2 to 5 µs gap time. When this gap has been detected, the transmitter circuitry shall be initiated to respond on the bus. The transmitter circuitry shall then wait to ensure a 2 to 5 µs gap time and shall then begin transmission of the Status Word. The requested number of Data Words, as requested by the transmit command word, shall be transmitted immediately. These words shall be fetched from main memory sequential address, starting at the established BAP address value. A 2 µs gap time will be inserted after the Last Data Word has been transmitted.

3.6.2.2.1.2.3.2.3 Mode command word. This is much like the Transmit Command Word sequence discussed above. After receiving the Mode Command Word, the SIM shall monitor for a 2 µs gap time on the bus. When this gap has been detected, the transmitter circuitry shall be initiated to respond with a Status Word on the bus. The transmitter shall again wait to ensure a 2 to 5 µs gap time and shall then begin transmission of the Status Word. The SIM shall next generate an EII interrupt to the processor and I/O chaining activity shall again be re-initiated.

3.6.2.2.1.2.3.3 Bus monitor mode. When the SIM is in the RT mode, and initiated via an IM or IO instruction with bit 6 a logic 1 value, the Bus monitoring function shall begin. The SIM shall receive the first word activity on the Bus, and store the received word at address BAP in main memory. The BAP value in control memory location 5 shall be incremented by 1 for each word received (commands, data, and status) during Bus traffic activity as these words are stored in main memory. If a 7 µs gap is detected during this time, I/O chaining activity shall be re-initiated. The Bus monitoring function, however, shall not be terminated. This provides for I/O channel instruction executions between each message monitored on the Bus. The BAP value may be changed, the processor interrupted with OCI interrupt, and then another IM or IO instruction executed again.

3.6.2.2.1.2.3.4 SIM test capabilities. The SIM I/O channel type shall provide for a wrap-around test capability. Note that this shall define an OFF mode for the SIM and shall present a continuous high impedance output to both Bus A and Bus B. This wrap-around shall be internal to the SIM I/O channel type and shall connect the Bus transmitter inputs to the circuitry normally fed by the Bus receiver outputs. It shall exercise the message handling section, digital transmitter logic, digital receiver logic, and data word buffers of the SIM. External bus inputs shall not be accepted in the test mode.

3.6.2.2.1.2.4 Bus interface requirements. The clarifications to MIL-STD-1553 transmitter/receiver requirements specified in 3.6.2.2.1.2.4.1 through 3.6.2.2.1.2.4.3 shall apply.

3.6.2.2.1.2.4.1 Output waveform. The output of valid Manchester code, observed at any point along a properly terminated cable up to 300 feet in length, shall have zero crossings that deviate from an ideal waveform by no more than ± 15 nanoseconds. This requirement shall apply across word boundaries, with the exception that the long term stability of the subsystem clock shall be as defined in MIL-STD-1553A. Peak amplitude variation in the waveform within a message shall be less than ± 15% of the average peak value for the entire message as shown in Figure 18. This requirement shall also

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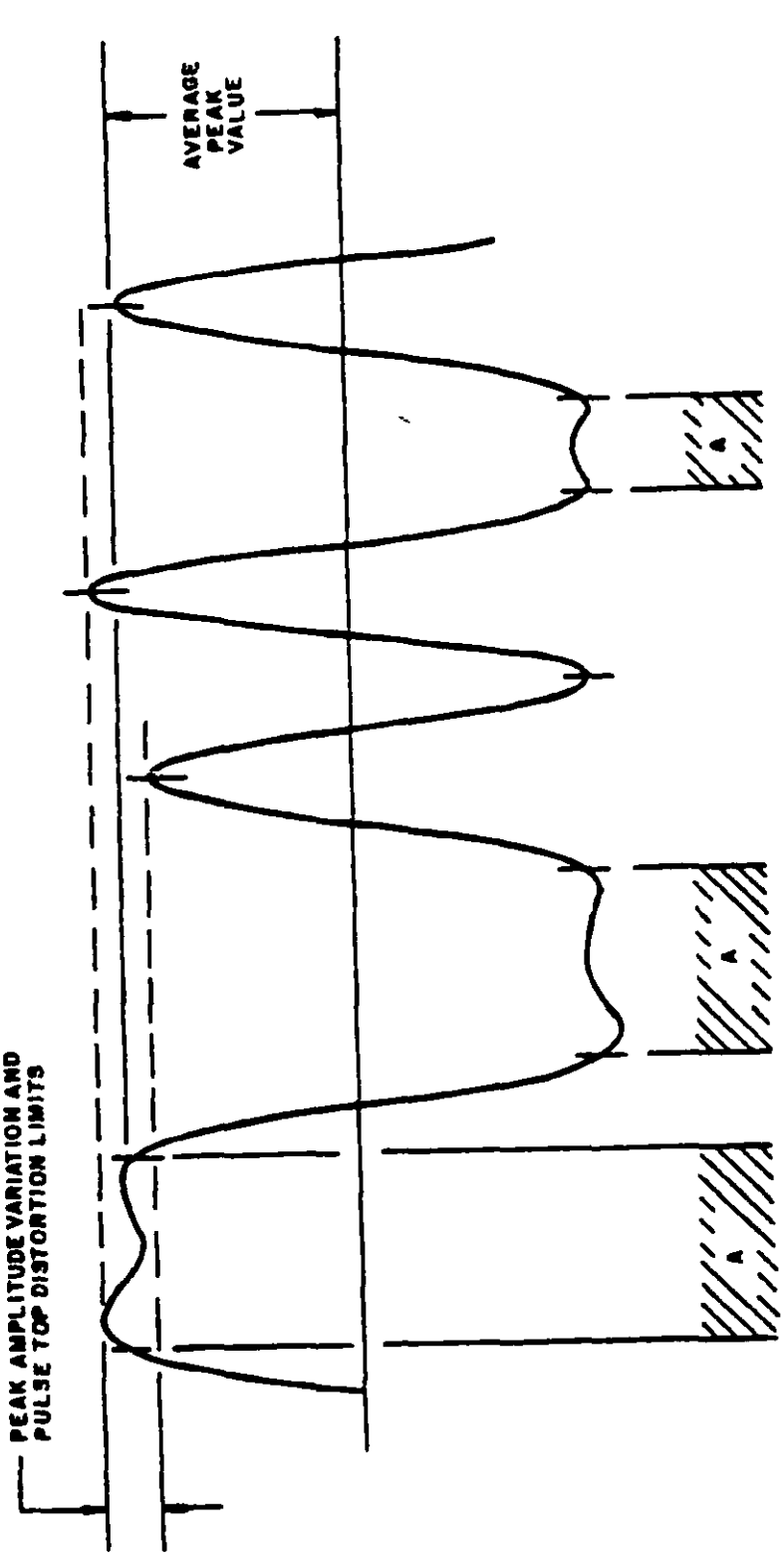


FIGURE 18. Waveform anomaly limits for SIM interface.

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apply to pulse top distortion phenomena (droop and ripple) occurring in areas of the waveform other than leading and trailing edge periods. These limits shall be met at the data bus driving point with a properly terminated cable up to 300 feet in length, and also with a 35 ohm load in place of the cable. The waveform shall be symmetrical such that an energy buildup does not occur on the transmission line. Harmonics above 1.0 MHz shall essentially be removed from the waveform prior to insertion onto the data bus. Harmonic content of the waveform shall be limited such that frequency components from 0.01 to 1.0 MHz do not deviate more than ± .6 db referenced to average peak value. Output signal components at 1.5 MHz shall be attenuated at least 3 db referenced to average peak value. Similarly, the component at 2.5 MHz shall be attenuated at least 13.5 db. Attenuation of spectral components above 4.0 MHz shall be 25.5 db or greater. Filtering shall be implemented so as to minimize phase distortion of the output waveform. Harmonic content and phase distortion requirements shall include the effects of the output driver and shall be met with its output loaded with a purely resistive 35 ohm load.

3.6.2.2.1.2.4.2 Common mode rejection. RT receiver design (and component selection) shall preserve common mode rejection beyond the differential response frequency limit of the data detector. Balanced line-to-ground impedance to the data bus conductor pair shall be maintained. Filtering shall be employed at the receiver input to decrease noise susceptability in the presence of externally caused imbalances in line-to-ground impedances.

3.6.2.2.1.2.4.3 Cable coupling and isolation resistors. An isolation resistor (56.2 ohms ± 1%) shall be installed in series with each output lead of the coupling transformer as required by MIL-STD-1553A. Data bus interface electronics shall have two sets of connections from their input/output transformers, one set connected directly to the transformer, the other set connected through the isolation resistors. The input/output transformer shall be center tapped on the bus side, with provisions for optionally strapping the center tap directly to chassis ground.

3.6.2.2.1.3 Discrete interface module. The discrete I/O interface module (DIM) shall provide the following I/O interfaces:

a. Eight external interrupts (INT) with individual mask bits and program selectable priority.

b. 32 bi-directional differential discretes (DIO) program selectable in groups of four as input or output signals.

An additional 32 input discretes (16 differential (DID) and 16 switch closures (DIS) shall be provided via the DIM connector (see 3.6.2.2.1.3.3). With no signal applied (either interfacing equipment not present or not powered), the sixteen input differential discrete input receivers shall be biased to the logic zero state. The 16 switch closure discrete inputs shall be biased such that an open is interpreted as a logic 0 and a short circuit (0 ± 0.5 volts DC) shall be interpreted as a logic 1. Bi-directional discretes programmed as inputs shall also be biased to the logic 0 state. Discrete outputs shall be held at a logic 0 state during power up or recovery from a power interrupt. The discrete channel SRA shall also provide the capability of accepting eight externally generated interrupts, each seperately maskable under software control. The eight externally generated interrupts shall have eight levels of

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priority settable under software control. All discretes and external interrupts shall be TTL compatible. The discrete lines shall be capable of operating with a device located at a distance of up to 300 feet (total cable length) from the computer.

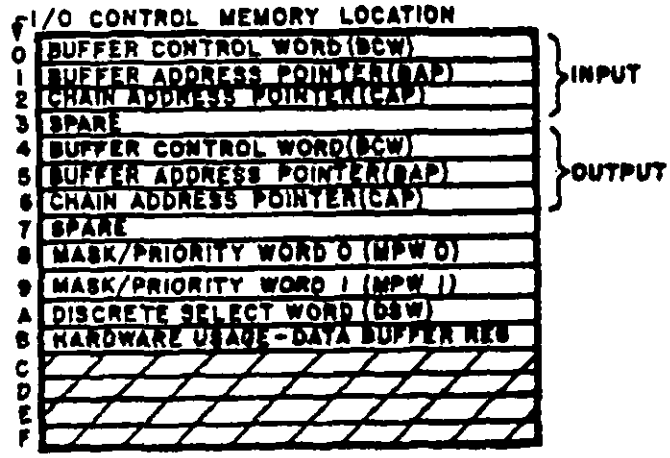
3.6.2.2.1.3.1 DIM control memory. Figure 19 specifies the control memory format and usage for the DIM I/O channel type. Locations 0, 1, and 2 shall be associated with input activity; locations 4, 5, and 6 with output activity; locations 8 and 9 shall define the priorities and mask bits for the INT signals; and location A hexadecimal shall define the enable/disable state of the DIO signal drivers. Figures 19 and 20 define the formats of control memory words with definitions unique to DIM. Locations 0 and 4 are termed "buffer control words" and define the word count for input and output data transfer, respectively. The maximum useful count for input is 4 (64 input discretes) and for outputs is 2 (32 output discretes). The mask/priority words (MPW 0 and MPW 1) shall contain eight 4-bit groups with each group corresponding to an interrupt priority level. Within each group, the most significant bit shall be a mask bit and the other three are a binary code to be selected at that priority level. A logic 1, in a given bit position in the Discrete Select Word (see Figure 19), shall enable the transmitters for the associated group of 4 DIO signals and selects that group as outputs, rather than inputs. DIM channel status word formats are defined in Figure 21.

3.6.2.2.1.3.2 DIM test capabilities. The DIM I/O channel type shall provide complete functional wrap-around testing capability for all 32 of the DIO signals. This testing shall include the transmitter and receiver. The circuit used to implement the DIO signals and DIS signals shall feed directly into an input bus on the DIM module where the DIO receivers also feed. No additional testing of these DIO and DIS receivers shall be provided, but the remainder of the logic associated with them shall be tested via the DIO signal testing. The interrupt logic shall also be entirely testable with the exception of the INT signal receivers. The interrupt actives may be individually set or cleared as a group. During testing, inputs to the DIO and DIS receivers are ignored.

3.6.2.2.1.3.3 DIM cable connector. The DIM shall provide a connector compatible with the DIM cable specified in 3.6.2.5 to allow connection of the 32 additional input dicretes to the computer front panel, when the DIM SRA is used in specific locations as specified in 3.6.2.5.

3.6.2.2.1.4 PROTEUS interface module. The computer shall provide a PROTEUS Interface Module (PIM) which shall contain two PROTEUS Digital Channels (PDC) for full-duplex operation at a peak data transfer rate of 10 megabits per second, as specified in NAC TP-591. The PDC which transmits information shall be defined as the source, and the receiving PDC shall be defined as the sink. The source shall transmit control frames, data words, and a sampling clock to an external sink. The control frames and data words shall be transmitted on the same pair of differential lines, and the clock shall be transmitted on separate pair of differential lines. The external sink shall respond on its own two pairs of differential lines: one shall transmit control frames; and the other shall transmit a sampling clock signal. The internal sink shall receive control frames and data words from an external source and transmit control frame responses back to the source. The control frames and data words shall be received on one pair of differential lines,

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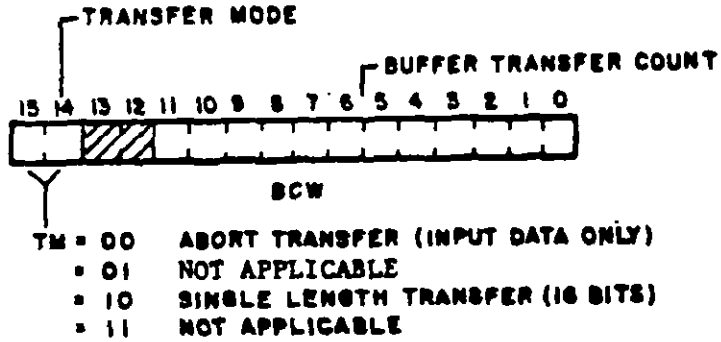
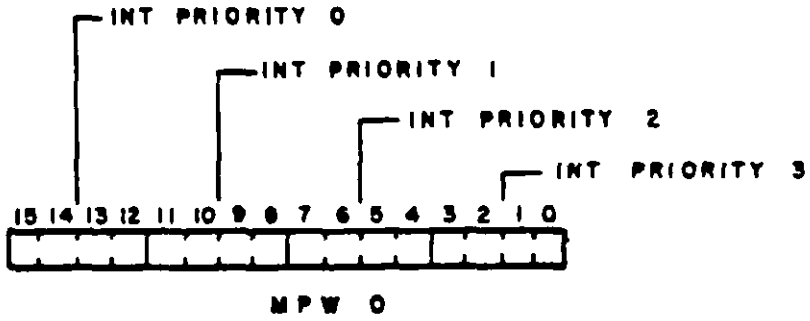
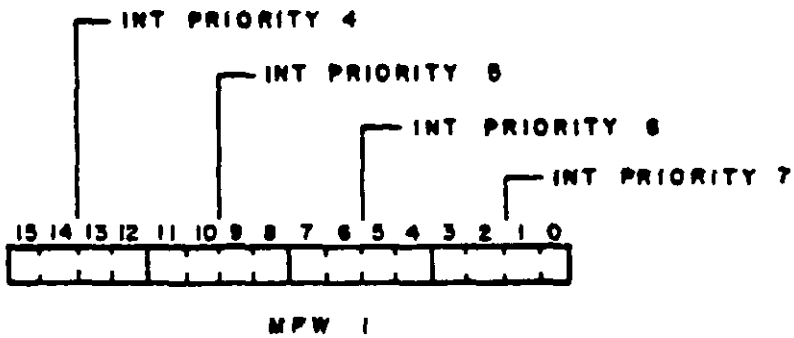


FIGURE 19. DIM control memory usage.

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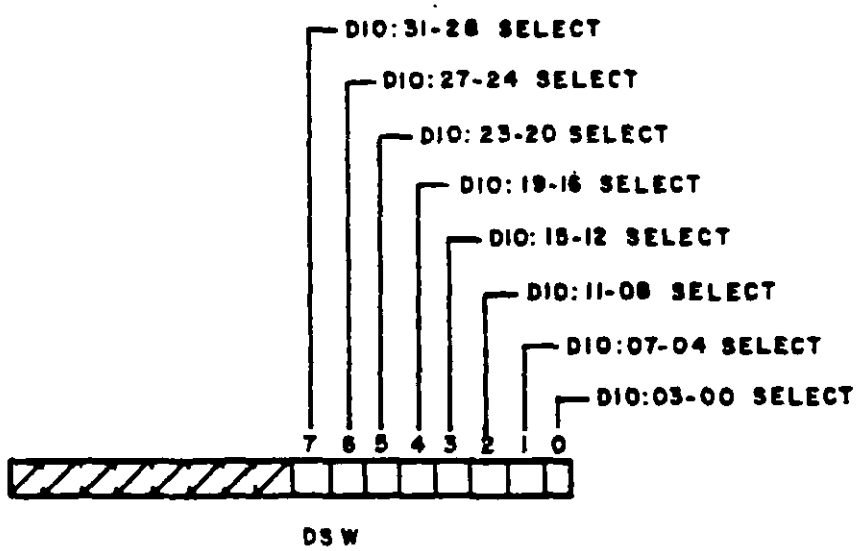
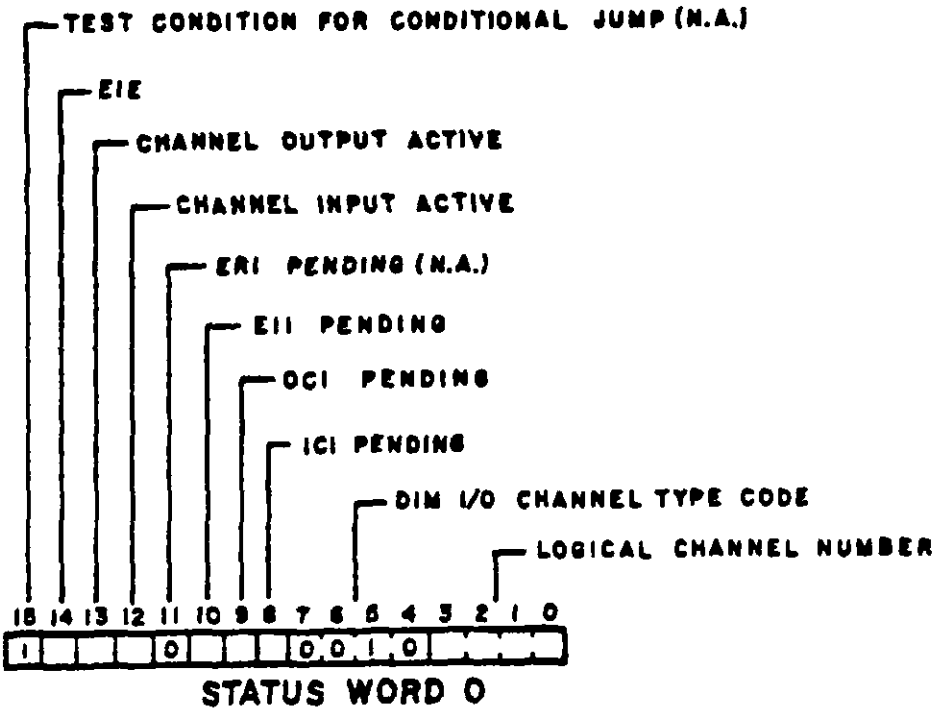


FIGURE 20. DIM MPW and DSW formats.

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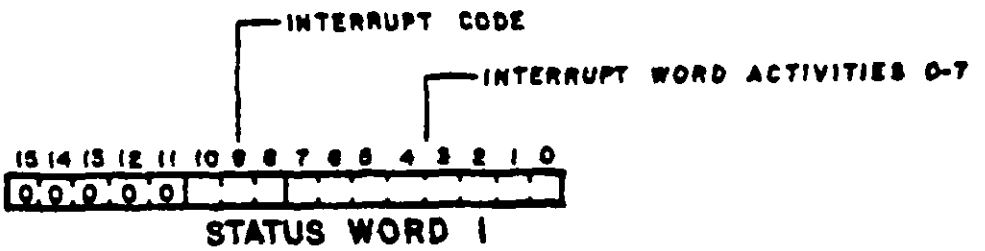


FIGURE 21. DIM channel status word formats.

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and the sampling clock on another pair. The sink shall respond with control frames on one pair of lines and a sampling clock on a separate pair. Messages include 6-bit control frames and 34-bit data words. Each has an odd parity bit length bit to distinguish control frames from data words. A sequence is a series of transmissions and receptions which follow a predetermined channel protocol. There shall be sequences for control, data transfer, error indication, and error correction as shown in Table VIII.

TABLE VIII. PROTEUS channel sequences.

Hex Trans Code	Source to Sink Trans	Sink to Source Trans
0	Control interrupt word active (CIWA)	Request control interrupt word (RCIW)
1	Normal interrupt word active (NIWA)	Request normal interrupt word (RNIW)
2	Control word active (CWA)	Request control word (RCW)
3	Data word active (DWA)	Request data word (RDW)
4	Repeat, error detected (RED)	Repeat, error detected (RED)
5	WAIT	Repeat, sink busy (RSB)
6	STOP	ENDS
7	Sequence error (SEQE)	Sequence error (SEQE)
8-D	Undefined	
E	System reset and load (SRL)	SRL acknowledge (SRLA)
F	Channel reset (CRS)	CRS acknowledge (CRSA)

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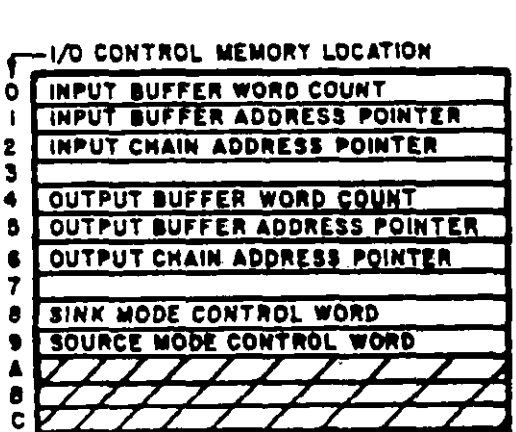
Synchronizing control frames (Trans Code 5) shall be used in word sequences to delay word transfer, while maintaining proper transmission rates. The PIM control memory shall be as defined in Figure 22. Since two 16-bit words are transferred between the PIM and memory for each 32-bit PROTEUS transmission, the word count must be set to an even number. Figure 22 also shows the formats of the sink mode control word and source mode control word. The PIM shall have three channel status words, as defined in Figure 23. The PIM shall be capable of ERI, EII, OCI, and ICI interrupts. Both input and output chains shall be capable of being simultaneously active.

3.6.2.2.1.4.1 PROTEUS test capabilities. The PROTEUS test mode shall provide an internal wrap-around capability which connects the output circuitry to the input circuitry ahead of the transmitters and receivers. This allows loop testing of this PROTEUS module without transmitting to an external sink or receiving from an external source.

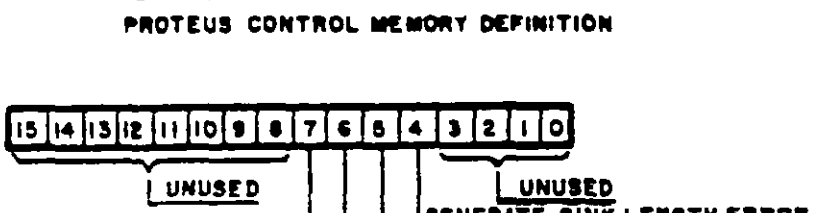
3.6.2.2.1.5 RS-232 interface module. The computer shall provide an EIA-STD-RS-232-C Interface Module (RIM) capable of full-duplex operation at selectable baud rates from 50 to 9600 baud for the asynchronous mode. In synchronous mode, the baud rate is determined by the external equipment and is limited to 20K baud. An input signal shall be provided in the front panel connector to allow external selection of synchronous or asynchronous modes. In the asynchronous mode, two additional modes (AYK-14 mode and UYK-20 mode) shall be provided. These modes shall also be externally selectable via a front panel connector signal. The AYK-14 mode shall provide programmable selection of thirteen baud rates as specified in Table IX and Figure 24. In the UYK-20 mode, any four of ten possible baud rates (Table IX) shall be selectable via signals in the front panel connector. Selection of any of these four baud rates shall be programmable (see Figure 24). Note that the RIM operates in a synchronous (i.e., non-handshaking control) manner during message sequences. Once a message sequence has been initiated, data transfer must be continuous within that message. Therefore, attention must be given to the priority of the RIM channels and their simultaneous utilization with other I/O channel utilization in a given configuration. Both input and output chains shall be capable of being simultaneously active.

3.6.2.2.1.5.1 RS-232 control memory. The RS-232-C control memory shall be as defined by Figure 24. Separate input and output control words shall provide for simultaneously active input and output chain programs. In the monitor mode, the monitor word (location 8) shall be compared to each character as received. Upon a match, the character shall be stored in memory and the monitor flag set. The suppress word (location 9) shall be used to compare to received characters, to prevent storage of these suppressed characters in memory. Figure 24 shows the format of the serial mode information. The RS-232-C channel programs shall use the two channel status words defined in Figure 25. The channel shall be capable of EII, OCI, and ICI interrupts. Also shown is the external interrupt word format. When the interrupt is generated, this word shall be stored in main memory at the assigned external interrupt word for that channel. The EII interrupt shall be generated when the receive line signal detector goes from an on condition to an off condition indicating data is no longer being received and when the ring indicator goes to an on condition indicating that data is being received. The interrupt word format is shown in Figure 25.

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PROTEUS CONTROL MEMORY DEFINITION



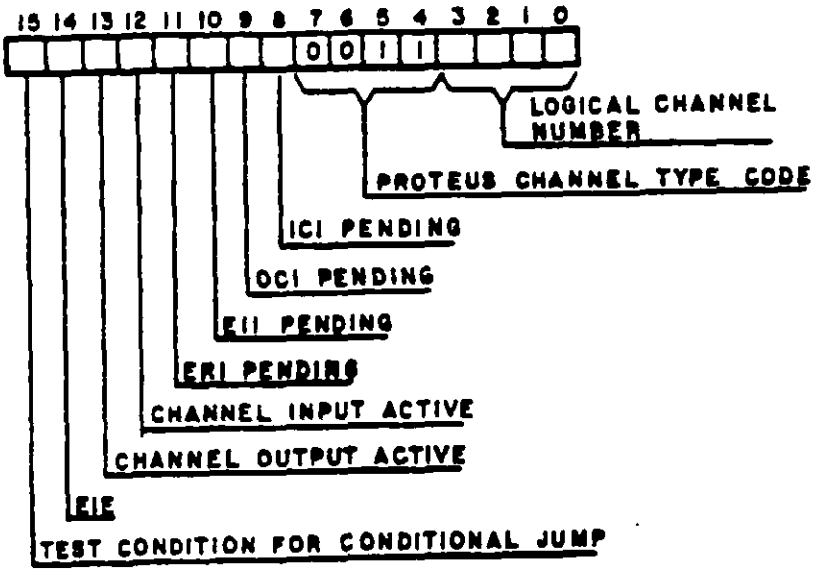
SINK MODE CONTROL WORD



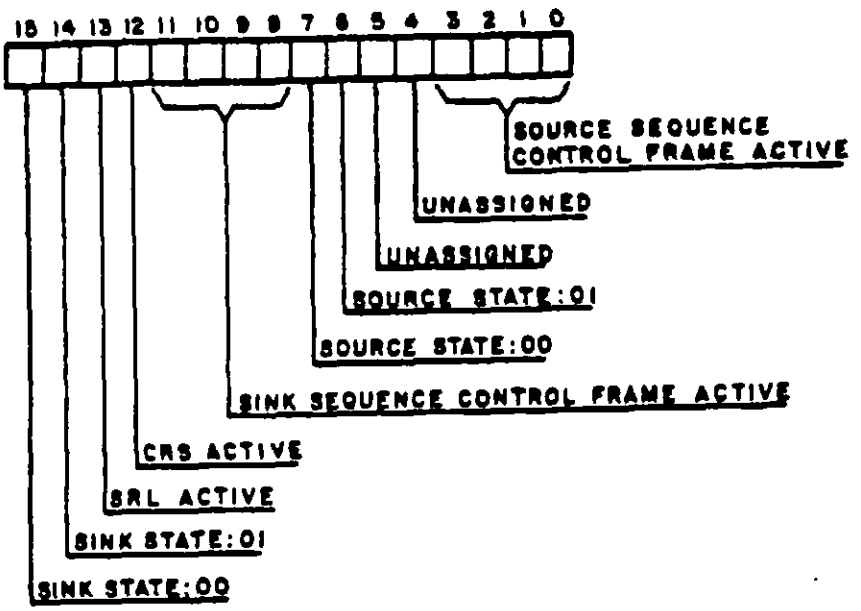
SOURCE MODE CONTROL WORD

FIGURE 22. PROTEUS control memory usage.

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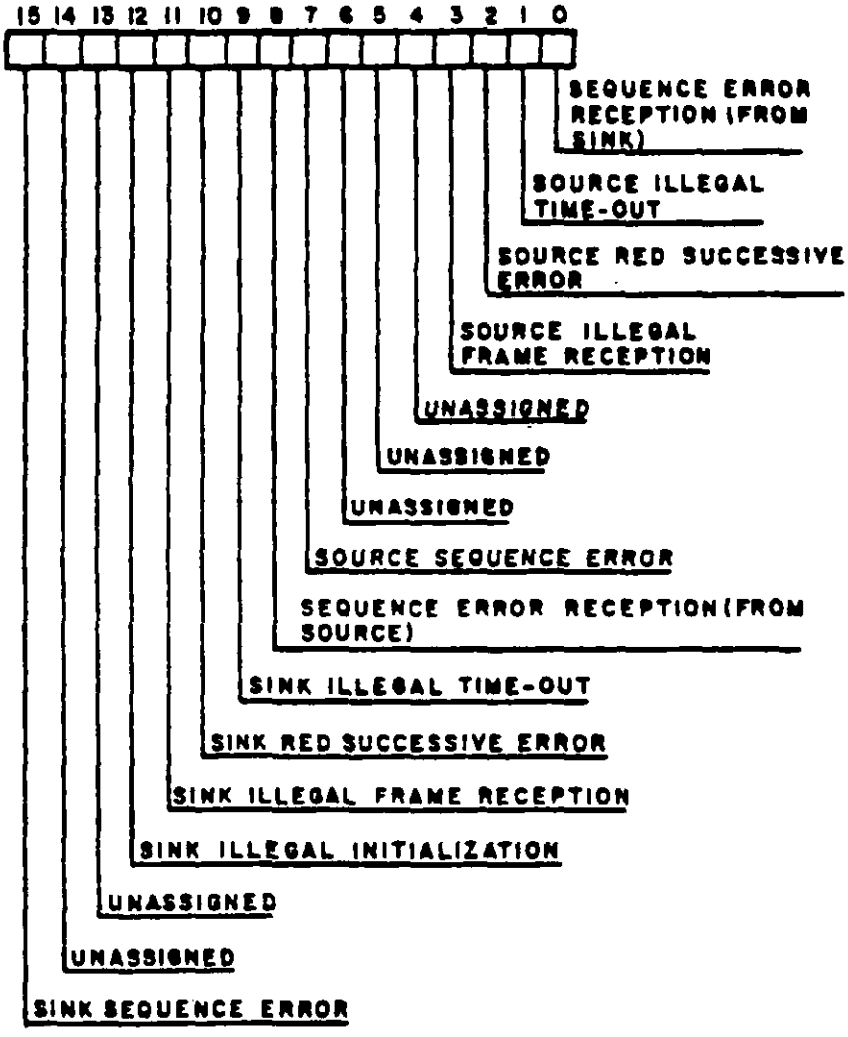
PROTEUS CHANNEL STATUS WORD 0



PROTEUS CHANNEL STATUS WORD 1

FIGURE 23. PROTEUS channel status words (Sheet 1 of 2).

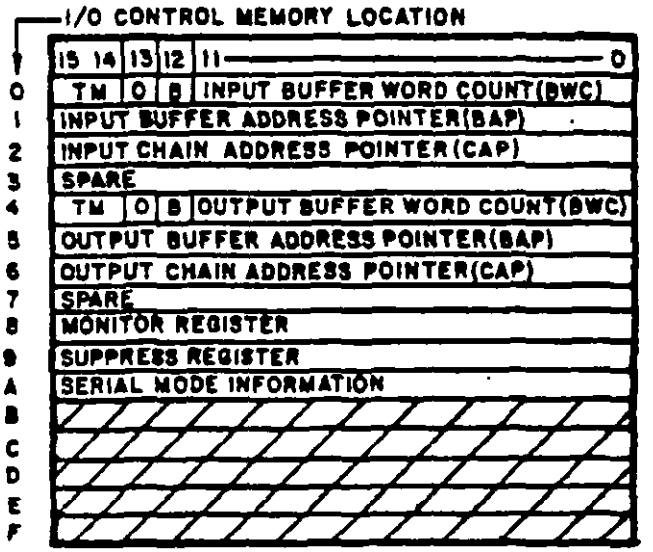
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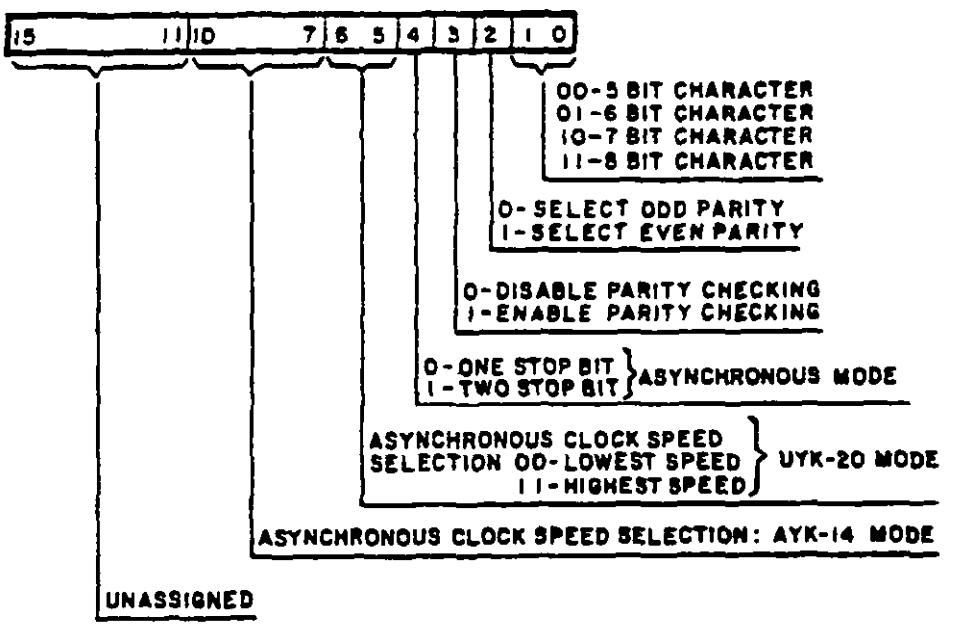
PROTEUS CHANNEL STATUS 2

FIGURE 23. PROTEUS channel status words (Sheet 2 of 2).

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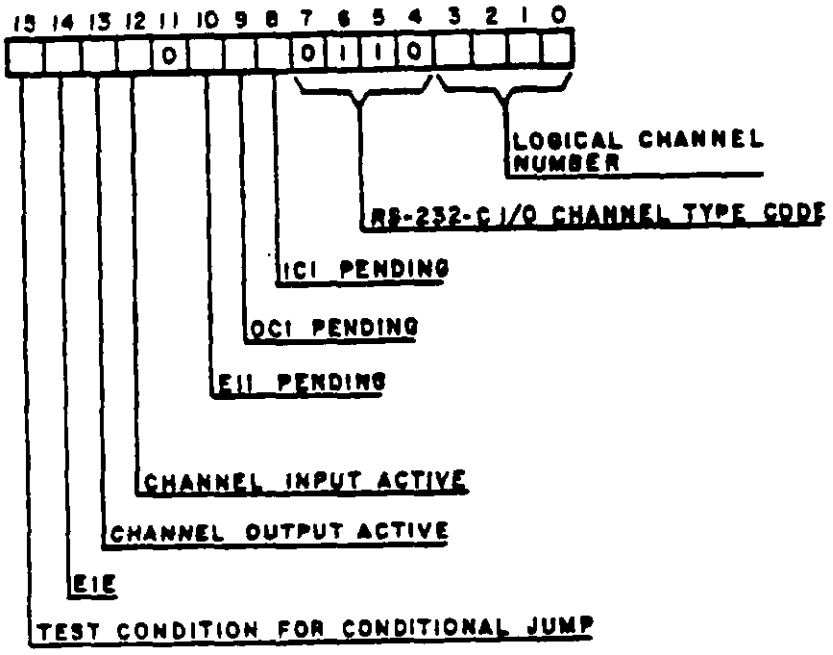
RS-232-C CONTROL MEMORY DEFINITION



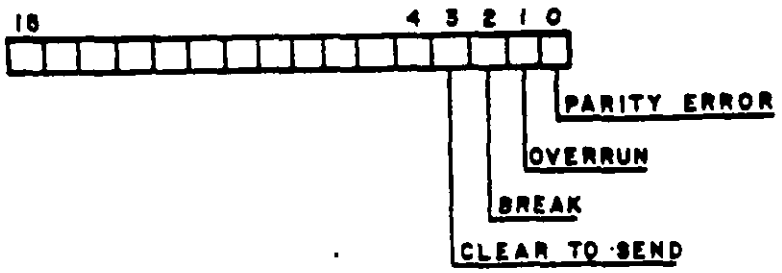
RS-232 SERIAL MODE INFORMATION

FIGURE 24. RS-232-C control memory usage.

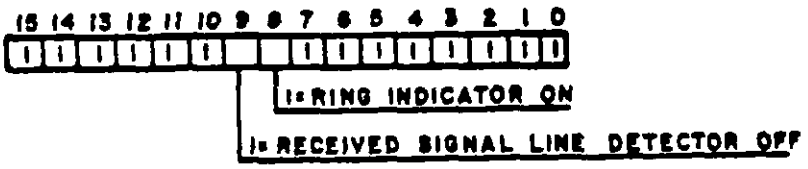
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RS-232-C I/O CHANNEL STATUS WORD 0 FORMAT



RS-232-C I/O CHANNEL STATUS WORD 1 FORMAT



RS-232-C I/O CHANNEL INTERRUPT WORD FORMAT

FIGURE 25. RS-232-C channel status and interrupt word formats.

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TABLE IX. RS-232-C channel speed selection for asychronous operation.

AYK-14 PROGRAMMABLE BAUD RATE

SERIAL MODE BITS				BAUD RATE	UYK-20 Mode Available Baud Rates
10	9	8	7		
0	0	0	0	-	-
0	0	0	1	-	75
0	0	1	0	50	110
0	0	1	1	75	150
0	1	0	0	134.5	300
0	1	0	1	200	600
0	1	1	0	600	1200
0	1	1	1	2400	1800
1	0	0	0	9600	2400
1	0	0	1	4800	4800
1	0	1	0	1800	9600
1	0	1	1	1200	
1	1	0	0	2400	
1	1	0	1	300	
1	1	1	0	150	
1	1	1	1	110	

3.6.2.2.1.5.2 RS-232 test capabilities. The RS-232 I/O channel shall provide for a wrap-around test capability. This wrap-around shall be internal to the RS-232 I/O channel and shall connect the data transmitter input to the circuitry normally fed by the data receiver output circuitry. It shall exercise the data handling logic without receiving data from external equipment.

3.6.2.2.1.6 Special parallel input (PIC), parallel output (POC), and serial output (SOC) channels. The computer shall provide a special PIC/POC/SOC SRA (PPSM). Two logical channels shall be required for this SRA - one for the PIC/POC and one for the SOC. The PIC/POC channel shall each be capable of operating at two different voltage levels as specified in 3.6.2.2.1.6.4. These levels shall be determined via back-panel wiring in a given configuration. Each of these two logical channels shall provide the capabilities specified in 3.6.2.2.1 except that item (1) which defines I/O channel type SRA interchangeability shall not be required and there shall be a capability to clear each of the parallel input and parallel output interfaces separately. Channel logical number and priority number shall be defined identically for

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the PIC/POC channel and for the SOC channel. In addition, the channel numbers for the PIC/POC channel and the SOC channel shall be fixed relative to each other.

3.6.2.2.1.6.1 Parallel input channel (PIC).

3.6.2.2.1.6.1.1 Parallel input (PIC) characteristics. Each PIC/POC/SOC SRA shall have one parallel input (PIC) capable of receiving a data word of 32-bits in length. PIC data input transfer activity shall be initiated via an INITIATE I/O TRANSFER instruction encountered in an I/O channel program. The PIC I/O channel program and input data buffer shall be controlled by locations 0 through 2 of the PIC/POC channel control memory. Refer to Figure 26. PIC signal levels and characteristics shall conform to the limits of 3.6.2.2.1.6.4.

a. The PIC I/O channel program shall be capable of generating an ICI interrupt.

b. The PIC data input control logic on the PIC/POC/SOC SRA shall be of the request-acknowledge type, with a maximum word transfer rate of TBD words per second. The actual rate, however, shall be established by the external equipment and internal control logic to prevent input data loss or input of erroneous data.

c. The PIC shall be capable of being reset (or cleared) at any time via the execution of an IOCR processor instruction. After this clear operation, the PIC may be orderly re-initiated to execute a new I/O channel program.

d. When input transfer activity on the PIC is active, the channel stop (CBX) signal associated with that PIC shall be false "high"; when the data input transfer activity is not active, the channel stop (CBX) signal associated with the PIC shall be true "low".

3.6.2.2.1.6.1.2 Parallel input timing and control signals. The timing of the parallel input channels shall be as shown in Figure 27. (The signal definition X represents the associated channel, i.e., A through C. The signal definition YY represents the associated channel data bit line 0 through 31.) Signal functions shall be as defined below:

a. Data input strobe (IAX) - The data input strobe (from the computer) shall go "low" when true, indicating that the channel is ready to accept data on the input data line. This signal shall go high after receipt of response (IBX) from the device attached to the channel and shall be "low" true when data input transfer activity on the PIC is not active as shown in Figure 27.

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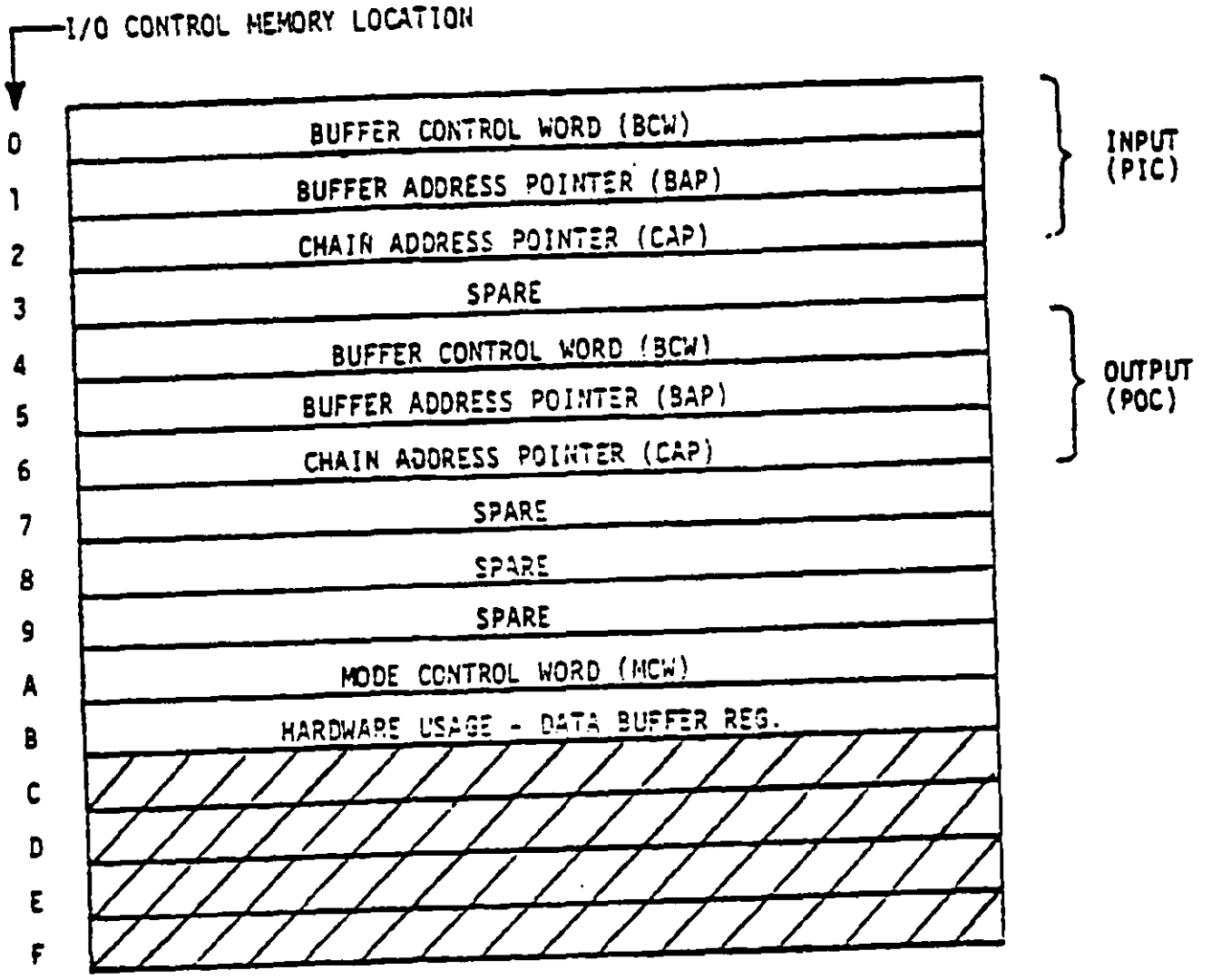
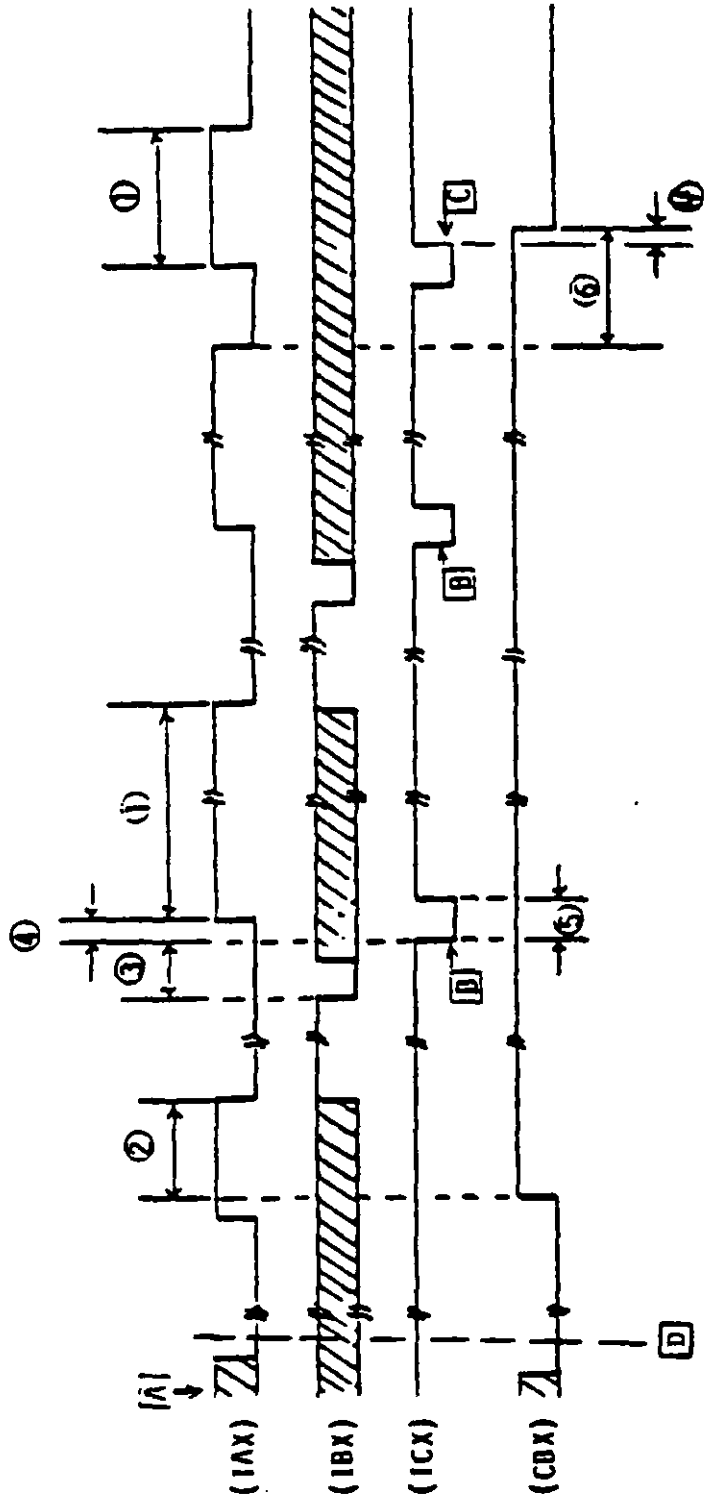


FIGURE 26. PIC/POC control memory definition map.

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Notes:
- A / Shown from hardware master clear.
- B / Input of two words is shown in this diagram.
- C / After the last word in the buffer is input, and addition (IAX)-(ICX) sequence occurs, regardless of (IBX).
- D / The combination of signal levels along this dashed line defines the normal idle state.

Times
- (1) / ≥ 3250 ns
- (2) / ≥ 1000 ns
- (3) / 450 ns to 1000 ns
- (4) / 125 ns nominal
- (5) / 500 ns nominal
- (6) / ≥ 750 ns

FIGURE 27. PIC timing.

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b. Response (IBX) - This line (to the computer) shall normally be high, and shall be true "low" when the attached device has valid data for the channel. This line shall become true only after receipt of a data input strobe (IAX).

c. Response acknowledge (ICX) - This line from the computer shall normally be high, and shall become true "low" after the channel receives a response (IBX) from the transmitting device as shown in Figure 27.

d. Input/data line (IDXYY) - The input data lines shall normally be high, and each line shall be true "low" when the particular input line is true prior to the receipt of the response (IBX) from the device. When the response (IBX) becomes true, the input data shall be stable.

e. Zero indicator line (CBX) - When all the data has been received (chain is complete), this line shall be true "low". This signal from the computer shall be false "high" when all the data has not been received (data input transfer activity is not complete).

3.6.2.2.1.6.2 Parallel output (POC).

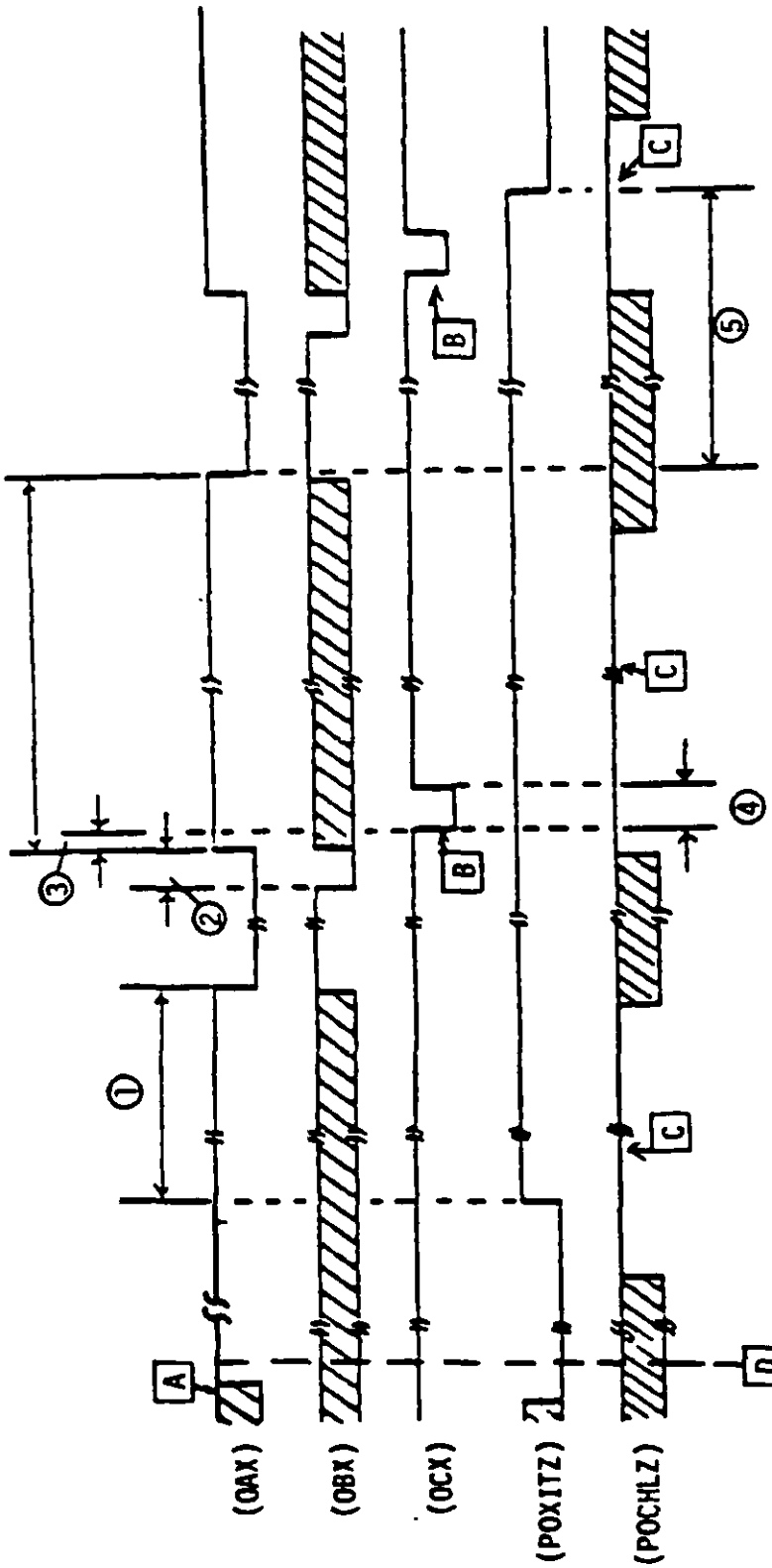
3.6.2.2.1.6.2.1 Parallel output (POC) characteristics. Each PIC/POC/SOC SRA shall have one parallel output (POC) capable of transmitting a data word of 32 bits in length. POC data output transfer activity shall be initiated via an INITIATE I/O TRANSFER instruction encountered on an I/O channel program. The POC I/O channel program and output data buffer shall be controlled by locations 4 through 6 of the PIC/POC channel Control Memory. (See Figure 28) POC signal levels, shall conform to the limits specified in 3.6.2.2.1.6.4.

a. The POC I/O channel program shall be capable of generating an OCI interrupt.

b. The POC data output control logic on the PIC/POC/SOC SRA shall be of the request-acknowledge type, with a maximum word transfer rate of TBD words per second. The actual rate, however, shall be established by the external equipment and the internal control logic to prevent output data loss or output of erroneous data.

c. The POC shall be capable of being reset or cleared at any time via the execution of an IOCR processor instruction. After this clear operation, the POC may be orderly re-initiated to execute a new I/O channel program.

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Notes

A Shown from hardware master clear

B Output of two 32-bit words is shown in this diagram.

C The (POCHLZ) signal is sampled prior to the rising edge of POXITZ and periodically while (OAX) is "high".

D The combination of signal levels along this dashed line defines the normal idle state.

Times

① ≥1250 ns

② 325 ns to 925 ns

③ 125 ns nominal

④ 500 ns nominal

⑤ ≥3000 ns

FIGURE 28. POC timing.

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d. When data output transfer activity on the POC is active, the channel zero indicator line (POXITZ) signal associated with that POC shall be false "high"; when the data output transfer activity is not active, the channel zero indicator line (POXITZ) signal associated with that POC shall be true "low"

3.6.2.2.1.6.2.2 Parallel output channel timing and control signals. The timing of the parallel output channels shall be as shown in Figure 28. (The signal definition X represents the associated channel i.e., A through C. The signal definition YY represents the associated channel data bit lines 0 through 31.) Signal functions shall be as defined below:

a. Data output strobe (OAX) - The data output strobe (from the computer) shall go true "low" to indicate valid data on the output lines (ODXYY). When the device connected on the channel sends a response (OXB), this line shall go "high" and shall remain "high" when the data transfer output activity on the POC is not active as shown in Figure 28.

b. Response (OXB) - This response line to the computer shall normally be high, and shall become true "low" when the device attached to the channel can accept the data on the output lines.

c. Response acknowledge (OCX) - This line (from the computer) shall normally be high, and shall become true "low" after the channel receives a response (OBX) from the device attached to the channel.

d. Output data lines (ODXYY) - The output data lines shall be true "low" and contain valid data during the data output strobe time (OAX).

e. Zero indicator line (POXITZ) - When all the data has been transmitted (chain is complete), this line shall be true "low". This signal from the computer shall be false "high" when all the data has not been transmitted (data output transfer activity is not complete).

f. Channel halt (POXHLZ) - This signal shall normally be false "high", and shall go true "low" when the receiving device desires to terminate the transmittal of data. An EII interrupt shall be generated to the program. An EIE shall also be implemented to enable this EII interrupt.

3.6.2.2.1.6.3 Serial output channel (SOC).

3.6.2.2.1.6.3.1 Serial output channel characteristics. Each PIC/POC/SOC SRA shall have one (1) serial output (SOC) capable of transmitting serial NRZ data at one of two rates, either 200 KHz or 1 MHz, selectable under program

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control. The SOC shall be capable of transmitting a data transmission consisting of a single word of 16 bits, followed by any number of 16-bit words. SOC data output transfer activity shall be initiated via an INITIATE I/O TRANSFER instruction encountered in an I/O channel program. The SOC Control Memory definition shall be as defined in Figure 29. The SOC portion of the PPSM SRA shall provide for both single-ended and differential signals on its two output signals and its input signal.

a. The SOC shall be capable of generating an OCI interrupt.

b. The SOC gated output clock (SOA) and output data (SOD) signals may experience periods of suspension between data words transmitted due to regulating control logic on the PIC/POC/SOC SRA (i.e., channel traffic and conflicts on internal computer buses) or due to a channel halt (SOALE) signal from the external receiving equipment. These controls shall be provided to prevent output data loss or output of erroneous data.

c. The SOC mode control word (MCW) shall have the format shown in Figure 30 and shall define the transmission rate.

3.6.2.2.1.6.3.2 *Serial output channel timing control signals.* The timing of the SOC channels shall be as shown in Figure 31. Signal functions shall be as specified below:

a. Gated output clock (SOA) - The gated clock shall go "low" when true, indicating that valid data is available on the output data line. The gated clock shall be coincident with leading edge of the data. The clock shall maintain a 45 percent to 55 percent of period duty cycle.

b. Output data (SOD) - The output data line shall normally be high, and shall go "low" when true and contain valid data during the gated output clock time SOA.

c. Channel halt (SOALE) - This signal shall normally be false "high", and shall go true "low" when the associated external device desires to stop the transmittal of data. Serial transmission of data (SOD) and the grated clock SOA shall be suspended at the end of the transmission of the current word. Normal transmission shall resume when SOALE returns to a false "high" state. An interrupt shall not be generated for this condition.

3.6.2.2.1.6.4 PIC/POC/SOC signal characteristics. The signal characteristics of the PIC/POC/SOC channels shall be as listed in Table X.

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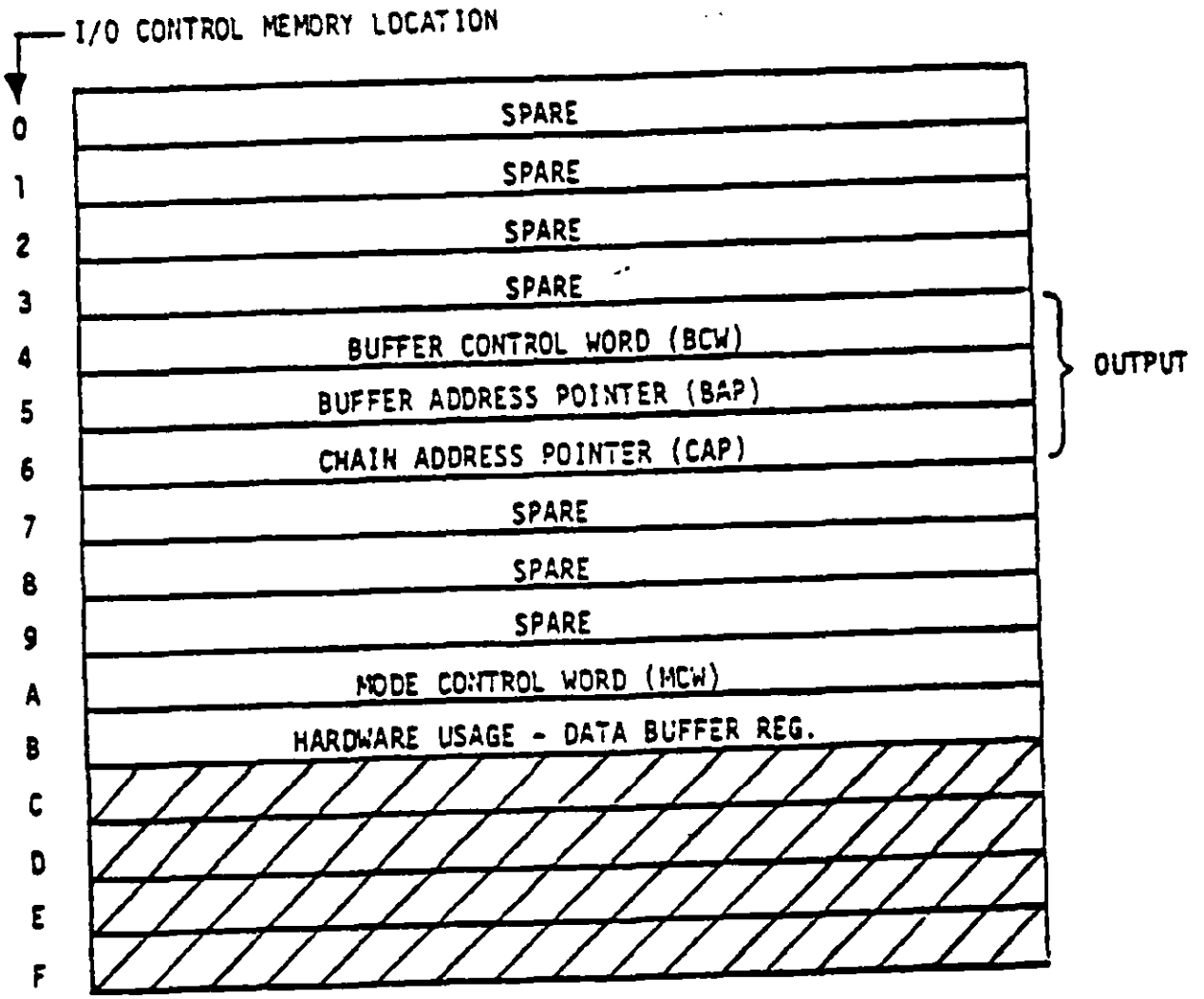
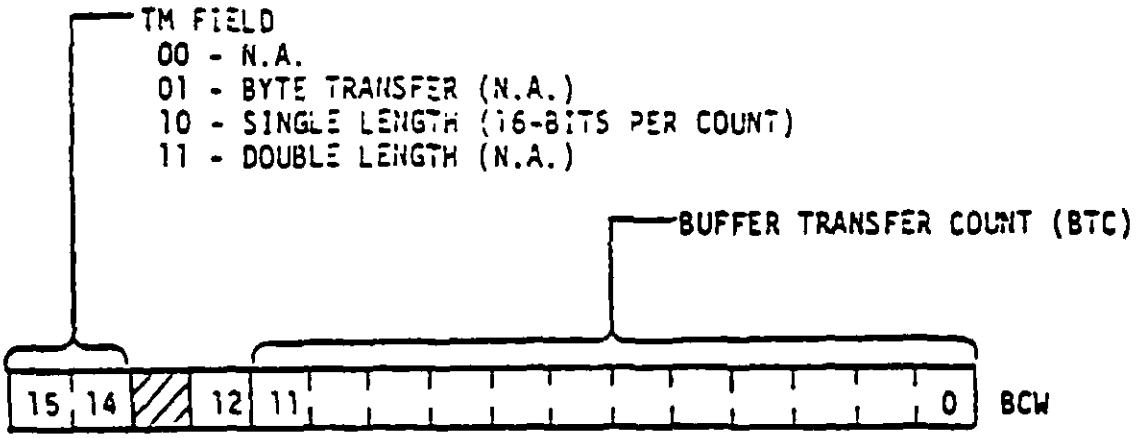


FIGURE 29. SOC control memory definition map.

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SOC Buffer Control Word

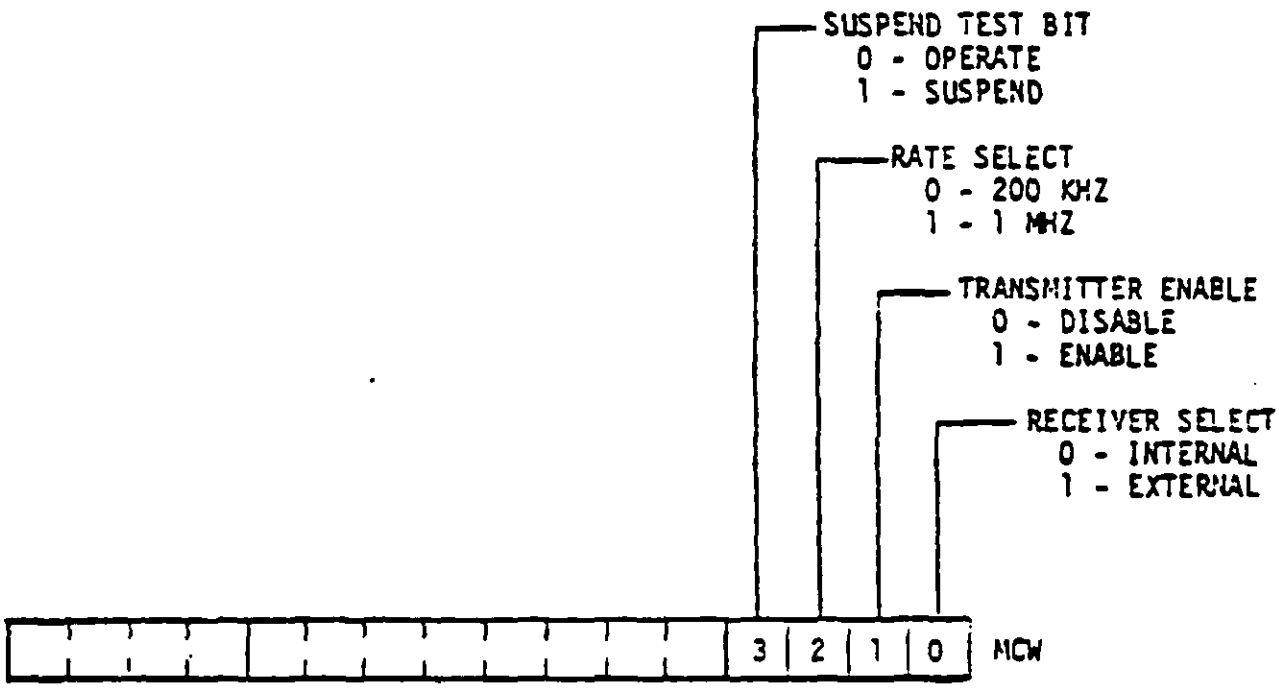
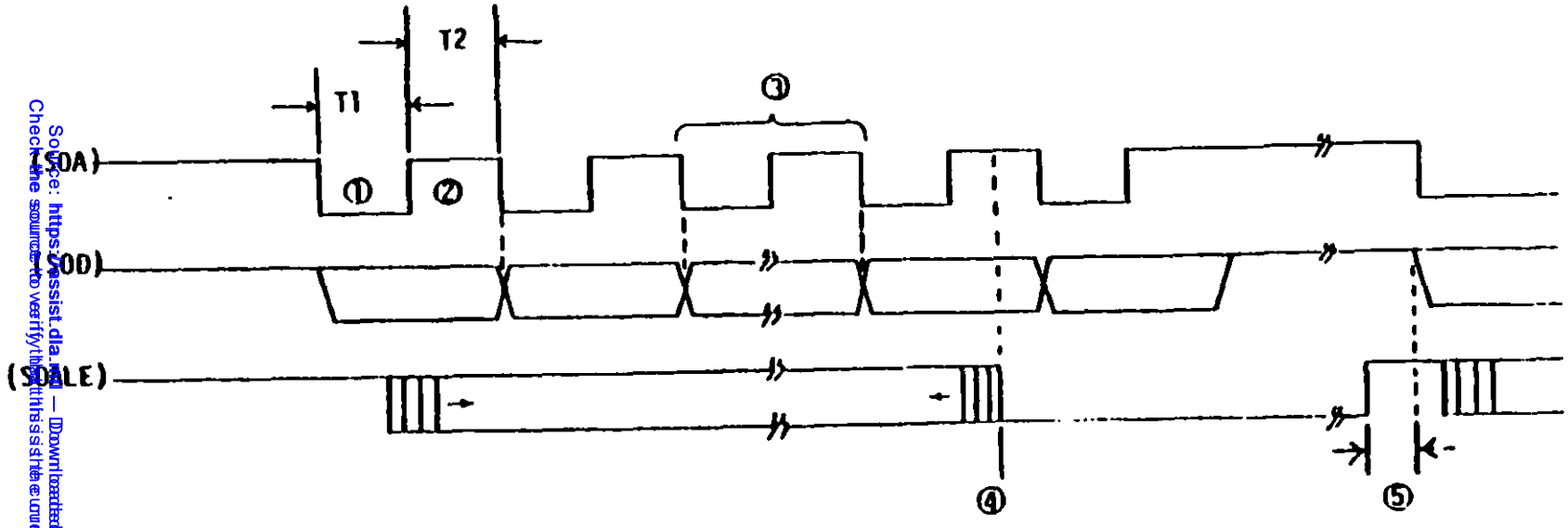


FIGURE 30. SOC control words.

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NOTES:

①/ At 1 Mlz, T1 = 0.5 microseconds; at 200 Klz, T1 = 2.5 microseconds.

②/ T2 = T1

③/ Data (SOD) switches at T2 falling edges.

④/ Suspend line (SOALE) must be low by the midpoint of T2 in the second last bit time to insure suspend action.

⑤/ If additional words are pending, the next bit time will begin 0.5 ±2.5 microseconds (at 200 Klz rate), or 1.9 ±0.5 microseconds (at 1 Mlz rate) after the suspend line is high.

FIGURE 31. SOC timing.

TABLE X. PIC/POC/SOC channels signal characteristics.

+15 V Interface	Min	Max
Driver Characteristics		
Output Logic 0		
No line load (open circuit)	+13.5V	+15.75V
Sourcing 8 ma to line	+10.0V	+12.5V
Output Logic 1		
No line load (open circuit)	+ 0.0V	+ 0.4V
Sinking 8 ma from line	+ 0.1V	+ 0.65V
Receiver Characteristics		
Input Logic 0 threshold	+ 8.0V	----
Input Logic 1 threshold	----	+ 4.0V
Input voltage	- 1.0V	+ 15.0V
Input DC impedance	2KΩ	----
Input AC impedance	100Ω	----
+5 Interface		
Driver Characteristics		
Output Logic 0		
No line load (open circuit)	+4.5V	+5.5V
Sourcing 4 ma to line	+2.7V	+3.6V
1/ Sourcing 40 ma to line	+2.5V	----
Output Logic 1		
No line load (open circuit)	0.0V	+0.4V
Sinking 20 ma from line	+0.1V	+0.5V
1/ Sinking 40 ma from line	----	+0.5V
Receiver Characteristics		
Input Logic 0 threshold	+ 2.3V	----
Input Logic 1 threshold	----	+1.0V
Input voltage	- 1.0V	+ 5.5V
Input DC impedance	2KΩ	----
Input AC impedance	100Ω	----

1/ SOC +5V Driver Characteristics only.

3.6.2.2.1.6.5 PIC/POC/SOC testing. The PIC/POC I/O channels shall be capable of being internally loop-tested under software control. In the test mode, the POC outputs (prior to the interface driver) shall feed the PIC input register (instead of the interface receiver) to test all internal data and control paths. In this test mode, the control signal transmitter and receivers shall be in their normal idle state (see Figures 27 and 28). The SOC I/O channels shall be capable of being internally loop tested under software control. The serial bit stream transmitted shall be internally recirculated for reading and checking, via software, to test all internal data and control paths. When the SOC channel is in the test mode, the transmitters and receivers shall be disabled.

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3.6.2.2.1.6.6 PIC/POC/SOC status and control words. PIC/POC Buffer and Mode Control Words are shown in Figure 32. PPSM Status Words are shown in Figure 33 and 34.

3.6.2.2.1.7 Special DIM I/O and interrupt channel (DIOM). The computer shall provide a single SRA special discrete I/O and interrupt channel module. This channel shall provide the capabilities specified in 3.6.2.2.1, except that item which defines I/O channel type SRA interchangeability shall not be required. The channel logical number and priority number shall also be hardwired.

3.6.2.2.1.7.1 Discrete inputs.

3.6.2.2.1.7.1.1 Discrete input characteristics. The DIOM shall have the capability of accepting 48 discrete inputs. These discrete inputs shall be "leading edge detected" on the logic 0 to logic 1 transition by capture logic on the DIOM. Minimum pulse width shall be 2.5 microseconds on these discrete inputs to ensure capture. The logic level of each of these 48 discrete inputs shall be readable under program control in maximum groups of 16. In addition, captured values associated with each of these 48 discrete inputs shall be readable under program control in maximum groups of 16, and shall be capable of being cleared (reset) as a group or individually under software control. This data input transfer activity of discrete input levels and capture values shall be initiated via an INITIATE I/O TRANSFER instruction encountered in an I/O channel program.

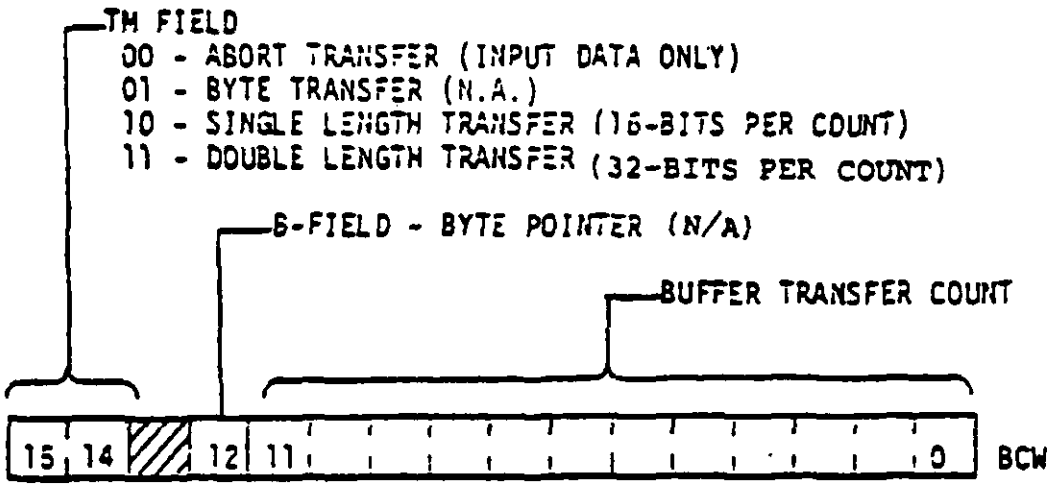
3.6.2.2.1.7.1.2 Discrete input signal characteristics. The signal characteristics of the discrete inputs shall be as follows:

	Logic 0	Logic 1
a. Inputs 0 - 23	+5.2 V max +2.3 V min	+0.65 V max -0.2 V min
b. Inputs 24 - 47	+15.0 V max + 9.0 V min	+1.0 V max -0.5 V min

3.6.2.2.1.7.2 Discrete outputs.

3.6.2.2.1.7.2.1 Discrete output characteristics. The computer shall have the capability of transmitting 144 discrete outputs. The discrete outputs shall be capable of being set and reset individually under program control. The discrete output channel shall have a minimum transfer rate of 100K bits per second. The discrete outputs will remain in the command state until altered under program control. The computer shall have the capability of setting/resetting the discretes individually. The actual status of the discrete outputs shall be capable of being read under program control in groups of 16. Discretes shall be organized in memory in groups of 16 bits and transferred to the SRA in that manner. Any of the nine groups of 16 outputs shall be capable of being read individually. Three additional discretes shall be implemented on this module. One shall provide GO/NO-GO status as specified in 3.4.19.12 and 3.4.7.2. This discrete shall be of the +15.0V maximum type. The second additional discrete shall provide a thermal overtemperature (95°C ±

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PIC/POC Buffer Control Word

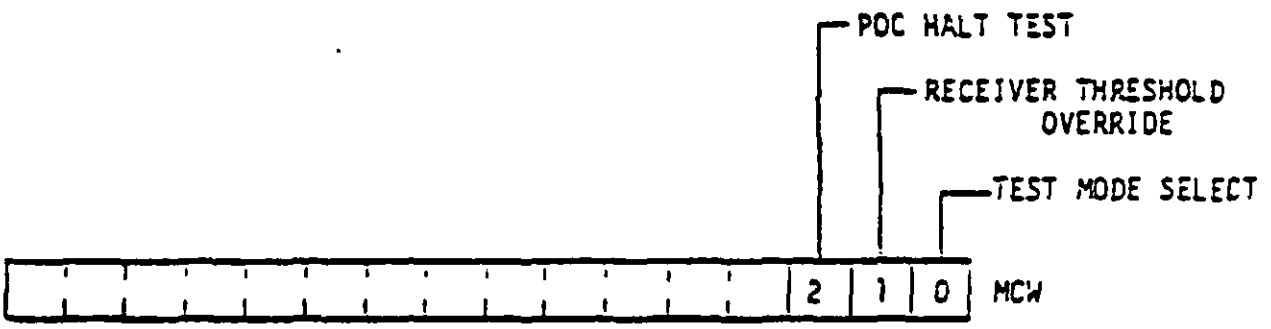
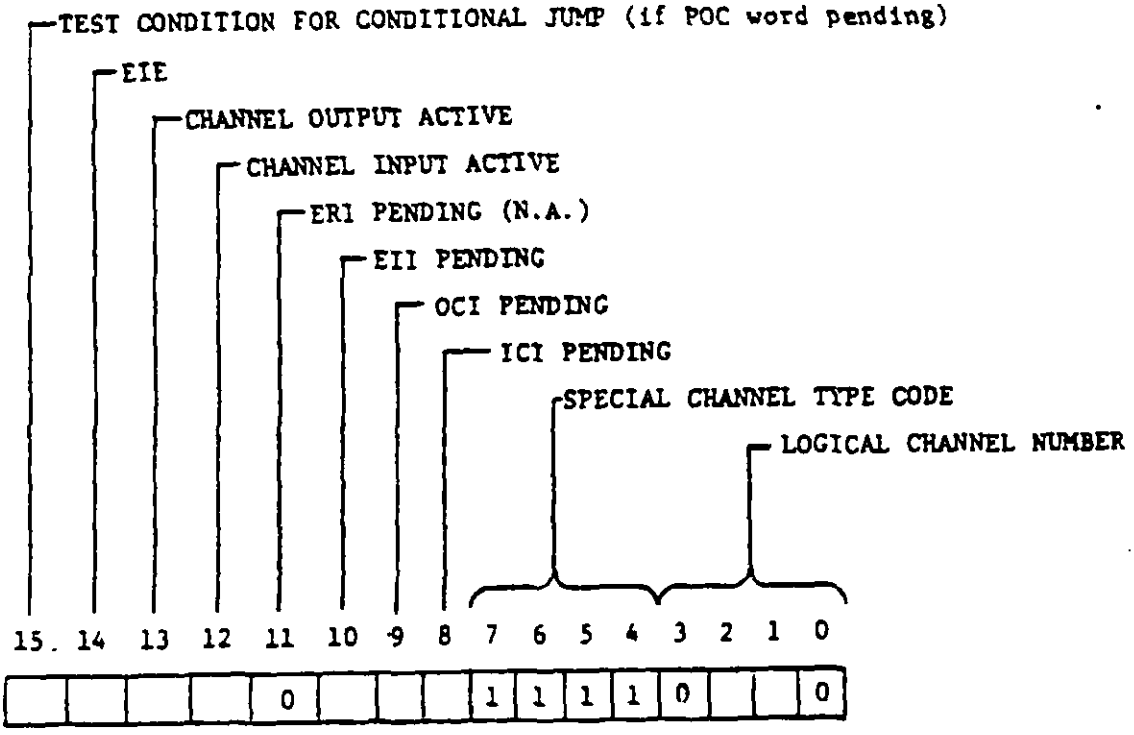


FIGURE 32. PIC/POC control words.

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PIC/POC I/O Channel Status Word 0 Format

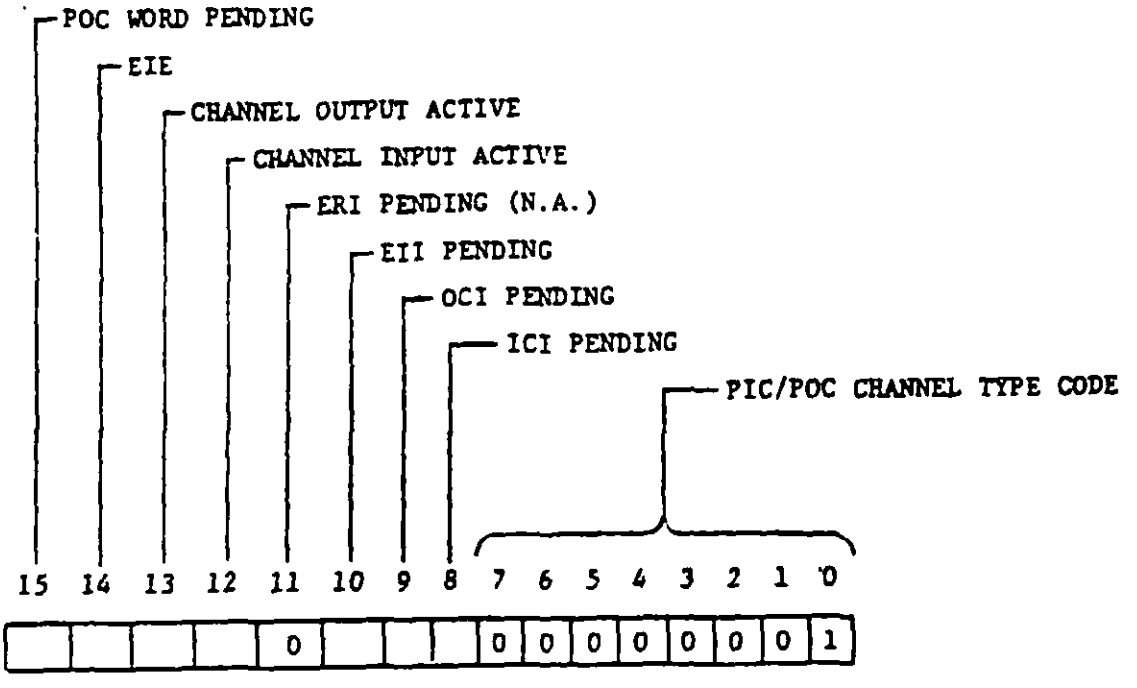
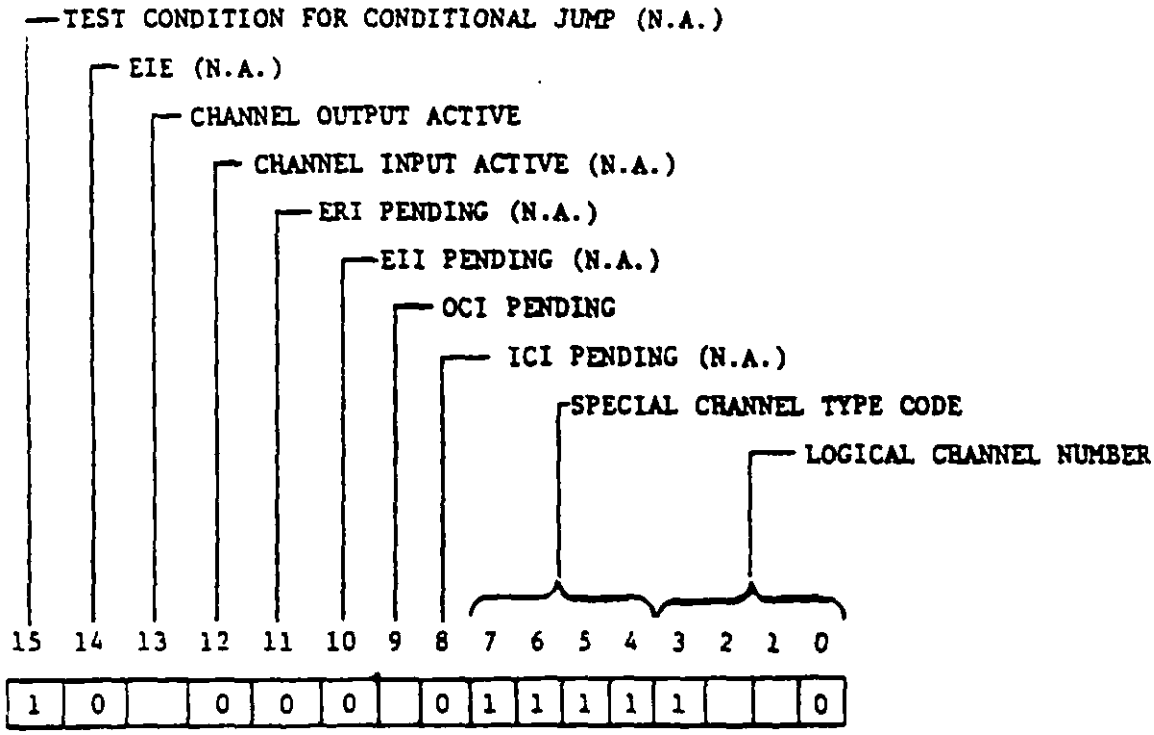


FIGURE 33. PIC/POC I/O channel status words.

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SOC I/O Channel Status Word 0 Format

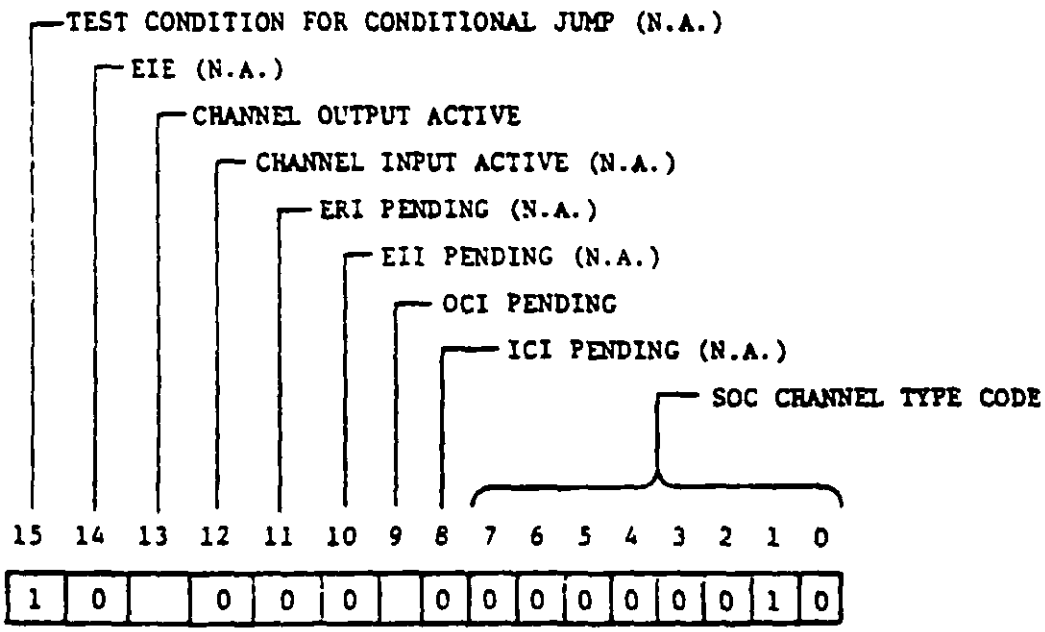


FIGURE 34. SOC I/O channel status words.

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5°C) indication as specified in 3.4.19.8 and 3.4.10. The output of this discrete shall be capable of sinking 250 mA steady state from a +28V source when the overtemperature condition exists and shall present a high impedance under normal temperature conditions. The third discrete shall provide a switch closure input for the System Reset input discrete specified in 3.4.19.3.

3.6.2.2.1.7.2.2 Discrete output signal characteristics. The signal characteristics of the discrete outputs shall be as listed in Table XI.

TABLE XI. Signal characteristics of the discrete outputs.

Output 0 - 113 +5 V Interface	Min	Max
Driver Characteristics Output Logic 0 No line load (open circuit) Sourcing 0.75 ma to line Output Logic 1 No line load (open circuit) Sinking 8 ma =from line	 +3.3V +2.4V +0.10V +0.10V	 +5.5V +4.2V +0.4V +0.5V
Output 114 - 143 +15V Interface		
Driver Characteristics Output Logic 0 No line load (open circuit) Sourcing 8 ma to line Output Logic 1 No line load (open circuit) Sinking 8 ma from line	 +13.5V +10.0V 0.0V + 0.1V	 +15.75V +12.5V + 0.4V + 0.65V

3.6.2.2.1.7.3 External interrupts. The DIOM shall provide the capability of generating an external interrupt (EII) upon the occurrence of any of the 48 discrete inputs. The capture values of the 48 discrete inputs (see 3.6.2.2.1.7.1.1) shall be organized into 48 individual levels of fixed priority with discrete 0 highest priority and discrete 47 lowest priority. Each of these capture values shall have associated with it an interrupt mask bit which shall be definable under program control. In addition to the 48 individual interrupt mask bits, an External Interrupt Enable (EIE) bit shall be provided to enable/disable the entire group of 48 discrete inputs into the interrupt system. Also, a channel Class III mask bit shall be implemented in the DIOM to mask a pending EII. When a given interrupt mask bit is set and the associated capture value is set, that associated discrete input shall become an active interrupt into the DIOM interrupt logic. When a given mask bit is cleared, no interrupt shall be generated by that associated discrete

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put regardless of its logic level or capture value. If a given interrupt sk bit is cleared when the associated discrete input occurs (i.e., a logic 0 logic 1 transition occurs and causes the associated capture value to set) d the mask bit is later set, that discrete input shall then become an active terrupt into the DIOM interrupt logic. If the DIOM EIE is set and one or re interrupts are active, and EII shall be generated along with an interrupt rd defining the highest priority group of active interrupt(s).

3.6.2.2.1.7.4 DIOM testing. The DIOM channel shall also provide for ternal wrap-around testing of all input discretes, output discretes, and terrupt logic paths internal to the module. This testing will not disable tput transmitter nor test the input receivers during this test mode.

3.6.2.2.1.7.5 DIOM control memory. Figures 35, 36 and 37 define the ntrol memory format and usage for the special DIOM I/O channel type.

3.6.2.2.1.7.6 DIOM status word usage. Figures 38 and 39 define DIOM atus words zero and one.

3.6.2.2.2 BUSEXTEND module (BEM). The computer shall provide parallel s extension channels for the IOBUS, Memory Buses, CPUBUS and EBUS, ysically located on a single SRA. The BEM shall provide electrical olation between the computer mainframe and points outside the computer. The M shall be capable of operating with an external device located at a stance of up to fifteen feet (total cable length) from the computer. All ltage levels shall be TTL compatible and employ differential line ivers/receivers for all input/output lines.

3.6.2.3 Memory subsystem. The memory subsystem shall provide main orage for programmed instructions and data. It shall also provide access d control interfacing between the memory and the IOBUS and CPUBUS as shown Figure 40. The subsystem shall consist of combinations of the following ve subelements:

a. Core Memory Module (CMM) - Main Memory

b. Semiconductor Memory Module (SMM) - Main Memory

c. Memory Controller (MCM)

d. Read/Write Expansion Memory (RXM)

e. RXM Adapter

3.6.2.3.1 Main memory modules. The main memory may be comprised of core d semiconductor modules.

3.6.2.3.1.1 Core memory modules (CMM). The 32K Core Memory Module MM-32) shall be 32,768 words by 18 bits, random access, non-volatile, and ectrically alterable. The 64K Core Memory Module (CMM-64) shall be 65,536 rds by 18 bits, random access, non-volatile, and electrically alterable. e CMM block diagram is shown in Figure 41.

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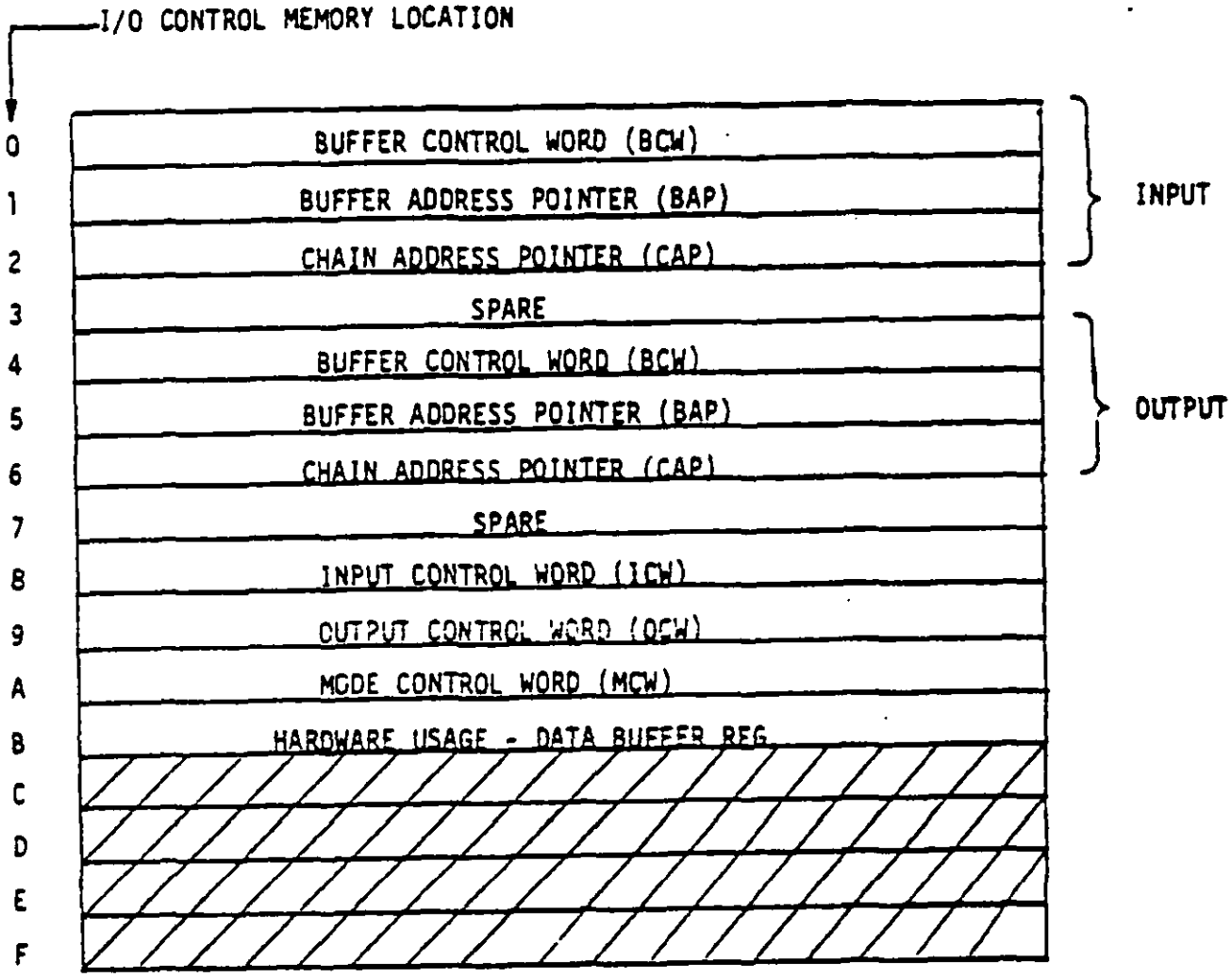
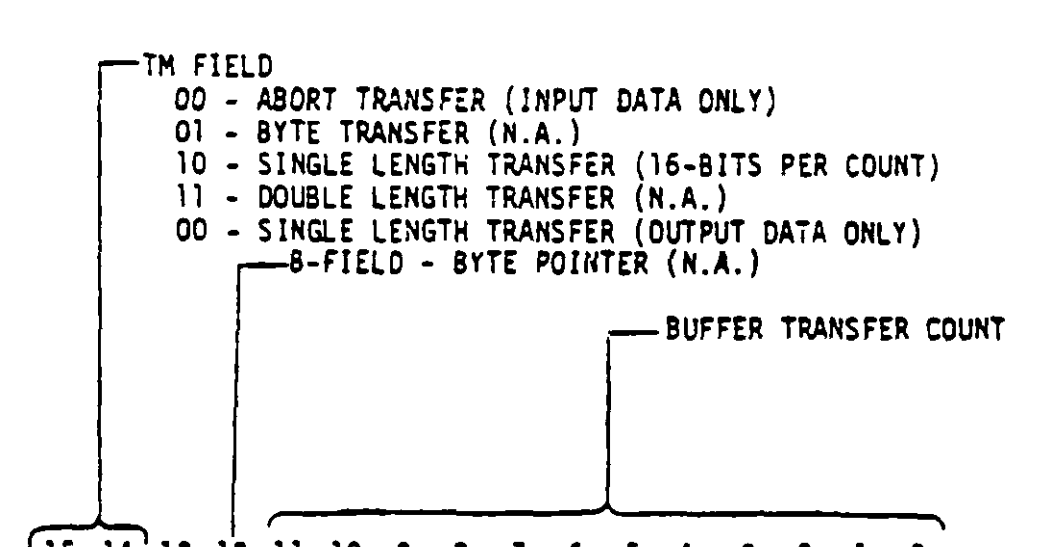


FIGURE 35. DIOM control definition map.

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DIOM BUFFER CONTROL WORD



DIOM MODE CONTROL WORD

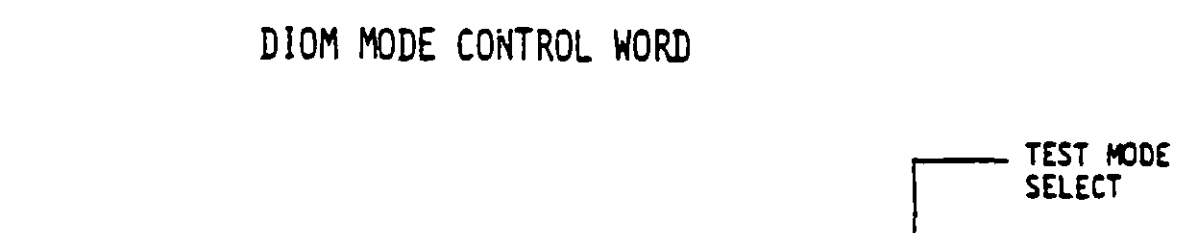
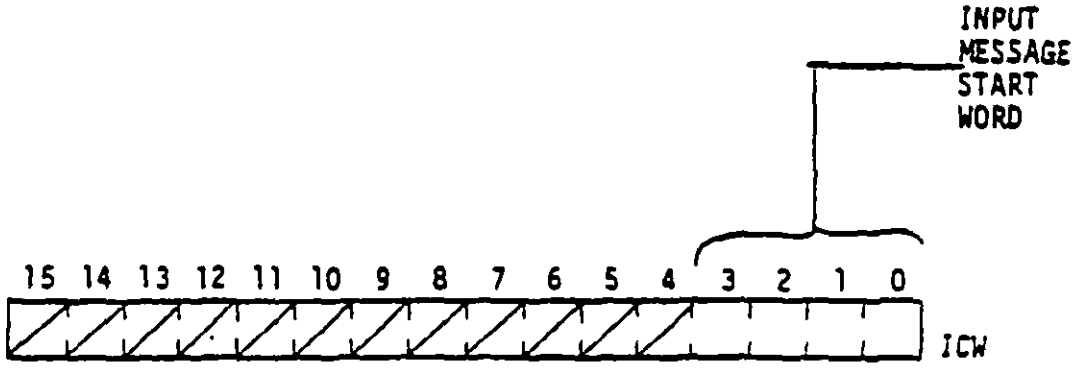


FIGURE 36. DIOM buffer control word and DIOM mode control word.

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DIOM INPUT CONTROL WORD



DIOM OUTPUT CONTROL WORD

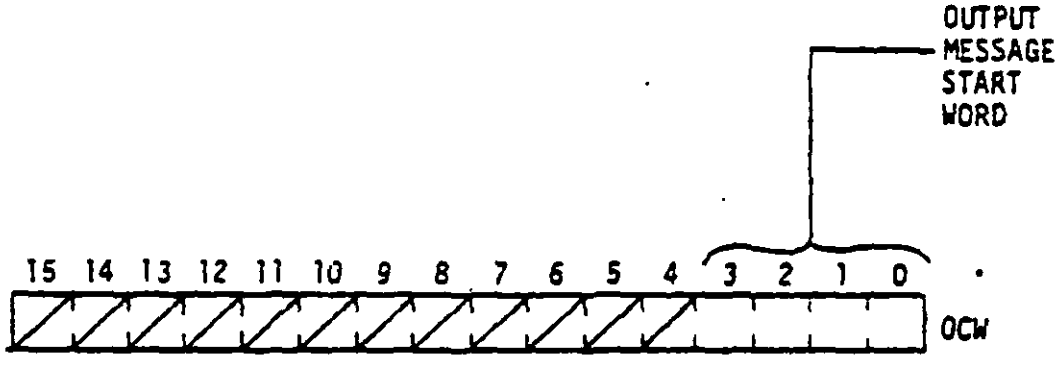


FIGURE 37. DIOM input control word and DIOM output control word.

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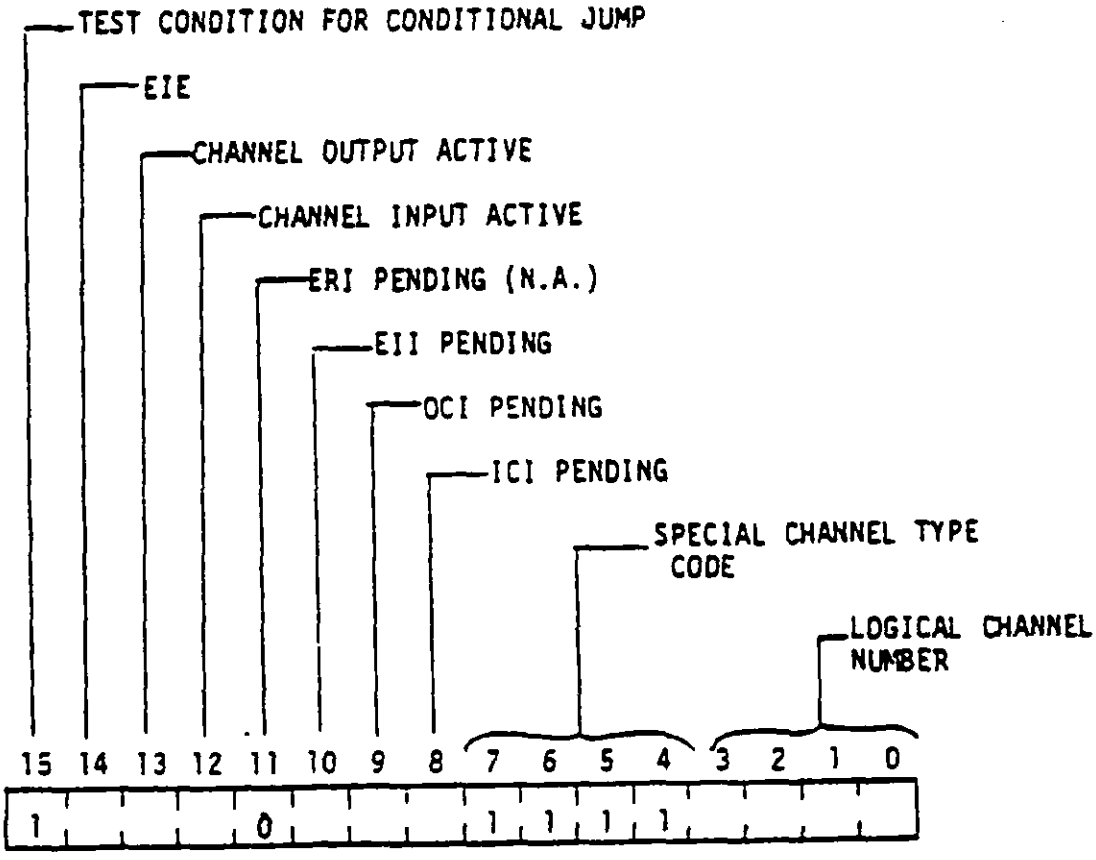


FIGURE 38. DIOM I/O channel status word 0 format.

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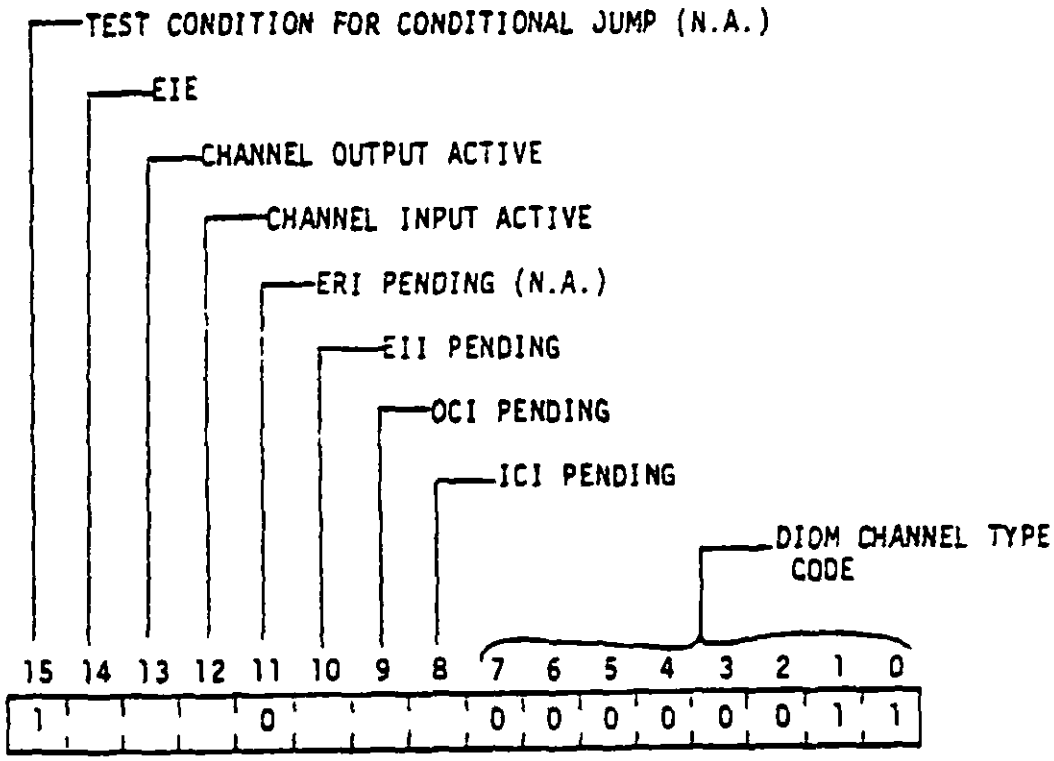


FIGURE 39. DIOM I/O channel status word 1 format.

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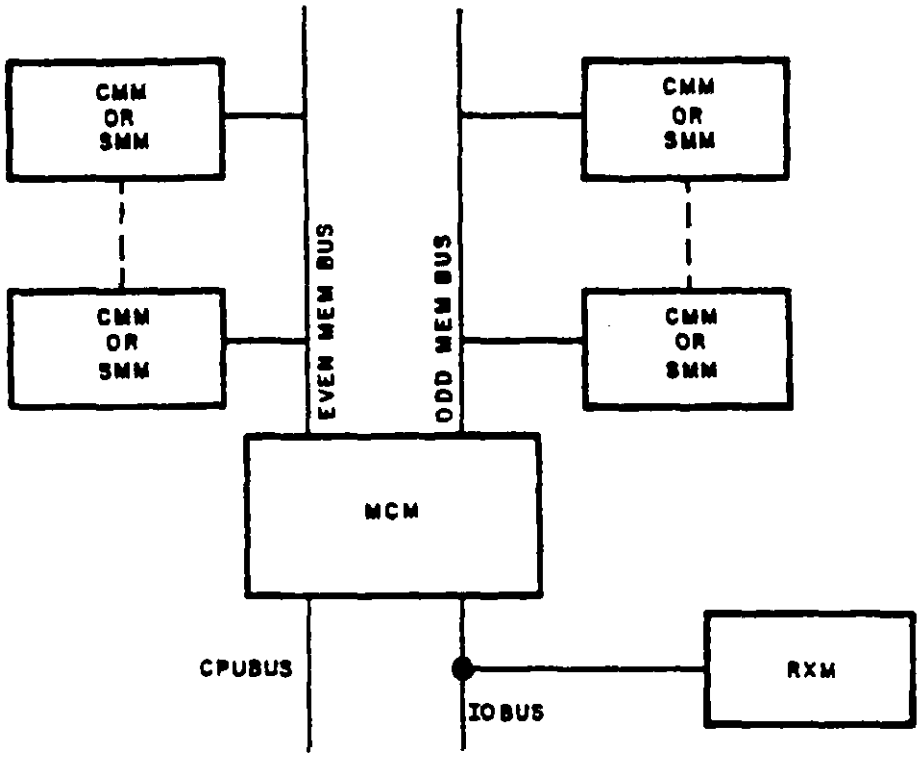


FIGURE 40. Memory subsystem.

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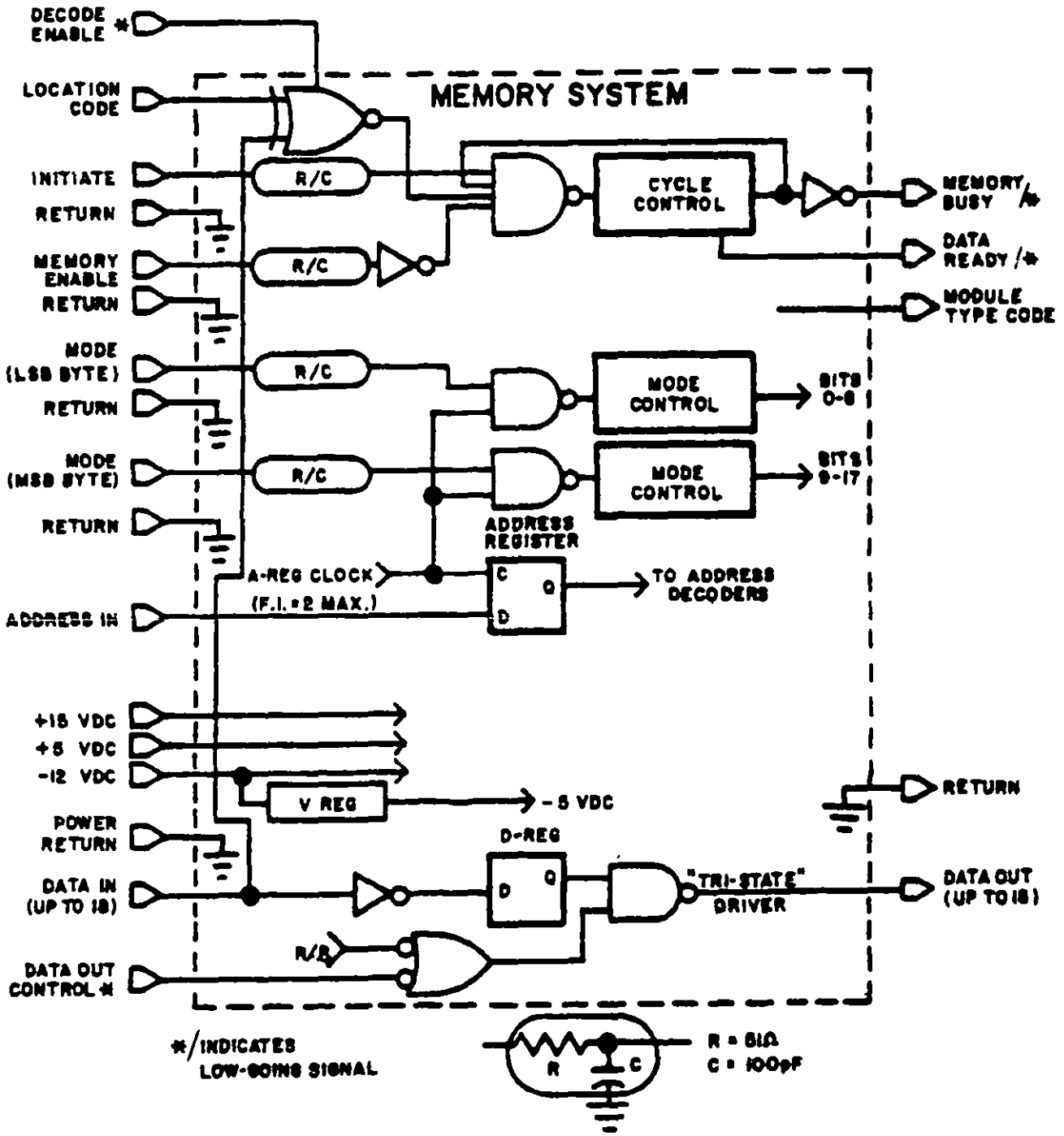


FIGURE 41. Core memory module block diagram.

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3.6.2.3.1.1.1 Core memory speed. The core memory module shall have a read/write cycle time of 900 nanoseconds or less, and an access time of 400 nanoseconds or less from cycle initiate to data ready. The memory timing is shown in Figure 42.

3.6.2.3.1.2 Semiconductor memory module (SMM). The 32K Semiconductor Memory Module shall be 32,768 words by 18 bits, random access, volatile, electrically alterable, and shall be compatible and interchangeable with the core memory module. Semiconductor and Core Memory Modules shall be capable of operating together in a single computer. The SMM block diagram is shown in Figure 43.

3.6.2.3.1.2.1 Semiconductor memory speed. The semiconductor memory modules shall have a read/write cycle time of 400 nanoseconds or less and an access time of 200 nanoseconds or less from cycle initiate to data ready. The memory timing is shown in Figure 44.

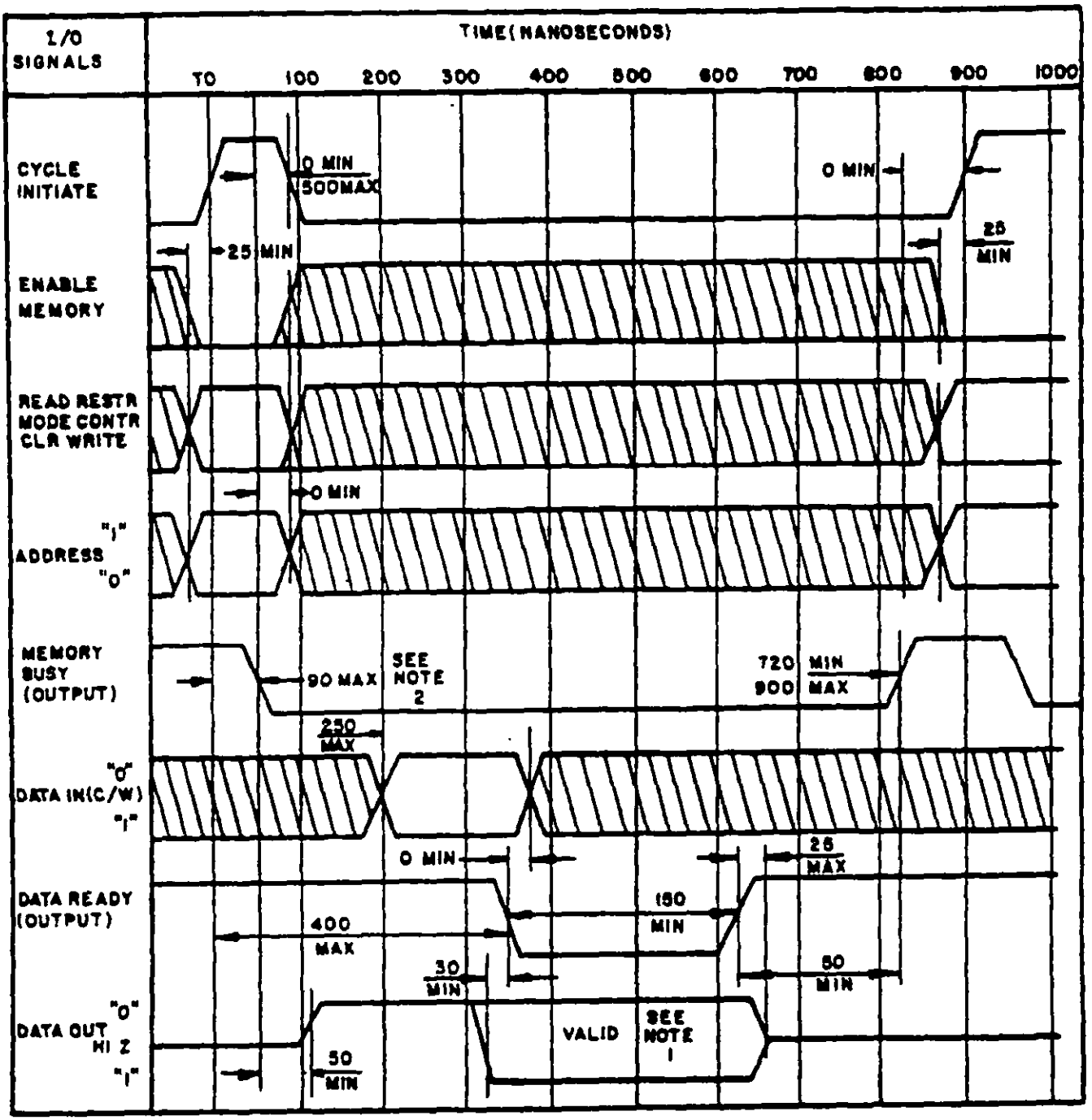
3.6.2.3.1.3 Memory organization. The memory shall be configured in 18-bit word lengths. There shall be one bit available for use as a parity bit for each eight bits of data.

3.6.2.3.1.4 Memory modularity. Each memory module shall contain all necessary drive and sense electronics required to read or write the memory. All memory modules shall contain all electronics necessary to interface with the memory controller.

3.6.2.3.2 Memory bus. Two memory buses (odd and even) shall interface the memory modules with the memory controller. The memory bus shall be capable of operating with the interface characteristics depicted in Figure 41 and 43 for all core and semiconductor memories with timing characteristics as depicted in Figures 42 and 44. All voltage and impedance levels shall be TTL compatible.

3.6.2.3.3 Memory controller. The memory controller (MCM) shall interface the memory bus with the IOBUS and the CPUBUS. The memory controller shall provide a two-port paged interface to the independent memory channels (odd and even), allowing simultaneous access by two users. Memory channel operation shall be as illustrated in Figure 45. The memory controller shall be capable of locking out access to a 1024-word (18 bits) memory page designated by the CPU to allow valid operation of the Stack, Queue, and Biased Fetch instructions for multiprocessor configurations. The memory controller shall contain sufficient hardware to provide read and write protection for blocks of memory in increments of 1K words maximum, starting with the first location in the memory. The memory controller shall provide a relative (16-bit address from the CPU) to physical (19-bit address to Memory) memory translation mechanism using page registers as shown in Figure 46.

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1/ DATA OUT VALID TTL LEVEL ONLY DURING READ RESTORE CYCLE UNLESS HELD VALID ON-LINE BY LOW-LEVEL "DATA OUT CONTROL" SIGNAL AT I/O.

2/ 90 NANOSECONDS VALID ONLY FOR ISOLATED REFERENCES. THIS MAY BE 270 MAX WHEN MEMORY IS REFERENCED AT MAXIMUM REF RATE.

FIGURE 42. Interface timing for 900 nanosecond core memory.

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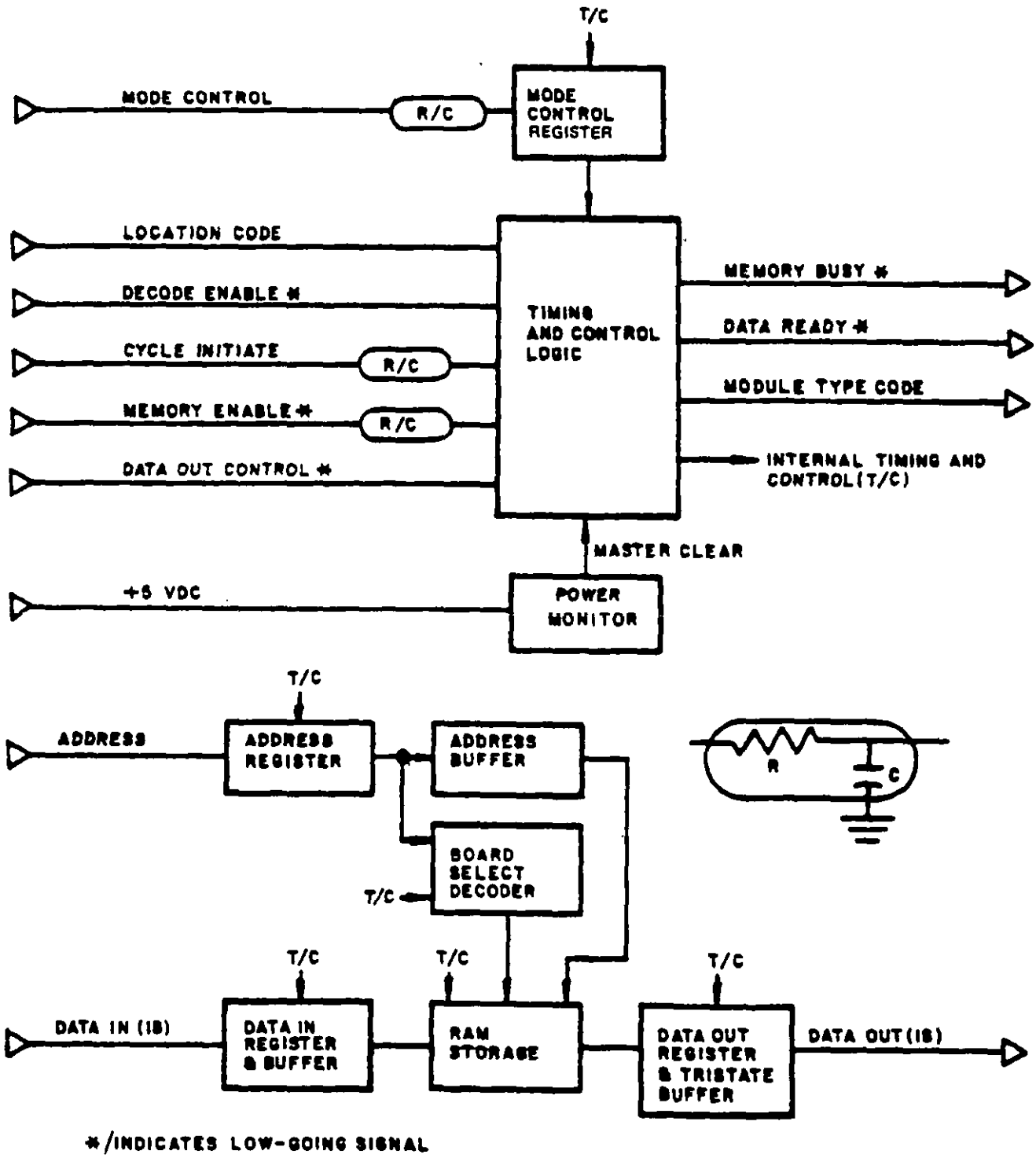
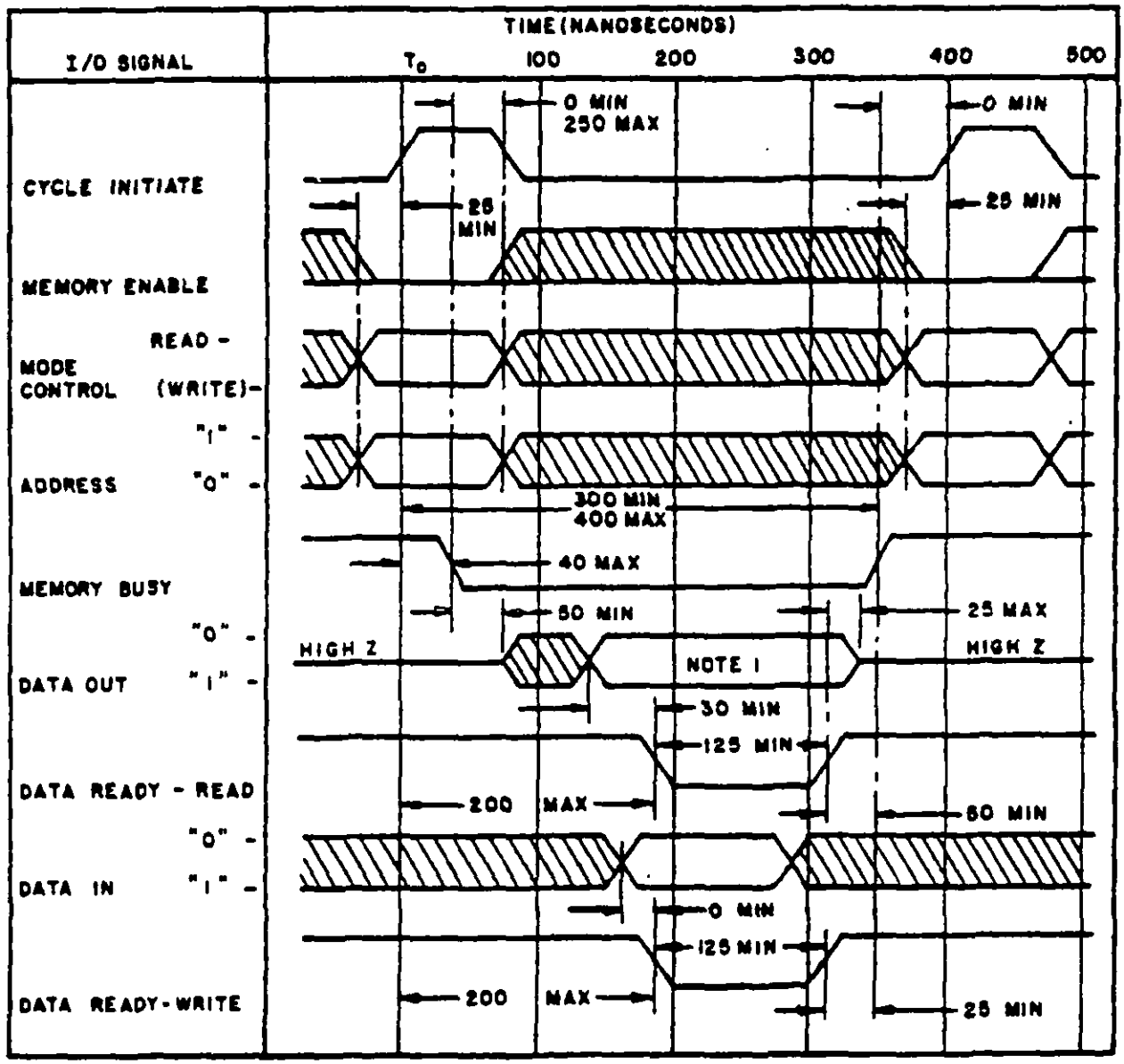


FIGURE 43. Block diagram for semiconductor memory.

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1/ DATA OUT AT VALID TTL LEVEL ONLY DURING READ CYCLE UNLESS HELD VALID ON-LINE BY LOW-LEVEL ON THE "DATA OUT CONTROL" I/O LINE.

FIGURE 44. Interface timing for semiconductor memory.

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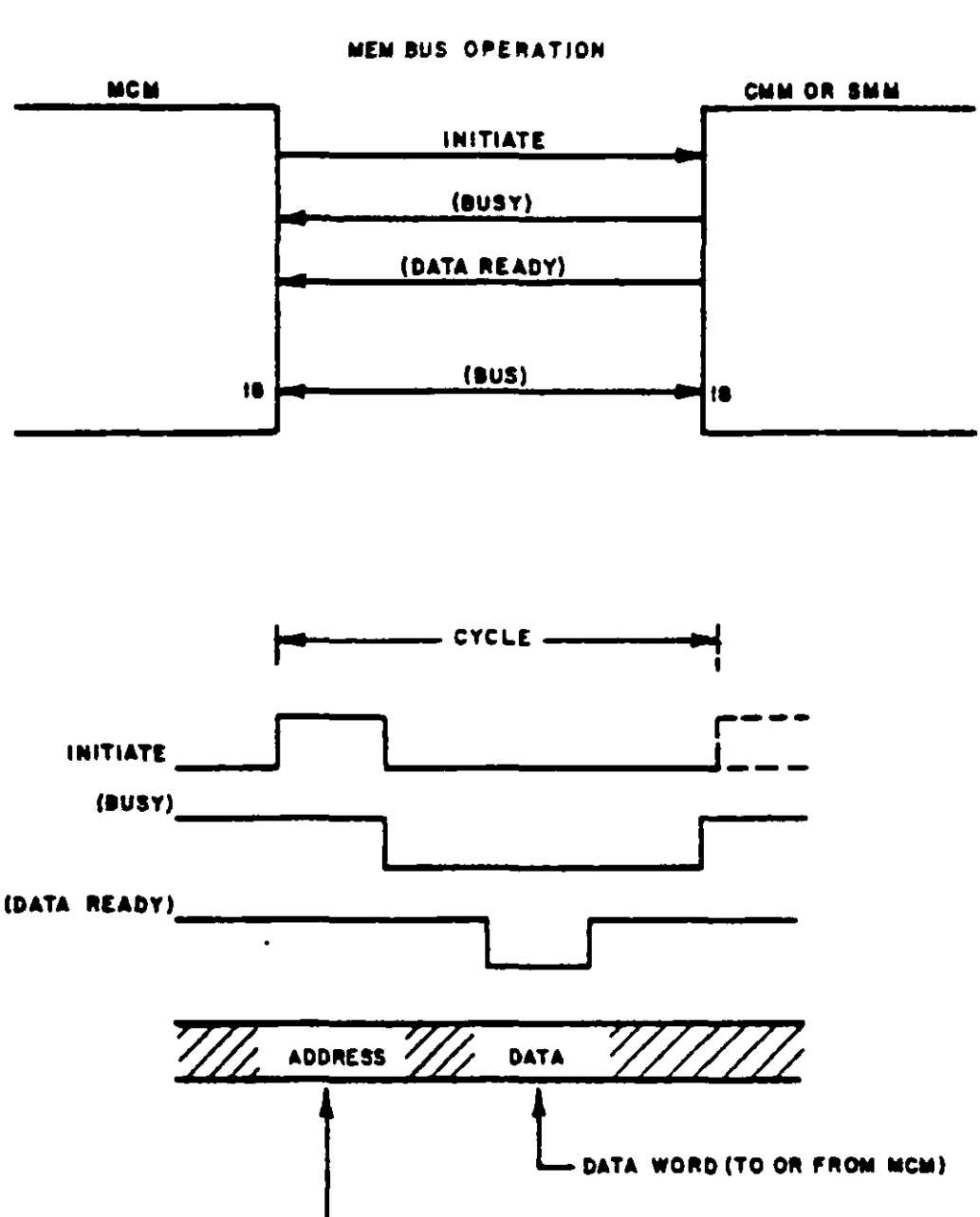


FIGURE 45. Memory module interface.

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The memory controller shall detect memory parity errors, and shall provide a Memory Parity interrupt to the CPU after detection. The memory controller shall be capable of generatinng a Memory Resume interrupt if Data Ready is not received from the accessed memory module. The Memory controller shall automatically word interleave the memory when an even number of like memory modules are present except when the BIFI signal (see 3.4.19.14) is set. When an odd number of memory modules or certain combinations of different word capacity memory modules are present in the main chassis, or the BIFI signal is set, the memory controller shall automatically bank interleave (consecutive addresses shall be in the same memory module). Memory protection features on a 1024-word by 18-bit page basis shall be provided. Three types of protection shall be implemented via bits stored in the page register as shown in Figure 46.

a. Execute protection shall generate a Memory Protect Fault interrupt if instruction execution is attempted from the protected page.

b. Write protection shall generate a Memory Protect Fault interrupt if a write operation is attempted in a protected page.

c. Read protection shall generate the Memory Protect Fault interrupt if a read operation is attempted in a protected page. In addition, a bit in each page register shall indicate that a write operation was made in the associated page.

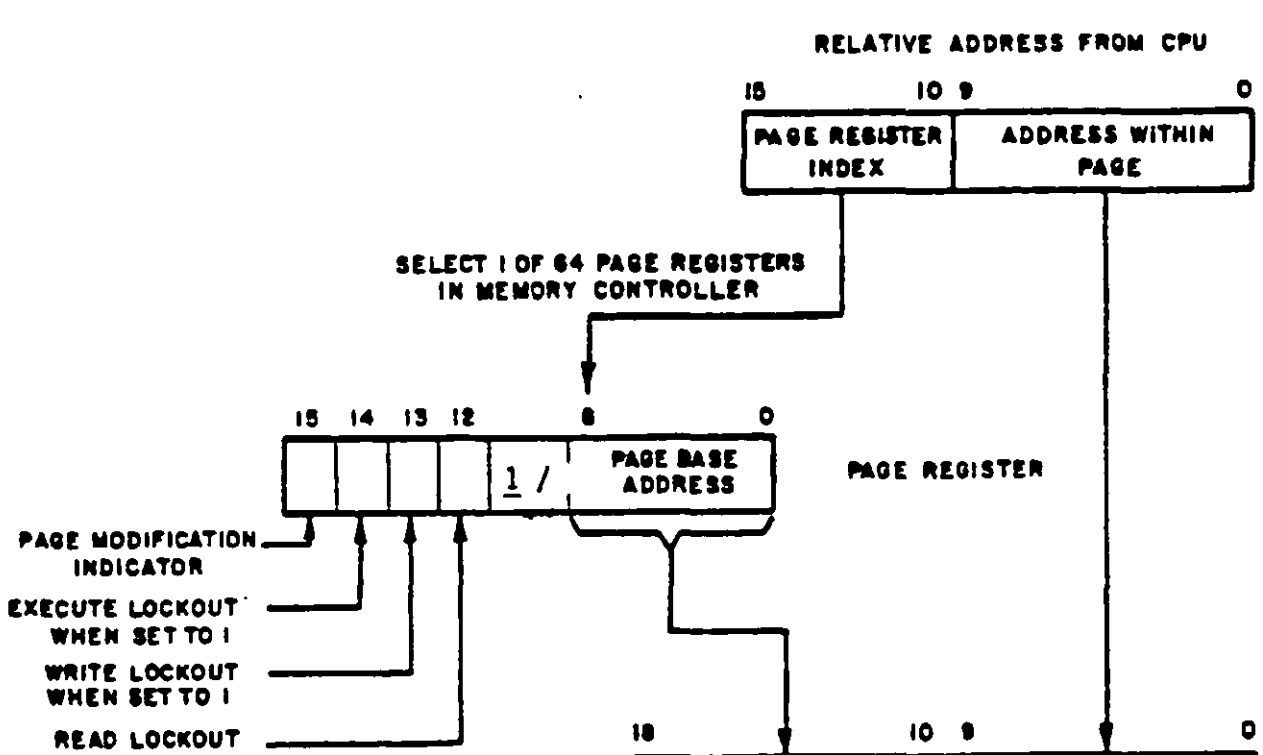
3.6.2.3.4 RXM memory. The RXM memory shall be configured on a single SRA. The memory shall contain 4096 x 18-bit words of random access (RAM), electrically alterable, read/write semiconductor memory and shall have growth space available for 4096 x 16-bit words of ROM or PROM memory. The RXM shall be used to provide high-speed memory for the IOP and shall be expandable in 4096-word increments to a maximum of 65,536 (32K RAM, 32K PROM). The RAM portion of this memory shall be capable of being loaded from the main memory, beginning at any location, by the IOP under program control. In a dual processor computer with an RXM installed, the computer chassis wiring shall determine whether an IOP memory reference is to be serviced by the RXM or MCM.

3.6.2.3.4.1 RXM speed. The RXM memory shall have a cycle time of 400 nanoseconds or less.

3.6.2.3.4.2 RXM organization. The read/write portion of the RXM memory shall be configured in 18-bit word lengths. There shall be one parity bit included in each 8-bit byte. The ROM or PROM portion of the RXM shall be configured in 16-bit word lengths. No parity bits shall be included for the ROM or PROM.

3.6.2.3.4.3 RXM adapter. The RXM adapter shall be to allow the installation of the RXM in place of one of the main memory modules. (See 3.6.2.5.1.1).

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1 / Reserved for Firmware.

FIGURE 46. Memory address generation.

3.6.2.4 Power supply subsystem. The power supply subsystem shall be modularly designed and shall provide the power requirements for the various computer configurations. The input power characteristics shall be as specified in 3.4.12. The input power factor of any phase shall be 0.80 or greater. The power supply shall monitor the AC input power for out-of-tolerance conditions. When the power supply senses that AC power is going down (or out of tolerance), or that one or more phases has been lost, a Power Fault interrupt shall be generated using the event system specified in 3.6.1.3. A Master Clear signal shall be sent to the computer, a minimum of 280 microseconds after an out-of-tolerance condition has been sensed. During this time, the power supply shall maintain its output voltages to allow storing of register data into non-volatile memory under program control, if available. After this time, an orderly shutdown of output voltages shall be performed to assure no alteration of memory data. The power supply shall provide power-on sequencing to assure memory protection and proper initialization. The Master Clear signal shall be supplied by the power supply until all output voltages are within tolerance. The power off and on sequencing shall assure no alteration of non-volatile memory contents due to power oscillations.

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3.6.2.4.1 Power conversion module - 1 (PCM-1). PCM-1 shall be capable of providing the following nominal voltages and currents:

a.	+ 5 VDC	51A min
b.	- 5 VDC	1.0A min
c.	-12 VDC	5.0A min
d.	+15 VDC	3.3A min

3.6.2.4.2 Power conversion module - 2 (PCM-2). PCM-2 shall be capable of providing the following nominal voltages and current:

a.	+ 5 VDC	78A min
b.	- 5 VDC	1.0A min
c.	-12 VDC	5.0A min
d.	+15 VDC	3.3A min

3.6.2.5 Chassis subsystem. The chassis subsystem shall consist of the combinations of the following SRAs:

a. Chassis
   XN-1
   XN-3
   XN-4
   XN-5

b. Fan

c. Cooling Plate

3.6.2.5.1 Chassis - Maximum chassis configurations shall be as specified in 3.6.2.5.1.1 through 3.6.2.5.1.4. Additional requirements for specific configurations are specified in the appendices. Connectors not used in a particular configuration shall be capped. All connector covers (caps), where used, shall be captivated to the front panel in all configurations. Capped connectors shall not extend beyond two inches from the front panel.

3.6.2.5.1.1 Type 1 chassis. The Type 1 chassis shall provide the necessary interconnection circuitry and space to support and contain a total of four main memory modules (or three main memory modules and one RXM module with RXM adapter), one MCM, one EAU, one GPM, one PSM, one IOP or IOP Jumper, five I/O channels, and one BEM and a DIM cable for connection to the DIM connector specified in 3.6.2.2.1.3.3. I/O channel logical number, priority assignment, and associated front panel connectors shall be fixed in the back panel wiring as shown in Table XII.

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TABLE XII. I/O channel logical number, priority assignment and provided front panel connectors for Type 1 chassis.

Location	Logical No.	Priority Assigned	Associated Connector
A02	F (hex)	F (Lowest)	J1 and J3
A03	3	3	J2
A04	2	2	J8
A05	1	1	J9
A06	0	0 (Highest)	J10

Note that J3 is associated with the extra input discrete capabilities of the DIM I/O channel type, when it is configured in location A02. J3 is not used for any I/O channel signals for other I/O channel types in location A02. Chassis configuration options shall be as shown in Figure 47.

3.6.2.5.1.2 Type 3 chassis. The Type 3 chassis shall provide the necessary interconnection circuitry to support and contain a total of two main memory modules, one IOP or IOP jumper, one RXM, six I/O channels, and one BEM. Chassis configuration options shall be as shown in Figure 48. I/O channel logical number, priority assignment, and associated front panel connectors shall be selectable as listed in Table XIII.

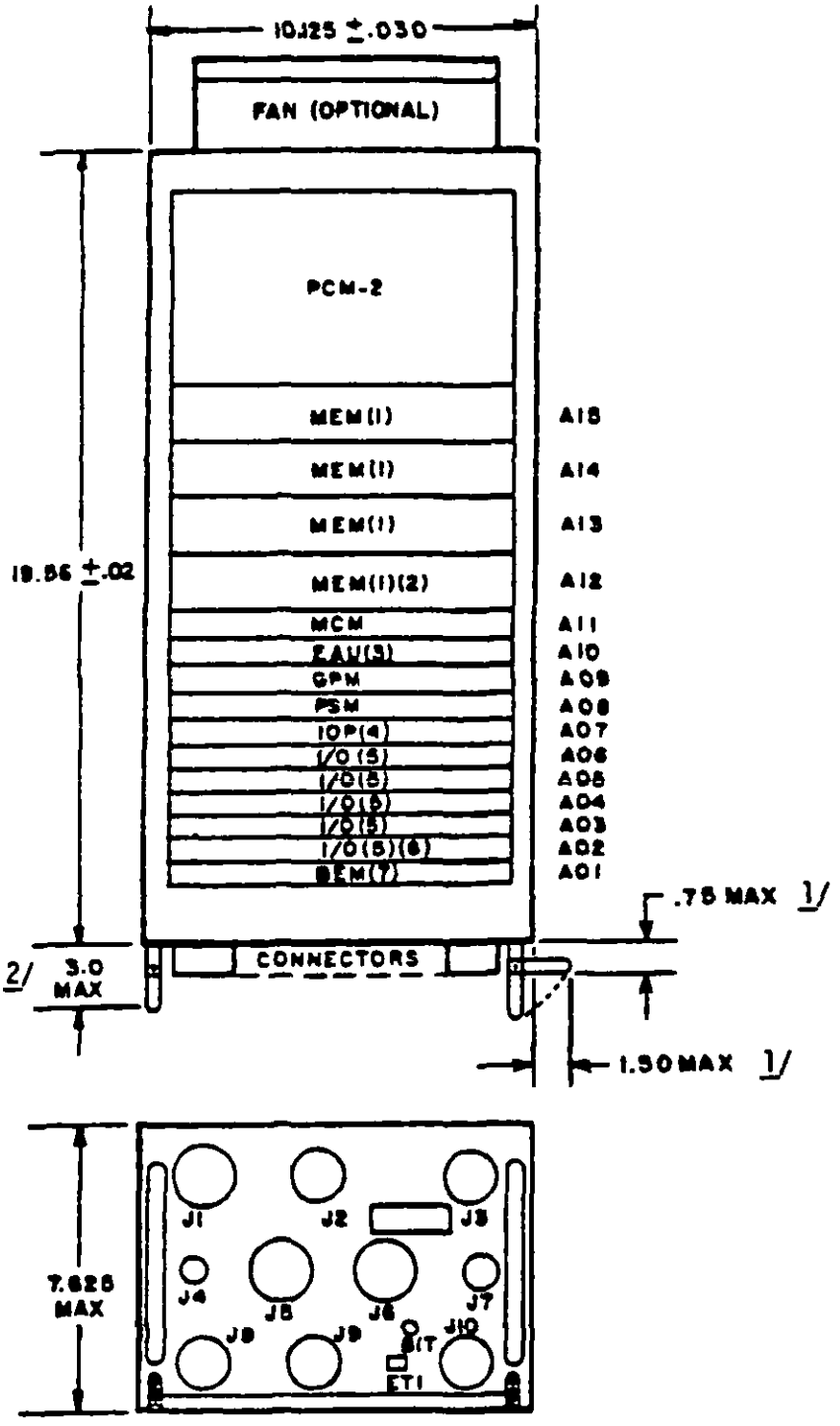
3.6.2.5.1.2.1 Memory extension unit. When the Type 3 chassis is used as a memory extension unit, it shall be possible to cascade units to provide a maximum of 524,288 total system words of memory. It shall also be possible to cascade units to provide a total of 16 system I/O channels. When the Type 3 chassis is used to extend memory or I/O, the IOP jumper SRA shall be located in the Type 3 chassis.

TABLE XIII. I/O channels logical number, priority assignment and front panel connector for Type 3 chassis.

Location	Logical No.	Priority Assigned	Associated Connector
A02	D/D/F(hex)	D/D/F(hex)(Lowest)	J1
A03	C/C/E(hex)	C/C/E(hex)	J2
A04	9/9/B(hex)	9/9/B(hex)	J3
A05	8/8/A(hex)	8/8/A(hex)	J8
A06	1/5/7	1/5/7	J9
A07	0/4/6	0/4/6 (Highest)	J10

Note that there are three possible logical numbers and priority assignments for each location. Via appropriate jumpering of pins in connector J3, either set may be defined. A Type 3 chassis, therefore, may contain channel logical numbers 4, 5, 8, 9, C and D; 6, 7, A, B, E, and F; or 0, 1, 8, 9, C and D. When the Type 3 chassis is used as an extension unit the channel number selection shall not be identical to any channel used in the main computer chassis.

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1/ Typical of handle when in "folded" position.

2/ Typical of handle when in "straight" position.

FIGURE 47. Type 1 chassis configuration options.

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STANDARD The Type 1 Chassis shall contain the following basic module set:
- Power Conversion Module (PCM-2)
- General Processor Module (GPM)
- Processor Support Module (PSM)
- Memory Control Module (MCM)

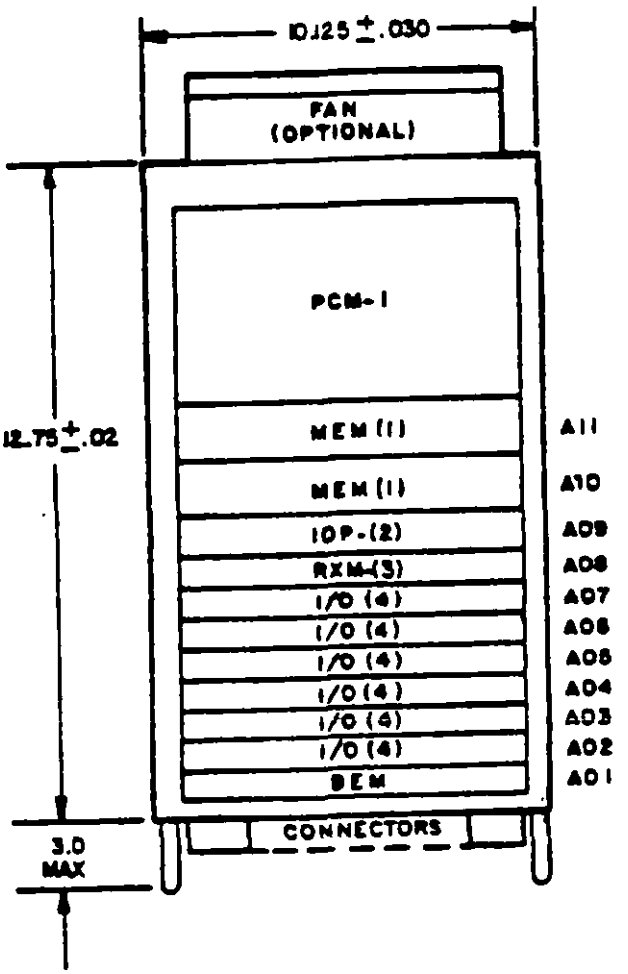
OPTIONS

1. Combinations of main memory modules, up to four total.
2. 4K x 18-bit word Read/Write Memory (RXM), optionally used withthe IOP (additional 4K PROM optional).
3. Extended Arithmetic Unit (EAU).
4. Input/Output Processor (IOP) or IOP jumper.
5. Any combination of the following I/O modules, up to five total:
   - 1553A Serial (SIM)
   - NTDS Slow (NIM-A)
   - NTDS Fast (NIM-B)
   - NTDS ANEW (NIM-C1)
   - NTDS Serial (NIM-S)
   - PROTEUS (PIM)
   - RS-232 (RIM)
   - Discrete I/O (DIM) with 32 discretes and 8 interrupts wired to the front panel
6. Discrete I/O (DIM) with 64 discretes and 8 interrupts wired to the front panel (utilizing DIM cable).
7. Bus Extension Module (BEM).

Connector	*Type*	*Use*
J1	MS27656T25B35SB	I/O
J2	MS27656T23B35S	I/O
J3	MS27656T23B35SA	I/O
J4	MS27468T11B35P	Power
J5	MS27656T25B35SD	BEM
J6	MS27656T25B35S	BEM
J7	MS27656T17B35S	Maint.
J8	MS27656T23B35SB	I/O
J9	MS27656T23B35SC	I/O
J10	MS27656T23B35SD	I/O

FIGURE 47. Type 1 chassis configuration options - Continued

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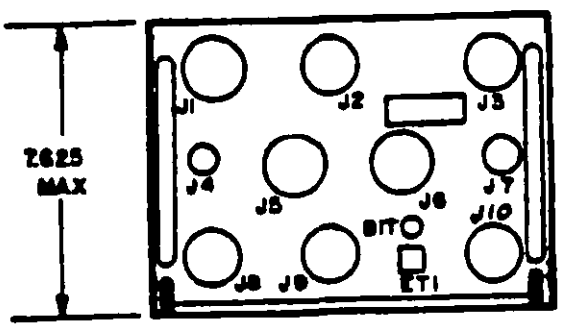


FIGURE 48. Type 3 chassis configuration options.

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STANDARD The Type 3 Chassis shall contain:
A Power Conversion Module (PCM-1)

OPTIONS

1. Combinations of main memory modules, up to two total.
2. Input/Output Processor (IOP) or IOP jumper.
3. Read/Write Expansion Memory (RXM)
4. Any combination of the following I/O modules, up to six total:
   - 1553A Serial (SIM)
   - PROTEUS (PIM)
   - RS-232 (RIM)
   - NTDS Fast (NIM-B)
   - NTDS Slow (NIM-A)
   - NTDS ANEW (NIM-C1)
   - NTDS Serial (NIM-S)
   - Discrete I/O (DIM) with 32 discretes and 8 interrupts wired to the front panel

NOTES:

A. As an expansion unit
   - System I/O expandable up to 16 I/O channels
   - System memory expandable up to 512K words
   - IOP jumper required
   - BEM required

B. As a standalone processor
   - IOP required
   - Can address only one 32K main memory module plus optional RXM.

Connector	Type	Use
J1	MS27656T25B35SB	I/O
J2	MS27656T23B35S	I/O
J3	MS27656T23B35SA	I/O
J4	MS27468T11B35P	Power
J5	MS27656T25B35SD	BEM
J6	MS27656T25B35S	BEM
J7	MS27656T17B35S	Maint.
J8	MS27656T23B35SB	I/O
J9	MS27656T23B35SC	I/O
J10	MS27656T23B35SD	I/O

FIGURE 48. Type 3 chassis configuration options - Continued

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3.6.2.5.1.3 Type 4 chassis. As shown in Figure 49, the Type 4 chassis shall provide the necessary interconnection circuitry and space to support and contain a total of four main memory modules, one MCM, one EAU, one GPM, one PSM, one IOP, two SIM, three PIC/POC/SOC, one DIOM, two standard serial channels in accordance with Figure 49, and two standard channels, in accordance with Figure 49, of any type except PIC/POC/SOC and DIOM. The two expansion serial I/O channels can be any serial I/O type except RS-232. The maximum complement of modules which can be installed in the Type 4 chassis shall be limited by the output power capacity of the power conversion module. This limitation may be reached before all module slots are populated. I/O channel logical number and priority assignment shall be fixed in the back panel wiring as listed in Table XIV.

TABLE XIV. I/O channel logical number and priority assignment for Type 4 chassis.

Location	Hex Channel No.	Usage	Associated Connector
A10	0(highest priority)	SIM Expansion	J1
A09	1	SIM	J1
A06	2	PIC B (5V)/POC A (5V)	J6/J2
A08	3	I/O Expansions-Any serial except RS-232	J1
A05	4	PIC C (5V)/POC C (15V)	J4/J10
A07	5	I/O Expansion - Standard	J5
A04	6	PIC A (15V)/POC B (15V)	J14/J9
A03	7	I/O Expansion-Any Serial	J1
A06	A	SOC (Single Ended)	J3
A05	C	SOC (Differential)	J1
A02	D	I/O Expansion - Standard	J12
A04	E	SOC (Differential)	J1
A01	F (lowest priority)	DIOM	J2/J3/J7/J9/J13

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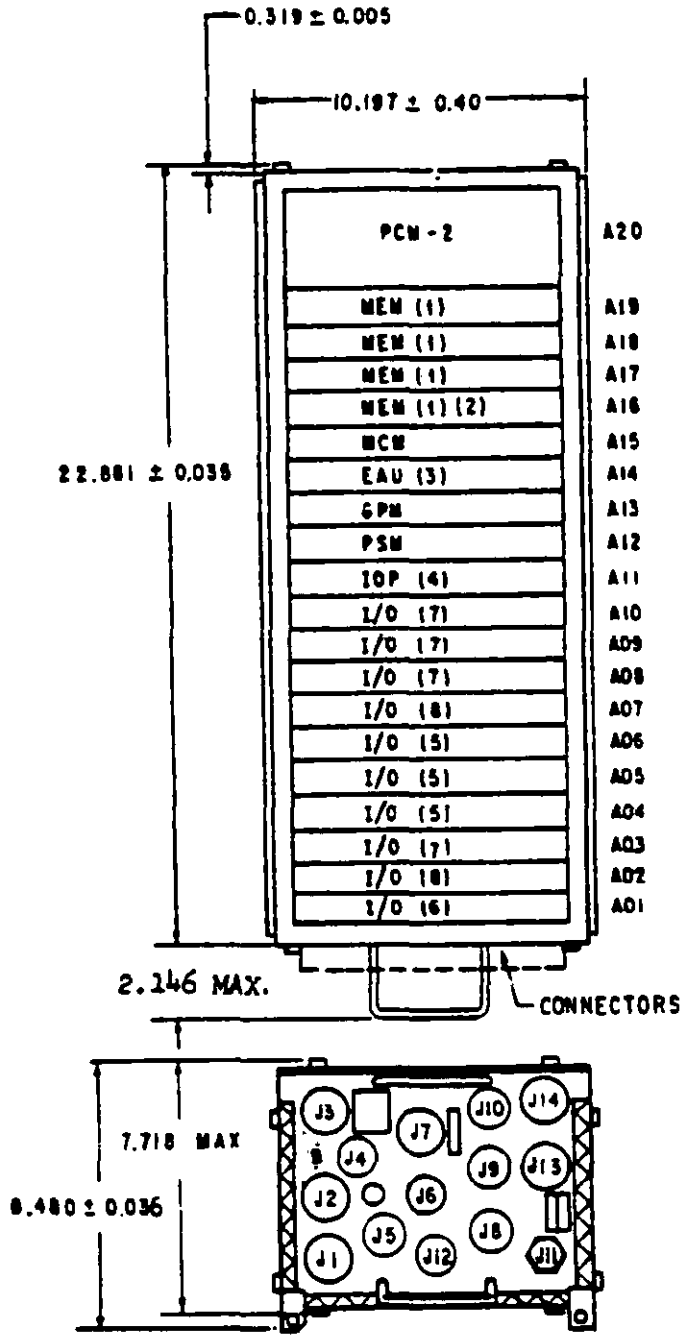


FIGURE 49. Type 4 chassis configuration options.

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STANDARD Type 4 Chassis shall contain the following basic module set:
- Power Conversion Module (PCM-2)
- General Processor Module (GPM)
- Processor Support Module (PSM)
- Memory Control Module (MCM)

OPTIONS
1. Combinations of main memory modules, up to four total.
2. 4K x 18-bit word Read/Write Memory (RXM), optionally used withthe IOP (additional 4K PROM optional).
3. Extended Arithmetic Unit (EAU).
4. Input/Output Processor (IOP) or IOP jumper.
5. PIC/POC/SOC module (PPSM).
6. Discrete Input/Output Module (DIOM).
7. Any combination of the following I/O modules:
   - 1553A Serial (SIM)
   - PROTEUS (PIM)
   - NTDS SERIAL (NIM-S)
8. Any combination of the following I/O modules with the exception noted in 3.5.2.5.14.
   - 1553A SERIAL (SIM)
   - NTDS SLOW (NIM-A)
   - NTDS FAST (NIM-B)
   - NTDS ANEW (NIM-C1)
   - NTDS SERIAL (NIM-S)
   - PROTEUS (PIM)
   - RS-232 (RIM)
   - DISCRETE I/O (DIM) with 32 discretes and 8 interrupts wired to front panel.

CONNECTOR	TYPE	USE
J1	M81511/41WF01S4	I/O
J2	M81511/41WF01P2	I/O
J3	M81511/41WF01P1	I/O
J4	M81511/41WE01P1	I/O
J5	M81511/41WF01S1	I/O
J6	M81511/41WE01P2	I/O
J7	M81511/41WF01P3	I/O
J8	MS27656T17B35S	MAINT.
J9	M81511/41WE01S2	I/O
J10	M81511/41WE01S3	I/O
J11	MS3474W10-6P	POWER
J12	M81511/41WF01S2	I/O
J13	M81511/41WF01S3	I/O
J14	M81511/41WE01P4	I/O

FIGURE 49. Type 4 chassis configuration options. - Continued

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The dimensions of the Type 4 chassis shall be as shown in Figure 49. The bolt-on fan specified in 3.6.2.5.5 cannot be installed on the Type 4 chassis. Input/output connectors shall be in accordance with MIL-C-81511, except the maintenance connector shall be in accordance with MIL-C-38999. The input power shall be provided on a separate connector in accordance with MIL-C-26482. The Thermal Safety Shutdown signal (see 3.4.19.9) shall not be provided on an operational connector. The Thermal Overtemperature signal (see 3.4.19.8) function shall be provided as specified in 3.6.2.2.1.7.2.1).

3.6.2.5.1.4 Type 5 chassis. The Type 5 chassis shall provide the necessary interconnection circuitry and space to support a total of two main memory modules, one MCM, one GPM, one PSM, one IOP or IOP Jumper, three I/O channels one of which is limited to SIM, PIM, or NTDS serial and one DIM utilizing its full capability and DIM cable for connection to the DIM connector specified in 3.6.2.2.1.3.3. Chassis configuration options shall be as shown in Figure 50. The handles(s) provided on the Type 5 chassis shall not protrude more than 2-1/2 inches from the front panel and shall not be located within the bottom 2 inches of the front panel. I/O channel logical number, priority assignment, and associated front panel connectors shall be fixed in the back panel wiring as listed in Table XV.

TABLE XV. I/O channel logical number, priority assignment and associated front panel connectors for Type 5 chassis.

Location	Logical No.	Priority Assigned	Associated Connector
A01	F (hex)	F (Lowest)	J1 and J3
A02	3	3	J4, J5, J8
A03	2	2	J4, J5, J6
A04	1	1 (Highest)	J4 and J5

3.6.2.5.2 Fan. A bolt-on fan shall be provided for applications specified in the specification sheets which require ambient air cooling. Sea level maximum power requirements for this fan shall be as follows: 225 VA (170 watts), three-phase, 115 VAC, 400 Hz in accordance with MIL-STD-704. Power shall be supplied to the fan through an M24308/4-2 connector located on the rear of the unit. The fan shall extend no more than 3.25 inches from the rear of the chassis. Environmental limits using this fan shall be as specified in the applicable specification sheets.

3.6.2.5.3 Cooling plate. A cooling adapter plate shall be provided for those configurations (except Type 4 chassis configurations) defined in the applicable appendices which utilize aircraft cooling. The cooling plate shall extend a maximum of 0.08 inch from the rear of the chassis. The cooling plate plus mounting hardware shall extend no more than 0.2 inch from the rear of the chassis.

3.7 Workmanship. The computer set shall conform in every respect to the workmanship requirements of this specification and MIL-STD-454, Requirement 9.

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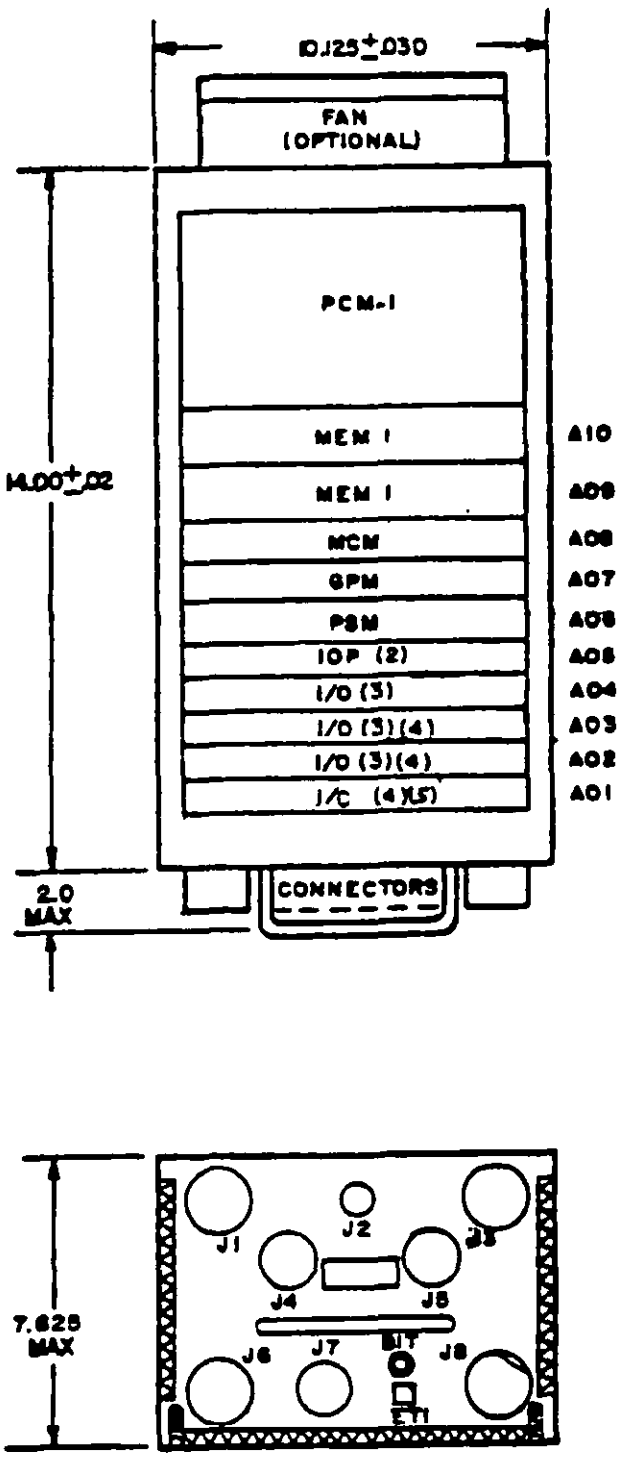


FIGURE 50. Type 5 chassis configuration options.

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STANDARD Type 5 Chassis shall contain the following basic module set:
- Power Conversion Module (PCM-1)
- General Processor Module (GPM)
- Processor Support Module (PSM)
- Memory Control Module (MCM)

OPTIONS
1. Combinations of main memory modules, up to two total.
2. Input/Output Processor (IOP) or IOP Jumper.
3. Any of the following I/O modules:
   - 1553A Serial (SIM)
   - PROTEUS (PIM)
   - NTDS Serial (NIM-S)
4. Any of the following I/O modules:
   - 1553A Serial (SIM)
   - NTDS Slow (NIM-A)
   - NTDS Fast (NIM-B)
   - NTDS ANEW (NIM-C1)
   - NTDS Serial (NIM-S)
   - PROTEUS (PIM)
   - RS-232 (RIM)
   - Discrete I/O (DIM with 32 descretes and 8 interrupts wired to the front panel (utilizing DIM cable).
5. Discrete I/O (DIM) with 64 discretes and 8 interrupts wired to the front panel (utilizing DIM cable).

Connector	Type	Use
J1	MS27656T25B35SA	I/O
J2	MS27468T11B35P	Power
J3	MS27656T25B35SB	I/O
J4	MS27656T15B35S	I/O
J5	MS27656T15B35SA	I/O
J6	MS27656T25B35SD	I/O
J7	MS27656T17B35S	Maint.
J8	MS27656T25B35S	I/O

FIGURE 50. Type 5 chassis configuration options - Continued

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.2.1).

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2)

b. Initial Production Tests (see 4.3)

c. Quality Conformance Inspection (see 4.4)

4.2 First article inspections. First article inspections shall be conducted by the contractor on an equipment representative of the production equipment to be supplied under the contract. First article tests shall be accomplished under the approved test procedure of 4.6. No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity (see 6.2.2).

4.2.1 First article test data. The contractor shall retain and make available all data collected in conducting these tests to the procuring activity for review and acceptance (see 6.2.2).

4.2.2 Scope of first article tests. First Article Tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article tests shall include Environmental Tests in accordance with the procedures of MIL-T-5422, a Random Vibration Test as specified herein and an Electromagnetic Compatibility Test in accordance with MIL-STD-461 and MIL-STD-462 as modified herein. The Sinusoidal Vibration Tests shall consist of the tests of MIL-T-5422, Procedure I, using the levels defined by Curve III(b) of MIL-E-5400, and a sweep test consisting of one sweep at the levels defined by Curve III(b) in the frequency range 5 Hz to 33 Hz and levels defined by Curve IV(a) in the frequency range 33 Hz to 2000 Hz. The sweep rate shall be logarithmic over a period of 20 minutes.

4.2.3 First article approval. Approval of the first article sample will be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the first article sample. The approved first article sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article sample shall not be considered as one of the equipment under the contract.

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4.2.4 Electromagnetic compatibility tests. An Electromagnetic Compatibility Test shall be conducted on the equipment in accordance with MIL-STD-461A, with Notice 3, for Class A1 equipment, as modified herein (see Table XVIII) with all I/O signal connectors capped.

4.2.4.1 Radiated susceptibility tests (RS03). The Radiated Susceptibility Test (RS03) parameters of MIL-STD-461 shall be replaced by those listed in Table XVI.

TABLE XVI. Radiated susceptibility test (RS03).

Frequency (MHz)	Power Density (mW/CM^2)	
	Peak	Average
.014 - 30	.106102	.0265
30 - 200	.106102	.000265
200 - 225	20	.09
225 - 400	1.12	.000265
400 - 450	1.12	.01
450 - 1215	.106102	.00265
1215 - 1365	7	.05
1365 - 2900	2.65	.00265
2900 - 3110	130	.09
3110 - 3235	100	.09
3235 - 3400	2.65	.000265
3400 - 3600	4	.09
3600 - 3700	9	.09
3700 - 5400	2.65	.000265
5400 - 5900	250	.09
5900 - 6000	2.65	.09
6000 - 7000	2.65	.000265
7000 - 7900	2.65	.000265
7900 - 8400	2.65	.09
8400 - 8500	2.65	.000265
8500 - 9600	140	.05
9600 - 12000	2.65	.000265
12000 - 40000	.00106	.00106

Pulse modulation shall be used with pulse widths in the range of 1 to 10 microseconds.

4.2.4.2 Conducted RF susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461, the following RF Susceptibility Test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment, when the RF signal levels listed in Table XVII are coupled into the interconnecting wiring.

TABLE XVII. Susceptibility test signals.

Frequency	Required Signal Levels in One-Turn Loops
2 to 30 MHz	1000 mv rms

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The RF signal shall be modulated 80 percent with 400 Hz and with that frequency and/or type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment.

TABLE XVIII. Electromagnetic compatibility test requirements.1/

Test	Limits
1. CE01	Figure 51
2. CE03	Figures 52, 53
3. CS01	MIL-STD-461A w/Notice 3
4. CS02	MIL-STD-461A w/Notice 3
5. CS06	MIL-STD-461A w/Notice 3
6. RE02 (Narrowband)	Figure 54
7. RE02 (Broadband)	MIL-STD-461A w/Notice 3
8. RS02	MIL-STD-461A w/Notice 3
9. RS03	4.2.4.1

1/ The equipment shall also pass the Conducted RF Susceptibility Requirement specified in 4.2.4.2 herein.

The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown in Figure 55. The signal generator shall have an output impedance of 50 ohms or less and a maximum open-circuit voltage of at least 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under the test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of response) obtained in (b) above. Also include in the test report the corrective action required for each wire that does not meet the required level.

Wire bundles may be tested in sections, if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

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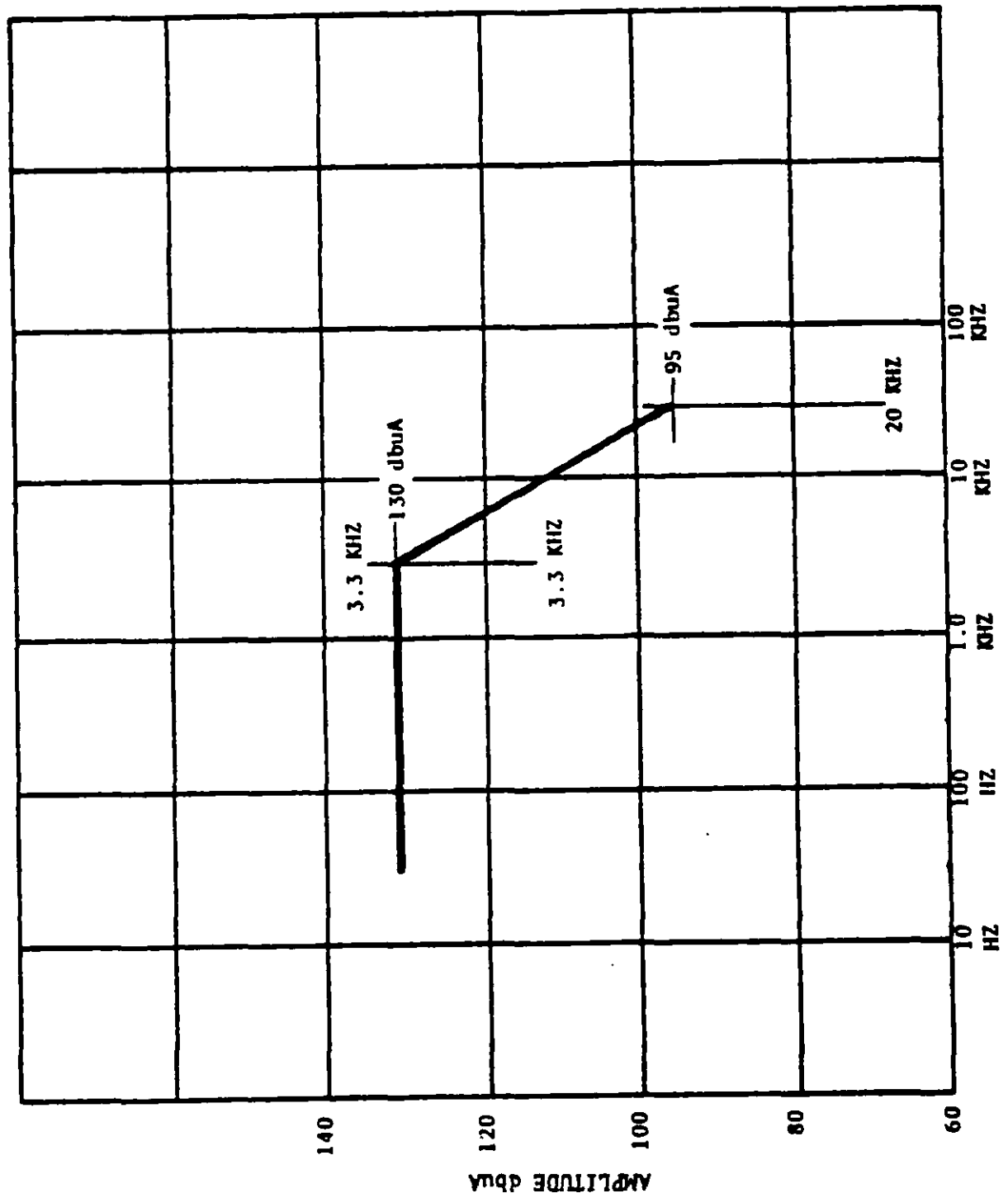


FIGURE 51. CE01, conducted emission, narrow band limits.

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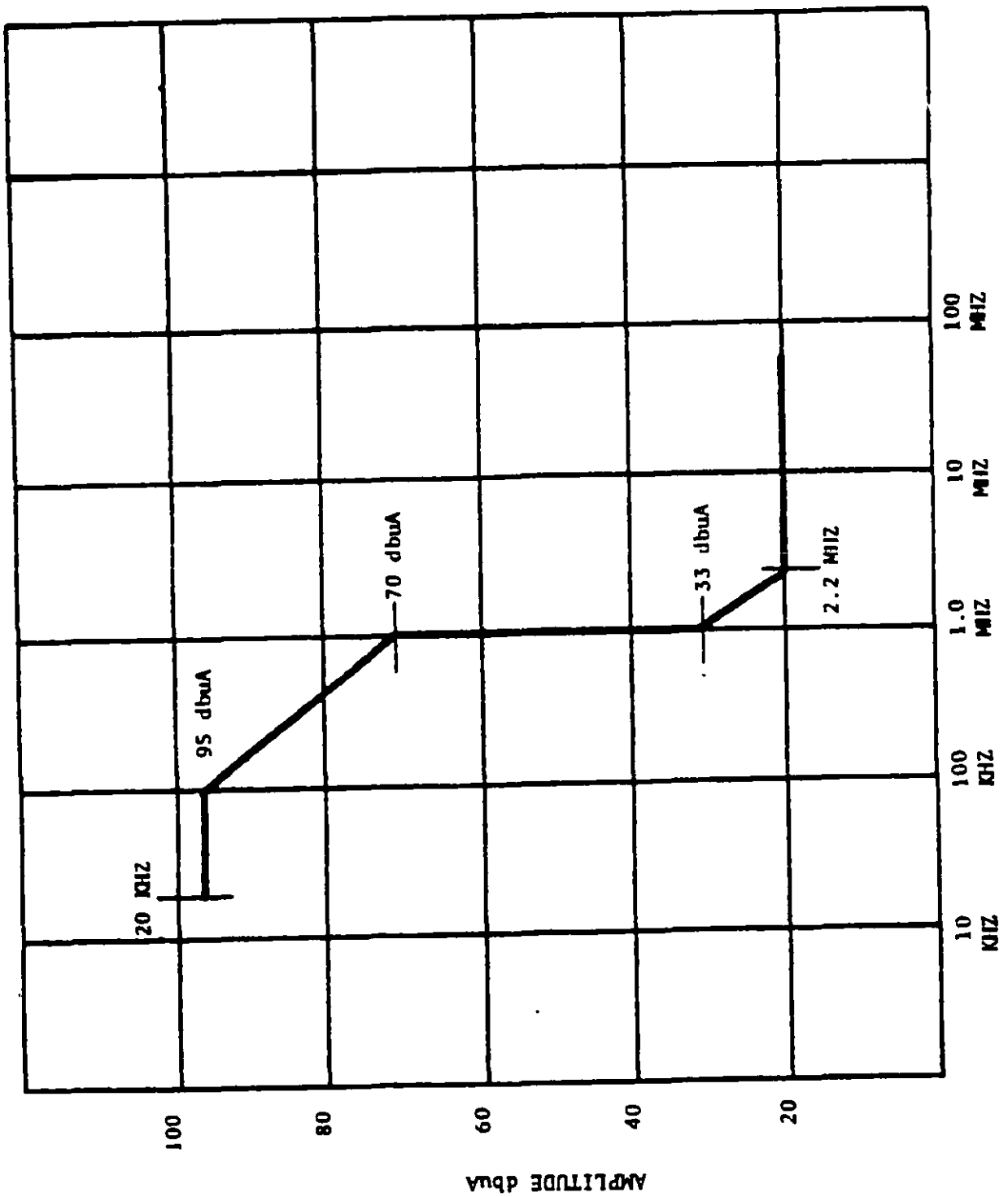


FIGURE 52. CE03, conducted emission, narrow band limits.

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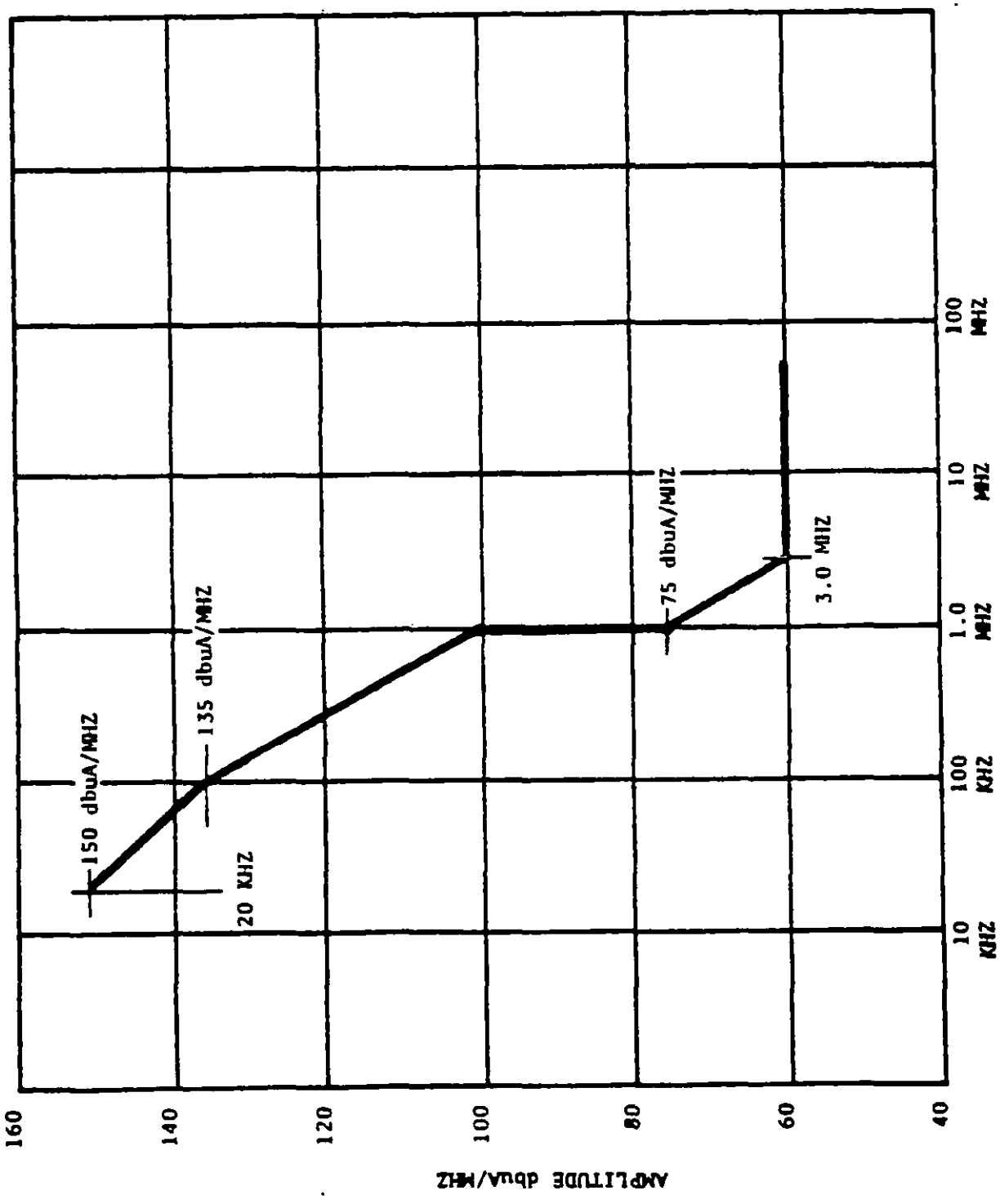


FIGURE 53. CE03, conducted emission, broad band limit.

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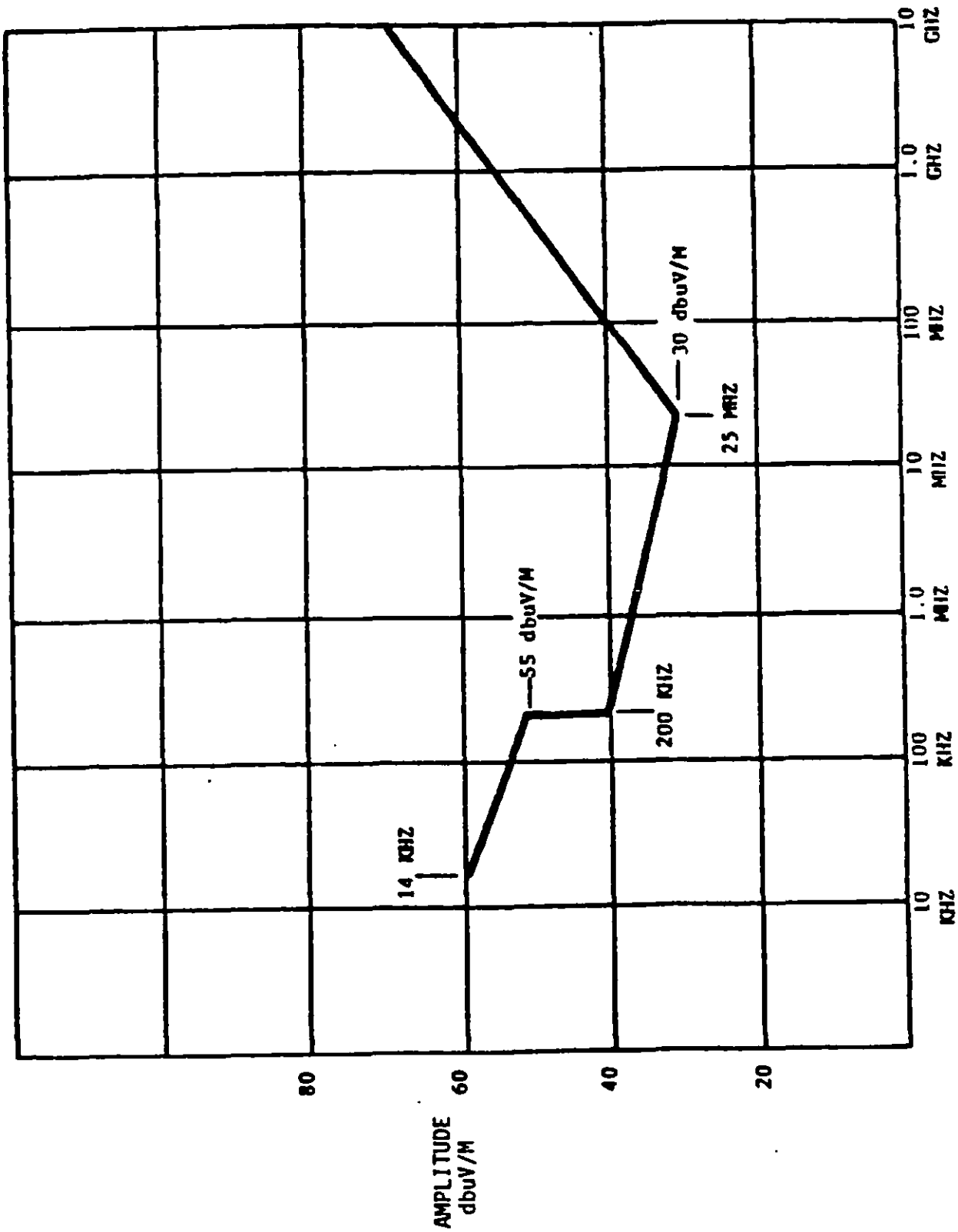


FIGURE 54. RE02, radiated emission, narrow band limits.

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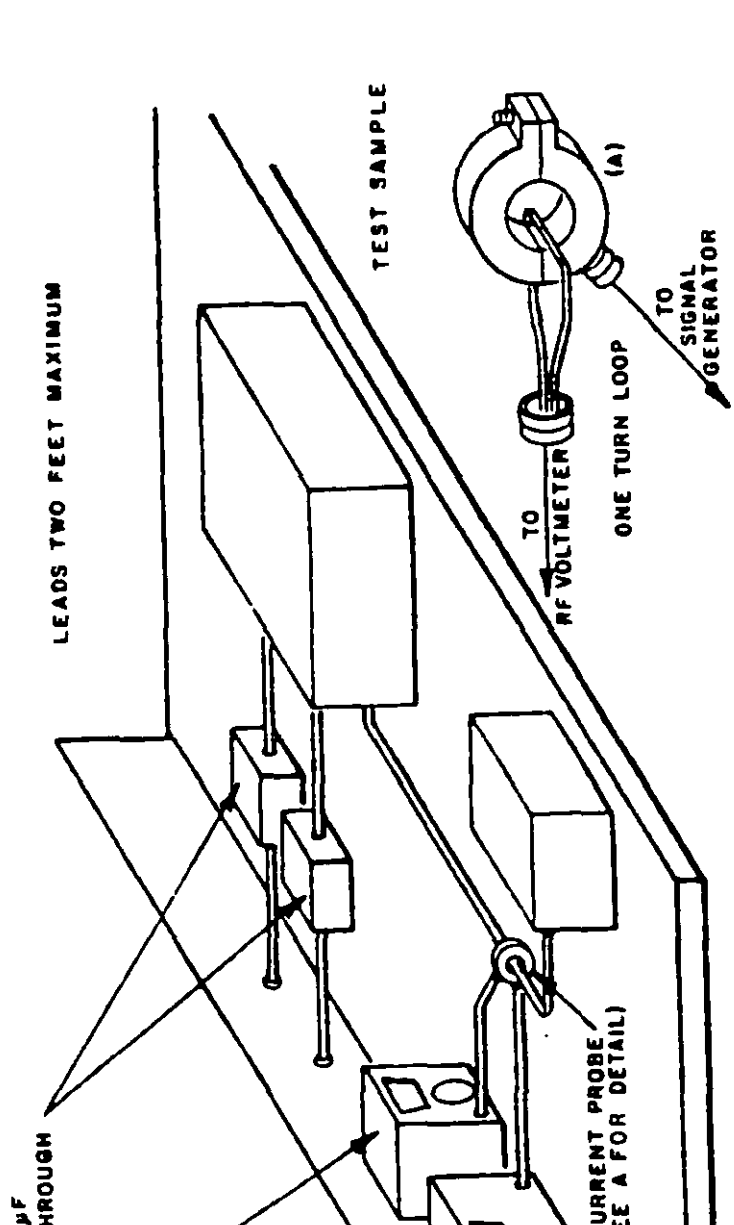


FIGURE 55. EMI test setup.

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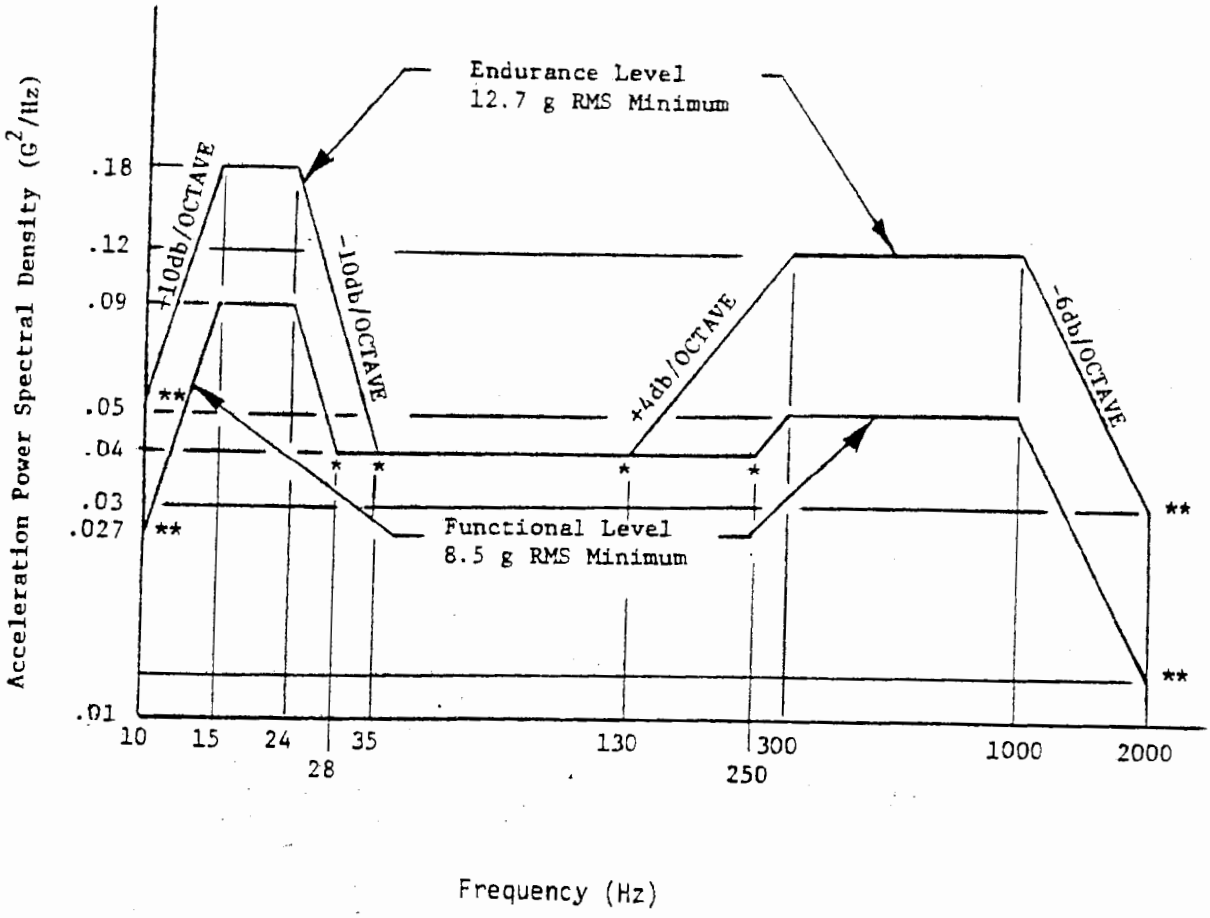
4.2.5 Random vibration test. A Random Vibration Test shall be conducted as specified in 4.2.5.1 through 4.2.5.3.

4.2.5.1 Test item operation. Unless otherwise specified herein, the test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. Both functional and endurance vibration tests shall be conducted. The test item shall meet specified performance requirements while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.2.5.2 Mounting techniques. The test item shall be attached by its normal mounting means, either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.2.5.3 Performance of test. The individual equipment test item shall be subject to broadband random vibration excitation with the power spectral density levels and power spectral density envelopes shown in Figure 56 for the vertical axis and Figure 57 for the major and minor horizontal axes. The test item shall be attached to the vibration exciter according to 4.2.5.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required: a functional level, and an endurance level. For each axis, one-half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the equipment specification operating requirements as specified in 4.2.5.1. The acceleration power spectral density (G^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in the appropriate figures. Test times shall, for each axis, be one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than 100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2000 Hz, except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2000 Hz. Tolerance levels in terms of dB are defined as:

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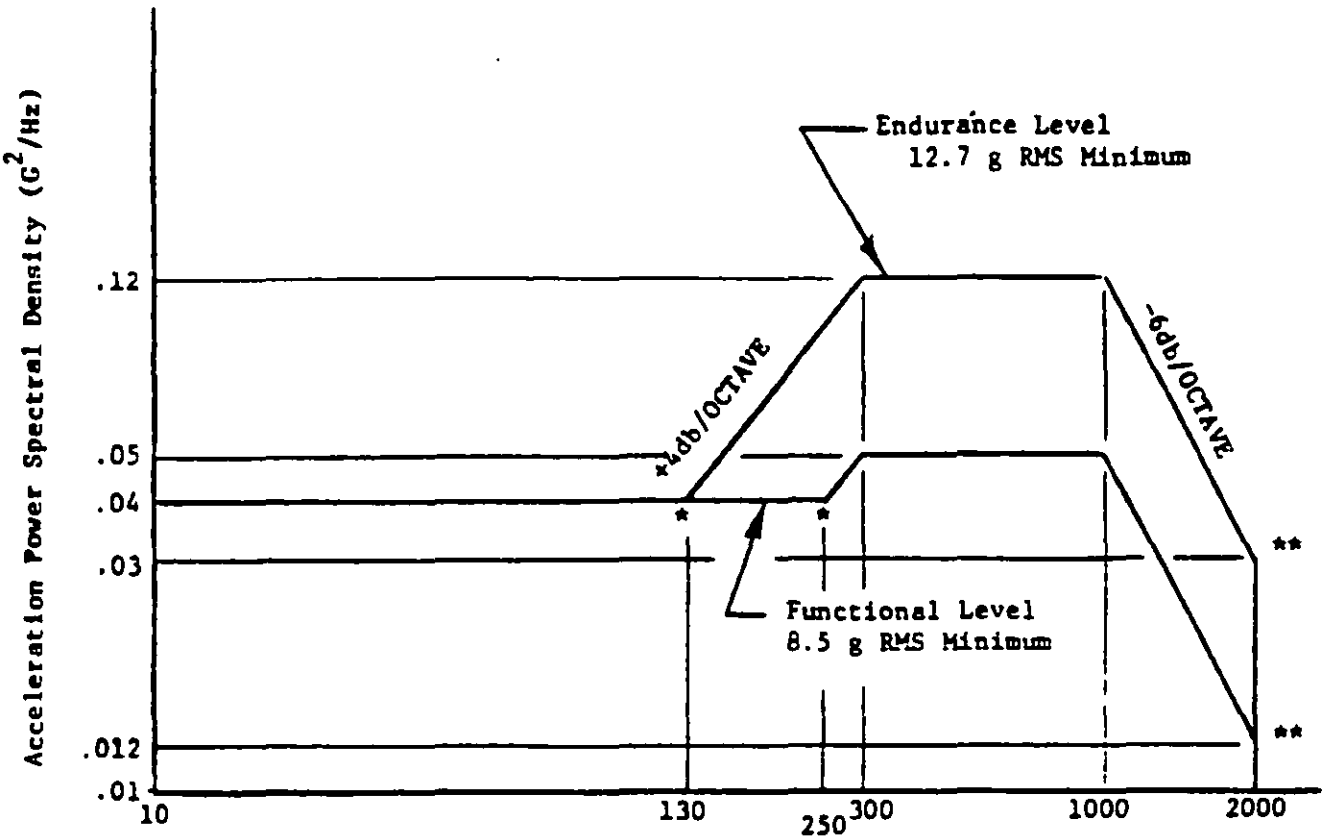


* Approximated frequency point. Actual frequency determined by where slope intersects power level.

** Approximated power level point. Actual power level determined by where slope intersects frequency.

FIGURE 56. Random vibration envelope (vertical plane).

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* Approximated frequency point. Actual frequency determined by where slope intersects power level.

** Approximated power level point. Actual power level determined by where slope intersects frequency.

FIGURE 57. Random vibration envelope (major/minor axes).

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$$dB = 10 \log_{10} \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density in G^2/HZ units. The term W_0 defines the specified level in G^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product. BT = 50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT = 50 minimum limitation.

a. On-line, contiguous filer, equalization/analysis system having a bandwidth as follows:

B = 25 Hz, maximum between 20 and 200 Hz B = 50 Hz, maximum between 200 and 1000 Hz
B = 100 Hz, maximum between 1000 and 2000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant Bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 25 Hz, maximum between 20 and 200 Hz
B = 50 Hz, maximum between 200 and 1000 Hz
B = 100 Hz, maximum between 1000 and 2000 Hz

(b) Analyzer averaging time = T = 2 RC = 1 second, minimum, where T = True averaging time constant.

(c) Analysis sweep rate (linear) = R = $\frac{B}{4\ RC}$ or $\frac{B^2}{8}$ (Hz/second)

(2) Constant percentage bandwidth analyzer.

(a) Filter bandwidth = pf_c = one-tenth of center frequency maximum ($0.1f_c$) where p = percentage and f_c = analyzer center frequency.

(b) Analyzer averaging time = T = $\frac{50}{pf_c}$, minimum.

(c) Analysis sweep rate (logarithmic) = R = $\frac{pf_c}{4RC}$ or (pf_c^2) (Hz/second), maximum, whichever is smaller.

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c. Digital power spectral density analysis system employing quantization techniques providing accuracies corresponding to the above approach.

4.2.6 Production equipments. Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample (see 6.3). Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production inspections. One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for test. This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The first article sample shall not be selected for this test.

4.3.1 Scope of initial production tests. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved first article sample in design, construction, workmanship, performance and quality and that it meets all requirements of this specification (see 6.2.1).

4.3.2 Accessory material. In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also have available such accessory material and data necessary to test the equipment (see 6.2.2).

4.3.3 Initial production sample approval. Approval of the initial production sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during the test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted, when specified in the contract (see 6.2.1).

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the Quality Conformance Tests. All inspection and testing shall be to the approved acceptance test procedures under the supervision of the Government representative (see 6.2.2). Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test reports showing quantitative results for all Quality Conformance Tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality Conformance Tests shall consist of the following:

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a. Individual Tests (see 4.4.1)

b. Sampling Tests (see 4.4.2)

c. Reliability Assurance Tests (see 4.4.3)

d. Special Tests (see 4.4.4)

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the Individual Tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product (see 4.4.1.1)

b. Operational Test (see 4.4.1.2)

c. Manufacturing Run-In Test (see 4.4.1.3)

d. Random Vibration Individual Test (see 4.4.3.3)

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified equipment operation.

4.4.1.3 Manufacturing run-in test. Each equipment (WRA) shall be operated under the conditions specified herein for a period of 50 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. This test shall not be required, if the Reliability Test includes a test on each equipment which consumes at least 50 hours of operation.

a. Temperature Ambient Room

b. Humidity Ambient Room

The equipment shall be operated during this period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of required performance is mandatory. During the 50 hour period of operation, the equipment shall be cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures (see 6.2.2).

4.4.1.3.1 Spare SRA run-in testing. Each SRA not subjected to run-in testing as an integral part of a computer shall be operated under the conditions specified in 4.4.3.4.1 for a period of 50 consecutive hours without

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failure. A failure shall be defined as anything which causes malfunctioning of the equipment. This test shall not be required if the reliability test includes a test on each equipment which consumes at least 50 hours of operation. Should a failure occur, it should be repaired and the test started over. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures (see 6.2.2).

4.4.2 Sampling tests. Equipment selected for Sampling Tests shall first have passed the Individual Tests. Sampling Tests are not required when Reliability Assurance Tests are conducted. Equipment will be selected for Sampling Tests by the Government representative, in accordance with Table XIX.

TABLE XIX. Sampling tests.

Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1 1/
Next 50	1
Next 100	1 for each additional 200 or fraction thereof

1/ One out of first ten need not be selected and tested, if Initial Production Tests are conducted.

4.4.2.1 Scope of sampling tests. As a minimum, each equipment selected for Sampling Tests shall be subjected to the following tests.

a. Complete Operational Test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational Test at certain environmental conditions. The conditions may vary for each equipment test and shall be based on results of the Preproduction, Initial Production, Individual and Special Tests.

c. Manufacturing Run-In Test specified in 4.4.1.3, except that the test duration shall be 120 hours, with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment accepted by the Government.

4.4.3 Reliability assurance test. Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

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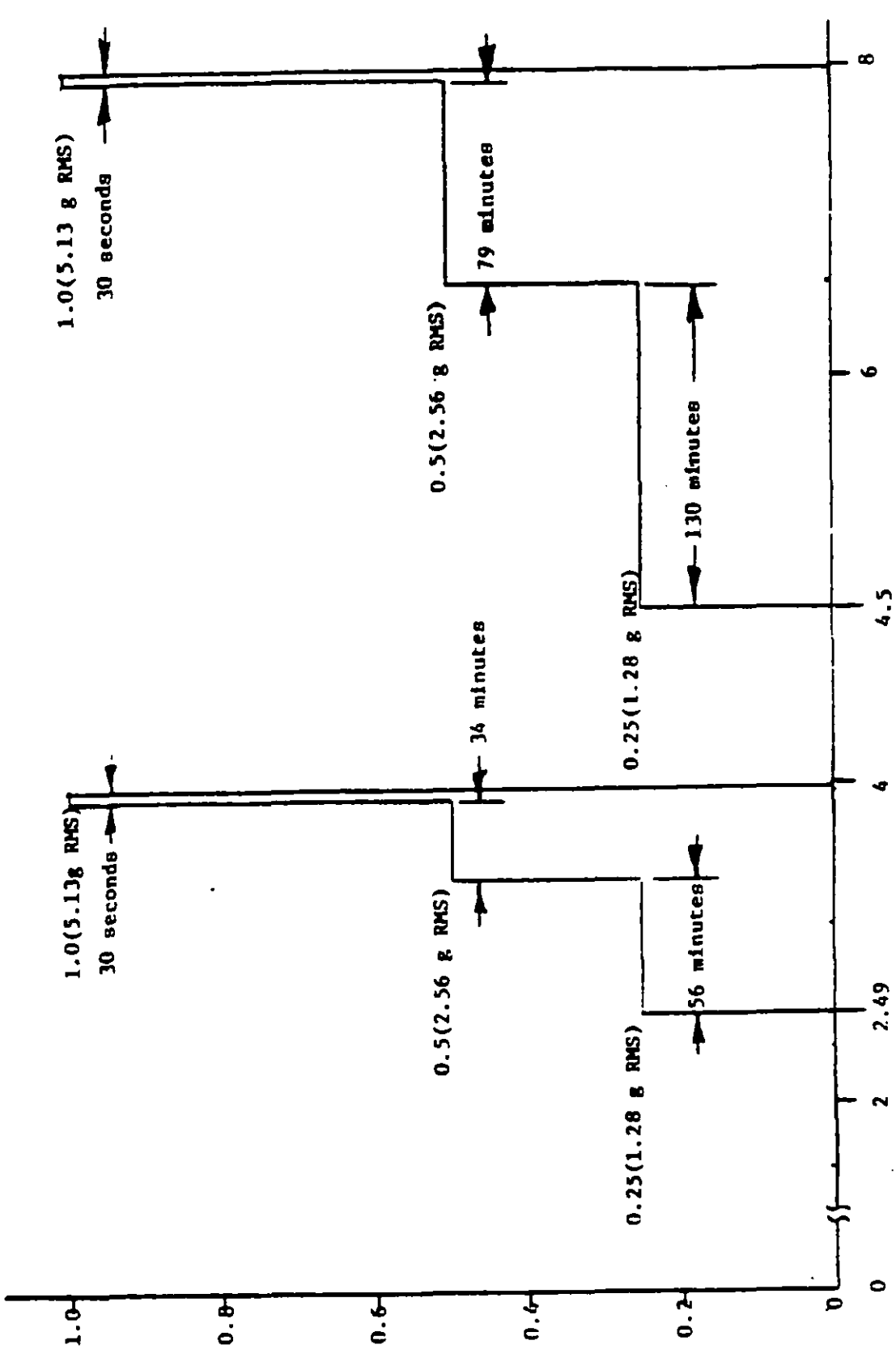
4.4.3.1 Qualification phase. Prior to the acceptance of equipment specified in the contract or order, the equipment shall be tested as in accordance with in MIL-STD-781, under the section entitled "First Article Reliability Qualification Test." The minimum number of equipments of each chassis type and configuration will be specified in the contract and the maximum number of equipment to be used shall be those listed in MIL-STD-781 (see 6.2.1). For the Qualification Phase, thermal cycling between -54C and +71C shall be used, and random vibration applied throughout the test in accordance with Figure 58. The power spectral density envelope shall be as shown in Figure 59. Mounting Techniques and Performance of Test (except for vibration levels and duration) shall be as specified in 4.2.5. Endurance test levels are not required. The Accept Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability production acceptance (sampling) phase tests. The equipment, throughout production, shall be tested as in accordance with in MIL-STD-781 (as modified below) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Thermal cycling between -54C and +71C shall be used, and random vibration applied through the test in accordance with Figure 58. The power spectral density envelope shall be as shown in Figure 59. Mounting Techniques and Performance of Test (except for vibration levels and duration) shall be as specified in 4.2.5. Endurance test levels are not required.

4.4.3.3 Random vibration individual test. During acceptance testing and prior to the All Equipment Test, each equipment shall operate failure free for a minimum of 10 minutes, while subjected to random vibration. The power spectral density envelope shall be as shown in Figure 60. Vibration isolators shall not be used.

4.4.3.4 All equipment test (AET). Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for 250 operating hours. MIL-STD-781 as modified by Figure 61 shall be used. The last five temperature cycles shall be failure free during the All Equipment Test; vibration in accordance with MIL-STD-781 is not required. Prior to the 250-hour test on each equipment, a burn-in period may be used. If the burn-in period is to be used, the details thereof must be included in the approved test procedures and must be the same for all equipment, including spare WRAs. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures from the beginning of the contract therein (not counting burn-in failures and burn-in operating time) shall be plotted in accordance with MIL-STD-781 (Test Plan XVIII). The reject line of Test Plan IIC shall be used as the reject line and the accept line of Test Plan IIC shall be used as the boundary line (these lines shall be extended as necessary to accomodate the data. At the conclusion of each month, the test results shall be made available to the procuring activity (see 6.2.2). At any time the cumulative totals of test hours and test failures plotted in accordance with the above methodology show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists, pending a review of the contractor's efforts to improve the equipment, the equipment parts, and the equipment workmanship, so that the entire compilation will show other than a reject decision.

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NOTES:
1/ Power applied to computer from hour 2 to hour 8.
2/ Chamber temperature cycle begins downward excursion at hour zero and upward excursion at hour 4.
3/ Cycle repeats after hour 8.

FIGURE 58. Vibration profile (RQT and sampling tests).

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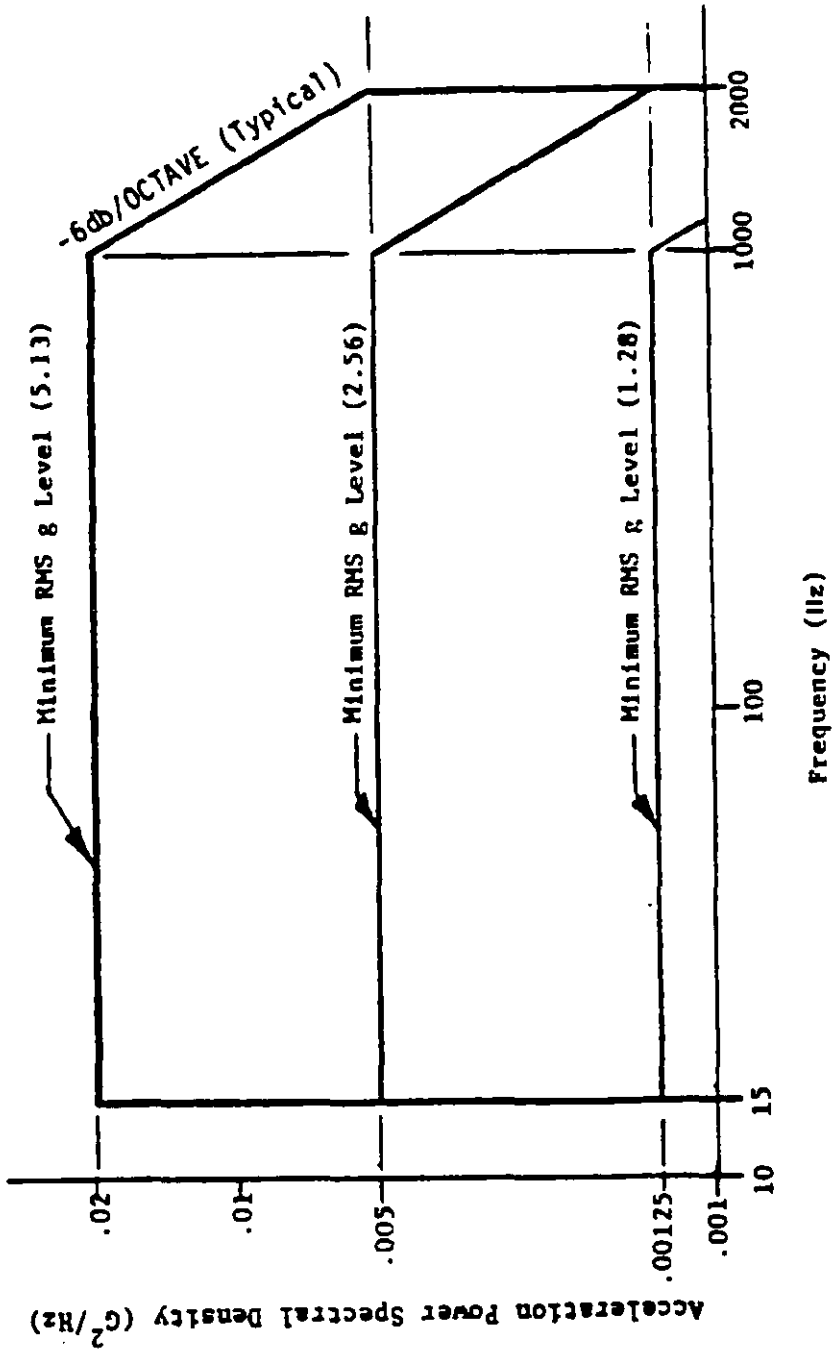


FIGURE 59. Random vibration power spectral density envelopes (RQT and sampling tests).

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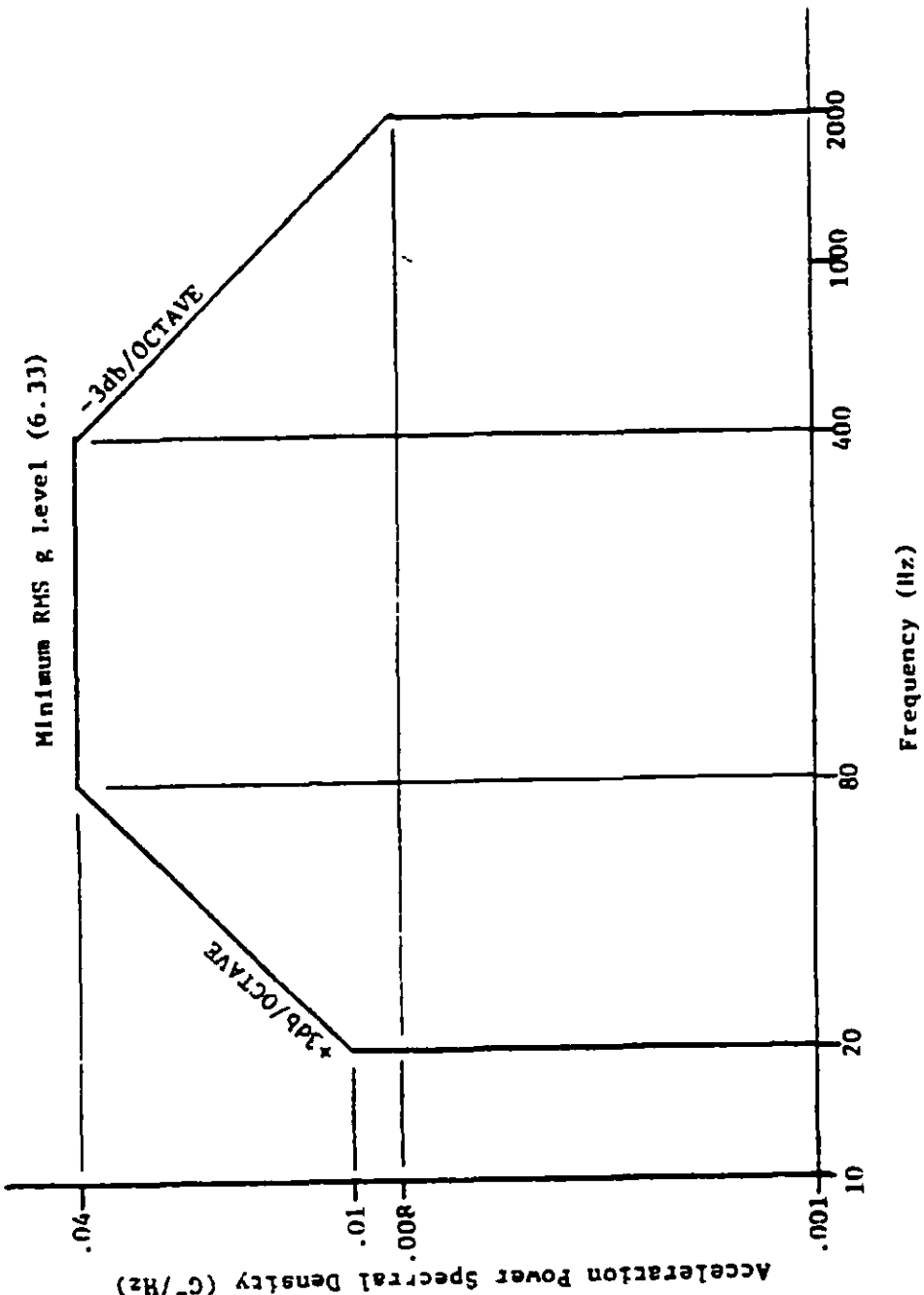
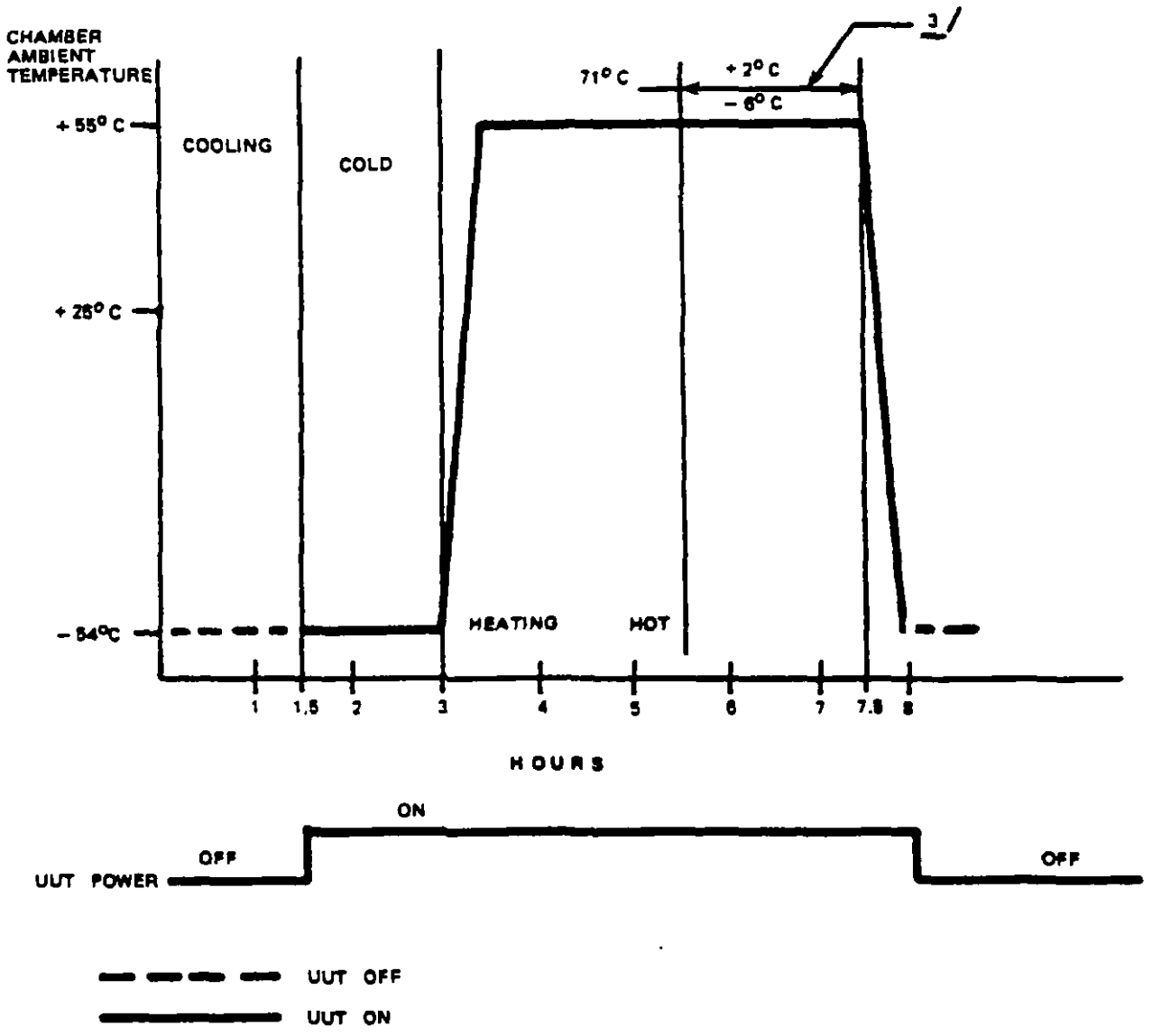


FIGURE 60. Random vibration envelope.

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NOTES:

1. The unit under test (UUT) shall be continuously and dynamically tested using a software test program during UUT power-on time.

2. Test chamber air may be used for cooling. Cooling airflow may be left on during the cool down cycle.

3. During the last 2 hours (minimum) of the hot cycle the exhaust air of the UUT shall be regulated to 71 +2/-6°C. For UUT configurations using the Type 3 chassis the exhaust air shall be 71 +2/-10°C.

FIGURE 61. AET temperature cycle.

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4.4.3.4.1 GFE SRA and spare SRA testing. GFE SRAs and spare SRAs shall be tested in either of the following methods, except for chassis and fans.

a. The SRA shall be tested in an AN/AYK-14(V) chassis for 250 hours of AET as specified in 4.4.3.4 with the last five temperature cycles failure free.

b. The SRA under test shall be installed in a test fixture in a manner approximating its installation in a computer for 250 hours of AET in accordance with 4.4.3.4 with the last five temperature cycles failure free.

(1) Chassis shall be tested for five temperature cycles of AET, power on, non-operating.

(2) Fans do not require burn-in or AET testing.

(3) Non-operational SRA hardware shall not be tested, i.e., RXM adapter, fan plate, etc.

(4) The IOP jumper shall be functionally tested only.

4.4.3.5 Test details. The test details, such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be prepared for acceptance by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests (see 6.2.2).

4.4.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change of the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to the Sampling or Reliability Assurance Tests.

4.4.4.1 Special test schedule. Selection of equipment for special tests shall be made as follows:

a. On an early equipment, after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of special tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests (see 6.2.2).

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4.4.5 Equipment failure. Should a failure occur during either the Sampling, Reliability Assurance, or Special Tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure.

b. Determine if the failure is an isolated case or design defect.

c. Make available for procuring activity acceptance, proposed corrective action intended to reduce the possibility of the same component or BIT failure(s) occurring in future tests, or documented justification where no corrective action has been proposed (see 6.2.2).

d. Where practical, include a test in the Individual Test to check all equipment for this requirement, until assurance is obtained that the defect has been corrected.

4.5 Maintainability demonstration. A maintainability demonstration for intermediate level MTTR shall be conducted in accordance with MIL-STD-2084 and MIL-STD-471. Test Method 1, Test Plan B or Method 8 of MIL-STD-471 shall be used to demonstrate the MTTR. The maintainability demonstration for the computer diagnostic programs specified in 3.4.7.4 shall include verification of the accuracy and completeness of the SRA diagnostic test programs. This shall include insertion of faults into the computer of the type and magnitude specified in MIL-STD-2077. The ability of the SRA diagnostic program to detect and identify the failure shall be verified.

4.6 Test procedures. The procedures used for conducting all tests specified in Section 4 shall be prepared by the contractor and submitted to the procuring activity for review and acceptance (see 6.2.2). The conformance test procedures shall include verification of the accuracy and completeness of all test programs developed for computer performance monitoring and end-to-end testing, and used to demonstrate specified computer performance during Quality Conformance Testing. Documentation of the Quality Conformance Test Programs shall be compatible with the documentation requirements of MIL-STD-2077. The right is reserved by the procuring activity or the Government representative to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided by the procuring activity and may be used. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.7 Memory module and SRA interchangeability demonstration test. The contractor shall conduct a Memory Module and SRA Interchangeability Test on all multiple sourced memory modules, and SRAs. The test procedure shall be made available for acceptance by the procuring activity (see 6.2.2). The test shall include, but not be limited to:

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a. Module/SRA Performance Tests

b. Substitution of second source modules/SRAs

c. Substitution of semiconductor memory module for core memory module

4.8 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all requirements of this specification.

4.9 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of MIL-E-17555.

5. PACKAGING

5.1 Preservation-packaging. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555, MIL-STD-794, and MIL-STD-129. In the event the equipment is not covered in MIL-E-17555, the method preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and Method 11A of MIL-P-116.

6. NOTES

6.1 Intended use. The equipment specified in this specification is intended for use as a general purpose airborne Navigation, Fire Control, Communications, Sensor Monitor and Control, System Test functions, and any tactical use within a weapon system.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and the appropriate "slash" sheet defining the configuration. Configurations not contained in a "slash" sheet may be ordered by specifying the chassis and modules required. Individual chassis or modules may be ordered (see 1.2).

b. Requirements for a reliability program (see 3.4.3).

c. EMI test results and test cable to be furnished by procuring activity (see 3.4.6).

d. Requirements for a Human Performance/Human Engineering Program (see 3.4.14).

e. Selection of test requirements (see 4.1. and 4.3.1).

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f. Corrective action on previously accepted equipment, if required (see 4.3.3).

g. Minimum number of equipments of each chassis type and configuration (see 4.4.3.1).

h. When test procedures are to be expanded or modified (see 4.6).

i. Selection of applicable levels of packaging and packing (see 5.1).

j. When a first article is required for inspection and approval (see 6.3).

k. When maintainability demonstration test is required (see 40.3 of appendix B).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR-7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.
3.3	*Engineering Change Proposal (ECP's) and* Requests for Deviations and Waivers (Long Form)	DI-E-2037
4.2, 4.6	Procedure, First Article Inspection	DI-T-4901
4.2.1, 4.2.6	Report, First Article Inspection	DI-T-4902
4.3.2, 4.4, 4.4.4.2, 4.6,	Procedures, Initial Production/ Acceptance Inspection	DI-T-4903
4.4	Reports, Production Inspection	DI-T-4904
4.4.3.5, 4.6, 40.2.1, 40.2.3	Procedures, Reliability Tests and Demonstrations	DI-R-7035
4.4.3.4, 4.7, 30.2	Report, Reliability Tests and Demonstrations	DI-R-7034
4.4.5c, 40.2.2.3, 40.4	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
4.4.5c	Plan, Corrective Action	DI-R-7038

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Paragraph No.	Data requirement title	Applicable DID No.
4.4.5c, 4.4.1.3, 4.4.1.3.1, 40.4.2, 40.4.3	Report, Failure	UDI-R-21141A
4.7	Procedures, Test	UDI-T-21347
30.1, 30.1.2, 30.1.3, 30.1.4, 30.3, 30.4.2.4, 30.5.5, 30.8, 40.4.2	Plan, Reliability Program	DI-R-7079
40.6.1	Report, Reliability Status	DI-R-7080
30.4.1	Report, Reliability Prediction	DI-R-7082
30.4.2.1	Report, Failure Modes, Effects and Criticality Analysis Plan	DI-R-7086
30.4.2.1	Report, Failure Modes, Effects and Criticality Analysis (FMECA)	DI-R-7085
30.4.2.3	Report, Sneak Circuit Analysis	DI-R-7083
30.4.2.4	Report, Electronic Parts/Circuits Tolerance Analysis	DI-R-7084
30.4.2.5	Report, Thermal Survey	DI-R-7036
30.4.2.6	Report, Vibration Survey	DI-R-7037
30.4.2.7	Analysis, Worst Case	DI-R-22953
30.5.1	Plan, Parts Control Program	DI-E-7026
30.5.2	Drawings, Engineering and Associated Lists	DI-E-7031
30.5.2	Specification Requirement Sheets	DI-E-1133
30.5.3	Nonstandard Parts Approval Requests/ Proposed Additions to PPSL	DI-E-7028
30.5.4	Report, Reliability Stress Analysis	UDI-R-21423
30.9	Minutes, Formal Reviews, Inspections and Audits	DI-E-3118
30.10.2, 30.10.3, 30.10.4, 30.10.6	Plan, Maintainability Program	DI-R-7103

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Paragraph No.	Data requirement title	Applicable DID No.
30.10.3, 30.10.7	Report, Maintainability Predictions	DI-R-7108
40.1, 40.2.2.1,	Plan, Reliability Test	DI-R-7033
40.6.2	Report, Maintainability Status	DI-R-7104

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article requirement. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and test approval of the documents first article.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.5 Type designations. The type designation (nomenclature) may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.13. The correct type number should be used on nameplates, shipping records and instruction books, as applicable.

6.6 Associated equipment. The computer is used with the AN/ASM-607(V)7 Memory Loader Verifier of MIL-T-85413/7.

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APPENDIX A

INSTRUCTION REPERTOIRE FOR COMPUTER SET, STANDARD AIRBORNE AN/AYK-14(V)

10. SCOPE

10.1 Scope. This appendix details the instruction repertoire for the Computer Set, Standard Airborne AN/AYK-14(V). This Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. COMPUTER INSTRUCTION REQUIREMENTS

30.1 Instruction repertoire for AN/AYK-14(V). Refer to Table XX for instruction repertoire for AN/AYK-14(V).

30.2 AN/UYK-20(V) instructions modified for AN/AYK-14(V). Refer to Table XXI for AN/UYK-20(V) instructions modified for AN/AYK-14(V).

30.3 Unique AN/AYK-14(V) processor instructions. Refer to Table XXII for unique AN/AYK-14(V) processor instructions.

30.4 IOP instruction repertoire for AN/AYK-14(V). Refer to Table XXIII for IOP instruction repertoire for AN/AYK-14(V).

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TABLE XX. Instruction repertoire for AN/AYK-14(V).

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
*00 2 a m	02 a m	SPT a,y,m	Stack Put Top	(Y)→(Ra), (Ra)→Y		-NA-		8.50
*00 3 a m	03 a m	BL a,y,m	Byte Load	(Y) byte → Ra	0	0	X	*2.03-2.18
*01 0 a m	04 a m	LR a,m	Load (Register)	(Rm) → Ra	0	0	X	.79
*01 1 a m	05 a m	LI a,m	Load (Indirect)	(Y*) → Ra	0	0	X	1.49
*01 2 a m	06 a m	LK a,y,m	Load (Constant)	Y → Ra	0	0	X	1.58
*01 3 a m	07 a m	L a,y,m	Load	(Y) → Ra	0	0	X	1.80
*02 0 a 00	08 a 0	PR a	Make Positive	If (Ra) < 0, (Ra)'→Ra	X	X	X	.90
*02 0 a 01	08 a 1	NR a	Make Negative	If (Ra) > 0, (Ra)'→Ra	X	0	X	.90
02 0 a 02	08 a 2	RR a	Round	(Ra) + (Ra ⊕ 1):15→Ra	X	X	X	.90
*02 0 a 03	08 a 3	IPI a	·Initiate Processor Int	Set processor interrupt a		-NA-		1.75
*02 0 a 04	08 a 4	TCR a	Two's Complement	(Ra)' → Ra	X	X	X	.81
02 0 a 05	08 a 5	TCDR a	Two's Complement Double	(Ra, Ra ⊕ 1)'→Ra, Ra ⊕ 1	X	X	X	.93
*02 0 a 06	08 a 6	OCR a	One's Complement	(Ra) bit-by-bit complement → Ra	0	0	X	.78
*02 0 a 07	08 a 7	SCI a	Support Channel Input	Support channel input → Ra	0	0	X	.93
*02 0 a 10	08 a 8	IRDR a	Increase Ra by 1	(Ra) + 1 → Ra	X	X	X	.78
*02 0 a 11	08 a 9	DRDR a	Decrease Ra by 1	(Ra) - 1 → Ra	X	X	X	.78
*02 0 a 12	08 a A	IRTR a	Increase Ra by 2	(Ra) + 2 → Ra	X	X	X	.80
*02 0 a 13	08 a B	DRTR a	Decrease Ra by 2	(Ra) - 2 → Ra	X	X	X	.78
*02 0 - 14	08 - C	IPLF	·IPL Failed	Set IPLF discrete		-NA-		1.50
*02 0 a 15	08 a D	DJ a	·Diagnostic Jump	R15 → uP		-NA-		TBD
*02 0 - 16	08 - E	RBT	·Reset Bit Timer	0 → Bit Timer		-NA-		3.75
*02 0 - 17	08 - F	SBT	·Set BIT Indicator	Set BIT Indicator		-NA-		4.25
02 1 a m	09 a m	LDI a,m	Load Double (Indirect)	(Y*, Y* ⊕ 1) → Ra, Ra ⊕ 1	0	0	X	2.02
02 3 a m	0B a m	LD a,y,m	Load Double	(Y, Y ⊕ 1) → Ra,Ra ⊕ 1	0	0	X	2.40
*03 0 a 00	0C a 0	ER a	Executive Return	Generate Interrupt: P-1 → Ra	0	0	X	.93
*03 0 a 01	0C a 1	SSOR a	Store SR1	(SR1) → Ra	0	0	X	.96
*03 0 a 02	0C a 2	SSTR a	Store SR2	(SR2) → Ra	0	0	X	.80
*03 0 a 03	0C a 3	SCR a	Store Clock	(RTC register):15-0 → Ra	0	0	X	1.25
03 0 a 04	0C a 4	LPR a	Load P	(Ra) → P		-NA-		1.70
*03 0 a 05	0C a 5	LSOR a	·Load SR1	(Ra) → SR1		-NA-		6.02
*03 0 a 06	0C a 6	LSTR a	·Load SR2	(Ra) → SR2		-NA-		.90
*03 0 a 07	0C a 7	LCR a	·Load RTC lower	(Ra) → RTC register: 15-0		-NA-		5.40
*03 0 - 10	0C - 8	ECR	·Enable Clock and Interrupt	Enable RTC register and Interrupt		-NA-		3.51
*03 0 - 11	0C - 9	DCR	·Disable Clock	Disable RTC register		-NA-		1.48
03 0 a 12	0C a A	LEM a	·Load and Enable Monitor Clock	(Ra) → Monitor clock register: enable count-down		-NA-		5.28
03 0 - 13	0C - B	DM	·Disable Monitor Clock	Disable monitor clock register		-NA-		2.78
03 0 a 14	0C a C	LCRD a	·Load Double and Enable Clock	(Ra, Ra ⊖ 1) → RTC. enable count up		-NA-		6.00
*03 0 a 15	0C a D	SCRD a	Store Clock Double	(RTC register) → Ra, Ra ⊖ 1	0	0	X	2.29
*03 0 - 16	0C - E	ECIR	·Enable Clock Interrupt	Enable RTC overflow interrupt		-NA-		1.62
*03 0 - 17	0C - F	DCIR	·Disable Clock Interrupt	Disable RTC overflow interrupt		-NA-		1.32

*/IOP Instructions ·/Executive Mode Instructions
**/byte 0-byte 1
**/With 900 nsec core memory, word interleaved, no IOP or EAU

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C OV CC	CPU** Time in usec
#03 1 a m	0D a m	SCIO a,m	Support Channel I/O	(Ra) → support channel buffer, m → I/O code, Set channel busy	-NA-	TBD
03 3 a m	0F a m	LM a,y,m	Load multiple	(Y...Y+m-a) → Ra...Rm	-NA-	***2.12+1.2n
04 0 a 01	10 a 1	RVR a	Reverse Register	Reverse (Ra)	0 0 X	7.26
04 0 a 02	10 a 2	CNT a	Count Ones	Number of binary ones in Ra → Ra ⊙ 1	-NA-	6.96
04 0 a 03	10 a 3	SFR a	Scale Factor	Shift (Ra, Ra ⊙ 1) left until (Ra):15 ≠ (Ra):14; shift count → Ra ⊙ 1+1 ⑪	-NA-	⑫3.48-8.82
04 0 a 04	10 a 4	SMC a	Store Monitor Clock	(Mon) → Ra	-NA-	1.75
#04 2 a m	12 a m	QPT a,y,m	Queue Put Top	(Y) → (Ra); (Ra) → Y; if (Y) was = 0 then (Ra) → Y ⊙ 1	-NA-	8.28
#04 3 a m	13 a m	BLX a,y,m	Byte Load and Index by 1	(Y) byte → Ra; (Rm)+1 → Rm	0 0 X	*4.19-2.56
#05 0 a m	14 a m	SBR a,m	Set Bit	1 → (Ra):m	0 0 X	.78
#05 1 a m	15 a m	LXI a,m	Load and Index by 1 (Indirect)	(Y*) → Ra; (Rm)+1 → Rm	0 0 X	1.29
#05 2 a m	16 a m	QPB a,y,m	Queue Put Bottom	(Ra) → (Y ⊙ 1); (Ra) → Y ⊙ 1, 0 → (Ra)	-NA-	10.34
#05 3 a m	17 a m	LX a,y,m	Load and Index by 1	(Y) → Ra; (Rm)+1 → Rm	0 0 X	1.92
#06 0 a m	18 a m	ZBR a,m	Zero Bit	0 → (Ra):m	0 0 X	.80
06 1 a m	19 a m	LDXI a,m	Load Double Index by 2 (Indirect)	(Y*, Y* ⊙ 1) → Ra, Ra ⊙ 1; (Rm)+2 → Rm	0 0 X	2.25
#06 2 a m	1A a m	SGT a,y,m	Stack Get Top	(Y) → Ra; if (Y) ≠ 0 then ((Y)) → Y and P+3 → P; if (Y) = 0 then P+2 → P	-NA-	9.80
06 3 a m	1B a m	LDX a,y,m	Load Double, Index by 2	(Y, Y ⊙ 1) → Ra, Ra ⊙ 1; (Rm)+2 → Rm	0 0 X	2.42
#07 0 a m	1C a m	CBR a,m	Compare Bit	Compare bit m of Ra with zero	0 0 X	.78
#07 1 - m	1D - m	LPI m	Load PSW (Indirect)	(Y*, Y*+1, Y*+2) → P, SR1, SR2	-NA-	7.92
#07 2 a m	1E a m	QGT a,y,m	Queue Get Top	(Y) → Ra; if (Y) = 0 then P+2 → P; if (Y) ≠ 0 then P+3 → P, ((Y)) → Y; if ((Y)) = 0 then Y → Y ⊙ 1	-NA-	9.80
#07 3 - m	1F - m	LP y,m	Load PSW	(Y, Y+1, Y+2) → P, SR1, SR2	-NA-	7.78
#10 0 a m	20 a m	LRSR a,m	Logical Right Shift (Register)	Shift (Ra) right (Rm): 5-0 places; zero fill	0 0 X	⑬2.28+.20n
#10 2 a m	22 a m	LRS a,y,m	Logical Right Shift	Shift (Ra) right Y:5-0 places; zero fill	0 0 X	3.10+.22n
#10 3 a m	23 a m	BS a,y,m	Byte Store	(Ra):7-0 → Y byte	-NA-	*4.25-2.39
#11 0 a m	24 a m	ARSR a,m	Algebraic Right Shift (Register)	Shift (Ra) right (Rm): 5-0 places; sign fill	0 0 X	2.28+.20n
#11 1 a m	25 a m	SI a,m	Store (Indirect)	(Ra) → Y*	-NA-	1.40
#11 2 a m	26 a m	ARS a,y,m	Algebraic Right Shift	Shift (Ra) right Y:5-0 places; sign fill	0 0 X	3.06+.24n
#11 3 a m	27 a m	S a,y,m	Store	(Ra) → Y	-NA-	1.91
12 0 a m	28 a m	LRDR a,m	Logical Right Double Shift (Register)	Shift (Ra, Ra ⊙ 1) right (Rm):5-0 places; zero fill	0 0 X	2.61+.21n

#/ IOP Instructions ·/Executive Mode Instructions

**/ byte 0 - byte 1

**/ With 900 nsec core memory, word interleaved, no IOP or EAU

***/1.2 usec additional per word over 1

⑪/ Count = 31 for all zeros or all ones

⑫/ varies with # of bits to scale

⑬/ n equals # of shifts

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
12 1 a m	29 a m	SDI a,m	Store Double (Indirect)	(Ra, Ra ⊕ 1) → Y*, Y* ⊕ 1		-NA-		2.06
12 2 a m	2A a m	LRD a, y, m	Logical Right Double Shift	Shift (Ra, Ra ⊕ 1) right Y:5-0 places; zero fill	0	0	X	①3.39+.22n
12 3 a m	2B a m	SD a,y,m	Store Double	(Ra, Ra ⊕ 1) → Y, Y ⊕ 1		-NA-		2.40
13 0 a m	2C a m	ARDR a,m	Algebraic Right Double Shift (Register)	Shift (Ra, Ra ⊕ 1) right (Rm):5-0 places; sign fill	0	0	X	2.61+.21n
13 2 a m	2E a m	ARD a,y,m	Algebraic Right Double Shift	Shift (Ra, Ra ⊕ 1) right Y:5-0 places, sign fill	0	0	X	3.42+.20n
13 3 a m	2F a m	SM a,y,m	Store Multiple	(Ra...Rm) → Y...Y+ m-a		-NA-		⑤2.50-18.0
*14 0 a m	30 a m	ALSR a,m	Algebraic Left Shift (Register)	Shift (Ra) left (Rm): 5-0 places; zero fill	0	X	X	2.64+.21n
*14 2 a m	32 a m	ALS a,y,m	Algebraic Left Shift	Shift (Ra) left Y:5-0 places; zero fill	0	X	X	3.25+.21n
*14 3 a m	33 a m	BSX a,y,m	Byte Store, Index by 1	(Ra):7-0 → Y byte; (Rm)+1 → Rm		-NA-		**2.44-2.60
*15 0 a m	34 a m	CLSR a,m	Circular Left Shift (Register)	Shift (Ra) circularly left (Rm):5-0 places	0	0	X	2.66+.21n
*15 1 a m	35 a m	SXI a,m	Store, Index by 1 (Indirect)	(Ra) → Y*; (Rm)+1 → Rm		-NA-		1.14
*15 2 a m	36 a m	CLS a,y,m	Circular Left Shift	Shift (Ra) circularly left Y:5-0 places	0	0	X	3.26+.21n
*15 3 a m	37 a m	SX a,y,m	Store, Index by 1	(Ra) → Y; (Rm)+1 → Rm		-NA-		1.74
16 0 a m	38 a m	ALDR a,m	Algebraic Left Double Shift (Register)	Shift (Ra, Ra ⊕ 1) left (Rm):5-0 places; zero fill	0	X	X	3.10+.21n
16 1 a m	39 a m	SDXI a,m	Store Double, Index by 2 (Indirect)	(Ra, Ra ⊕ 1) → Y*, Y* ⊕ 1; (Rm)+2 → Rm		-NA-		2.10
16 2 a m	3A a m	ALD a,y,m	Algebraic Left Double Shift	Shift (Ra, Ra ⊕ 1) left Y:5-0 places; zero fill	0	X	X	3.86+.20n
16 3 a m	3B a m	SDX a,y,m	Store Double, Index by 2	(Ra, Ra ⊕ 1) → Y, Y ⊕ 1; (Rm)+2 → Rm		-NA-		2.63
17 0 a m	3C a m	CLDR a,m	Circular Left Double Shift (Register)	Shift (Ra, Ra ⊕ 1) circularly left (Rm): 5-0 places	0	0	X	2.61+.21n
17 1 - m	3D - m	SZI m	Store Zeros (Indirect)	0 → Y*		-NA-		1.17
17 2 a m	3E a m	CLD a,y,m	Circular Left Double Shift	Shift (Ra, Ra ⊕ 1) circularly left Y:5-0 places	0	0	X	3.39+.20n
17 3 - m	3F - m	SZ y,m	Store Zeros	0 → Y		-NA-		2.85
*20 0 a m	40 a m	SUR a,m	Subtract (Register)	(Ra) - (Rm) → Ra	X	X	X	.80
*20 1 a m	41 a m	SUI a,m	Subtract (Indirect)	(Ra) - (Y*) → Ra	X	X	X	1.30
*20 2 a m	42 a m	SUK a,y,m	Subtract (Constant)	(Ra) - Y → Ra	X	X	X	1.60
*20 3 a m	43 a m	SU a,y,m	Subtract	(Ra) - (Y) → Ra	X	X	X	1.83
21 0 a m	44 a m	SUDR a,m	Subtract Double (Register)	(Ra, Ra ⊕ 1) - (Rm, Rm ⊕ 1) → Ra, Ra ⊕ 1	X	X	X	1.11
21 1 a m	45 a m	SUDI a,m	Subtract Double (Indirect)	(Ra, Ra ⊕ 1) - (Y*, Y* ⊕ 1) → Ra, Ra ⊕ 1	X	X	X	2.15

*/ IOP Instructions
**/byte 0 - byte 1
**/With 900 nsec core memory, word interleaved, no IOP or EAU
①/n equals # of shifts
⑤/varies with # words to store

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
21 3 a m	47 a m	SUD a,y,m	Subtract Double	(Ra, Ra ⊙ 1) - (Y, Y ⊙ 1) → Ra, Ra ⊙ 1	X	X	X	2.40
#22 0 a m	48 a m	AR a,m	Add (Register)	(Ra) + (Rm) → Ra	X	X	X	.78
#22 1 a m	49 a m	AI a,m	Add (Indirect)	(Ra) + (Y*) → Ra	X	X	X	1.50
#22 2 a m	4A a m	AK a,y,m	Add (Constant)	(Ra) + Y → Ra	X	X	X	1.58
#22 3 a m	4B a m	A a,y,m	Add	(Ra) + (Y) → Ra	X	X	X	1.80
23 0 a m	4C a m	ADR a,m	Add Double (Register)	(Ra, Ra ⊙ 1) + (Rm, Rm ⊙ 1) → Ra, Ra ⊙ 1	X	X	X	1.08
23 1 a m	4D a m	ADI a,m	Add Double (Indirect)	(Ra, Ra ⊙ 1) + (Y*, Y* ⊙ 1) → Ra, Ra ⊙ 1	X	X	X	2.10
23 3 a m	4F a m	AD a,y,m	Add Double	(Ra, Ra ⊙ 1) + (Y, Y ⊙ 1) → Ra, Ra ⊙ 1	X	X	X	2.43
#24 0 a m	50 a m	CR a,m	Compare (Register)	(Ra) - (Rm)	X	X	X	.80
#24 1 a m	51 a m	CI a,m	Compare (Indirect)	(Ra) - (Y*)	X	X	X	1.30
#24 2 a m	52 a m	CK a,y,m	Compare (Constant)	(Ra) - Y	X	X	X	1.60
#24 3 a m	53 a m	C a,y,m	Compare	(Ra) - (Y)	X	X	X	1.83
25 0 a m	54 a m	CDR a,m	Compare Double (Register)	(Ra, Ra ⊙ 1) - (Rm, Rm ⊙ 1)	X	X	X	1.08
25 1 a m	55 a m	CDI a,m	Compare Double (Indirect)	(Ra, Ra ⊙ 1) - (Y*, Y* ⊙ 1)	X	X	X	1.93
25 3 a m	57 a m	CD a,y,m	Compare Double	(Ra, Ra ⊙ 1) - Y, Y ⊙ 1)	X	X	X	2.40
#26 0 a m	58 a m	MR a,m	Multiply (Register)	(Ra ⊙ 1) · (Rm) → Ra, Ra ⊙ 1	0	0	X	4.14
#26 1 a m	59 a m	MI a,m	Multiply (Indirect)	(Ra ⊙ 1) · (Y*) → Ra, Ra ⊙ 1	0	0	X	4.50
#26 2 a m	5A a m	MK a,y,m	Multiply (Constant)	(Ra ⊙ 1) · Y → Ra, Ra ⊙ 1	0	0	X	4.96
#26 3 a m	5B a m	M a,y,m	Multiply	(Ra ⊙ 1) · (Y) → Ra, Ra ⊙ 1	0	0	X	4.98
#27 0 a m	5C a m	DR a,m	Divide (Register)	(Ra, Ra ⊙ 1)/(Rm) → Ra ⊙ 1; remainder → Ra	0	X	X	8.40
#27 1 a m	5D a m	DI a,m	Divide (Indirect)	(Ra, Ra ⊙ 1)/(Y*) → Ra ⊙ 1; remainder → Ra	0	X	X	9.03
#27 2 a m	5E a m	DK a,y,m	Divide (Constant)	(Ra, Ra ⊙ 1)/Y → Ra ⊙ 1; remainder → Ra	0	X	X	9.50
#27 3 a m	5F a m	D a,y,m	Divide	(Ra, Ra ⊙ 1)/(Y) → Ra ⊙ 1; remainder → Ra	0	X	X	10.66
#30 0 a m	60 a m	ANDR a,m	AND (Register)	(Ra) ⊙ (Rm) → Ra	0	0	X	.80
#30 1 a m	61 a m	ANDI a,m	AND (Indirect)	(Ra) ⊙ (Y*) → Ra	0	0	X	1.30
#30 2 a m	62 a m	ANDK a,y,m	AND (Constant)	(Ra) ⊙ Y → Ra	0	0	X	1.60
#30 3 a m	63 a m	AND a,y,m	AND	(Ra) ⊙ (Y) → Ra	0	0	X	1.82
#31 0 a m	64 a m	ORR a,m	OR (Register)	(Ra) ⊕ (Rm) → Ra	0	0	X	.81
#31 1 a m	65 a m	ORI a,m	OR (Indirect)	(Ra) ⊕ (Y*) → Ra	0	0	X	1.30
#31 2 a m	66 a m	ORK a,y,m	OR (Constant)	(Ra) ⊕ Y → Ra	0	0	X	1.57
#31 3 a m	67 a m	OR a,y,m	OR	(Ra) ⊕ (Y) → Ra	0	0	X	1.84
32 0 a m	68 a m	XORR a,m	Exclusive OR (Register)	(Ra) $\bar{\oplus}$ (Rm) → Ra	0	0	X	.81
32 1 a m	69 a m	XORI a,m	Exclusive OR (Indirect)	(Ra) $\bar{\oplus}$ (Y*) → Ra	0	0	X	1.30
32 2 a m	6A a m	XORK a,y,m	Exclusive OR (Constant)	(Ra) $\bar{\oplus}$ Y → Ra	0	0	X	1.57
32 3 a m	6B a m	XOR a,y,m	Exclusive OR	(Ra) $\bar{\oplus}$ (Y) → Ra	0	0	X	1.84

*/IOP Instructions

**/With 900 nsec core memory, word interleaved, no IOP or EAU

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
33 0 a m	6C a m	MSR a,m	Masked Substitute (Register)	If (Ra ⊖ 1):n = 1; (Rm):n → Ra:n	0	0	X	1.08
33 1 a m	6D a m	MSI a,m	Masked Substitute (Indirect)	If (Ra ⊖ 1):n = 1; (Y*):n → Ra:n	0	0	X	1.73
33 2 a m	6E a m	MSK a,y,m	Masked Substitute (Constant)	If (Ra ⊖ 1):n = 1; Y:n → Ra:n	0	0	X	1.88
33 3 a m	6F a m	MS a,y,m	Masked Substitute	If (Ra ⊖ 1) = 1; (Y):n → Ra:n	0	0	X	2.28
#34 0 a m	70 a m	CMR a,m	Compare Masked (Register)	[(Ra) ⊙ (Ra ⊖ 1)]: [(Rm) ⊙ (Ra ⊖ 1)]	0	0	X	1.08
#34 1 a m	71 a m	CMI a,m	Compare Masked (Indirect)	[(Ra) ⊙ (Ra ⊖ 1)]: [(Y*) ⊙ (Ra ⊖ 1)]	0	0	X	1.73
#34 2 a m	72 a m	CMK a,y,m	Compare Masked (Constant)	[(Ra) ⊙ (Ra ⊖ 1)]: [Y ⊙ (Ra ⊖ 1)]	0	0	X	1.87
#34 3 a m	73 a m	CM a,y,m	Compare Masked	[(Ra) ⊙ (Ra ⊖ 1)]: [(Y) ⊙ (Ra ⊖ 1)]	0	0	X	2.28
#35 0 - -	74 - -	IOCR	•Input/Output Command	Execute (C Cell): 0 → C Cell:15-14		-NA-		TBD
35 1 - m	75 - m	BFI m	Biased Fetch (Indirect)	(Y*):set CC: 1 → (Y*):15,14	0	0	X	①5.50-5.70
35 2 - m	76 - m	REX y,m	Remote Execute	Execute (Y): P + 2 → P unless jump		-NA-		TBD
35 3 - m	77 - m	BF y,m	Biased Fetch	(Y): set CC; 1 → (Y): 15,14	0	0	X	6.10
#40 0 00 m	80 0 m	JER m	Jump Equal	If CC indicates = or 0, (Rm) → P		-NA-		1.75
#40 0 01 m	80 1 m	JNER m	Jump Not Equal	If CC indicates ≠ or not 0, (Rm) → P		-NA-		1.75
#40 0 02 m	80 2 m	JGER m	Jump Greater or Equal	If CC indicates ≥ or =, (Rm) → P		-NA-		1.75
#40 0 03 m	80 3 m	JLSR m	Jump Less	If CC indicates < or -, (Rm) → P		-NA-		1.75
#40 0 04 m	80 4 m	JOR m	Jump Overflow	If overflow set, (Rm) → P		-NA-		1.75
#40 0 05 m	80 5 m	JCR m	Jump Carry	If carry set, (Rm) → P		-NA-		1.75
#40 0 06 m	80 6 m	JPTR m	Jump Power out of Tolerance	If power out of tolerance, (Rm) → P		-NA-		1.75
40 0 07 m	80 7 m	JBR m	Jump Bootstrap 2 Selected	If bootstrap 2 selected, (Rm) → P		-NA-		TBD
#40 0 10 m	80 8 m	JR m	Jump	(Rm) → P		-NA-		1.70
#40 0 11 m	80 9 m	JSR m	•Jump after Stop	Stop; (Rm) → P		-NA-		2.28
40 0 12 m	80 A m	JKSR 1,m	•Jump. If Key Set - Stop, then Jump (Register)	If key 1 set, stop; (Rm) → P		-NA-		2.28
40 0 13 m	80 B m	JKSR 2,m	•Jump. If Key Set - Stop, then Jump (Register)	If key 2 set, stop; (Rm) → P		-NA-		2.28
#40 0 14 m	80 C m	JSCR m	Jump Support Channel Busy - RR	If support channel busy, (Rm) → P		-NA-		1.11
#40 1 d	81 d	LJ xD	Local Jump	P + xD → P		-NA-		1.65
#40 2 00 m	82 0 m	JE y,m	Jump Equal	If CC indicates = or 0, Y → P		-NA-		1.95
#40 2 01 m	82 1 m	JNE y,m	Jump Not Equal	If CC indicates ≠ or not 0, Y → P		-NA-		1.95
#40 2 02 m	82 2 m	JGE y,m	Jump Greater than or Equal	If CC indicates ≥ or =, Y → P		-NA-		1.95

#/IOP Instructions ·/Executive Mode Instructions ①/ Operand; positive-negative

**/With 900 nsec core memory, word interleavec, no IOP or EAU
Note: Times are for jumps taken

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C OV CC	CPU** Time in usec
#40 2 03 m	82 3 m	JLS y,m	Jump Less	If CC indicates < or ., Y → P	-NA-	1.95
#40 2 04 m	82 4 m	JO y,m	Jump on Overflow	If overflow set, Y → P	-NA-	1.95
#40 2 05 m	82 5 m	JC y,m	Jump on Carry	If carry set, Y → P	-NA-	1.95
#40 2 06 m	82 6 m	JPT y,m	Jump of Power out of Tolerance	If power out of tolerance, Y → P	-NA-	1.95
40 2 07 m	82 7 m	JB y,m	Jump if Bootstrap 2 Selected	If bootstrap 2 selected, Y → P	-NA-	2.13
#40 2 10 m	82 8 m	J y,m	Jump	Y → P	-NA-	1.75
#40 2 11 m	82 9 m	JS y,m	·Jump after Stop	Stop; Y → P	-NA-	2.30
40 2 12 m	82 A m	JKS 1,y,m	·Jump. If Key Set - Stop, then Jump	If key 1 set, stop; Y → P	-NA-	2.30
40 2 13 m	82 B m	JKS 2,y,m	·Jump. If Key Set - Stop, then Jump	If key 2 set, stop; Y → P	-NA-	2.30
#40 2 14 m	82 C m	JSC y,m	Jump Support Channel Busy - RX	If support channel busy, Y → P	-NA-	TBD
#40 3 00 m	83 0 m	JE *y,m	Jump Equal	If CC indicates = or 0, (Y) → P	-NA-	2.54
#40 3 01 m	83 1 m	JNE *y,m	Jump Not Equal	If CC indicates ≠ or not 0, (Y) → P	-NA-	2.54
#40 3 02 m	83 2 m	JGE *y,m	Jump Greater or Equal	If CC indicates ≥ or ., (Y) → P	-NA-	2.54
#40 3 03 m	83 3 m	JLS *y,m	Jump Less	If CC indicates < or ., (Y) → P	-NA-	2.54
#40 3 04 m	83 4 m	JO *y,m	Jump on Overflow	If overflow set, (Y) → P	-NA-	2.54
#40 3 05 m	83 5 m	JC *y,m	Jump on Carry	If carry set, (Y) → P	-NA-	2.54
#40 3 06 m	83 6 m	JPT *y,m	Jump if Power out of Tolerance	If power out of tolerance, (Y) → P	-NA-	2.54
40 3 07 m	83 7 m	JB *y,m	Jump if Bootstrap 2 Selected	If bootstrap 2 selected, (Y) → P	-NA-	3.21
#40 3 10 m	83 8 m	J *y,m	Jump	(Y) → P	-NA-	2.48
#40 3 11 m	83 9 m	JS *y,m	·Jump after Stop	Stop; (Y) → P	-NA-	3.24
40 3 12 m	83 A m	JKS 1,*y,m	·Jump. If Key Set - Stop, then Jump	If key 1 set, stop; (Y) → P	-NA-	3.24
40 3 13 m	83 B m	JKS 2,*y,m	·Jump. If Key Set - Stop, then Jump	If key 2 set, stop; (Y) → P	-NA-	3.24
#40 3 14 m	83 C m	JSC *y,m	Jump Support Channel Busy - RX	If support channel busy; (Y) → P	-NA-	TBD
#41 0 a m	84 a m	XJR a,m	Index Jump Register	If (Ra) ≠ 0, (Ra) - 1 → Ra; (Ra) → P	-NA-	⑤.90,1.69
#41 1 d	85 d	LJI xD	Local Jump (Indirect)	[P + xD] → P	-NA-	2.46
#41 2 a m	86 a m	XJ a,y,m	Index Jump	If (Ra) ≠ 0, (Ra) - 1 → Ra; Y → P	-NA-	⑤1.75,1.80
#41 3 a m	87 a m	XJ a,*y,m	Index Jump	If (Ra) ≠ 0, (Ra) - 1 → Ra; (Y) → P	-NA-	⑤1.75,2.49
#42 0 a m	88 a m	JLRR a,m	Jump, Link Register (Register)	(P) + 1 → Ra; (Rm) → P	-NA-	1.86
#42 2 a m	8A a m	JLR a,y,m	Jump, Link Register	(P) + 2 → Ra; Y → P	-NA-	1.86
#42 3 a m	8B a m	JLR a,*y,m	Jump, Link Register	(P) + 2 → Ra; (Y) → P	-NA-	1.86
#43 1 d	8D d	LJLM xD	Local Jump, Link Memory	(P) + 1 → P + xD; P + xD + 1 → P	-NA-	1.89

*/IOP Instructions ·/Executive Mode Instructions ⑤/no jump, jump

**/With 900 nsec core memory, word interleaved, no IOP or EAU
Note: Times are for jumps taken

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C OV CC	CPU** Time in usec
#43 2 - n	8E - n	JLM y,n	Jump, Link Memory	(P) + 2 → Y; Y + 1 → P	-NA-	2.08
#43 3 - n	8F - n	JLM *y,n	Jump, Link Memory	(P) + 2 → (Y); (Y) + 1 → P	-NA-	3.28
#44 0 a n	90 a n	JZR a,n	Jump Zero (Register	If (Ra) = 0, (Rn) → P	-NA-	1.63
#44 1 d	91 d	LJE xD	Local Jump Equal	If CC indicates = or 0, (P) + xD → P	-NA-	1.95
#44 2 a n	92 a n	JZ a,y,n	Jump Zero	If (Ra) = 0, Y → P	-NA-	2.10
#44 3 a n	93 a n	JZ a,*y,n	Jump Zero	If (Ra) = 0, (Y) → P	-NA-	2.76
#45 0 a n	94 a n	JNZR a,n	Jump Not Zero (Register)	If (Ra) ≠ 0, (Rn) → P	-NA-	1.63
#45 1 d	95 d	LJNE xD	Local Jump Not Equal	If CC indicates ≠ or not 0, (P) + xD → P	-NA-	1.95
#45 2 a n	96 a n	JNZ a,y,n	Jump Not Zero	If (Ra) ≠ 0, Y → P	-NA-	2.10
#45 3 a n	97 a n	JNZ a,*y,n	Jump Not Zero	If (Ra) ≠ 0, (Y) → P	-NA-	2.75
#46 0 a n	98 a n	JPR a,n	Jump Positive (Register)	If (Ra ≥ 0, (Rn) → P	-NA-	1.65
#46 1 d	99 d	LJGE xD	Local Jump Greater or Equal	If CC indicates ≥ or =, (P) + xD → P	-NA-	1.98
#46 2 a n	9A a n	JP a,y,n	Jump Positive	If (Ra) ≥ 0, Y → P	-NA-	2.11
#46 3 a n	9B a n	JP a,*y,n	Jump Positive	If (Ra) ≥ 0, (Y) → P	-NA-	2.77
#47 0 a n	9C a n	JNR a,n	Jump Negative (Register)	If (Ra) < 0, (Rn) → P	-NA-	1.64
#47 1 d	9D d	LJLS xD	Local Jump Less	If CC indicates < or =, (P) + xD → P	-NA-	1.95
#47 2 a n	9E a n	JN a,y,n	Jump Negative	If (Ra) < 0, Y → P	-NA-	2.10
#47 3 a n	9F a n	JN a,*y,n	Jump Negative	If (Ra) < 0, (Y) → P	-NA-	2.77
50 0 a n	A0 a n	FSUR a,n	Floating Point Subtract (Register)	(Ra, RA ⊙ 1) - (Rn, RN ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
50 1 a n	A1 a n	FSUI a,n	Floating Point Subtract (Indirect)	(Ra, Ra ⊙ 1) - (Y*, Y* ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
50 3 a n	A3 a n	FSU a,y,n	Floating Point Subtract	(Ra, Ra ⊙ 1) - (Y, Y ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
51 0 a n	A4 a n	FAR a,n	Floating Point Add (Register)	(Ra, Ra ⊙ 1) + (Rn, Rn ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
51 1 a n	A5 a n	FAI a,n	Floating Point Add (Indirect)	(Ra, Ra ⊙ 1) + (Y*, Y* ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
51 3 a n	A7 a n	FA a,y,n	Floating Point Add	(Ra, Ra ⊙ 1) + (Y, Y ⊙ 1) → Ra, Ra+1; Res.→ Ra+2, Ra+3	X X	TBD
52 0 a n	A8 a n	FMR a,n	Floating Point Multiply (Register)	(Ra, Ra ⊙ 1) · (Rn, Rn ⊙ 1) → Ra, Ra+1, Res. → Ra+2, Ra+3	X X	TBD
52 1 a n	A9 a n	FMI a,n	Floating Point Multiply (Indirect)	(Ra, Ra ⊙ 1) · (Y*, Y* ⊙ 1) → Ra, Ra+1, Res. → Ra+2, Ra+3	X X	TBD
52 3 a n	AB a n	FM a,y,n	Floating Point Multiply	(Ra, Ra ⊙ 1) · (Y, Y ⊙ 1) → Ra, Ra+1, Res. → Ra+2	X X	TBD
53 0 a n	AC a n	FDR a,n	Floating Point Divide (Register)	(Ra, Ra ⊙ 1)/(Rn, Rn ⊙ 1) →Ra,Ra+1; Res.→Ra+2,Ra+3	X X	TBD

*/ IOP Instructions
**/With 900 nsec core memory, word interleaved, no IOP or EAU
Note: Times are for jumps taken.

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexidecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
53 1 a m	AD a m	FDI a,m	Floating Point Divide (Indirect)	(Ra, Ra⊖1)/(Y*, Y*⊖1) → Ra, Ra+1; Res. → Ra+2, Ra+3		X	X	TBD
53 3 a m	AF a m	FD a,y,m	Floating Point Divide	(Ra, Ra⊖1)/(Y, Y⊖1) → Ra, Ra+1; Res.→ Ra+2, Ra+3		X	X	TBD
54 0 a m	B0 a m	LARR a,m	·Load Address Register (Register)	(Rm) → ARr		-NA-		3.16
54 1 a m	B1 a m	LARI a,m	·Load Address Register (Indirect)	(Y*) → ARr		-NA-		3.57
54 3 a m	B3 a m	LARM a,y,m	·Load Address Register Multiple	(Y...,Y + u) → ARr..., ARr+u		-NA-		7.01+2.0w
55 0 a m	B4 a m	SARR a,m	Store Address Register (Register)	(ARr) → Rm		-NA-		1.82
55 1 a m	B5 a m	SARI a,m	Store Address Register (Indirect)	(ARr) → Y*		-NA-		2.96
55 3 a m	B7 a m	SARM a,y,m	Store Address Register Multiple	(ARr,...,ARr+u) → Y,..., Y+u		-NA-		5.03+1.8w
56 0 a m	B8 a m	MDR a,m	Multiply Double (Register)	(Ra, Ra⊖1) · (Rm, Rm⊖1) → Ra, Ra+1, Ra+2, Ra+3	0	0	X	8.06
56 1 a m	B9 a m	MDI a,m	Multiply Double (Indirect)	(Ra, Ra⊖1) · (Y*, Y*⊖1) → Ra, Ra+1, Ra+2, Ra+3	0	0	X	8.58
56 3 a m	BB a m	MD a,y,m	Multiply Double	(Ra, Ra⊖1) · (Y, Y⊖1) → Ra, Ra+1, Ra+2, Ra+3	0	0	X	8.31
57 0 a m	BC a m	DDR a,m	Divide Double (Register)	(Ra, Ra+1, Ra+2, Ra+3)/(Rm, Rm⊖1) → Ra+2, Ra+3; Rem.→ Ra, Ra+1	0	X	X	44.76
57 1 a m	BD a m	DDI a,m	Divide Double (Indirect)	(Ra, Ra+1, Ra+2, Ra+3)/(Y*, Y*⊖1) → Ra+2, Ra+3; Rem.→ Ra, Ra+1	0	X	X	44.96
57 3 a m	BF a m	DD a,y,m	Divide Double	(Ra, Ra+1, Ra+2, Ra+3)/(Y, Y⊖1) → Ra+2, Ra+3; Rem.→ Ra, Ra+1	0	X	X	45.60
*60 0 a m	C0 a m	LLRS a,m	Literal Logical Right Shift	Shift (Ra) right m places, zero fill	0	0	X	1.32+.21n
*60 1 a m	C1 a m	LARS a,m	Literal Algebraic Right Shift	Shift (Ra) right m places, sign fill	0	0	X	1.32+.21n
60 2 a m	C2 a m	LLRD a,m	Literal Logical Right Double Shift	Shift (Ra, Ra⊖1) right m places; zero fill	0	0	X	1.47+.21n
60 3 a m	C3 a m	LARD a,m	Literal Algebraic Right Double Shift	Shift (Ra, Ra⊖1) right m places, sign fill	0	0	X	1.47+.21n
*61 0 a m	C4 a m	LALS a,m	Literal Algebraic Left Shift	Shift (Ra) left m places, zero fill	0	X	X	1.68+.21n
*61 1 a m	C5 a m	LCLS a,m	Literal Circular Left Shift	Shift (Ra) left circular m places	0	0	X	1.32+.21n
61 2 a m	C6 a m	LALD a,m	Literal Algebraic Left Double Shift	Shift (Ra, Ra⊖1) left m places, zero fill	0	X	X	2.10+.20n
61 3 a m	C7 a m	LCLD a,m	Literal Circular Left Double Shift	Shift (Ra, Ra⊖1) left circular m places	0	0	X	1.47+.21n

*/IOP Instructions ·/Executive Mode Instructions

**/With 900 nsec core memory, word interleaved, no IOP or EAU

n equals # of shifts

w equals # of words

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	CPU** Time in usec
#62 0 a m	C8 a m	LSU a,m	Literal Subtract	(Ra) - m → Ra	X	X	X	.80
62 1 a m	C9 a m	LSUD a,m	Literal Subtract Double	(Ra, Ra ⊕ 1) - m →Ra, Ra ⊕ 1	X	X	X	1.12
#62 2 a m	CA a m	LA a,m	Literal Add	(Ra) + m → Ra	X	X	X	.80
62 3 a m	CB a m	LAD a,m	Literal Add Double	(Ra, Ra ⊕ 1) + m → Ra, Ra ⊕ 1	X	X	X	.91
#63 0 a m	CC a m	LL a,m	Literal Load	m → Ra	0	0	X	.80
#63 1 a m	CD a m	LC a,m	Literal Compare	(Ra) - m	X	X	X	.81
63 2 a m	CE a m	LMUL a,m	Literal Multiply	(Ra ⊕ 1) · m → Ra, Ra ⊕ 1	0	0	X	4.15
63 3 a m	CF a m	LDIV a,m	Literal Divide	(Ra, Ra ⊕ 1)/m → Ra ⊕ 1; remainder → Ra	0	X	X	8.53
64 3 a m	D3 a m	BSU a,y,m	Byte Subtract	(Ra) - (Y) byte → Ra	X	X	X	2.07
65 3 a m	D7 a m	BA a,y,m	Byte Add	(Ra) + (Y) byte → Ra	X	X	X	2.06
66 3 a m	DB a m	BC a,y,m	Byte Compare	(Ra) - (Y) byte	X	X	X	2.04
67 3 a m	DF a m	BCX a,y,m	Byte Compare and Index by 1	(Ra) - (Y) byte; (Rm) + 1 → Rm	X	X	X	2.22
			COMMAND/CHAIN INSTRUCTIONS					
#70 0 00 00	E0 0 0	ACR	Channel Control	Master clear all channels				4.67/7.12
#70 0 00 04	E0 0 4	ACR 4 CCR 0,4	Channel Control	Enable external interrupts, all channels				4.67/7.12
#70 0 00 05	E0 0 5	ACR 5 CCR 0,5	Channel Control	Disable external interrupts, all channels				4.67/7.12
#70 0 00 06	E0 0 6	ACR 6 CCR 0,6	Channel Control	Enable Class III Interrupts, Priorities 2, 3, 4				4.67/7.12
#70 0 00 07	E0 0 7	ACR 7 CCR 0,7	Channel Control	Disable Class III Interrupts, Priorities 2, 3, 4				4.67/7.12
#70 0 a 10	E0 a 8	CCR a,8	Channel Control	Master clear channel a				3.82/6.90
#70 0 a 14	E0 a C	CCR a,12	Channel Control	Enable channel a external interrupts				3.82/6.90
#70 0 a 15	E0 a D	CCR a,13	Channel Control	Disable channel a external interrupts				3.82/6.90
#70 0 a 16	E0 a E	CCR a,14	Channel Control	Enable channel a, Class III, Priorities 2, 3, 4				3.82/6.90
#70 0 a 17	E0 a F	CCR a,15	Channel Control	Disable channel a, Class III, Priorities 2, 3, 4				3.82/6.90
			COMMAND INSTRUCTIONS					
#71 2 a m	E6 a m	LCMK m,y	Write control memory	(Y) → Channel a C&m				TBD
#71 2 a 02	E6 a 2	ICK a,y	Initiate Input Chain	Y → Channel a Chain Pointer, initiate input chain				6.69
#71 2 a 06	E6 a 6	OCK a,y	Initiate Output Chain	Y → Channel a Chain Pointer, initiate output chain				6.69
#71 3 a m	E7 a m	WIM a,y,m	Write Control Memory	(Y) → Channel a C&m				6.12
#72 3 a m	EB a m	RIM a,y,m	Read Control Memory	Channel a (C&m) → Y				4.78

#/IOP Instructions

**/With 900 nsec core memory, word interleaved, no IOP or EAU

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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C OV CC	CPU** Time in usec
			COMMAND INSTRUCTIONS (Cont)			
#76 0 a m	F8 a m	SICR a,m	Set and Clear Discretes	Set or clear channel a discrete function		3.95
#76 3 a m	FB a m	SST a,y,m	Store Status	Channel a Status bits per m → Y		5.09
#77 0 - m	FC - m	SIOP m,y	Start IOP ③	m:0 → IOP SR1:12, Y → IOP P if m = 0 or 1		4.09
#77 2 a m	FE a m	XIM a,y,m	Exchange Control Memory ④	Channel a (CMm) → Y; (Y ⊙ 1) → Channel a CMm		5.15
			CHAIN INSTRUCTIONS			
#70 2 a m	E2 a m	IM a,y,m	Initiate Message	Y → CMm; Initiate message activity		TBD
#70 3 a 00	E3 a 0	IO a,y	IO Function a	(Y, Y ⊙ 1) → BCW, BAP; Initiate transfer		TBD
#71 2 00 m	E6 0 m	LCMK m,y	Load Control Memory	Y → CMm		6.75-7.50
#71 3 00 m	E7 0 m	LCM m,y	Load Control Memory	(Y) → CMm		6.58-7.39
#72 3 00 m	EB 0 m	SCM m,y	Store Control Memory	(CMm) → Y		5.75
#73 0 00 00	EC 0 0	HCF	Halt Chain	Halt chaining, a even		5.61
#73 0 01 00	EC 1 0	IPR	Interrupt Processor	Generate chain interrupt, a odd		5.61
#73 3 00 00	EF 0 0	ZF y	Zero Flag	0 → Y:15,14, a even		6.75
#73 3 01 00	EF 1 0	SF y	Set Flag	1 → Y:15,14, a odd		6.75
#74 2 00 00	F2 0 0	SJMC 0,y	Serial Jump on Met Condition	Unconditional, Y → CAP; clear flag		TBD
#74 2 01 00	F2 1 0	SJMC 1,y	Serial Jump on Met Condition	If suppress flag not set, Y → CAP; clear flag		TBD
#74 2 02 00	F2 2 0	SJMC 2,y	Serial Jump on Met Condition	If monitor flag set, Y → CAP; clear flag		TBD
#75 0 00 m	F4 0 m	SFSC m	Search for SYNC	Perform function(s) assigned to m-bits		5.25
#76 0 00 m	F8 0 m	CSIR m	Serial Interface Control	Set or clear discrete function		5.61
#76 3 - m	FB - m	CSST y,m	Store Status	Status bits per m → Y		6.30
#77 1 - m	FD - m	BJ m,y	Bit Jump	(Y) → CAP if (CM3):m = 1		6.75
#77 2 - m	FE - m	XCM m,y	Exchange Control Memory	(CMm) → Y; (Y ⊙ 1) → CMm		7.45

*/IOP Instructions ②/m = 2 or 6 ③/no operation unless IOP
**With 900 nsec memory, word interleaved, no IOP or EAU

1/ CPU Times are nominal
2/ IOP instructions
3/ BYTE 0-BYTE 1
4/ With 900 μsec core memory, word interleaved, no IOP or EAU
5/ 1.2 μsec additional per word over 1
6/ Executive mode instruction
7/ Count equals 31 for all zeros all ones
8/ Varies with number of bits to scale
9/ N equals number of shifts
10/ Varies with number of words to store
11/ Operand; positive-negative
12/ Times are for jumps
13/ No jump, jump
14/ W equals number of words

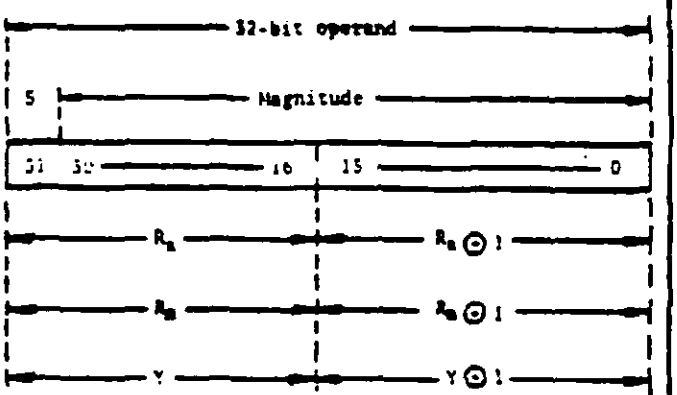
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TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued

LEGEND	
B/	Byte pointer, 0 = Upper, 1 = Lower
C,	Carry
CC	Condition Code
OV/	Overflow
IW	Indirect Word
j/	Designator Field in IW
x/	General Register Designator in IW1
y/	Contents of Second Instruction Word or IW2
Y/	Effective Operand Address or Constant
Y* /	Effective Operand Address in Rm
TM/	I/O Transfer Mode 00 = Abort Input Transfer 01 = 8-bit Byte Transfer 10 = 16-bit Word Transfer 11 = 32-bit Dual word Transfer
BWC /	Buffer Word Count
BAF /	Buffer Address Pointer
CM /	Control Memory Word
CAF /	Chain Address Pointer
RTC /	Real-Time Clock
( ) :	Contents of Register or Address
r /	(R_n) 5-0
u :	(R_n) 13-8
· /	Multiply
⊙ /	Or
⊕ /	Exclusive or
⊙ /	And
SR1 /	Status Register No. 1
SR2 /	Status Register No. 2
: /	Bit number follows
' /	Two's Complement
C Cell	I/O Command Cell

FORMAT	OPERAND FORMATION
RR	Operand = (R_m)
RI-1	Local Jump Address Y = (P) + xD
RI-2	Operand at Y* = (R_m)
RX	Operand Y = y + (R_m) if m ≠ 0 Operand Y = y if m = 0
*RX Word	Operand at Y = y if m = 0 Operand at Y = y + (R_m) if m ≠ 0,8,A,C,E Operand at indirect address if m = 8,A,C,E
*RX Byte	Operand at Y upper if m = 0 Operand at Y = (R_m)/2 + y if m ≠ 0,8,A,C,E B = (R_m):0 Operand at indirect address if m = 8,A,C,E
RL	Operand = m (an absolute literal)

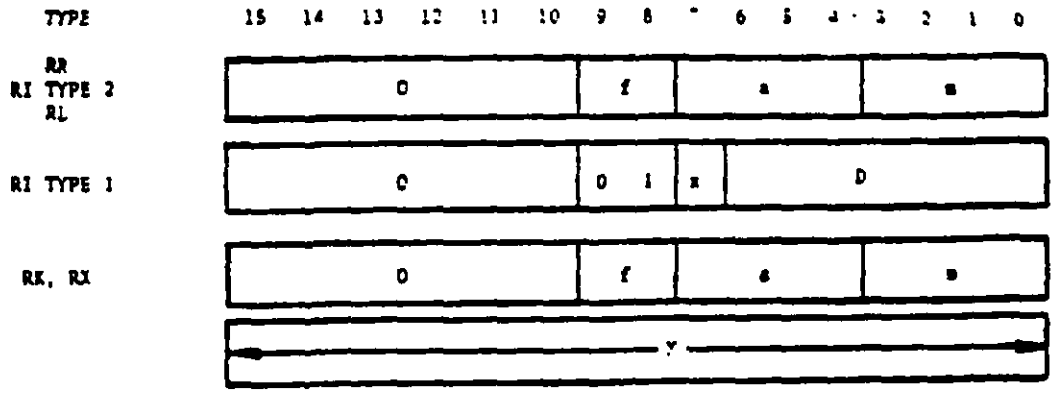
*No indirect addressing in IOP



Double Length Operands

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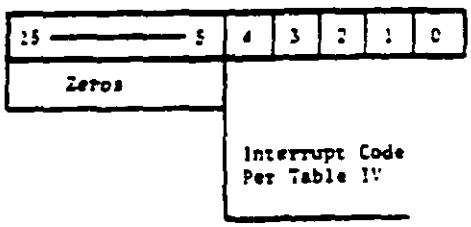
TABLE XX. Instruction repertoire for AN/AYK-14(V). - continued



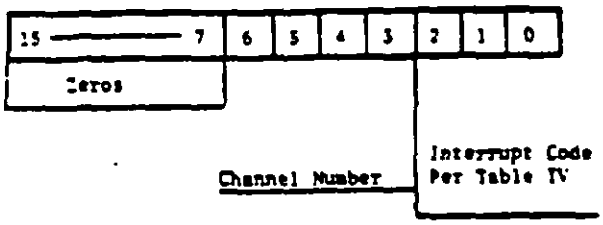
DEFINITION OF FIELDS

O	Operation (Function) Code
f	Format Designator
a	General Register or Subfunction Designator
m	General Register or Subfunction Designator 4-bit Unsigned Literal Constant in RL Format
xD	Signed Deviation Value (Two's Complement)
Y	Address or Arithmetic Constant

Instruction Word Format



Class I & II Interrupts



Class III Interrupts

Interrupt Entrance Address Index

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TABLE XXI. AN/UYK-20(V) instructions modified in AN/AYK-14(V).

Octal Format	Hexadecimal Format	Coding Format	Instruction	AN/AYK-14 Difference
00 0 a m 03 0 a 03	00 a m 0C a 3	-- SCR a	Diagnostic Return Store Real-Time Clock	CP Instruction fault Internal RTC frequency is 1 MHz.
03 0 a 07	0C a 7	LCR a	Load RTC (Lower)	Internal RTC frequency is 1 MHz.
03 0 a 10	0C a 8	ECR	Enable Real-Time Clock	Internal RTC frequency is 1 MHz.
03 0 a 12	0C a A	LEM a	Load Enable Monitor Clock	Internal Monitor Clock frequency is 10 KHz.
03 0 a 14	0C a C	LCRD a	Load RTC Double and Enable Count	Internal RTC frequency is 1 MHz.
03 0 a 15	0C a D	SCRD a	Store Real-Time Clock Double	Internal RTC frequency is 1 MHz.
52 0 a m 52 1 a m 52 3 a m	A8 a m A9 a m Ab a m	FMR a,m FMI a,m FM a,y,m		Floating point multiply does not require normalization of operands.
53 0 a m 53 1 a m 53 3 a m	AC a m AD a m AF a m	FDR a,m FDI a,m FD a,y,m		Floating point divide shall overflow if divisor is not normalized and its absolute magnitude is less than or equal to the absolute magnitude of the dividend divided by 16.
54 0 a m	B0 a m	LARR a,m	Load Address Register RR	Address (page) registers contain 3 more bits in AN/AYK-14 allowing addressing of up to 512K words of memory.
54 1 a m	B1 a m	LARI a,m	Load Address Register RR	
54 3 a m	B3 a m	LARM a,y,m	Load Address Register RR	
55 0 a m	B4 a m	SARR a,m	Store Address Register RR	
55 1 a m	B5 a m	SARI a,m	Store Address Register RI	
55 3 a m	B7 a m	SARM a,y,m	Store Address Register RX	

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TABLE XXI. AN/UYK-20(V) instructions modified in AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	AN/AYK-14 Difference
56 0 a m 56 1 a m 56 3 a m	B8 a m B9 a m BB a m	MDR a,m MDI a,m MD a,y,m	Multiply Double RR Multiply Double RI Multiply Double RX	Multiply by zero is allowed and not an abnormal case.

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TABLE XXII. Unique AN/AYK-14(V) processor instructions.

Operation Code		Description
Octal	Hex	
00 2 a m	02 a m	STACK PUT TOP (Y)⟶(Ra), (Ra)⟶Y This instruction stores the contents of memory address Y at the memory address specified by the contents of Ra. Then (Ra) is stored at memory address Y. NOTE: Memory address Y is the stack pointer and (Ra) is the address of the new top of stack.
02 0 a 03	08 a 3	INITIATE EXTERNAL INTERRUPT This instruction sets an external interrupt (Class II, Priorities 11, 12) as specified by the least significant bit (LSB) of the a-field: LSB(a) = 0 set external interrupt 0 LSB(a) = 1 set external interrupt 1 If set by the CPU, an external interrupt will interrupt IOP processing if an IOP is provided. Otherwise, the interrupt is recognized by the CPU itself. Similarly, an external interrupt set by an IOP will interrupt a CPU if provided, or itself, if no CPU is provided. This instruction is a privileged mode instruction fault if not in privileged mode.
02 0 a 07	08 a 7	SUPPORT CHANNEL INPUT Support channel buffer⟶Ra This instruction stores the contents of the computer support channel buffer into Ra.
02 0 - 13	08 - D	DIAGNOSTIC JUMP (R15)⟶uP This instruction transfers the contents of R15 to uP if executing in privileged mode. It is a privileged instruction fault if not in privileged mode.
02 0 - 14	08 - C	IPL FAILED This instruction sets the IPL failed discrete, indicating initial program load failure by the bootstrap loader. It is a privileged instruction fault if not in privileged mode.
02 0 - 16	08 - E	RESET BIT TIMER 0⟶BIT Timer if privileged mode This instruction resets the BIT timer if executing in privileged mode. The BIT indicator FF is also cleared. Note that this clears only the signal and not the mechanical indicator. It is a privileged instruction fault if not in privileged mode.

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TABLE XXII. Unique AN/AYK-14(V) processor instructions. - continued

Operation Code		Description
Octal	Hex	
02 0 - 17	08 - F	SET BIT INDICATOR This instruction sets the BIT indicator which denotes a computer failure if executing in privileged mode. It is a privileged instruction fault if not in privileged mode.
03 1 a m	0D a m	INITIATE SUPPORT I/O (Ra)→Support channel buffer, m→I/O Code, set channel busy. This instruction loads (Ra) into computer support channel buffer, uses m as I/O codes and sets the computer support channel busy. m 00XX initiate data input 01XX initiate data output 10XX initiate status input 11XX initiate function output m XX00 word mode XX10 byte mode XXX0 software user XXX1 SRA isolation diagnostic
04 0 a 04	10 a 4	STORE MONITOR CLOCK (MONITOR CLOCK) ⟶Ra if privileged mode. This instruction stores the contents of the monitor clock into Ra if executing in privileged mode. It is a privileged instruction fault if not privileged mode.
04 2 a m	12 a m	QUEUE PUT TOP (Y)→(Ra) (Ra)→Y, if (Y) = 0 then (Ra)→Y ⊕ 1 This instruction stores the contents of memory address Y at the memory address specified by the contents of Ra. Next (Ra) is stored at memory address Y. Then if (Y) = 0, the queue was empty and (Ra) is stored at memory address Y ⊕ 1. Address Y must be even. NOTE: Memory address Y is the queue's top pointer, Y ⊕ 1 is the queue's bottom pointer, and (Ra) is the address of the new top of queue.
05 2 a m	16 a m	QUEUE PUT BOTTOM (Ra)→(Y ⊕ 1), (Ra)→Y ⊕ 1, 0→(Ra) This instruction stores (Ra) at the memory address specified by the contents of Y ⊕ 1. Next (Ra) is stored at memory address Y ⊕ 1. Then 0 is stored at the memory address specified by (Ra). Address Y must be even. NOTE: Memory address Y ⊕ 1 is the queue's bottom pointer and (Ra) is the address of the new bottom of queue.

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TABLE XXII. Unique AN/AYK-14(V) processor instructions. - continued

Operation Code		Description
Octal	Hex	
06 2 a m	1A a m	STACK GET TOP (Y)→Ra, if (Y) ≠ 0 then ((Y))→Y and skip next instruction, if (Y) = 0 then execute next instruction. This instruction loads the contents of memory address Y into Ra. If (Y) does not equal zero, then the content of the address specified by the contents of address Y is stored at memory address Y, and the next instruction is skipped by executing the next instruction from (P) + 3. If (Y) equals zero, then the next instruction is from (P) + 2. The instruction following this instruction must be a 16-bit instruction. NOTE: Memory address Y is the stack pointer and if (Y) = 0 the stack is empty.
07 2 a m	1E a m	QUEUE GET TOP (Y)→Ra; if (Y) = 0 execute the next instruction; if (Y) ≠ 0 then skip next instruction and ((Y))→Y; if ((Y)) = 0 then Y→Y ⊕ 1. This instruction loads the contents of memory address Y into Ra. If (Y) equals zero then the next instruction if from (P) + 2. If (Y) does not equal zero, then the next instruction is skipped by executing the next instruction from (P) + 3. In addition the contents of the address specified by the contents of address Y is stored at memory address Y and if the new contents of address Y is zero then Y is stored at memory address Y + 1. Address Y must be even. NOTE: Memory address Y is the queue's top pointer and if (Y) = 0 the queue is empty.
40 0 14 m	80 C m	JUMP SUPPORT CHANNEL BUSY - RR Rm→P
40 2 14 m	83 C m	JUMP SUPPORT CHANNEL BUSY - RK If busy, Y→P If the computer support channel is busy, this instruction jumps to the instruction located at the address specified by the operand Y. If the computer support channel is not busy, the next instruction is executed.

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TABLE XXII. Unique AN/AYK-14(V) processor instructions. - continued

Operation Code		Description
Octal	Hex	
40 3 14 m	83 C m	JUMP SUPPORT CHANNEL BUSY - RX If busy, (Y)→P If the computer support channel is busy, this instruction jumps to the instruction located at the address specified by the contents of memory address Y. If the computer support channel is not busy, the next instruction is executed.
70 2 a m	E2 a m	INITIATE MESSAGE (Chaining Instruction) When encountered in a SIM I/O channel chain program, this instruction shall store the value y into the control memory location specified by the a field.
76 3 - m	FB - m	STORE STATUS (Chaining Instruction) Status bits per m→y. This stores channel status as specified by m field in (y). Channel number is specified by the a field.
77 0 - m	FC - m	START IOP (Command Instruction)
77 1 - m	FD - m	BIT JUMP (Chaining Instruction) This instruction tests bit m of control memory location 3 for the active I/O channel. If bit m is set, the chain address pointer is replaced with y.
77 2 - m	FE - m	EXCHANGE CONTROL MEMORY (Command Instruction) This instruction stores the contents of the control memory location specified by the m field for the I/O channel specified by the a field into y, then loads the contents of location y⊕1 into this control memory location. If m ≠ 2, 6, an I/O instruction fault is generated.
77 2 - m	FE - m	EXCHANGE CONTROL MEMORY (Chaining Instruction) This instruction stores the contents of the control memory location specified by the m field for the active I/O channel into y, then loads this control memory location with the contents of y⊕1.

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V).

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	IOP Time in usec
00 2 a m	02 a m	SPT a,y,m	Stack Put Top	(Y) → (Ra); (Ra) → Y		-NA-		8.36
00 3 a m	03 a m	BL a,y,m	Byte load	(Y) byte → Ra	0	0	X	2.93
01 0 a m	04 a m	LR a,m	Load (Register)	(Rm) → Ra	0	0	X	1.01
01 1 a m	05 a m	LI a,m	Load (Indirect)	(Y*) → Ra	0	0	X	2.28
01 2 a m	06 a m	LK a,y,m	Load (Constant)	Y → Ra	0	0	X	1.94
01 3 a m	07 a m	L a,y,m	Load	(Y) → Ra	0	0	X	2.92
02 0 a 00	08 a 0	PR a	Make positive	If (Ra) < 0, (Ra)' → Ra	X	X	X	1.71
02 0 a 01	08 a 1	NR a	Make negative	If (Ra) > 0, (Ra)' → Ra	X	0	X	1.71
02 0 a 03	08 a 3	IPI a	Initiate Processor Int	Set processor interrupt a		-NA-		5.63
02 0 a 04	08 a 4	TCR a	Two's Complement	(Ra)' → Ra	X	X	X	1.51
02 0 a 06	08 a 6	OCR a	One's Complement	(Ra) bit-by-bit complement → Ra	0	0	X	1.51
02 0 a 07	08 a 7	SCI a	Support Channel Input	Support channel input → Ra	0	0	X	1.76
02 0 a 10	08 a 8	IROR a	Increase Ra by 1	(Ra) + 1 → Ra	X	X	X	1.50
02 0 a 11	08 a 9	DROR a	Decrease Ra by 1	(Ra) - 1 → Ra	X	X	X	1.50
02 0 a 12	08 a A	IRTR a	Increase Ra by 2	(Ra) + 2 → Ra	X	X	X	1.50
02 0 a 13	08 a B	DRTR a	Decrease Ra by 2	(Ra) - 2 → Ra	X	X	X	1.50
02 0 - 14	08 - C	IPLF	IPL Failed	Set IPLF discrete		-NA-		2.13
02 0 a 15	08 a D	DJ a	Diagnostic Jump	R15 → uP		-NA-		TBD
02 0 - 16	08 - E	RBT	Reset Bit Timer	0 → Bit Timer		-NA-		4.87
02 0 - 17	08 - F	SBT	Set BIT Indicator	Set BIT Indicator		-NA-		TBD
02 0 a 00	0C a 0	ER a	Executive Return	Generate interrupt: P+1 → Ra	0	0	X	TBD
03 0 a 01	0C a 1	SSOR a	Store SR1	(SR1) → Ra	0	0	X	2.26
03 0 a 02	0C a 2	SSTR a	Store SR2	(SR2) → Ra	0	0	X	1.51
03 0 a 03	0C a 3	SCR a	Store Clock	(RTC register):15-0 → Ra	0	0	X	1.76
03 0 a 05	0C a 5	LSOR a	Load SR1	(Ra) → SR1		-NA-		8.73
03 0 a 06	0C a 6	LSTR a	Load SR2	(Ra) → SR2		-NA-		1.50
03 0 a 07	0C a 7	LCR a	Load RTC lower	(Ra) → RTC register: 15-0		-NA-		1.52
03 0 - 10	0C - 8	ECR	Enable Clock and Interrupt	Enable RTC register and interrupt		-NA-		3.30
03 0 - 11	0C - 9	DCR	Disable Clock	Disable RTC register		-NA-		2.52
03 0 a 15	0C a D	SCRD a	Store Clock Double	(RTC register) → Ra, Ra ⊙ 1, SYSTEM CLOCK→ Ra ⊙ 1	0	0	X	2.00
03 0 - 16	0C - E	ECIR	Enable Clock Interrupt	Enable RTC overflow interrupt		-NA-		2.80
03 0 - 17	0C - F	DCIR	Disable Clock Interrupt	Disable RTC overflow interrupt		-NA-		1.50
03 1 a m	0D a m	SCIO a,m	Support Channel I/O	(Ra) → support channel buffer,m → I/O code. Set channel busy		-NA-		TBD
03 2 a m	12 a m	QPT a,y,m	Queue Put Top	(Y) → (Ra), (Ra) → Y: if (Y) was = 0 then (Ra) → Y ⊙ 1		-NA-		8.60
04 3 a m	13 a m	BLX a,y,m	Byte Load and Index by 1	(Y) byte → Ra; (Rm)+1 → Rm	0	0	X	3.04
05 0 a m	14 a m	SBR a,m	Set Bit	1 → (Ra):m	0	0	X	2.51
05 1 a m	15 a m	LXI a,m	Load and Index by 1 (Indirect)	(Y*) → Ra; (Rm)+1 → Rm	0	0	X	2.51
05 2 a m	16 a m	QPB a,y,m	Queue Put Bottom	(Ra) → (Y ⊙ 1); (Ra) → Y ⊙ 1, 0 → (Ra)		-NA-		10.45

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	IOP Time in usec
05 3 a m	17 a m	LX a,y,m	Load and Index by 1	(Y) → Ra; (Rm)+1 → Rm	0	0	X	3.26
06 0 a m	18 a m	ZBR a,m	Zero Bit	0 → (Ra):m	0	0	X	2.77
06 2 a m	1A a m	SGT a,y,m	Stack Get Top	(Y) → Ra, if (Y) ≠ 0 then ((Y)) → Y and P+3 → P; if (Y) = 0 then P+2 → P		-NA-		9.45
07 0 a m	1C a m	CBR a,m	Compare Bit	Compare bit m of Ra with zero	0	0	X	2.51
07 1 - m	1D - m	LPI m	Load PSW (Indirect)	(Y*, Y*+1, Y*+2) → P, SR1, SR2		-NA-		9.00
07 2 a m	1E a m	QGT a,y,m	Queue Get Top	(Y) → Ra: if (y) = 0 then P+2 → P; if (Y) ≠ 0 then P+3 → P, ((Y)) → Y; if ((Y)) = 0 then Y → Y ⊙ 1		-NA-		9.70
07 3 - m	1F - m	LP y,m	Load PSW	(Y, Y+1, Y+2) → P, SR1, SR2		-NA-		8.68
10 0 a m	20 a m	LRSR a,m	Logical Right Shift (Register)	Shift (Ra) right (Rm): 5-0 places; zero fill	0	0	X	3.26+.25n ⊙n
10 2 a m	22 a m	LRS a,y,m	Logical Right Shift	Shift (Ra) right Y:5-0 places; zero fill	0	0	X	4.08+.25n
10 3 a m	23 a m	BS a,y,m	Byte Store	(Ra):7-0 → Y byte		-NA-		4.40
11 0 a m	24 a m	ARSR a,m	Algebraic Right Shift (Register)	Shift (Ra) right (Rm): 5-0 places; sign fill	0	0	X	3.28+.25n
11 1 a m	25 a m	SI a,m	Store (Indirect)	(Ra) → Y*		-NA-		1.98
11 2 a m	26 a m	ARS a,y,m	Algebraic Right Shift	Shift (Ra) right Y:5-0 places; sign fill	0	0	X	4.08+.25n
11 3 a m	27 a m	S a,y,m	Store	(Ra) → Y		-NA-		2.91
14 0 a m	30 a m	ALSR a,m	Algebraic Left Shift (Register)	Shift (Ra) left (Rm): 5-0 places; zero fill	0	X	X	4.08+.25n
14 2 a m	32 a m	ALS a,y,m	Algebraic Left Shift	Shift (Ra) left Y:5-0 places; zero fill	0	X	X	4.83+.25n
14 3 a m	33 a m	BSX a,y,m	Byte Store, Index by 1	(Ra):7-0 → Y byte; (Rm)+1 → Rm		-NA-		4.55
15 0 a m	34 a m	CLSR a,m	Circular Left Shift (Register)	Shift (Ra) circularly left (Rm):5-0 places	0	0	X	3.26+.25n
15 1 a m	35 a m	SXI a,m	Store, Index by 1 (Indirect)	(Ra) → Y*; (Rm)+1 → Rm		-NA-		2.23
15 2 a m	36 a m	CLS a,y,m	Circular Left Shift	Shift (Ra) circularly left Y:5-0 places	0	0	X	4.08+.25n
15 3 a m	37 a m	SX a,y,m	Store, Index by 1	(Ra) → Y; (Rm)+1 → Rm		-NA-		2.92
20 0 a m	40 a m	SUR a,m	Subtract (Register)	(Ra) - (Rm) → Ra	X	X	X	1.25
20 1 a m	41 a m	SUI a,m	Subtract (Indirect)	(Ra) - (Y*) → Ra	X	X	X	2.26
20 2 a m	42 a m	SUK a,y,m	Subtract (Constant)	(Ra) - Y → Ra	X	X	X	2.07
20 3 a m	43 a m	SU a,y,m	Subtract	(Ra) - (Y) → Ra	X	X	X	2.92
22 0 a m	48 a m	AR a,m	Add (Register)	(Ra) + (Rm) → Ra	X	X	X	1.25
22 1 a m	49 a m	AI a,m	Add (Indirect)	(Ra) + (Y*) → Ra	X	X	X	2.26
22 2 a m	4A a m	AK a,y,m	Add (Constant)	(Ra) + Y → Ra	X	X	X	2.07
22 3 a m	4B a m	A a,y,m	Add	(Ra) + (Y) → Ra	X	X	X	2.92
24 0 a m	50 a m	CR a,m	Compare (Register)	(Ra) - (Rm)	X	X	X	1.25
24 1 a m	51 a m	CI a,m	Compare (Indirect)	(Ra) - (Y*)	X	X	X	2.26
24 2 a m	52 a m	CK a,y,m	Compare (Constant)	(Ra) - Y	X	X	X	2.07
24 3 a m	53 a m	C a,y,m	Compare	(Ra) - (Y)	X	X	X	2.92
26 0 a m	58 a m	MR a,m	Multiply (Register)	(Ra ⊙ 1) · (Rm) → Ra, Ra ⊙ 1	0	0	X	6.53

⊙n equals # of shifts

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	IOP Time in µsec
26 1 a m	59 a m	MI a,m	Multiply (Indirect)	(Ra ⊙ 1) · (Y*) → Ra, Ra ⊙ 1	0	0	X	7.79
26 2 a m	5A a m	MK a,y,m	Multiply (Constant)	(Ra ⊙ 1) · Y → Ra, Ra ⊙ 1	0	0	X	7.35
26 3 a m	5B a m	M a,y,m	Multiply	(Ra ⊙ 1) · (Y) → Ra, Ra ⊙ 1	0	0	X	7.56
27 0 a m	5C a m	DR a,m	Divide (Register)	(Ra, Ra ⊙ 1)/(Rm) → Ra ⊙ 1; remainder → Ra	0	X	X	14.16
27 1 a m	5D a m	DI a,m	Divide (Indirect)	(Ra, Ra ⊙ 1)/(Y*) → Ra ⊙ 1; remainder → Ra	0	X	X	14.57
27 2 a m	5E a m	DK a,y,m	Divide (Constant)	(Ra, Ra ⊙ 1)/Y → Ra ⊙ 1; remainder → Ra	0	X	X	15.41
27 3 a m	5F a m	D a,y,m	Divide	(Ra, Ra ⊙ 1)/(Y) → Ra ⊙ 1; remainder → Ra	0	X	X	14.75
30 0 a m	60 a m	ANDR a,m	AND (Register)	(Ra) ⊙ (Rm) → Ra	0	0	X	1.26
30 1 a m	61 a m	ANDI a,m	AND (Indirect)	(Ra) ⊙ (Y*) → Ra	0	0	X	2.25
30 2 a m	62 a m	ANDK a,y,m	AND (Constant)	(Ra) ⊙ Y → Ra	0	0	X	2.07
30 3 a m	63 a m	AND a,y,m	AND	(Ra) ⊙ (Y) → Ra	0	0	X	2.91
31 0 a m	64 a m	ORR a,m	OR (Register)	(Ra) ⊕ (Rm) → Ra	0	0	X	1.26
31 1 a m	65 a m	ORI a,m	OR (Indirect)	(Ra) ⊕ (Y*) → Ra	0	0	X	2.25
31 2 a m	66 a m	ORK a,y,m	OR (Constant)	(Ra) ⊕ Y → Ra	0	0	X	2.07
31 3 a m	67 a m	OR a,y,m	OR	(Ra) ⊕ (Y) → Ra	0	0	X	2.91
34 0 a m	70 a m	CMR a,m	Compare Masked (Register)	[(Ra) ⊙ (Ra ⊙ 1)]: [(Rm) ⊙ (Ra ⊙ 1)]	0	0	X	2.01
34 1 a m	71 a m	CMI a,m	Compare Masked (Indirect)	[(Ra) ⊙ (Ra ⊙ 1)]: [(Y*) ⊙ (Ra ⊙ 1)]	0	0	X	3.26
34 2 a m	72 a m	CMK a,y,m	Compare Masked (Constant)	[(Ra) ⊙ (Ra ⊙ 1)]: [Y ⊙ (Ra ⊙ 1)]	0	0	X	2.83
34 3 a m	73 a m	CM a,y,m	Compare Masked	[(Ra) ⊙ (Ra ⊙ 1)]: [(Y) ⊙ (Ra ⊙ 1)]	0	0	X	3.53
35 0 - -	74 - -	IOCR	Input/Output Command	Execute (C Cell); 0 → C Cell:15-14		-NA-		TBD
40 0 00 m	80 0 m	JER m	Jump Equal	If CC indicates = or 0, (Rm) → P		-NA-		2.89
40 0 01 m	80 1 m	JNER m	Jump Not Equal	If CC indicates ≠ or not 0, (Rm) → P		-NA-		2.89
40 0 02 m	80 2 m	JGER m	Jump Greater or Equal	If CC indicates ≥ or =, (Rm) → P		-NA-		2.89
40 0 03 m	80 3 m	JLSR m	Jump Less	If CC indicates < or -: (Rm) → P		-NA-		2.89
40 0 04 m	80 4 m	JOR m	Jump Overflow	If overflow set, (Rm) → P		-NA-		2.89
40 0 05 m	80 5 m	JCR m	Jump Carry	If carry set, (Rm) → P		-NA-		2.89
40 0 06 m	80 6 m	JPTR m	Jump Power Out of Tolerance	If power out of tolerance, tolerance: (Rm) → P		-NA-		TBD
40 0 10 m	80 8 m	JR m	Jump	(Rm) → P		-NA-		2.60
40 0 11 m	80 9 m	JSR m	Stop	Stop		-NA-		3.10
40 0 14 m	80 C m	JSCR m	Jump Support Channel Busy - RR	If support channel busy, (Rm) → 0		-NA-		2.01
40 1 m	81 d	LJ xD	Local Jump	P + xD → P		-NA-		2.64
40 2 00 m	82 0 m	JE y,m	Jump Equal	If CC indicates = or 0, Y → P		-NA-		2.87
40 2 01 m	82 1 m	JNE y,m	Jump Not Equal	If CC indicates ≠ or not 0, Y → P		-NA-		2.87
40 2 02 m	82 2 m	JGE y,m	Jump Greater than or Equal	If CC indicates ≥ or =, Y → P		-NA-		2.87
40 2 03 m	82 3 m	JLS y,m	Jump Less	If CC indicates < or -, Y → P		-NA-		2.87

Note: Times are for jumps taken.

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	IOP Time in usec
40 2 04 m	82 4 m	JO y,m	Jump on Overflow	If overflow set, Y → P		-NA-		2.87
40 2 05 m	82 5 m	JC y,m	Jump on Carry	If carry set, Y → P		-NA-		2.87
40 2 06 m	82 6 m	JPT y,m	Jump if power out of tolerance	If power out of tolerance: Y → P		-NA-		TBD
40 2 10 m	82 8 m	J y,m	Jump	Y → P		-NA-		2.39
40 2 11 m	82 9 m	JS y,m	Stop	Stop		-NA-		2.89
40 2 14 m	82 C m	JSC y,m	Jump Support Channel Busy - RX	If support channel busy, Y → P		-NA-		TBD
40 3 00 m	83 0 m	JE *y,m	Jump Equal	If CC indicates = or 0, (Y) → P		-NA-		3.86
40 3 01 m	83 1 m	JNE *y,m	Jump Not Equal	If CC indicates ≠ or not 0, (Y) → P		-NA-		3.86
40 3 02 m	83 2 m	JGE (y,m	Jump Greater or Equal	If CC indicates ≥ or +, (Y) → P		-NA-		3.86
40 3 03 m	83 3 m	JLS *y,m	Jump Less	If CC indicates < or -, (Y) → P		-NA-		3.86
40 3 04 m	83 4 m	JO *Y,m	Jump on Overflow	If overflow set, (Y) → P		-NA-		3.86
40 3 05 m	83 5 m	JC *y,m	Jump on Carry	If carry set, (Y) → P		-NA-		3.86
40 3 06 m	83 6 m	JPT *y,m	Jump if power out of tolerance	If power out of tolerance: (Y) → P		-NA-		TBD
40 3 10 m	83 8 m	J *y,m	Jump	(Y) → P		-NA-		3.36
40 3 11 m	83 9 m	JS *y,m	Stop	Stop		-NA-		4.10
40 3 14 m	83 C m	JSC *y,m	Jump Support Channel Busy - RX	If support channel busy, (Y) → P		-NA-		TBD
41 0 a m	84 a m	XJR a,m	Index Jump Register	If (Ra) ≠ 0, (Ra) - 1 → Ra; (Rm) → P		-NA-		1.50
41 1 d	85 d	LJI xD	Local Jump (Indirect)	[P + xD] → P		-NA-		3.11
41 2 a m	86 a m	XJ a,y,m	Index Jump	If (Ra) ≠ 0, (Ra) - 1 → Ra; Y → P		-NA-		2.38
41 3 a m	87 a m	XJ a,*y,m	Index Jump	If (Ra) ≠ 0, (Ra) - 1 → Ra; (Y) → P		-NA-		3.38
42 0 a m	88 a m	JLRR a,m	Jump, Link Register (Register	(P) + 1 → Ra; (Rm) → P		-NA-		2.04
42 2 a m	8A a m	JLR a,y,m	Jump, Link Register	(P) + 2 → Ra; Y → P		-NA-		2.04
42 3 a m	8B a m	JLR a,*y,m	Jump, Link Register	(P) + 2 → Ra; (Y) → P		-NA-		2.04
43 1 d	8D d	LJLM xD	Local Jump, Link Memory	(P) + 1 → P + xD; P + xD + 1 → P		-NA-		3.61
43 2 - m	8E - m	JLM y,m	Jump, Link Memory	(P) + 2 → Y; Y + 1 → P		-NA-		3.61
43 3 - m	8F - m	JLM *y,m	Jump, Link Memory	(P) + 2 → (Y); (Y) + 1 → P		-NA-		4.30
44 0 a m	90 a m	JZR a,m	Jump Zero (Register	If (Ra) = 0, (Rm) → P		-NA-		2.88
44 1 d	91 d	LJE xD	Local Jump Equal	If CC indicates = or 0, (P) + xD → P		-NA-		3.14
44 2 a m	92 a m	JZ a,y,m	Jump Zero	If (Ra) = 0, Y → P		-NA-		2.89
44 3 a m	93 a m	JZ a,*y,m	Jump Zero	If (Ra) = 0, (Y) → P		-NA-		3.85
45 0 a m	94 a m	JNZR a,m	Jump Not Zero (Register)	If (Ra) ≠ 0, (Rm) → P		-NA-		2.88
45 1 d	95 d	LJNE xD	Local Jump Not Equal	If CC indicates ≠ or not 0, (P) + xD → P		-NA-		3.14
45 2 a m	96 a m	JNZ a,y,m	Jump Not Zero	If (Ra) ≠ 0, Y → P		-NA-		2.89
45 3 a m	97 a m	JNZ a,*y,m	Jump Not Zero	If (Ra) ≠ 0, (Y) → P		-NA-		3.86
46 0 a m	98 a m	JPR a,m	Jump Positive (Register)	If (Ra) ≥ 0, (Rm) → P		-NA-		2.88
46 1 d	99 d	LJGE xD	Local Jump Greater or Equal	If CC indicates ≥ or +, (P) + xD → P		-NA-		3.14

Note: Times are for jumps taken

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C	OV	CC	IOP Time in usec
46 2 a m	9A a m	JP a,y,m	Jump Positive	If (Ra) ≥ 0, Y → P	-NA-			2.89
46 3 a m	9B a m	JP a,*y,m	Jump Positive	If (Ra) ≥ 0, (Y) → P	-NA-			3.85
47 0 a m	9C a m	JNR a,m	Jump Negative (Register)	If (Ra) < 0, (Rm) → P	-NA-			2.89
47 1 d	9D d	LJLS xD	Local Jump Less	If CC indicates < or -, (P) + xD → P	-NA-			3.14
47 2 a m	9E a m	JN a,y,m	Jump Negative	If (Ra) < 0, Y → P	-NA-			2.89
47 3 a m	9F a m	JN a,*y,m	Jump Negative	If (Ra) < 0, (Y) → P	-NA-			3.14
60 0 a m	C0 a m	LLRS a,m	Literal Logical Right Shift	Shift (Ra) right m places; zero fill	0	0	X	(1) 2.26+.25n
60 1 a m	C1 a m	LARS a,m	Literal Algebraic Right Shift	Shift (Ra) right m places; sign fill	0	0	X	2.26+.25n
61 0 a m	C4 a m	LALS a,m	Literal Algebraic Left Shift	Shift (Ra) left m places; zero fill	0	X	X	2.56+.25n
61 1 a m	C5 a m	LCLS a,m	Literal Circular Left Shift	Shift (Ra) left circular m places	0	0	X	2.26+.25n
62 0 a m	C8 a m	LSU a,m	Literal Subtract	(Ra) - m → Ra	X	X	X	1.51
62 2 a m	CA a m	LA a,m	Literal Add	(Ra) + m → Ra	X	X	X	1.51
63 0 a m	CC a m	LL a,m	Literal Load	m → Ra	0	0	X	1.26
63 1 a m	CD a m	LC a,m	Literal Compare	(Ra) - m	X	X	X	1.51
			COMMAND/CHAIN INSTRUCTIONS					
70 0 00 00	E0 0 0	ACR	Channel Control	Master clear all channels				11.10/7.94
70 0 00 04	E0 0 4	ACR 4 CCR 0,4	Channel Control	Enable external interrupts, all channels				11.10/7.94
70 0 00 05	E0 0 5	ACR 5 CCR 0,5	Channel Control	Disable external interrupts, all channels				11.10/7.94
70 0 00 06	E0 0 6	ACR 6 CCR 0,6	Channel Control	Enable Class III interrupts, Priorities 2, 3, 4				11.10/7.94
70 0 00 07	E0 0 7	ACR 7 CCR 0,7	Channel Control	Disable Class III interrupts, Priorities 2, 3, 4				11.10/7.94
70 0 a 10	E0 a 8	CCR a,8	Channel Control	Master clear channel a				11.00/7.69
70 0 a 14	E0 a C	CCR a,12	Channel Control	Enable channel a external interrupts				11.00/7.69
70 0 a 15	E0 a D	CCR a,13	Channel Control	Disable channel a external interrupts				11.00/7.69
70 0 a 16	E0 a E	CCR a,14	Channel Control	Enable channel a, Class III, Priorities 2, 3, 4				11.00/7.69
70 0 a 17	E0 a F	CCR a,15	Channel Control	Disable channel a, Class III, Priorities 2, 3, 4				11.00/7.69
			COMMAND INSTRUCTIONS					
71 2 a m	E6 a m	LCMK m,y	Write control memory	Y → Channel a CMm				TBD
71 2 a 02	E6 a 2	ICK a,y	Initiate Input Chain	Y → Channel a Chain Pointer; initiate input chain				14.04
71 2 a 06	E6 a 6	OCK a,y	Initiate Output Chain	Y → Channel a Chain Pointer; initiate output chain				14.04
71 3 a m	E7 a m	WIM a,y,m	Write Control Memory	(Y) → Channel a CMm				(3) 12.25-12.75
72 3 a m	EB a m	RIM a,y,m	Read Control Memory	Channel a (CMm) → Y				10.40

(1) n equals # of shifts
(2) m = 2 - m - 6

Note: Times are for jumps taken

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TABLE XXIII. IOP instruction repertoire for AN/AYK-14(V). - continued

Octal Format	Hexadecimal Format	Coding Format	Instruction	Operation	C OV CC	IOP Time in usec
			COMMAND INSTRUCTIONS (Cont)			
76 0 a m	FB a m	SICR a,m	Set and Clear Discretes	Set or clear channel a discrete function		10.40
76 3 a m	FB a m	SST a,y,m	Store Status	Channel a Status bits per m → Y		11.32
77 0 - m	FC - m	SIOP m,y	Start IOP ①	m:0 → IOP SR1:12, Y → IOP P if m = 0 or 1		TBD
77 2 a m	FE a m	XIM a,y,m	Exchange Control Memory ⑤	Channel a (CMm) → Y; (Y ⊕ 1) → Channel a CMm		TBD
			CHAIN INSTRUCTIONS			
70 2 a m	E2 a m	IM a,y,m	Initiate Message	Y → CMm: Initiate message activity		TBD
70 3 a 00	E3 a 0	IO a,y	IO Function a	(Y, Y ⊕ 1) → BCW, BAP; initiate transfer		TBD
71 2 00 m	E6 0 m	LCMK m,y	Load Control Memory	Y → CMm		9.19
71 3 00 m	E7 0 m	LCM m,y	Load Control Memory	(Y) → CMm		7.28-9.44 ⑤
72 3 00 m	EB 0 m	SCM m,y	Store Control Memory	(CMm) → Y		5.78
73 0 00 00	EC 0 0	HCR	Halt Chain	Halt chaining, a even		6.28
73 0 01 00	EC 1 0	IPR	Interrupt Processor	Generate chain interrupt, a odd		7.94
73 3 00 00	EF 0 0	ZF y	Zero Flag	0 → Y:15,14 a even		7.53
73 3 01 00	EF 1 0	SF y	Set Flag	1 → Y:15,14 a odd		7.32
74 2 00 00	F2 0 0	SJMC 0,y	Serial Jump on Met Condition	Unconditional, Y → CAP; clear flag		TBD
74 2 01 00	F2 1 0	SJMC 1,y	Serial Jump on Met Condition	If suppress flag not set, Y → CAP; clear flag		TBD
74 2 02 00	F2 2 0	SJMC 2,y	Serial Jump on Met Condition	If monitor flag set, Y → CAP: clear flag		TBD
75 0 00 m	F4 0 m	SFSC m	Search for SYNC	Perform function(s) assigned to m-bits		5.78
76 0 00 m	F8 0 m	CSIR m	Serial Interface Control	Set or clear discrete function		5.78
76 3 - m	FB - m	CSST y,m	Store Status	Status bits per m → Y		6.75
77 1 - m	FD - m	BJ m,y	Bit Jump	(Y) → CAP if (CM3):m = 1		8.28
77 2 - m	FE - m	XCM m,y	Exchange Control Memory	(CMm) → Y: (Y ⊕ 1) → CMm		9.44

⑤/ m = 2 or 6 ①/ no operation unless IOP ⑤/ m = 8 - m = 6

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APPENDIX B

RELIABILITY AND MAINTAINABILITY PROGRAM REQUIREMENTS FOR COMPUTER SET, STANDARD AIRBORNE AN/AYK-14(V)

10. SCOPE

10.1 Scope. This appendix details the reliability and maintainability program requirements for the Computer Set, Standard Airborne AN/AYK-14(V). This appendix supplements and modifies the requirements specified in MIL-STD-785, MIL-STD-470, MIL-STD-2084, and MIL-E-5400 and provides for: reliability and maintainability organizations; written procedures regarding the implementation of reliability and maintainability tasks; control of reliability and maintainability program objectives and activities in design, manufacturing, testing, and demonstration; comprehensive records of reliability and maintainability achievements; methods of detecting product reliability and maintainability deficiencies; timely and effective corrective actions; and reporting of reliability and maintainability program status. This appendix applies to both ongoing production of the AN/AYK-14(V) computer, and new developments of the AN/AYK-14(V). This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Standards, and handbooks. Unless otherwise specified, the following standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this appendix to the extent specified herein.

Standards

MIL-STD-756	Reliability Prediction
MIL-STD-1629	Procedures for Performing Failure Modes, Effects and Criticality Analysis

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MIL-STD-2068	Reliability Development Test for Avionic Equipment

Handbooks

MIL-HDBK-217	Reliability Stress and Failure Data for Electronics Equipment
MIL-HDBK-472	Maintainability Prediction

(Copies of specification, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Reliability. The contractor shall prepare and follow a reliability program in accordance with the requirements of MIL-STD-785 (see 6.2.2). The program plan shall be periodically updated and revised in accordance with contract requirements. The program plan shall, in addition to the requirements of MIL-STD-785, include all requirements specified in this document.

30.1.1 Subtier reliability requirements. The contractor is responsible for assuring that the reliability levels achieved by his suppliers are consistent with the overall reliability requirements for the program. The contractor is responsible for determining that portion of this reliability requirements document which is applicable to his suppliers and their products. As a minimum, the contractor's subtier reliability program shall be in accordance with the requirements of MIL-STD-785, Task 102. The contractor's subtier reliability assurance activities for purchased repairable and nonrepairable assemblies and sub-contract items shall include 30.5, where applicable.

30.1.2 Reliability organization. The contractor shall identify key personnel for managing the reliability program. The contractor shall also designate one organizational reliability function and individual responsible for the planning, implementation and evaluation of all reliability program activities. This individual shall be delegated sufficient authority to effectively implement the reliability program and shall interface directly to the Program Manager. The designated individual shall serve as the principal contact for the Navy and for all reliability subject matter. The contractor shall provide an organizational chart with the Reliability Program Plan showing organizational relationships and responsibilities for implementation of the reliability program elements (see 6.2.2).

30.1.3 Reliability documentation/data items. The contractor shall prepare and maintain all reliability documentation/data items (e.g., plans, procedures, reports, and policies) in accordance with the related CDRL and Reliability Program Plan (see 6.2.2). The absence from the CDRL of reliability documentation required by this document, does not alleviate the Contractor of the responsibility to prepare and maintain the documents on file and available for [illegible] activity audit and review.

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30.1.4 Reliability program plan. The contractor shall prepare a written Reliability Program Plan in accordance with Task 101 of MIL-STD-785 and shall also provide for and define the following tasks and techniques to be used:

a. Management. Organizational responsibilities, authority, relationship to Program tasks, technical direction and control techniques.

b. Schedule. Time phased plan to accomplish all reliability program tasks and their relationship to engineering design and other program milestones (e.g., design release, design reviews, procurement release).

c. Analytical tasks. The plan shall define the contractor's responsibilities and techniques used to fulfill analytical tasks such as reliability predictions and reliability stress analysis.

d. Control tasks. The plan shall define and describe procedures and techniques to be utilized in controlling the reliability design implementation and program. Specific attention shall be given to the parts control program and practices.

e. Evaluation tasks. The plan shall define the scope and implementation disciplines for the reliability test programs, procedures, plan, failure analysis and failure reporting.

f. Status reports. The plan shall define in detail both formal and informal status reporting. The existence of an approved reliability program plan and related documents shall not relieve the contractor of his responsibility to furnish articles which are compliant to specified performance requirements, including reliability or to continually pursue achievement of specified reliability levels. The contractor shall also provide, but not be limited to, the requirements of Task Description 101, paragraph 101.2 of MIL-STD-785 (see 6.2.2).

30.2 Records and reports. The contractor shall maintain verifiable objective evidence that all reliability program tasks have been performed in accordance with specified requirements. Contractor internal records and reports such as: Engineering logbooks, charts, materials and processes selection criteria; failure data and reports; and design review results shall be maintained in the contractor's format and made available for review upon request (see 6.2.2).

30.3 Reliability indoctrination and training. The contractor shall conduct appropriate reliability indoctrination and training program activities for personnel for the specific reliability requirements peculiar to the contracted article. The contractor shall include an outline of his proposed training program as part of his Reliability Program Plan (see 6.2.2).

30.4 Reliability analysis. This section shall apply to new subassemblies developed under the contract and changes made under the contract to existing subassemblies.

30.4.1 Reliability prediction. The contractor shall conduct a detailed design reliability prediction in accordance with MIL-STD-756 and Task 203 of

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MIL-STD-785, using the failure rate data specified in MIL-HDBK-217 (see 6.2.2). Part failure rates shall be consistent with the individual piece part procurement specification requirements for parts screening. Failure rates other than those listed in MIL-HDBK-217 shall be approved by the procuring activity prior to use. Failure rate justifications shall be based upon:

a. Similarity of individual part types (e.g., logic family, construction, processes.

b. Screening requirements of data base (e.g., less than or equal to proposed equipment.

c. Vintage of data base (e.g., data not more than 5-10 years old depending on part type.

d. Similarity of equipment usage environment (e.g., airborne).

The contractor shall simplify or make other corrections to the equipment design as necessary to meet the reliability requirement specified in the equipment specification. The specified reliability (θ_0 of MIL-STD-781) to be demonstrated in the all equipment test (paragraph 4.4.3.4) shall be 80 percent of the value predicted in accordance with the requirements of this paragraph.

30.4.2 Failure modes and fault tree analysis. This analysis shall be presented in two parts as indicated below. However, both analyses shall be performed at the same time and submitted as a single report, to be updated as required per CDRL.

30.4.2.1 FMECA. The contractor shall prepare a plan and conduct a FMECA in accordance with Task 204 of MIL-STD-785; and Tasks 101, 102, 103 and 105 of MIL-STD-1629 (see 6.2.2). This analysis shall include all equipment functional failure modes, their causes, effects, and criticality. The failure rates (available from the reliability analysis) shall be extended to provide the failure rates of the functional failure modes. The analysis shall indicate detection of these potential faults during in-house screening/ checkout or during subsequent testing.

30.4.2.2 FTA. This analysis shall result in a fault tree breakdown of all possible faults or anomalies which the computer may exhibit. The fault tree will include each individual platform configuration and relate each to whether or not each fault will be detectable, and isolatable, by software or BIT, as appropriate. Frequency of occurrence and/or failure rate considerations, need not be included.

30.4.2.3 Sneak circuit analyses. The contractor shall consider a sneak circuit analysis for incorporation in his Reliability Program Plan to identify non-apparent failure modes. The sneak circuit analysis, if conducted, shall be in accordance with, but not limited to, the requirements of MIL-STD-785, Task 205 and be made available to the procuring activity (see 6.2.2).

30.4.2.4 Circuit analysis. The contractor shall perform, and make available for procuring activity review, manual or computer aided circuit analyses or noise measurements to ensure reliable operation of the equipment

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under conditions of high electrical noise and variation of circuit values within allowed tolerances over the range of environmental conditions stated in the equipment specification. The methods to be used (calculations of noise margin/immunity, worst case analysis, etc.) and the extent of their application shall be described in the contractor's Reliability Program Plan (see 6.2.2).

30.4.2.5 Thermal analysis. The contractor's methods for analyzing the thermal aspects of his design shall be detailed in his Reliability Program Plan. The thermal analysis shall be made available to the procuring activity (see 6.2.2).

30.4.2.6 Dynamic analysis. Dynamic analysis of the response of the mechanical structure to vibration and shock shall be performed. Details shall be provided in the contractor's Reliability Program Plan (see 6.2.2).

30.4.2.7 Worst case analysis. The contractor shall conduct and make available to the procuring activity a worst case analysis or test as appropriate to assure that performance requirements will be met if all the constituent parts are at their extreme tolerances and combined in the worst possible manner at all combinations of required environmental conditions (see 6.2.2).

30.5 Parts and materials reliability requirements.

30.5.1 Parts and materials program. The contractor shall establish and maintain a Parts Program in accordance with the Parts Reliability requirements of Task 207 of MIL-STD-785 and as specified herein (see 6.2.2). The program shall contain a standardization plan and shall define the ground rules and criteria for the selection, application, and testing of parts to achieve and maintain the reliability required. The Parts Program shall be a part of the Reliability Program Plan. As a minimum, a Parts Program equivalent to that specified herein shall be required of subcontractors and suppliers.

30.5.2 General requirements. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Preference shall be given to the use of micro-electronic devices. Standard microelectronic devices shall be as specified in 3.3.

b. The selection and application of parts, materials and processes shall be in accordance with MIL-E-5400. Standard parts are required. Purchase specifications for all parts shall include reliability and quality assurance requirements consistent with requirements specified herein. Where adequate military specifications do not exist, controlling specifications shall be prepared and shall include reliability and quality assurance requirements in accordance with MIL-STD-965 and this document (see 6.2.2).

c. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts and materials shall be exercised. Parts selections shall be made in the following order of precedence:

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(1) Military standard established reliability parts (e.g., JAN TX, ER).

(2) Military standard parts in accordance with MIL-E-5400.

(3) Other Military standard parts as listed on QPL's.

(4) Nonstandard parts previously qualified for other programs. (Require nonstandard parts approval.)

(5) Other nonstandard parts (require nonstandard parts approval).

d. All electrical, electronic, and electromechanical parts shall be subjected to 100 percent screening. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similar military standard and/or standard-established reliability part. All integrated circuits and hybrids shall be screened to MIL-STD-883, Method 5004, Class B, as a minimum. Purchase specifications shall specifically delineate screening requirements.

30.5.3 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved as specified in MIL-STD-965, including interchangeability demonstration. Data conforming to requirements of MIL-STD-965 Procedure I shall be prepared. Thirty days prior to the delivery of the first equipment on the contract, the contractor shall furnish to the Procuring Activity a list of all parts and materials used in the deliverable equipment in accordance with the format specified in MIL-STD-965, and in accordance with Data requirements of the contract. This list shall be updated to reflect subsequent changes to the part complement of the computer (see 6.2.2).

30.5.4 Parts derating and application. The contractor shall establish a uniform policy regarding the derating of parts to enhance the reliability of his articles. This derating policy shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, fan out and duty cycles. Design deratings for electronic and electromechanical parts shall be provided in the contractor's Reliability Program Plan and shall reflect the worst case electrical and environmental stresses unless formal written approval is received from NAVAIRSYSCOM prior to incorporation into the design. Deratings of electronic parts listed in Table XXIV shall not be exceed. The contractor shall conduct a part-level stress analysis to verify derating compliance. This analysis shall verify that all parameter deratings are compliant under the following conditions:

a. Nominal Unit and module power dissipation.

b. Worst case piece part stress.

c. Worst case equipment thermal environment including worst case altitude and temperature.

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TABLE XXIV. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor, tantalum: foil	Applied voltage, ripple voltage	50% of rating 1/
Capacitor (except tantalum foil)	Applied voltage (ripple voltage for solid tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage	2/ 75% of rating 60% of rating
Transistors	Power Voltage	2/ 60% of rating
Integrated Circuits		
Digital	- Fan-out	80% of rating 2/
Linear	- Power Current Voltage	2/ 75% of rating 75% of rating
Transformers, Inductive Devices	Parts shall be used at a winding temperature at least 30°C below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall not be greater than 60 percent of rated voltage for the steady state conditions or 90 percent of rated voltage with transients. General purpose inductors, audio and power transformers shall operate at a maximum current not to be greater than 70 percent of rated current.	

1/ For solid tantalum and all electrolytic capacitors derate voltage such that the maximum applied stress does not exceed 50 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductors shall be mounted and/or operated in such a manner to ensure worst-case junction temperature shall not be greater than 110°C when operating the computer configuration with the exhaust air temperature maintained at 160°F (71°C).

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This analysis shall be submitted to the procuring activity prior to design release for fabrication and in accordance with the CDRL (see 6.2.2). All noncompliances to the derating criteria shall require written approval by the procuring activity. Non-compliances shall be identified for discussion at program review meetings via the meeting agenda submitted to the procuring activity. Within 30 days of the design review meeting, notification will be supplied as to whether or not approval will be granted.

30.5.4.1 Parts usage limitations. The following limitations on parts usage shall apply:

a. Prohibited Application

(1) Electron tubes

(2) Hot Carrier (Schottky) Rectifiers

(3) Non-Solid, Sintered-slug, Tantalum Capacitors

(4) Fixed paper and/or plastic dielectric capacitors in molded cases

(5) Plastic, epoxy, or other encapsulated semiconductors

(6) Germanium semiconductors

(7) Photon Couplers (photo transistors)

30.5.5 Parts reliability assurance. The contractor shall define in his Reliability Program Plan a parts reliability assurance program, including organizational responsibilities and relationships, to assure uniform part reliability throughout the program (see 6.2.2). To the extent possible, the contractor shall use his standard policies, practices and operating procedures to the extent that program control can be maintained. The part reliability assurance program tasks, as a minimum, shall:

a. Provide receiving test and inspection to assure uniform part quality as established and identified in the program plan.

b. Establish in-house and vendor surveillance over the application of parts, materials and processes during fabrication and assembly to insure sources of degradation and variability are isolated and controlled.

c. Maintain visibility over failed parts analysis to insure proper corrective action has been implemented.

d. Determine and evaluate problem areas related to parts, materials, and processes, and initiate corrective action.

e. Review and analyze proposed changes in parts, materials, and processes resulting from test, manufacturing, and field experience.

30.6 Life limit. The contractor shall identify all items that change significantly with age (have a calendar life limit that would fall within

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100,000 operating hours and/or a 20-year period) and whose change would degrade the reliability of the equipment. This effort shall be coordinated with the limited life/age environment control requirement as delineated in the contractor's quality program.

30.7 Long life assurance. When new design is required by the contract, the contractor shall use the design guidelines set forth in the Long Life Assurance Study for Manned Spacecraft Long Life Hardware (Vols. I and V). In evaluating the guidelines set forth in this study document, care should be taken in comparing the environments of aircraft vs spacecraft.

30.8 Spares reliability provisions. The contractor shall include in his Reliability Program Plan provisions for reliability of spares and spare parts at all levels of repairable asembly (see 6.2.2).

30.9 Reliability/maintainability design reviews. When new design is required by the contract, separate reliability/maintainability design reviews shall examine each of the following areas to establish confidence in the reliability and maintainability design of the equipment: circuit design and partitioning; component selection and application; environmental and packaging designs and the reliability and maintainability design features and characteristics. The contractor shall follow the requirements of MIL-STD-785, Task 103 for reliability design reviews and the requirements of MIL-STD-470, Task 103 for the maintainability design reviews (see 6.2.2). It is the intent of this requirement that contractor design review preparation be informal and minimal and that each of the above areas be reviewed independently in small working groups. Four sets of design review documentation, adequate to allow independent assessment, (e.g., schematics, block diagrams, mechanical layouts, descriptive data, etc.) shall be provided a minimum of three weeks prior to the scheduled review. This data package may consist of preliminary drawings or sketches. All such documents shall be legible and clean. The contractor's reliability/maintainability program plan shall delineate the methodology and schedule to achieve the design reviews. Reliability/Maintainability design reviews will be held as the procuring activity deems necessary.

30.10 Maintainability. (This section shall apply when the development of new subassemblies are required by the contract.)

30.10.1 Maintainability program. The contractor's maintainability program shall comply with the requirements of MIL-STD-470 and MIL-STD-2084 with the stated exceptions, additions, or clarification, specified herein.

30.10.1.1 Subtier maintainability requirements. The contractor is responsible for assuring that the maintainability levels achieved by his suppliers are consistent with the overall maintainability requirements of the program. The contractor is responsible for determining that portion of this maintainability requirement document which is applicable to his suppliers and their products.

30.10.1.2 Maintainability considerations. The contractor shall apply the following maintainability considerations in the design and implementation of his maintainability program.

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a. The equipment shall be designed to meet the functional modularity requirements of the equipment specification and MIL-STD-2084.

b. The equipment shall be designed to meet the MTTR requirements of the equipment specification and organizational and intermediate maintainability requirements of the MIL-STD-2084.

c. The equipment shall be designed to meet the ATE compatibility requirements of the equipment specification and MIL-STD-2076.

d. The equipment shall be designed to meet the specified MTBF requirements without need for scheduled maintenance with the exception of air filter cleaning and fan replacement. Fan replacement shall be scheduled at time intervals of no less than 10,000 hours. Insofar as it is possible, all air filters shall be a quick disconnect type, thus eliminating the need for removal of screws. Scheduled maintenance shall not be required.

e. The equipment shall be designed to use the existing ATE and support equipment for the AN/AYK-14(V) to the maximum extent practicable.

Human factors must be recognized in all operations during manufacturing, testing, maintenance, and operation of articles to minimize human induced failures.

30.10.2 Maintainability organization. The contractor shall designate one organizational maintainability function and individual responsible for the planning, implementation and evaluation of all maintainability program activities. This individual shall be delegated sufficient authority to effectively implement the maintainability program and shall interface directly to the Program Manager. The designated individual shall serve as the principal contact for the Navy for all maintainability subject matter. The contractor shall provide an organizational chart with the Maintainability Program Plan showing organizational relationships and responsibilities for implementation of the maintainability program elements (see 6.2.2).

30.10.3 Maintainability documentation/data items. The contractor shall prepare, submit and maintain all maintainability documentation/data items (e.g., plans, procedures, reports, policies) in accordance with the related CDRL and Maintainability Program Plan. The absence from the CDRL of maintainability documentation required by this document does not alleviate the contractor of the responsibility to prepare and maintain the documents on file and available for procuring activity review (see 6.2.2).

30.10.4 Maintainability program plan. The contractor shall prepare a written Maintainability Program Plan in accordance with MIL-STD-470, paragraph 101.2 of Task 101 (see 6.2.2) and shall also provide for and define the following tasks and techniques to be used:

a. Management. Organizational responsibilities, authority, relationship to program tasks, technical direction and control techniques.

b. Schedule. Time phased plan to accomplish all maintenance to engineering design and other program milestones (e.g., design release, design reviews, procurement release)

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c. Analytical tasks. The plan shall define the contractor's responsibilities and techniques used to fulfill analytical tasks such as maintainability predictions. The plan shall include coordination between the analysis required for the design phase and that required for the logistic support effort.

d. Control tasks. The plan shall define and describe procedures and techniques to be utilized in controlling the maintainability design implementation and program. Specific attention shall be given to Built in Test, modularity and test point requirements.

e. Evaluation tasks. The plan shall define the scope and implementation disciplines for the maintainability demonstration program, procedures, plan, reporting, etc.

f. Status reports. The plan shall define in detail both formal and informal status reporting.

The existence of an approved maintainability program plan and related documents shall not relieve the contractor of his responsibility to furnish articles which are compliant to specified performance requirements, including maintainability, or to continually pursue achievement of specified maintainability levels.

30.10.5 Records and reports. The contractor shall maintain verifiable objective evidence that all maintainability program tasks have been performed in accordance with specified requirements. Contractor internal records and reports such as: engineering, logbooks; data; reports; design review results, etc., shall be maintained in the contractor's format and made available for review upon request.

30.10.6 Maintainability indoctrination and training. The contractor shall conduct appropriate maintainability indoctrination and training program activities for personnel for the specific maintainability requirements peculiar to the contracted article. The contractor shall include an outline of his proposed training program as part of his Maintainability Program Plan (see 6.2.2).

30.10.7 Maintainability prediction and analysis. The contractor shall conduct a maintainability prediction and analysis in accordance with MIL-HDBK-472 and MIL-STD-2084 (see 6.2.2).

40. QUALITY ASSURANCE PROVISIONS

40.1 Performance verification and test planning. The contractor shall prepare an integrated system test plan encompassing all provisions of Section 4 of the basic specification and this appendix. The plan shall include a verification matrix that addresses all requirements of Section 3 of the basic specification and includes the verification technique (e.g., inspection, demonstration, analysis or test and test program). The plan shall be submitted to the procuring activity for approval in accordance with the CDRL (see 6.2.2).

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40.2 Reliability testing.

40.2.1 Design approval tests. Design approval tests shall be performed on all newly developed assemblies and subassemblies. Design approval tests shall be conducted in accordance with the equipment specification in order to verify environmental characteristics and functional performance of the design. The design approval tests shall be started prior to beginning the Reliability Development Test. Details of the test shall be part of the test procedures to be submitted and approved in accordance with the equipment specification and the CDRL (see 6.2.2).

40.2.2 Reliability development test. A Reliability Development Test (RDT) shall be performed on all newly developed assemblies and subassemblies. The RDT shall be conducted in accordance with MIL-STD-2068, and the requirements of this document. The test articles shall be representative of production equipments in materials, configuration and workmanship. This program is intended to identify, analyze and correct all deficiencies (both hardware and software) and to bring the equipment MTBF to its specified value. All faults will require failure analysis identification of corrective action, verification of corrective action effectiveness and corrective action implementation. Temporary and/or permanent design modifications may be added to the equipment to verify that the proposed corrective actions are adequate.

40.2.2.1 Test requirements. The detail requirements for the RDT implementation shall be in accordance with MIL-STD-2068 as modified herein. The environmental test profile, and test stress levels, shall be a part of the Reliability Test Plan to be prepared by the contractor and submitted to the procuring activity for acceptance in accordance with CDRL, prior to the commencement of the RDT tests (see 6.2.2). The vibration profile shall be that specified in 4.4.3.1 for the Reliability Qualification Test except that the normalized vibration levels of 0.25, 0.5, and 1.0 shall correspond to 1.93g rms, 3.85g rms, and 7.7g rms respectively. The tests shall culminate with a complete and thorough inspection for fault and/or failure of all articles. Equipments as specified by the contract shall be subjected to this testing. The operating test time for each equipment shall be a minimum of 1000 hours. The total operating test time shall be as specified in the contract.

40.2.2.2 Accept criteria. The equipment shall be judged to have satisfactorily completed the RDT when the total operating time has been completed. Acceptance shall be upon procuring activity approval of corrective action on all failures occurring during testing. Corrective actions shall be accepted/rejected within 30 days after the critical R&M Design Review.

40.2.2.3 RDT fault reporting, analysis and corrective action. In addition to the requirements of MIL-STD-2068, all faults shall be reported to the procuring activity cognizant engineer or his appointed designate within 24 hours of the incident. All faults shall be analyzed to determine the cause of the fault. Positive corrective actions shall be developed for all faults to prevent recurrence. Necessary modifications shall be incorporated into all RDT equipments for the purpose of corrective action verification. At the option of the procuring activity, corrective action verification may be accomplished by testing at a lower level of assembly or at the piece part level. For the purpose of the RDT, the definition of a fault shall include

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excessive wear, mechanical damage and BIT anomalies. The definition of an intermittent (recoverable fault) shall be submitted for approval with the RDT procedure. Corrective action shall be implemented to correct all recoverable fault conditions. Corrective action shall be incorporated in all systems submitted for acceptance and delivered systems shall be retrofitted per the statement of work. The contractor shall maintain a closed loop reporting system to insure follow-through from failure occurrence through failure analysis to corrective action (see 6.2.2).

40.2.3 Reliability qualification test. When specified in the contract, the contractor shall conduct a formal reliability demonstration test program in accordance with the requirements of MIL-STD-781 as specified in 4.4.3.1 to demonstrate compliance with the specified article reliability performance requirements. Reliability demonstration testing shall be integrated with other controlled test programs to the maximum extent practical to assure maximum economics in time and cost of the overall test program.

40.2.4 All equipment test. Each equipment submitted for procuring activity acceptance, except those submitted to Design Approval Test, Preproduction (First Article) Tests, Reliability Qualification Test and Reliability Development Tests, shall be subjected to the All Equipment Test.

40.3 Maintainability demonstration. When specified in the contract, the contractor shall conduct a maintainability demonstration in accordance with this specification and the Statement of Work.

40.3.1 Maintainability data/reports. All maintainability data/reports shall be maintained in accordance with the CDRL, the basic specification and this appendix.

40.4 Failure reporting, analysis and corrective action. The contractor shall, as part of the Reliability Program Plan, define and describe the implementation of a closed loop failure/fault reporting, analyses and corrective action program. The contractor shall provide for visibility and traceability of all reported failures/faults from discovery to closeout. The contractor's existing failure/fault reporting, analysis and corrective action system shall be utilized with minimum changes necessary to meet the contract requirements (see 6.2.2).

40.4.1 Failure definition. All unscheduled maintenance or adjustment shall constitute a failure. Failures/faults shall be specified and classified in accordance with MIL-STD-781, MIL-STD-2074 and this specification.

40.4.1.1 Failure review board (FRB). The contractor shall plan for participation on a Failure Review Board comprised of Navy and contractor personnel. The function of this board will be the classification of all failures and failure closeout during all testing specified in the equipment specification. The FRB procedure shall be contained in the R&M Program Plan and shall be in accordance with the requirements of Task 105 of MIL-STD-785.

40.4.2 Failure reporting. Failures (faults) or non-conformance experienced on end-item article during tests shall be recorded by the contractor in engineering failure reports or test log books which are subject to NAVAIRSYSCOM review upon request. Log book entries shall be complete with

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failure (fault) analysis and corrective actions to reduce or prevent repetition of failures. The contractor's Reliability Program Plan shall define the formal failure reporting, analysis and corrective action system to be implemented to meet the contract requirements. Failure reports and failure analyses shall be initiated for all Reliability and Acceptance Test faults and failures. The contractor's failure report and failure analysis forms shall be submitted to procuring activity for approval as part of the Reliability Program Plan (see 6.2.2).

40.4.3 Failure analysis. The contractor shall perform failure analysis, on all failures and faults reported to the procuring activity, to the in-depth level required to determine the cause of failure, define the failure mechanism and to develop corrective actions to limit their recurrence. In general, the analysis shall include a physics of failure analysis on electrical and electronic parts. The analysis of parts shall typically include electrical failure verification, dissection, microphotograph, and adequate chemical and metallurgical analysis to define the failure mechanism (e.g, most fundamental cause). Records of failure analyses, including causes and effects, shall be maintained by the contractor with data feedback to reliability and related design analysis functions (see 6.2.2).

40.4.4 Corrective actions. The contractor, in conjunction with failure analysis, shall develop and implement effective corrective actions to eliminate recurrence of the failure/fault mechanism and its effects. Corrective actions for those failures/faults reported to the procuring activity must be approved by the procuring activity. Approval of proposed contractor corrective actions shall be at FRB meeting. This requires that the contractor has:

a. Analytically or by test established the effectiveness of the corrective action to the satisfaction of the procuring activity.

b. Scheduled for incorporation of the corrective action as approved by the procuring activity.

40.5 Operating time controls. The contractor shall establish and maintain a system to accurately measure the operating and environmental time/cycles for each deliverable article and shall record such information in the articles log book.

40.6 Status reports.

40.6.1 Reliability status reports. The contractor shall prepare and make available to the procuring activity, monthly reliability status reports in accordance with the CDRL (see 6.2.2).

40.6.2 Maintainability status reports. The contractor shall prepare and make available to the procuring activity, maintainability status reports in accordance with the CDRL (see 6.2.2).

40.7 Reliability/maintainability assurance provisions. The contractor's articles which are designed, manufactured, assembled, inspected and tested, and delivered under this contract are subject to the pre-acceptance and

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acceptance requirements as specified in the contract and equipment specification.

40.8 Reliability/maintainability audits and surveillance. The procuring activity reserves the right to perform audits and surveillance inspections of contractor conformance to any or all contractual requirements, including the reliability and maintainability program, at any time during the performance of this contract. Due notice shall be provided to the contractor prior to conducting source audits or surveillance inspections by the procuring activity.

40.9 Notice of deficiencies, corrective actions. The contractor will be provided written notification within 30 days after the visit of deficiencies found during the procuring activity audits and surveillance inspections and will be given a fair and reasonable specified period of time to correct each deficiency to the procuring activity satisfaction.

40.10 Reliability/maintainability assurance residency. The procuring activity reserves the right to place in residency at the contractor's facility a reliability/maintainability assurance representative(s) for such periods as deemed necessary by the particular reliability/maintainability problem involved or need for reliability/maintainability program surveillance. Contractor arrangements for the procuring activity residents shall be as specified in the contract.

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APPENDIX C

SUPPORT SOFTWARE FOR COMPUTER SET, STANDARD AIRBORNE AN/AYK-14(V)

10. SCOPE

10.1 Scope. This appendix details the AN/AYK-14(V) Support Software items in conjunction with Navy Standard 16-Bit Computer Support Software. The software items shall consist of support software required to code, assemble, debug, test, revise, and support the microcode programs, application programs, bootstrap loader software, and maintenance and diagnostic programs for AN/AYK-14(V), Standard Airborne Computer Set. The AN/AYK-14(V) Support Software items specified herein shall be designed to meet the support software requirements for various Naval airborne platforms employing the AN/AYK-14(V) Standard Airborne Computer Set. This appendix is a mandatory part of this specification. The information contain herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specification, standards and handbooks. Unless otherwise specified, the followign standards of the issue llsted in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this appendix to the extent specified herein.

Standards

MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
DoD-STD-1679	Software Development
MIL-STD-1815	Ada Programming Language

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturer's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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20.1.2 Other publications. The following documents form a part of this appendix to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

PUBLICATIONS

American National Standards Institute (ANSI)

ANSI X3.9-1966	FORTRAN
ANSI X3.5-1970	Flowchart Symbols for Information Processing
ANSI X3.12-1970	Vocabulary for Information Processing

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

30. REQUIREMENTS

30.1 Development. This appendix makes provisions for first article testing.

30.2 Software materials. For the purposes of this appendix, software materials consist of supporting documentation, software media, and listings for each program.

30.2.1 AN/AYK-14(V) support software documentation. As specified in the Contract Data Requirements List, all AN/AYK-14(V) Support Software Documentation shall be documented in accordance with DoD-STD-1679 or a commercial equivalent and samples of its implementation shall be submitted to the procuring activity for certification of its equivalence and approval for its use.

30.2.2 AN/AYK-14(V) support software media. All AN/AYK-14(V) Support Software Media shall be provided magnetic tapes and as specified in the Contract Data Requirements List.

30.2.3 Reproducible copy.

30.2.3.1 Software documentation. A reproducible copy for software documentation, also known as Camera Ready Copy, is defined as final typed copy (one side only) ready in all respects for photographic reproduction (litho negatives) used in offset printing. Paper shall be 50-pound coated white offset, inked in black, printed on one side, no holes drilled, and 8-1/2 x 11

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inches in size except for fold-outs. Reproducible copies shall be submitted for all delivered documentation.

30.2.3.2 Software media. A reproducible copy for software media (disks, tapes, card decks, listings, etc.) is defined as verified copy produced from a master which has been accepted by the procuring activity.

30.2.4 Change pages. Change pages, as required, shall be submitted to previous documentation except when the number of change pages exceeds fifty percent of the original document in which case a completely revised document shall be submitted.

30.2.5 Flow charts. For documentation other than DoD-STD-1679, the contractor shall provide hierarchical, top-down flow charts with at least a high, intermediate and low level diagram(s) for each program module. All flow charts shall conform to ANSI X3.5-1970 standards and shall be accompanied by narrative description. Plain language flow charts shall be provided whenever possible. Computer drawn flow charts may be provided containing ANSI X3.5-1970 equivalent symbols. The equivalent symbols shall be submitted to the procuring activity for approval. When authorized by the procuring activity a program design language based on the Ada language may be used in lieu of flow charts as specified herein for the preparation of the program design specifications and the source program documentation. The program design language shall have the following characteristics:

a. The keywords, syntax, and semantics for data structures, control structures, and program structure shall be those of the Ada language;

b. All legal Ada constructs shall be legal program design language constructs;

c. Provision shall be made for the use of English narrative as a substitute for identifiers and in other appropriate situations; and

d. Provision shall be made for expressing design concepts and information by means of annotations which extend the Ada language.

The program design language shall be formally defined starting from the Syntax Summary of ANSI/MIL-STD-1815A and must have Government approval before it may be used.

30.2.5.1 General. The program functional flow diagram shall illustrate the general system flow of both data and control. If the program is designed to operate in more than one mode, each mode shall be clearly distinguishable in both text and flow diagram.

30.2.5.2 Detailed. The modular flow diagram shall illustrate detailed operations performed by the modules. The flow diagram shall specify all operations performed and shall include all equations used in mathematical computations. Further, logical conditions for entry into each mode shall be specified.

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30.2.6 Source list documentation.

30.2.6.1 Abstracts. Every module shall contain an abstract of comments which describes its inputs, outputs, functions or tasks, algorithms, and lists other modules called. The abstract shall cross-reference the applicable sections of the flow chart. The outline of each abstract shall be as shown in Table XXV.

TABLE XXV. Abstract outline.

I.	MODULE IDENTIFICATION NUMBER (ALPHA OR NUMERIC)
II.	DATA ORIGINALLY WRITTEN AND AUTHOR
III.	REVISION TABLE A. DOCUMENT REVISION NUMBER B. DATE WRITTEN C. PROGRAMMER D. REASON FOR REVISION
IV.	GENERAL DESCRIPTION A. FUNCTION DESCRIPTION OVERVIEW B. MODULE SIZE C. MODULE EXECUTION TIME D. TIMING CONSIDERATIONS E. INTERRUPT CONTROL
V.	SUBPROGRAMS/MODULES CALLED
VI.	SUBPROGRAMS REFERENCE. A complete list of all common data base items with a cross-reference which includes all referencing subprograms/modules.
VII.	INPUT/OUTPUT FORMAT. A brief description of each input and output messages (card format, tape format, record format, etc.) processed by the module.
VIII.	INITIATION CONDITIONS
IX.	PROCESSING DESCRIPTION
X.	RETURN CONDITIONS
XI.	SPECIAL CONSIDERATIONS. A brief description of any transferability, supportability, or testability constraints.
XII.	SYMBOL DEFINITION TABLE A. VARIABLES B. CONSTANTS C. FLAGS D. INDEXES AND POINTERS E. TABLES

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30.2.6.2 Comments. In order to facilitate computer software comprehension, in-line comment statements shall be used throughout the program code. Comment statements are non-executable statements (i.e., those which have no effect on computer operations) and are used provide documentation and clarification of the logic, variables, and algorithms. All listings provided shall be annotated, as a minimum, as follows: one pertinent comment for every three high level program statements, one pertinent comment for each assembly level program statement in the assembly level language comment field, one pertinent comment at the beginning of each new functional activity within each high-level and assembly level module, one pertinent comment for each high-level program transfer statement, and comments for all scaling operations within each high-level and assembly level module.

30.2.6.3 Listing format. When supported by the language used, all listings shall use indentation to indicate different levels of control. Indentation shall be uniform throughout the listings for all delivered software.

30.2.6.4 Data base description. The contractor shall provide in the source listings for each module, detailed descriptions for each variable, constant, flag, index, pointer, and table as applicable. Data names shall correspond to physical meaning or logical use of data whenever possible.

30.2.6.4.1 Variables. The description of each variable shall be presented in the source listing in alphabetical ordering of the mnemonic names of the variables. The minimum content for each variable shall include the variable name, purpose and type, size, binary point for fixed point variables, range of values and initial conditions, static/dynamic, and structure and bit layout.

30.2.6.4.2 Constants. The description of each constant shall be presented the source listing in alphabetical ordering of the mnemonic names of the constants. The minimum content for each constant shall include the constant name, purpose, initial condition, structure and bit layout, and binary point for fixed point constants.

30.2.6.4.3 Flags. The description of each flag shall be presented in the source listing in alphabetical ordering of the mnemonic names of the flags. The minimum content for each flag shall include flag name, purpose, initial condition, and structure and bit layout.

30.2.6.4.4 Indexes and pointers. The description of each index and pointer shall be presented in the source listing in alphabetical ordering of the mnemonic names. The minimum content for each index and pointer shall include name, purpose, initial condition, and range of values.

30.2.6.4.5 Tables. The description of each table shall be presented in alphabetical ordering of the mnemonic names of the table. The minimum content for each table shall include table name, purpose and type, and size and idnexing procedure. Each sub-item (field) making up each table shall further be described to include field name, purpose and type, size, binary point for fixed point data, range of values and initial condition, static/dynamic, and structure and bit layout.

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30.2.7 Software program storage and processing allocation. A description of the allocation of memory storage and processing time for each software program, subprogram, executive routine, module/subroutine or data base shall be provided in the Program Design Specification. In addition, the timing, sequencing requirements, and equipment constraints used in determining the allocations, shall be described.

30.2.8 Identification and marking. Deliverable software items shall be identified and marked in accordance with the following.

30.2.8.1 Software media. Software media (disks, tapes, card decks, etc.) shall be so identified and marked as to be visually and machine readable to provide a direct correlation with those documents which specify and identify their content.

30.2.8.2 Software documentation. All software documentation shall be so identified and marked as to provide direct correlation with controlling documentations and to provide direct correlation with the software media produced in conformance with it.

30.3 Software design. In the building of software, fulfillment of major design objectives shall be the prime consideration.

30.3.1 Software language.

30.3.1.1 Fortran. All software to be hosted on Host Computer Systems as specified in 30.3.13 and 30.3.14 shall be implemented in FORTRAN language and shall be written to be machine compiler level independent, and shall be in accordance with ANSI X3.9-1966 FORTRAN and restrictions set forth herein. Exceptions to these restrictions may be allowed by the government providing evidence of advantages gained. To ensure program reliability, supportability and transferability, it is necessary to place restrictions on the use of certain FORTRAN capabilities. The following restrictions, keyed to the American National Standards Institute (ANSI) FORTRAN Specification X3.9-1966, shall apply to the use of FORTRAN.

30.3.1.1.1 (Paragraph 3.3) statements. A statement shall consist of not more than 10 continuation lines.

30.3.1.1.2 (Paragraph 3.4) statement labels. Leading zeroes shall not be used.

30.3.1.1.3 (Paragraph 3.5) symbolic names. The conventions for names shall be as approved in the AN/AYK-14(V) Standards and Conventions Manual.

30.3.1.1.4 (Paragraph 4.2.2, 4.2.3, 4.2.4) real type, double precision and complex type. These types shall not be used.

30.3.1.1.5 (Paragraph 5.1.1.2, 5.1.1.3, 5.1.1.4) real constant, double precision constant and complex constant. These constant shall not be used.

30.3.1.1.6 (Paragraph 5.1.3) array. No three dimensional arrays shall be used.

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30.3.1.1.7 (Paragraph 6.1) arithmetic expressions. The exponentiation (**) operator shall not be used.

30.3.1.1.8 (Paragraph 6.4) evaluation of expressions. All expressions consisting of three or more elements shall use parenthetical grouping for all but the outer pair of elements.

30.3.1.1.9 (Paragraph 7.1) executable statements. All lines of an executable statement that will be continued must end with an operation.

30.3.1.1.10 Paragraph 7.1.1.3) GO TO assignment statement. The GO TO assignment statement shall not be used.

30.3.1.1.11 (Paragraph 7.1.2.1.1) unconditional GO TO statement. The unconditional GO TO statement may not reference a previous statement, except to implement structured programming constructions. In this case, the GO TO statement, the reference statement, and all statements between shall be indented.

30.3.1.1.12 (Paragraph 7.1.2.1.2) assigned GO TO statement. The assigned GO TO statement shall not be used.

30.3.1.1.13 (Paragraph 7.1.2.1.3) computed GO TO statement. The integer variable in a computer GO TO must be in the proper range, i.e., greater than zero and less than or equal to the number of statements referenced in the compared GO TO statement.

30.3.1.1.14 (Paragraph 7.1.2.5) RETURN statement. Each subroutine or function shall have only one RETURN statement, and the RETURN statement shall be located immediately preceding the subroutine or function END statement.

30.3.1.1.15 (Paragraph 7.1.2.7.1) STOP statement. There shall be only one STOP statement and it shall be in the main program.

30.3.1.1.16 (Paragraph 7.1.2.7.2) PAUSE statement. The PAUSE statement shall not be used.

30.3.1.1.17 (Paragraph 7.1.2.8) DO statement. The terminal statement shall be a CONTINUE statement. Only one DO shall terminate on any given CONTINUE. Nested DO loops shall be physically nested as shown in the following example:

```
      DO 20 I = 1, 10
          DO 10   IJ = 1, 10
              CALL X
10            CONTINUE
20        CONTINUE
```

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30.3.1.1.18 (Paragraph 7.1.2.8.2) extended range DO. The extended range of a DO shall not be used.

30.3.1.1.19 (Paragraph 7.1.3) input/output statements. All input/output to a specific logical unit shall be accomplished within one subroutine, and tha subroutine shall only reference one logical unit. Logical unit numbers shall be explicitly declared integer variables (i.e., INTERFER LU1).

30.3.1.1.20 (Paragraph 7.1.3.2.2) formatted read. All sequential files must contain an end-of-file record of fixed format.

30.3.1.1.21 (Paragraph 7.2.1) specification statements. Within each function or subroutine, the Specification Statements shall be placed in the following order: INTEGER, LOGICAL, COMMON, EQUIVALENCE, EXTERNAL, and DATA. An exception may be made to this to allow type statements for variables in a labeled common block to be placed immediately preceding the COMMON statement.

30.3.1.1.22 (Paragraph 7.2.1.1.2) adjustable dimension. Adjustable dimensions shall not be used.

30.3.1.1.23 (Paragraph 7.2.1.2) DIMENSION statement. The DIMENSION statement shall not be used.

30.3.1.1.24 (Paragraph 7.2.1.3) COMMON statements. The list of variable and array names shall be in alphabetical order. The name or type of variables or arrays must be identical in all subroutines containing a labeled common block. A common block name may occur only once in a subroutine, and no more than one common block name may occur in a single COMMON statement.

30.3.1.1.25 (Paragraph 7.2.1.4) equivalence statement. Arrays appearing in an equivalence statement shall always be subscripted.

30.3.1.1.26 (Paragraph 7.2.1.6) type statements. All variable and array names appearing in a subroutine shall appear in a type statement.

30.3.1.1.27 (Paragraph 8.3.1) defining function subprograms. The dummy arguments shall not be altered during execution of the functions.

30.3.1.1.28 (Paragraph 8.4.1) defining subroutine subprograms. The alteration of dummy arguments shall be minimized and clearly indicated by means of comments.

30.3.1.1.29 (Paragraph 9.1.7) program unit. A program unit shall not contain more than 200 executable statements.

30.3.1.1.30 (Paragraph B3) "Ω" character. The "Ω" character shall not be used to place two statements on one line (e.g., X = 21 Ω Y = 1).

30.3.1.1.31 Hollerith or packed bit. Hollerith or Packed Bit information shall use integer arrays.

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30.3.1.2 CMS-2M/assembly level/micro-code level. Software items that are specified to execute in the AN/AYK-14(V) shall be coded in CMS-2M.

core and execution time constraints and shall be compatible with the basic specification. The procuring activity shall have final approval for the language selected for implementing each on-board AN/AYK-14(V) software item.

30.3.1.3 INTEL 8080 assembly language. Software items that will execute in the Computer Control Unit (CCU) and/or the Memory Loader Verifier (L/V) shall be coded in the INTEL 8080 assembly language.

30.3.2 AN/UYK-20(V) support software. All applicable software items specified herein, excluding those used in the L/V, CCU, and microcode, shall be capable of being generated utilizing the MTASS Support Software as specified in NAVSEA 0967-LP-598-2010, -2020, -2030, -2040, and 2060, User's Handbook, MTASS/M Support Software. All software items generated utilizing the MTASS/M Suport Software shall execute in the AN/AYK-14(V) Computer.

30.3.3 Standard and conventions. Standards and conventions shall be employed to assure timely and cost-effective delivery of high quality software. The contractor shall ensure a commonality of programming standards, conventions, definitions, and techniques. Vocabulary in all documentation shall conform with ANSI X3.12-1970, or be specifically specified in in an acronym or definitions table included as part of the Programming Guidelines section of the Program Design Specification (PDS). The contractor shall provide and conform to standards for documentation and performance of in-process programming which provide management visibility of progress and potential impact areas. All Standards and Conventions to be utilized shall be submitted to the procuring activity for approval.

30.3.4 Modularity. The software shall be modularized to provide a uniform level of management control over all modules in order that design, test, and support problems are minimized. In determining the level of modularity, the prime considerations shall be module function, modular dependencies, high risk critical functions, and orderly utilization of the resulting software modules. Subprograms within the software shall be modular to the extent that modules can be easily added or removed from the software with a minimum of programming and documentation changes. The changes required shall be readily identifiable from the delivered documentation. All software modules which execute in the AN/AYK-14(V) computer shall utilize standard CMS-2M procedure calling conventions.

30.3.4.1 Module size. Higher level language module size shall be limited to no more than 200 single-line, high level executable statements. Assembly level language module size shall be limited to no more than 200 single-line assembly level executable statements. Micro-code programs are excluded from the module size limit; however, consideration shall be given in the design of the micro-code programs to minimize hardware changes required due to changes in micro-code programs. Exception to the above specification may be allowed by the government pending evidence of advantages gained.

30.3.4.2 Modular interface input/output. Each module shall have no more than one entry and one exit point. The contractor shall provide detailed explanation of input and output of each module. This shall include scaling and bit position information. Exceptions to the above specification may be allowed pending evidence of advantages gained.

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30.3.4.3 Module logic structure. The logic structure of each module shall permit software support personnel to easily understand and follow the module's implementation and shall be readily identifiable in the delivered documentation. Conditional branching instructions to implement the control logic shall be minimized to facilitate software support. The destinations of a branching instruction shall be forward of and as close as possible to the branching instruction. Micro-code programs are excluded from the above branching restriction. Exceptions to the above specification may be allowed pending evidence of advantages gained.

30.3.5 Transferability. All software items written in FORTRAN shall be transferable to any Host Computer specified in Table XXVI. Any program changes which must be made to effect the transfer must be isolated in specific modules, clearly documented, and approved by the procuring activity. The transfer from one Host Computer to any other Host Computer specified in Table XXVI shall not affect or degrade the software item's performance, with the exception of execution time as a result of different Host Computer throughput.

30.3.6 Supportability. The software items shall be built so that future support in the form of updates may be accomplished by the government. The operational independence afforded by good supportability will result from three factors:

a. Properly developed design that minimizes the complexity and interdependence of program units;

b. Preservation of the integrity of that design through coding and implementation; and

c. Providing Government control of all software required to operate, simulate, and support the computer programs.

This includes the delivery of a complete set of documentation required for the operation, maintenance, and modification of the software items specified herein.

30.3.7 Testability. The AN/AYK-14 support software items shall be built so that testing can be accomplished at the module, function, program, and system level. The software shall be verified, validated, and certified. This shall be achieved through comprehensive testing and evaluation against properly defined and established performance criteria. The testing and evaluation effort shall ensure that the probability of detecting an error during testing is maximized; so that the probability of experiencing an error during the program's operational use is minimized. A program package and documentation shall be provided for testing each deliverable item of software specified herein. All software required in the testing of the software items specified herein shall be documented and delivered to the Government.

30.3.8 Relocatability. All software items included herein shall be built in the form of relocatable object modules except as approved by the procuring activity. The resultant modules shall be capable of being linked to form absolute object programs, and of being relocated in memory and onto direct access devices. Micro-code programs are excluded from the relocatability requirement.

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TABLE XXVI. Alternative host computer system.

Host Machine	Host Operating System
UNIVAC 1100 Series	UNIVAC 1100 TIME SHARING EXEC EXEC 8 (level 28 or later)
IBM 360 Series IBM 370 Series	IBM Operating System SYSTEM/360 O.S. R21.8 OS/VS2 (SVS) R1.7 OS/VS2 (MVS) R3.7
CDC 6000 Series CYBER 70 Series CYBER 170 Series	Control Data Operating System KRONOS 2.1 NOS 1.3 (excluding NOS/BE 1.0)
DEC-10 KA 10 Series DEC-10 KI 10 Series DEC-10 KL 10 Series	TOPS-10 Operating System (7.01)
UNIVAC UYK-7	SHARE/7 Operating System (level 4 or later)
Honeywell 600 Series Honeywell 6000 Series Honeywell 66 Series	Honeywell General Comprehensive Operating Supervisor GCOS Level F or higher
IBM 370 Series	IBM Virtual Machine Conversational Monitor System VM/370 - CMS Version 03
DEC VAX-11/780	VAX/VMS Operating System (Version 2.4)
CDC 6000 Series CYBER 70 Series CYBER 170 Series	TRW Operating System TRW TIMESHARE SYSTEM (TRW/TSS)
CDC 6000 Series CYBER 70 Series CYBER 170 Series	Control Data Operating System SCOPE 3.4 NOS/BE 1.0
DECSYSTEM-20 Series	TOPS-20 Operating System (Monitor 4)

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30.3.9 Program size and timing. The requirements for allocation of memory shall be minimal for the software items. Dynamic module replacement or overlay techniques shall be employed wherever feasible to minimize resident memory requirements consistent with the performance, core, and timing requirements specified herein.

30.3.10 Human factors. Design criteria and principles of MIL-H-46855 shall be applied in the design of the software items. Particular attention shall be given to the man/machine interface function.

30.3.11 Safety. Design criteria and principles of MIL-STD-882 shall be applied in the design of the software items. Particular attention shall be given to the man/machine interface function.

30.3.12 Peripheral device adaptability. The design of input/output routines for all support software items shall remain device independent to the lowest possible level to enable peripheral devices and channel assignments to be changed with a minimum of programming and documentation changes.

30.3.13 Host computer. The prime Host Computer System shall be as directed by the procuring activity. This computer system shall be utilized and shall host all AN/AYK-14(V) support software items written in FORTRAN. All software items written in FORTRAN shall meet the requirements of 30.3.1.1.

30.3.14 Alternative host computer system. At the direction of the procuring activity, and when specified in the contract, the contractor shall host all AN/AYK-14(V) support software items written in FORTRAN on any of the alternate Host Computer Systems listed in Table XXVI at Government facilities. When the Alternate Host Computer Systems are specified, the contractor shall provide to the Government all software and documentation changes that are required for hosting the software specified herein on each of the alternative systems. For each of the Alternative Host Computer Systems, these changes should be independent, as much as possible, of different versions of the host's operating system. Acceptance by the Government of software hosted by the contractor on these computers is specified in 40.3.1.2.

30.3.15 FORTRAN coded software items. The AN/AYK-14(V) Support Software items to be coded in FORTRAN, as specified in 30.3.1.1 are:

a. AN/AYK-14(V) Micro-Code Cross-Assembler

b. AN/AYK-14(V) Micro-Code Simulator

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30.3.16 CMS-2M/assembly level/micro-code level coded software items. The AN/AYK-14(V) Support Software Items to be coded in CMS-2M, AN/AYK-14(V) Assembly Level, or AN/AYK-14(V) Micro-Code Level, as specified in 30.3.1.2 are:

a. AN/AYK-14(V) Micro-Code Programs.

b. AN/AYK-14(V) Bootstrap Loader Software.

c. AN/AYK-14(V) Maintenance and Diagnostic Software, excluding the Computer Control Unit Software and Loader/Verifier Unit Software.

d. AN/AYK-14(V) Software Test Programs.

30.3.17 INTEL 8080 assembly language coded software items. The AN/AYK-14(V) Support Software Item to be coded in the INTEL 8080 Assembly Level, as specified in 30.3.1.3 are:

a. Memory Loader Verifier Unit Software

b. Computer Control Unit

30.4 Detailed functional requirements. The following functional requirements specify the operation of the microcode assembler and microcode simulator.

30.4.1 AN/AYK-14(V) micro-code programs. Micro-Code Programs shall be developed for the Central Processing Unit (CPU) and Input-Output Processor (IOP), in response to the requirements of the basic specification.

30.4.2 AN/AYK-14(V) micro-code cross-assembler. The Micro-Code Cross-Assembler shall be coded in FORTRAN in accordance with the specifications and restrictions set forth in Appendix A. As a minimum, the Micro-Code Cross-Assembler shall provide the capabilities for:

a. Accepting micro-code instructions from source card images and producing program listings and absolute object code for the CPU and IOP onto magnetic tape, and disk. This absolute object code shall be executable by the CPU or IOP, as appropriate.

b. Assembling an average of 250 executable statements per minute from a source tape to an output magnetic tape.

c. Distinguishing and processing statement labels of one to eight characters in length.

d. The micro-code cross-assembler shall execute in no more than 35K decimal words of main memory on the CDC 6600 and no more than 43K decimal words on the Honeywell DPS-66. It shall be capable of assembling a micro-code program of at least 4,096 micro-code words.

e. Adapting to programming changes and documentation required for assembler to be updated for growth and specialized user-defined micro-code instructions.

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f. Allowing operands to consist of variables, fields, expressions, literals or constants.

g. Allowing programmer declaration of and use of decimal, octal, and/or hexadecimal notation for constants.

h. Allowing forward referencing of labels.

i. Allowing including or skipping over prespecified groups of statements based on parameter tests at assembly time.

j. Allowing inclusion of comments, both in and between instruction lines.

k. Enabling the user to change the Field Programmable Logic Arrays (FPLA) definition in the micro-code cross-assembler so that its outputted micro-code will properly interface with the reprogrammed FPLAs.

30.4.2.1 Directive capabilities. Directive capabilities shall have, as a minimum:

a. ALUREG. The ALUREG pseudo-op shall specify the symbol in the location field to have the value of the expression in the address field. The symbol shall then be accepted in the ALU expression in a command statement.

b. CFILE. The CFILE pseudo-op shall specify the symbol in the location field to have the value of the expression in the address field. The symbol shall then be accepted as a C file address in the symbol field of a command statement or the address field of pseudo-op.

c. CONST. The CONST pseudo-op shall specify the symbol in the location field to have the value of the expression in the address field. The symbol shall then be accepted in the V field of a command statement or the address field of a pseudo-op.

d. EJECT. The EJECT pseudo-op shall cause the listing to begin at top of form.

e. END. The END pseudo-op shall specify the end of the input for one assembly.

f. ENDIF. The ENDIF pseudo-op shall cause the assembler to unconditionally assemble the statements following until an IFN or IFZ pseudo-op is encountered.

g. EQU. The EQU pseudo-op shall specify the symbol in the location field to have the value of the expression in the address field. The symbol shall then be accepted in the symbol field of a command or in the address field of a pseudo-op.

h. FILE. The FILE pseudo-op shall specify the symbol in the location field to have the value of the expression in the address field. The symbol shall then be accepted as a file address in the symbol field of a command statement or in the address field of a pseudo-op.

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i. FINIS. The FINIS pseudo-op shall specify the end of all input for all assemblies.

j. IOP. The IOP pseudo-op shall specify the target machine to be the IOP. The absence of this pseudo-op shall specify the target machine to be the GPM.

k. IDENT. The IDENT pseudo-op shall name the assembly as the symbol in the address field. The assembler shall put this name on the deck identification card of the object module.

l. IFN. The IFN pseudo-op shall cause the assembler to not assemble the statements following it until an ENDIF pseudo-op if the address field evaluates to zero. If the address field evaluates to a nonzero value the statement shall be treated as a comment.

m. IFZ. The IFZ pseudo-op shall cause the assembler to not assemble the statements following it until an ENDIF pseudo-op if the address field evaluates to a nonzero value. If the address field evaluates to zero, the statement shall be treated as a comment.

n. NOLIST. The NOLIST pseudo-op shall suppress the assembly listing, including the summary information.

o. NOMAP. The NOMAP pseudo-op shall suppress the listing of the micro-memory map.

p. NOTABL. The NOTABL pseudo-op shall suppress the listing of all tables.

q. ORG. The ORG pseudo-op shall cause the statements following it to be assembled beginning at the location specified by the expression on the ORG statement until an ORGR, ORGS, or another ORG pseudo-op is encountered. If the previous ORG/ORGR/ORGS is an ORGS, then the current location counter is put into the auxiliary counter.

r. ORGR. The ORGR pseudo-op shall cause the leftmost four characters of the symbol in the address field to be used in the micro-memory map. If the previous ORG/ORGR/ORGS is an ORGS, then the current location counter is put into the auxiliary counter.

s. ORGS. The ORGS pseudo-op shall cause the statements following it to be assembled in the auxiliary micro-memory address region until an ORG or ORGR pseudo-op is encountered.

t. SET. The SET pseudo-op shall specify the auxiliary micro-memory address region to begin at the location specified by the expression in the address field.

u. SPACE. The SPACE pseudo-op shall cause the number of lines specified by the expression in the address field to be skipped on the listing.

v. SUBTTL. The SUBTTL pseudo-op shall use the 52 characters in columns 21 through 72 as the second line on the listing until replaced by a

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new SUBTTL statement. The SUBTTL statement shall also have the effect of an EJECT. If the SUBTTL pseudo-op immediately follows an EJECT, only one page eject shall occur.

w. TITLE. The TITLE pseudo-op shall use the 52 characters in columns 21 through 72 as the first line on the listing until replaced by a new TITLE statement. The TITLE statement shall also have the effect of an EJECT.

Exception to the above list of directives may be allowed by the Government pending evidence of advantages gained.

30.4.2.2 Inputs. The assembler shall have the capability of accepting variable-length input from source cards, paper tape, magnetic tape, disk, or from terminals.

30.4.2.3 Processing. The assembler shall receive the appropriate inputs as specified herein and produce the outputs specified herein. The Cross-Assembler shall continue to process even when errors are encountered and provide pertinent error notification.

30.4.2.4 Outputs. The Cross-Assembler shall have an output of program listing and an absolute code on magnetic tape, paper tape, disk, and terminal outputs. The following shall be provided:

a. A side-by-side listing, containing for each statement, error conditions (fatal, warning, and caution as a minimum), error type classifications, location counter, absolute code, the input statement and the input card number. In addition, the listing shall contain listing page numbers, the number of errors in this assembly, an alphabetized cross-reference listing, and a title and subtitle for each page.

b. Print an output listing.

c. Print Symbol Table.

d. Produce an output file, containing all information needed to produce an output listing and all information needed to produce the input to a standard process for the manufacturing process of PROMs/ROMs, and input to the Micro-Code Simulator, and ROM Simulator as described in the contract.

30.4.3 AN/AYK-14(V) micro-code simulator. The Micro-Code Simulator provides an independent host capability to enable a user to test the micro-code program assembled for the CPU and IOP of the AN/AYK-14(V). The simulator fully and completely simulates the micro-code instruction repertoire, interrupts and FPLA operations; and shall accept and execute assembled AN/AYK-14(V) macro-code except for the EAU trigonometric instructions. I/O timing and instruction timing are available by counters. The Micro-Code Simulator is coded in FORTRAN in accordance with the specifications and restrictions of 30.3.1.1. The micro-code executes in no more than 44K decimal words of main memory on the CDC 6600 and no more than 56K decimal words on the Honeywell DPS-66.

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30.4.3.1 Inputs.

a. Absolute formatted code generated by the AN/AYK-14(V) Micro-Code Asembler and resident on disk, magnetic tape, cards, or paper tape.

b. Absolute formatted AN/AYK-14(V) macro-code and resident on disk, magnetic tape, cards, or paper tape.

c. Debug capability as specified in 30.4.3.4.

d. Operating inputs for control of the Simulator shall be as a minimum:

(1) Address Breakpoints

(2) Micro-code instruction Breakpoints

(3) Debug capability

(4) Utility capability

30.4.3.2 Processing.

a. Simulate micro-instruction execution for AN/AYK-14(V) Production Unit in both timing and results.

b. Load absolute formatted code for AN/AYK-14(V) programs (Production Unit).

c. Allow Debug capability as specified in 30.4.3.4 to be employed by users.

d. Process operator inputs for control of Simulator for test purposes.

30.4.3.3 Outputs.

a. As provided by the Host Computer and the Debug capability as specified in 30.4.3.4.

b. Responses to operator inputs and Simulator results.

30.4.3.4 Debug capability. Provide the following modular debug capability to the operator/programmer to inspect, change, move, and dump selected memory areas and registers on printer/terminal/magnetic tape/disk during static and dynamic conditions of machine execution. Input and output devices and character codes of input/output data are selected at the discretion of the operator. Input and output formats shall be selectable by the operator to be decimal, signed decimal, octal or hexadecimal, in either instruction-field format or straight numerical format. Conversions on variables from binary/hexadecimal to decimal/signed decimal and floating point to decimal shall be included as additional operator option.

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30.4.3.4.1 Inspect and change. Capability shall be provided to inspect and change the contents of any memory location or registers.

30.4.3.4.2 Masked memory search. Capability shall be provided to search memory for a specified value as specified by a user-specified mask. All addresses and contents that are found may be displayed or printed.

30.4.3.4.3 Snapshot. Capability shall be provided to print/display on printer/magnetic tape/terminal the contents of registers and selected memory areas during execution of application programs.

30.4.3.4.4 Store constant. Capability shall be provided to allow a user to prestore a specified area of memory with zeroes or to change one memory cell at a time.

30.4.3.4.5 Dump. Provide capability for a user to list the contents of consecutive memory locations between user specified limits on disk/printer/ magnetic tape/terminal without zero-word suppress.

30.4.3.4.6 Trace. Provide an instruction trace capability to print/ display and time-tag program counter status/registers/effective memory/ addresses and contents thereof or record them on magnetic tape/terminal/disk every time an instruction execution occurs during execution of an application program within program counter limits specified by a user. Also provide a selectable additional capability for the same output whenever a branch instruction occurs, or whenever specific memory location references occur.

30.4.3.4.7 Move. Provide a capability to move the contents of an area of memory to another area.

30.4.3.4.8 Memory list. List off-line a memory dump that is on magnetic tape/disk onto printer/terminal with and without zero-word suppress at user's option.

30.4.4 AN/AYK-14(V) loader software. All AN/AYK-14(V) software, as specified herein and for future AN/AYK-14(V) application programs, generated in accordance with the requirements specified in 30.3.2 shall utilize the MTASS/M Loader/System Generator for the creation of AN/AYK-14(V) absolute load files.

30.4.4.1 Loader modules.

30.4.4.1.1 Bootstrap channel loader module. The Bootstrap Channel Loader Module shall be implemented as a hardware boot and shall meet the requirements of 3.6.2.1.1.12. The Bootstrap Channel Loader shall be initiated by the boot enable discrete input when the system is configured as in Figure 62D.

30.4.4.1.2 Support equipment interface (maintenance connector) loader modules. Capability shall be included in the software and firmware functions provided by 30.4.1 and 30.4.7 to load software into the AN/AYK-14 memory as shown in Figures 62A, 62B and 62C. The AN/AYK-14 interface will be as specified in 3.6.2.1.1.6.

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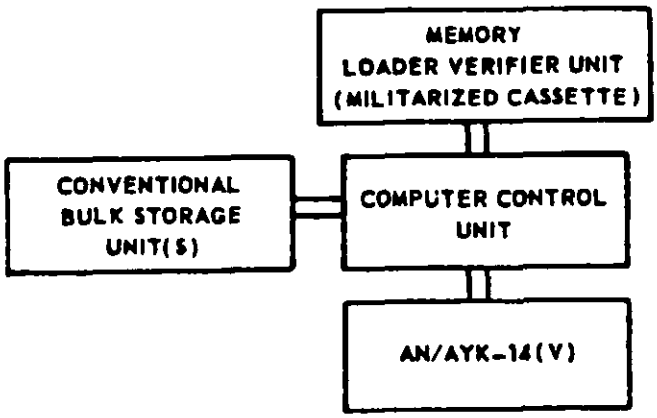


FIGURE A

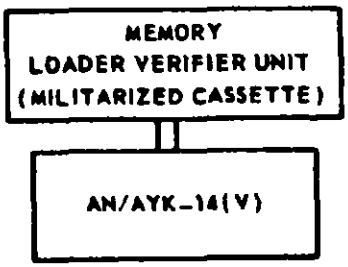


FIGURE B

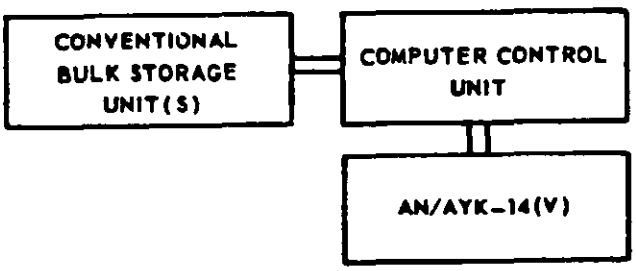


FIGURE C

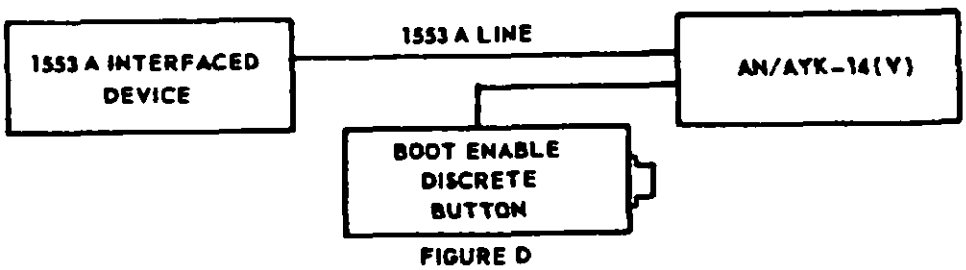


FIGURE D

NOTE: BOOT ENABLE DISCRETE BUTTON WILL BE BUILT INTO AIRCRAFT CONTROL PANEL (COCKPIT).

FIGURE 62. AN/AYK-14 system configurations.

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30.4.5 AN/AYK-14 maintenance and diagnostic software. The maintenance and diagnostic software, as a minimum, shall contain an In-Flight-Performance-Monitoring (IFPM) package and a Module Shop Replaceable Assembly (SRA) Isolation Diagnostic package, as specified herein, plus micro-code diagnostic software (see 30.4.5.1). All maintenance and diagnostic software shall conform with and be compatible with the hardware specified in the basic specification. The maintenance and diagnostic software shall perform self-test processing for real-time and off-line fault detection with zero or minimal operator intervention and shall provide informative notification of equipment status and fault location(s), as appropriate, to the operator. All maintenance and diagnostic software shall be coded in CMS-2M, Assembly language, micro-code, or combination thereof and shall be capable of operating within any AN/AYK-14(V) configuration. Measurement for rating effectivity of maintenance and diagnostic software shall be made by recognizing bona fide faults induced in specifically defined configurations using specifically defined support equipment.

30.4.5.1 AN/AYK-14(V) in-flight performance monitoring (IFPM) diagnostics. The In-Flight Performance Monitoring (IFPM) Diagnostics shall be provided to supplement the continuous hardware built-in-test equipment (BITE) to meet the equipment failure detection requirements shown in Table XXVII. Micro-code diagnostic software of not greater than 200 words of micro-memory, may be utilized. An IFPM module shall be provided to perform a CPU/Memory QUICK-LOOK Test. The CPU/Memory QUICK-LOOK Test shall be intended to be included in the user's hardware fault warning interrupt handler. The QUICK-LOOK Test shall perform a test of the interface between the CPU subsystem and the memory subsystem. The user's interrupt handler shall reset the BIT timer if no equipment faults are detected. The QUICK-LOOK Test shall be comprised of four separate modules - memory parity checker test, memory protect checker test, C-File test and page address register test. The page address registers to be tested shall be user selected. Additional IFPM modules shall be provided, as appropriate, to meet the equipment failure detection requirements. These modules shall provide In-Flight Performance failure detection of the Central Processing Unit (CPU), the Input/Output Processor (IOP), the Input/Output Subsystem, and the Memory Subsystem. The maximum sizes and execution times for the IFPM modules shall be as specified in Table XXVIII. These modules shall be designed to permit AN/AYK-14(V) users to build an IFPM program which is tailored to the user's AN/AYK-14(V) computer configuration for integration into an application program. These modules shall be implemented utilizing CMS-2M calling procedures. These modules shall require zero operator intervention and, given successful completion of the tests, shall return control to the calling application software program. Additionally, a 32-bit IFPM memory status word shall be provided which shall be used to record AN/AYK-14(V) functional status. The 32-bit IFPM memory status word shall be defined such that bit positions are utilized to store IFPM functional failure detection data. Failure of a given IFPM module test shall result in the appropriate IFPM memory status word bit(s) being set and an appropriate AN/AYK-14(V) interrupt being generated.

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TABLE XXVII. Failure detection requirements by configuration.

System	Minimum Valid Failure Detection	Nominal Words of Memory Used
CP-1424/AYK-14(V)	94%	2750
CP-1426/AYK-14(V) and CP-1428/AYK-14(V) (combination)	92%	4000
CP-1425/AYK-14(V)	93%	3100
CP-1427/AYK-14(V)	93%	2330
CP-1430A/AYK-14(V)	93%	4000
CP-1539/AYK-14(V)	93%	2930

TABLE XXVIII. IFPM module sizes and execution times.

IFPM Test Module	Maximum Words of Memory	Maximum Execution Time
IFPM Common	700	Included in times of other IFPM test modules
CPU/Memory QUICK-LOOK	350	3.8 ms (Note 1)
CPU (Including EAU)	1500 (Note 2)	54.0 ms
IOP	600	5.0 ms
Memory Subsystem	250	30.0 ms per 1024 words of memory tested
Input/Output Subsystem modules (unless otherwise specified)	300 each	6.0 ms each
DIOM	450	14.0 ms
PIM	325	6.5 ms
PPSM	525	300.0 ms
RIM	300	150.0 ms (Note 3)
SIM-A	300	9.0 ms

Notes:
1. With no page registers selected.
2. Including 300 words for EAU test. The EAU test shall be a user selectable option to the CPU test.
3. Due to the serial transfer rate of 9600 baud.

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For AN/AYK-14(V) configurations containing more than one of a given Input/Output module SRA type channel, duplicate Input/Output Subsystem module tests shall not be required. A Non-Destructive Read Out (NDRO) memory checksum test shall be provided in the memory test. Certain IFPM functions are provided for in a Common Module. The IFPM Common Module shall consist of:

a. Subroutines required for two or more of the IFPM module tests.

b. Flags and data areas including 64 words of scratch, required by the IFPM module tests.

The IFPM modules shall execute without interruption of other functions and shall be a subset of the Module SRA Isolation Diagnostics.

30.4.5.2 AN/AYK-14(V) module SRA isolation diagnostic. Diagnostic programs shall be provided that are modular and configurable for all possible AN/AYK-14(V) computer configurations. The diagnostic programs shall be capable of operating within a 64K word main memory AN/AYK-14(V) configuration. They shall detect and isolate hardware failures by means of such tests as:

a. Exercising all instructions, utilizing all general purpose and special purpose registers, and exercising all address modifications.

b. Performing memory tests by exercising address control logic and all memory locations. Tests such as the generation of all ones, all zeroes, checkerboard (125252), reverse checkerboard (052525), double checkerboard (146314), reverse double checkerboard (031463), as appropriate to the AN/AYK-14(V) memory, should be included. There shall be only one memory test module. This test module shall be able to perform tests on all configurations of memory in the computer.

c. Performing input/output and interrupt testing to verify data transfer and timing logic, utilizing I/O wraparound cables.

These diagnostics shall be provided to load, input, execute, output, and display with a stand-alone configuration of AN/AYK-14(V) Computer using the Memory Loader Verifier Unit (or functionally equivalent support equipment (Computer Control Unit)). Notification of GO/NO-GO decisions shall be reported for operator action. The SRA diagnostic programs shall operate in the AN/AYK-14(V) and shall be capable of automatic isolation of detected failure to a single faulty SRA 86% of the time and automatic isolation to two SRAs, one of which is faulty, 93% of the time excluding (a) a failed SRA that cannot be specifically identified due to ambiguities associated with BUS structure architecture (Stuck Bus) and (b) a case where a failed WRA Mother Board exists, provided that exclusions (a) and (b) do not consist of more than 7 percent of all predicted failures. SRA diagnostics shall be capable of being executed both with and without external wraparound cables attached. The isolation rating for the SRA diagnostics shall only be made when wraparound cables are present. The isolation provided by firmware and the L/V shall be included in rating the isolation ability of the SRA diagnostic software. Manual or semiautomatic procedures will be allowed for stuck bus isolation providing the procedure does not require SRA substitution (except for PSM, GPM, IOP, PCM and chassis) and can be isolated to a single SRA 86 percent of the time and two SRAs 93 percent of the time. A failed SRA condition which

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cannot be automatically isolated due to ambiguities associated with the bus structure architecture shall be identified and the equipment repaired within the Mean Time to Repair (MTTR) as specified in 3.4.7.5. The automatic fault isolation time shall be compatible with the MTTR as specified in 3.4.7.5, but in no event shall be greater than five minutes for SRA fault isolation as specified above. Maximum sizes and execution times of the SRAID modules shall be as specified in Table XXVIX.

TABLE XXVIX. SRA isolation diagnostic module sizes and execution times.

SRA Isolation Diagnostic Module	Maximum Words of Memory	Maximum Execution Time
Executive (including register configuring, and table lookup portions)	5700	1.0 second
All processor modules (e.g., CPU, IOP, etc.)	1700 each	10.0 seconds each
CPU with EAU	1900	10.0 seconds
Memory	3000	3.0 seconds per 32K words of memory tested
I/O and BEM modules (unless otherwise specified)	1500 each	0.1 second each
SIM-A	2000	0.1 second
DIOM	1900	0.1 second
PPSM	2000	0.1 second

30.4.6 AN/AYK-14(V) software test programs. The Software Test Programs shall be developed for the micro-code cross-assembler and the micro-code simulator to provide complete functional testing.

30.4.6.1 AN/AYK-14(V) micro-code cross-assembler test program. The Micro-Code Cross-Assembler Test Program shall provide a complete functional test of the assembler. This test program shall be written in a form suitable for input to the micro-code cross-assembler specified in 30.4.2. The test program shall completely exercise all functions of the assembler as specifid in 30.4.2. The results upon assembling the test program with the micro-code cross-assembler shall provide correlation between the generated code and predetermined correct results.

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30.4.6.2 AN/AYK-14(V) micro-code simulator test program. The Micro-Code Simulator Test Program shall provide a complete functional test of the simulator. The simulator test program shall be written in a form suitable for input to the micro-code simulator specified in 30.4.3. The simulator test program shall be executed with the micro-code simulator and shall, with proper operator's inputs, verify that the simulator is functionally correct.

30.4.7 AN/AYK-14(V) memory loader verifier software and equivalent support (computer control unit) software. The software modules specified below shall be implemented to meet the applicable requirements of NAC TP-579.

30.4.7.1 Memory loader verifier software. The Memory Loader Verifier Software shall allow the user of the Memory Loader Verifier (L/V) to select, load, verify, and execute programs stored in a loadable format on the L/V militarized cassette when the system is configured as in Figures 62A or 62D. The L/V Software program shall be developed and shall enable the L/V to meet the requirements of the support equipment interface of TP-579. The L/V Software shall enable the L/V to perform the L/V modes and AN/AYK-14(V) modes of operation specified in NAC TP-579. The L/V control program shall further permit the building of the L/V tape cassettes from programs/data contained in the AN/AYK-14(V) computer and on the external bulk storage unit of the Computer Control Unit (CCU). The L/V mode control software programs shall be modular and shall be hardware bootable into the L/V from the L/V militarized cassette. The hardware boot programs and other essential operational requirements shall be implemented as PROM hardware. The design of the L/V Software shall be such that new L/V functions can be added with a minimum of change and testing. It shall be possible, during program generation, to include or exclude any of the L/V modes of operation without affecting the operation of other included modes. The human factors requirements of 30.3.10 shall be met in the design of the man/machine interface. The format and content of data being provided to or accepted from the hexadecimal keyboard shall require approval of the procuring activity prior to implementation.

30.4.7.1.1 Inputs. The L/V Software shall accept inputs from the L/V control/indicator panel functional switches and the hexadecimal keyboard, the L/V tape cassette, the Computer Control Unit (CCU), and the AN/AYK-14(V) computer.

30.4.7.1.2 Processing. The L/V Software shall process input data as set forth in 30.4.7.1.1 and shall meet the requirements set forth in TP-579.

30.4.7.1.3 Outputs. The L/V Software shall output data and/or information to the L/V control/indicator panel hexadecimal keyboard display, the L/V BIT indicator, the CCU, the AN/AYK-14(V) computer, and the L/V tape cassette.

30.4.7.2 Computer control unit software. The Computer Control Software shall allow the user of the Computer Control Unit (CCU) to select, load, verify, and execute programs stored in a loadable format on any one of the CCUs conventional bulk storage units when the system is configured as in Figure 62A and as in Figure 62C. The CCU software shall be capable of transferring absolute load formatted files from a conventional bulk storage unit attached to the CCU to the militarized cassette unit attached to the

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Loader/Verifier. The CCU software shall also transfer load formatted files from the militarized cassette unit via the Loader/Verifier to any conventional bulk storage unit attached to the CCU. The format of the information on the militarized cassette produced by the CCU software shall be such that in the configuration shown by Figure 62B, the L/V can load AN/AYK-14(V) memory with the contents of a load file on the militarized cassette.

40. QUALITY ASSURANCE PROVISIONS

40.1 Responsibility for inspection. Unless otherwise specified, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to participate in and/or perform any of the tests/inspections set forth in this appendix where such participation/performance is deemed necessary to ensure that supplies and services conform to prescribed requirements.

40.2 Government verification. All quality assurance operations performed by the contractor shall be subject to Government verification at any time. Verification shall consist of the following functions.

40.2.1 Surveillance of operations. All operations shall be closely observed/monitored to determine that practices, methods, and procedures are being properly implemented and adhered to.

40.2.2 Government product inspection. Government (in-process) inspections shall be performed to ensure the quality of the product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product with prescribed criteria has been demonstrated.

40.3 Classification of inspections/tests. Items covered by this appendix shall be subjected to the following inspections to determine compliance with all applicable requirements:

a. Quality Conformance Tests

b. Special Tests

40.3.1 Quality conformance tests. Items covered by this appendix shall be subjected to the following Quality Conformance Tests:

a. Preliminary Qualification Testing (PQT)

b. Formal Qualification Testing (FQT)

40.3.1.1 Preliminary qualification testing. The contractor shall furnish all supporting software and shall be responsible for accomplishing the Preliminary Qualification Testing (PQT). All inspection and testing may be witnessed by a designated Government representative. The Government inspector and the procuring activity shall be advised, 30 days prior to start of testing, when tests are to be conducted so that a Government representative

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may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities for conducting all required tests shall engage the service(s) of a commerical testing laboratory acceptable to the Government. The contractor shall furnish test reports showing quantitative results for all Preliminary Qualification Testing. Such reports shall be signed by an authorized representative of the developer or laboratory, as applicable. Acceptance or approval of the testng during the course of software production shall not be construed as a guarantee of the acceptance of the finished product. Preliminary Qualification Testing shall consist of the following:

a. Algorithm Evaluation and Testing

b. Module Tests

c. Subprogram Tests

d. Program Performance Tests

e. Systems Integration Tests

40.3.1.1.1 Algorithm evaluation and test. Algorithm Evaluation and Testing shall be conducted by the contractor to support the design and development of the software at the discretion of the procuring activity.

40.3.1.1.1.1 Algorithm evaluation and test data. The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

40.3.1.1.1.2 Scope of tests. The criteria for determining the scope of tests shall be based upon design requirements and the need to demonstrate validity of a design concept prior to committing that concept to a design approach.

40.3.1.1.2 Module tests. Each module shall be subjected to the module tests in accordance with 5.8.1 of DoD-STD-1679. These tests shall be adequate to determine compliance with the applicable Program Design Specification (PDS). As a minimum, the contractor shall verify that each subprogram/module accepted has completed the following:

a. The requirements of 5.8.1 of DoD-STD-1679.

b. Verification of the coded subprogram/module to ensure that it actually reflects the Program Design Specification (PDS) requirements.

40.3.1.1.3 Subprogram tests. All modules submitted for subprogram tests shall have passed the module tests. The modules shall be integrated individually until all modules have been integrated into programs. These tests shall be in accordance with 5.8.2 of DoD-STD-1679 and shall be adequate to determine compliance with the applicable Program Performance Specification (PPS).

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40.3.1.1.4 Program performance tests. All subprograms shall have passed the subprogram tests prior to program performance testing. The subprograms shall be integrated individually until all subprograms have been integrated into the programs. The program performance tests shall be in accordance with 5.8.3 of DoD-STD-1679.

40.3.1.1.5 Systems integration tests. All subsystem program submitted for system integration tests shall have passed the program performance tests. The subsystem programs shall be integrated individually until all subsystem programs have been integrated into the system program. These tests shall be adequate to determine compliance with the applicable Program Performance Specifications. As a minimum, testing shall be performed to:

a. Verify the total man-machine, operating system, and file system interfaces.

b. Validate system initiation, data entries via peripheral devices, program loading, restarting, and the monitoring and controlling of system operation from display consoles and other control stations as applicable.

c. Verify system integration of equipment and subsystems.

40.3.1.2 Formal qualification testing. Formal Qualification Testing (FQT) shall certify that all requirements of each computer program configuration item have been met. FQT of the AN/AYK-14(V) Support Software Packages shall be conducted at the contractor facility using an AN/AYK-14(V) Laboratory Facilities and the Contractor's Host Computer System. The right shall be reserved by the Procuring Activity to require the FQT to be run at a Government facility, in lieu of contractor facilities, using an AN/AYK-14(V) Laboratory Facilities and Host Computer System. Tests for acceptance on Alternative Host Computer Systems shall be the smae or a subset of those tests conducted on the Prime Host Computer System. All programs submitted to be tested shall have passed Preliminary Qualification Testing. The FQT shall consist of the following tests:

a. System/Program Performance/Acceptance Test for AN/AYK-14(V) Host Support Software. The Host Support Software consists of the Micro-Code Cross-Assembler (see 30.4.2); and Micro-Code Simulator (see 30.4.3).

b. System/Program Performance/Acceptance Test for AN/AYK-14(V) On-Board Support Software. The On-Board Support Software consists of the 1553A Boot Loader Software (see 30.4.4.1.1); Maintenance and Diagnostic Software (see 30.4.5); and Micro-Code Programs (see 30.4.1).

c. AN/AYK-14(V) Benchmark Programs Test (see 40.3.1.2.3).

40.3.1.2.1 System/program performance/acceptance test for AN/AYK-14(V) host support software. All programs shall be compiled and/or assembled on the Host Computer System from the source tapes submitted as part of the Program Package (PP). An error-free source/object listing and object tape must be generated. Additional tests shall be performed to demonstrate that the object code that is generated for each Support Software program is identical, in content and format, to the object listing that is produced. Object Code shall also be inspected to insure that it agrees with the listing submitted with the

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PP. The programs shall be loaded and executed in accordance with approved tests procedures to verify that each program meets the performance requirements specified in Program Performance Specification (PPS). The contractor shall furnish all necessary supporting software required to conduct the tests. As a minimum, the tests shall verify that the programs properly process the specified inputs to produce the specified outputs and that the object modules are compatible with the AN/AYK-14(V) machine operation and other Support Software programs that are a part of this specification. A test to show that the output of the Micro-Code Assembler performs and is processed by the Micro-Code Simulator and the ROM Simulator shall be included in the testing.

40.3.1.2.2 System/program performance/acceptance test for AN/AYK-14(V) on-board support software. All programs submitted shall have passed Preliminary Qualification Testing. All programs shall be assembled using the MTASS/M support software from the source tapes submitted as part of the Program Package (PP). An error-free source/object listing and object tape must be generated. Additional tests shall be performed to demonstrate that the object code that is generated for each Support Software program is identical, in content and format, to the object listing that is produced. Object code shall also be inspected to insure that it agrees with the listing submitted with the PP. The programs shall be loaded and executed in the AN/AYK-14(V) in accordance with approved test procedure to verify that each program meets the performance requirements specified in the Program Performance Specification (PPS). For the micro-code programs of 30.4.1, the micro-code shall combine with the computer hardware to verify overall computer operation in accordance with this appendix. The contractor shall furnish all necessary supporting software required to conduct the tests.

40.3.1.2.3 AN/AYK-14(V) benchmark programs test. The AN/AYK-14(V) Benchmark Programs Test shall consist of assembling benchmark programs using the MTASS/M Support Software and the AN/AYK-14(V) Support Software and executing them on the AN/AYK-14(V) Laboratory Facilities. The benchmark programs will be provided by the procuring activity and will be comprised of supplementary test programs and representative application programs. An error-free source/object listing and object tape must be produced for a properly structured program with correct syntax. The results of Government simulated runs will be compared and tabulated with the results of the actual runs on the AN/AYK-14(V) by Government personnel; identical results must be obtained from a functional, performance, and timing standpoint prior to final acceptance. A set of benchmark programs will be written in AN/AYK-14(V) Assembly language and AN/AYK-14(V) micro-programming language.

40.3.1.2.4 Acceptance. The successful completion of the FQT shall provide for formal qualification and acceptance of the AN/AYK-14(V) Support Software by the procuring activity.

40.3.2 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design change on the performance of the software and to assure adequate quality control. The software selected for special tests may be selected from software previously subjected to the preliminary qualification testing or formal qualification testing.

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40.3.2.1 Special test schedule. Selection of software for special tests shall be made as follows:

a. On software after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

40.3.2.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the PQT or FQT shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

40.4 Test procedures. The procedures used for conducting preliminary qualification tests, formal qualification tests, and any special tests shall be prepared by the developer and submitted to the procuring activity for review and approval. The test procedures shall be approved by the contractor's quality assurance group and not the software implementors. The right shall be reserved by the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this appendix or the contract.

40.5 Test failure. Should a failure occur during either the PQT, FQT or Special Tests, the following actions shall be taken:

a. Notify the Government inspector immediately.

b. Determine whether the failure is a hardware failure or a software defect.

c. If the failure was a hardware failure:

(1) Submit to the procuring activity for approval, proposed hardware corrective action intended to reduce the possibility of the same hardware failure occuring in future tests.

(2) Where practical, include a test in the individual test to check for this hardware requirement until reasonable assurance is obtained that the hardware failure has been satisfactorily corrected.

d. If the failure was a software defect:

(1) Submit to the procuring activity, proposed software corrective action intended to correct the software defect. The submittal shall specify all information pertinent to the defect. As a minimum, the following shall be addressed:

(a) Software unit affected

(b) Description of defect

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(c) Time and location of occurrence

(d) Purpose of computer run in which defect was detected

(e) Qualification test and test procedure used

(f) Applicable console switch settings and registers, including exact hardware and software configuration.

(g) Person who detected defect

(h) Supporting computer program inputs and outputs to substantiate problem (i.e., documentation, memory dumps, description of inputs/outputs, etc.)

(i) Proposed software corrective action

(2) The procuring activity shall review such proposals and determine whether they are acceptable.

40.6 Test reports. The contractor shall provide test reports in accordance with the requirements of DoD-STD-1679 and all additional supporting test data as may be required by the procuring activity.

40.7 Presubmission testing. No module, program or complete software system shall be submitted to the procuring activity by the contractor until it has been previously tested, inspected, and certified by the developer and found to comply to the best of his knowledge and belief with all applicable requirements contain herein. With the consent of the contractor and at the discretion of the Goverment, this prior testing and inspection may be participated in or witnessed by the Government Inspector or procuring activity with the object of eliminating the necessity of repeating such test and inspection after the software has been formally submitted to the procuring activity.

40.8 Rejection and retest. Software which has been rejected and reworked to correct the defects must be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be provided to the procuring activity.

40.9 Physical inspection. The procuring activity may employ a physical inspection to determine the contractor's conformance with contractual requirements. As a minimum, areas of interest include development facilities, documentation controls, deliverable data items, Government imposed standards, and contractor's internal standards.

40.9.1 Facilities. The development and test facilities may be nspected for contractual conformity at any time during the life of the contract.

40.9.2 Products. The procuring activity will utilize inspection techniques in the acceptance of all deliverable products. Duplicate computer software deliverable media such as magnetic tapes, disks, cards, etc., will be produced from a master which has been accepted by the procuring activity.

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Quality assurance will be performed on each duplicate and will include as a minimum, compare and verify.

40.9.3 Filing system. The procuring activity may, at any time, inspect the contractor's filing and library system for completeness, traceability, security, and configuration management requirements conformance.

40.9.4 Configuration management. The contractor's conformance with the approved Configuration Management Plan may be audited through examination of records and Government attendance at contractor meetings on configuration control.

40.9.5 Internal standards. The procuring activity may audit the contractor's conformance with internal standards of software development and control.

50. PACKAGING

50.1 General. All major units and components of the software shall be preserved, packaged, packed and marked for the level of shipment specified in the contractor or purchase order.

60. NOTES

60.1 Intended use. The software delivered in accordance with this appendix is intended for use by both Government and industrial organizations in support and operation of the AN/AYK-14(V) Computer.

60.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. AN/AYK-14(V) Support Software Documentation (See 30.2.1)

c. AN/AYK-14(V) Support Software Media (See 30.2.2)

d. Identification and Marking (See 30.2.8)

e. Alternative Host Computers (See 30.3.14)

f. Responsibility for Inspection (See 40.1)

g. AN/AYK-14(V) Benchmark Program Test (See 40.3.1.2.3)

h. Test Procedures (See 40.4)

i. Test Reports (See 40.6)

j. Configuration Management Plan (See 40.9.4)

k. Selection of Applicable Levels of Packaging and Packing (See 50.1)

60.3 Government furnished equipment. The contracting officer should arrange to furnish, as required, the property listed below:

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60.3.1 Hardware. AN/UYK-20(V) Hardware Configuration as negotiated with AN/AYK-14(V) contractor.

60.3.2 Software. MTASS/M Support Software as negotiated with AN/AYK-14(V) contractor.

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b. ADDRESS *(Street, City, State, ZIP Code)*	

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